

PROGRAM/REMOTE FUNCTION MANUAL
FOR
AQ6319 OPTICAL SPECTRUM ANALYZER

Yokogawa Electric Corporation

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This manual has indicated the program function and remote control function which are one function of the AQ6319 Optical Spectrum Analyzer. Please read, in case you perform program function and remote control function.

Moreover, in case you use this unit, be sure to read AQ6319 Optical Spectrum Analyzer USER'S MANUAL in front of this manual.



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1.1 General

The program functions are a feature allowing the user to pre-register the measuring conditions, measuring procedures, etc. to run a measurement program.

Using these functions, the user can independently execute automatic measurements on the AQ6319, eliminating the need for using an external computer.

Also, the use of an input command for user parameters, provided with these functions, allows the user to enter parameters such as measuring conditions while running a program. Moreover, using an output command in relation to the screen, the user can output the analysis results of waveform and others to the OUTPUT WINDOW during execution of a program.

However, this function lets the user control an external device connected to the GP-IB2 port on the rear of this unit, a variety of applications can be coped with.

The size of programs that can be created is 200 steps per program, with a maximum capacity of 64 programs.

Created programs are stored in the internal HDD. It is also possible to store them on a floppy disk.

1.2 Editing a Program Name

To use the program functions, a program must be pre-registered in the AQ6319.

To create a program, first enter the name of the program.

To enter the program name, use the following keys:

- ① Press the [PROGRAM] switch.
- ② Press the <PROGRAM EDIT> key.

This causes a program list as shown in Figure 1-1 to appear.

- This program list displays 32 program names on the screen.
- The US list indicates "*" if a program has already been registered in the corresponding program number.
- The EXECUTE NO. list shows the registered key numbers for programs that have been registered to the <EXECUTE 1> to <EXECUTE 21> keys.

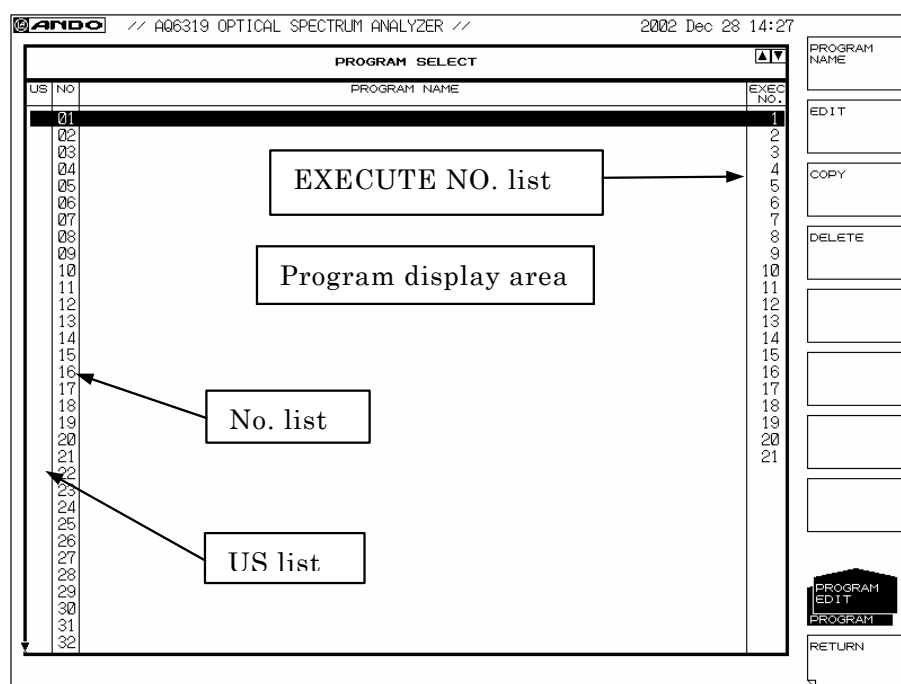


Figure 1-1: Screen Displayed when Pressing the <PROGRAM EDIT> Key

1.2.1 Entering and Modifying a Program Name

To enter or modify a program name, proceed as follows:

- ① On the screen as shown in Figure 1-1, press the <PROGRAM NAME> key. This causes the Label Input screen to appear in the same manner as inputting a label.
Enter the program name and press the <DONE> key to accept it.
The entered program name is simultaneously reflected in the program display area.

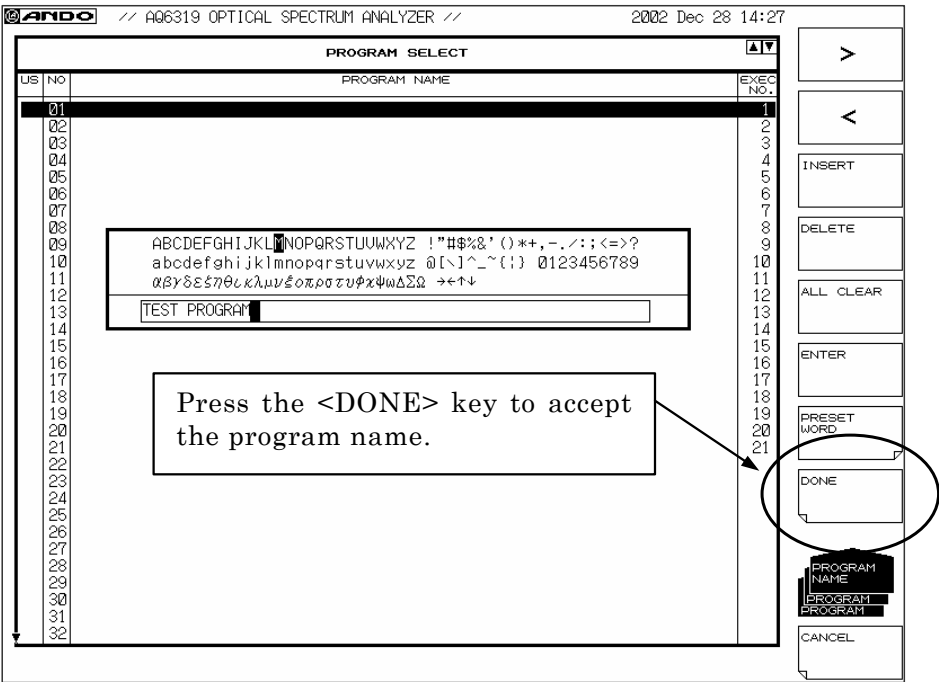


Figure 1-2: Screen for Entering a Program Name

NOTE

- To execute a registered program via GP-IB, specify a program number. This lets the program run.
- When you enter a program name to a program registered in any of the <EXECUTE 1*> to <EXECUTE 21> keys (soft keys displayed when the [PROGRAM] switch is pressed), the program name (of 11 characters) also appears on the relevant one of these soft keys.

1.3 Editing a Program

Select the program you wish to enter or modify the program name from the program list and press the <EDIT> key. This will move you to the PROGRAM EDIT screen.

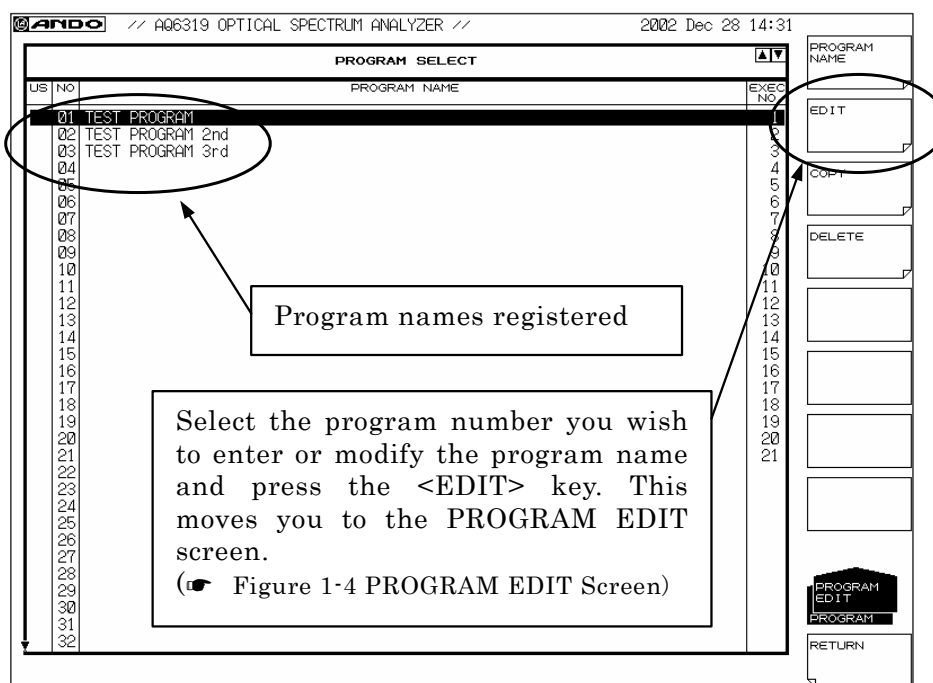


Figure 1-3: PROGRAM SELECT Screen

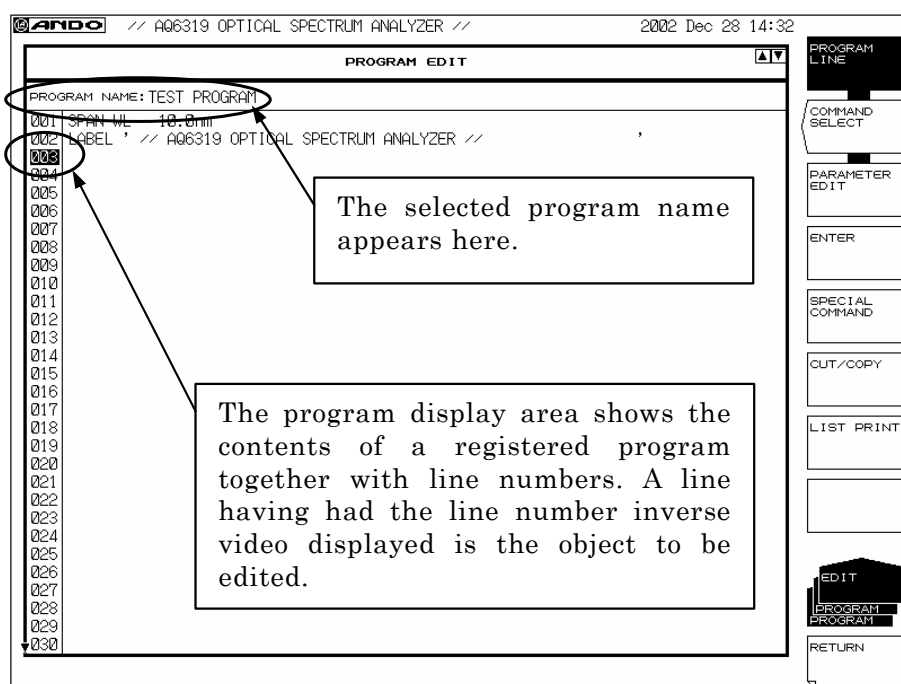


Figure 1-4: PROGRAM EDIT Screen

1.3.1 Entering a Program

There are broadly two types of commands that can be registered in the program area: the function commands and special commands.

- **Function Commands**

These commands execute the same function as a function switch (including the contents of a soft key).

(Commands corresponding to the soft keys such as <SINGLE> and)

- **Special Commands**

These commands include jump commands, program control commands for conditional decision, etc., control commands to an external device, and data output commands.

- **How to Input a Function Command**

① On the screen as shown in Figure 1-4, press a panel switch. This causes a list of commands corresponding to the pressed switch to appear in the right pane of the screen.

② Use the rotary knob or step keys to select a command to register and press the <ENTER> key.

The selected command will be registered in the program area.

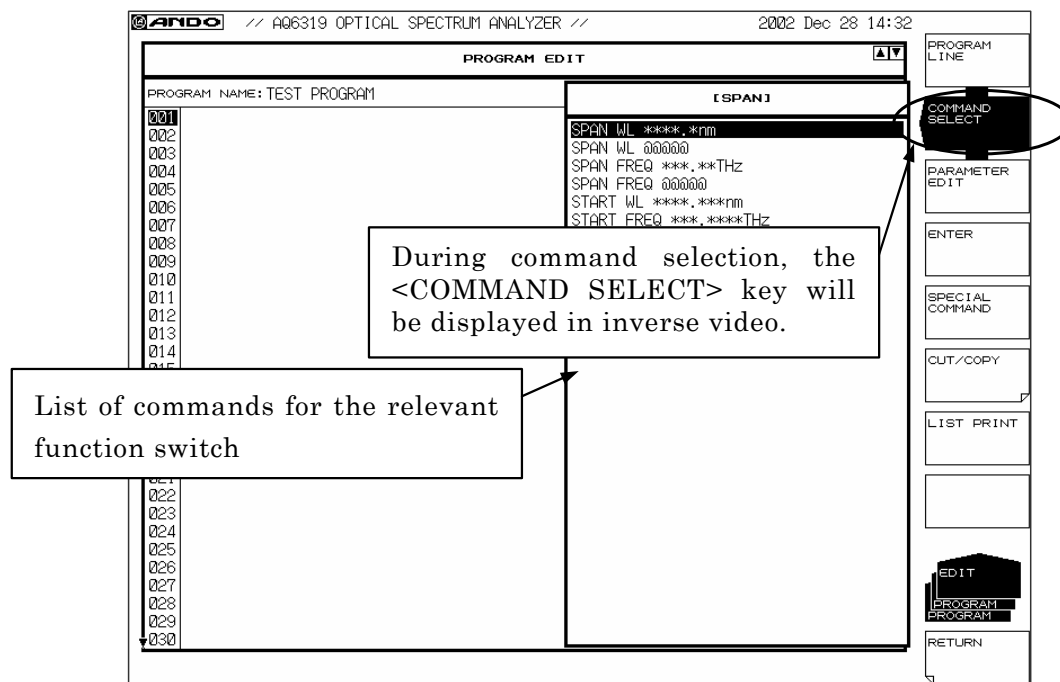


Figure 1-5: Screen Displaying a List of Function Commands

NOTE

- A function command can also be input using the mouse. Right clicking the mouse causes a list of hard keys to appear in the same way as mouse operations on the Waveform screen. Clicking on and selecting a hard key causes a list of the function commands corresponding to that hard key to appear.

- How to Input a Special Command
 - ① On the screen as shown in Figure 1-4, press the <SPECIAL COMMAND> key.

This causes a list of the special commands to appear on the screen (Figure 1-6).
 - ② Use the rotary knob or step keys to select the command to register and press the <ENTER> key. The selected command will be registered in the program area.

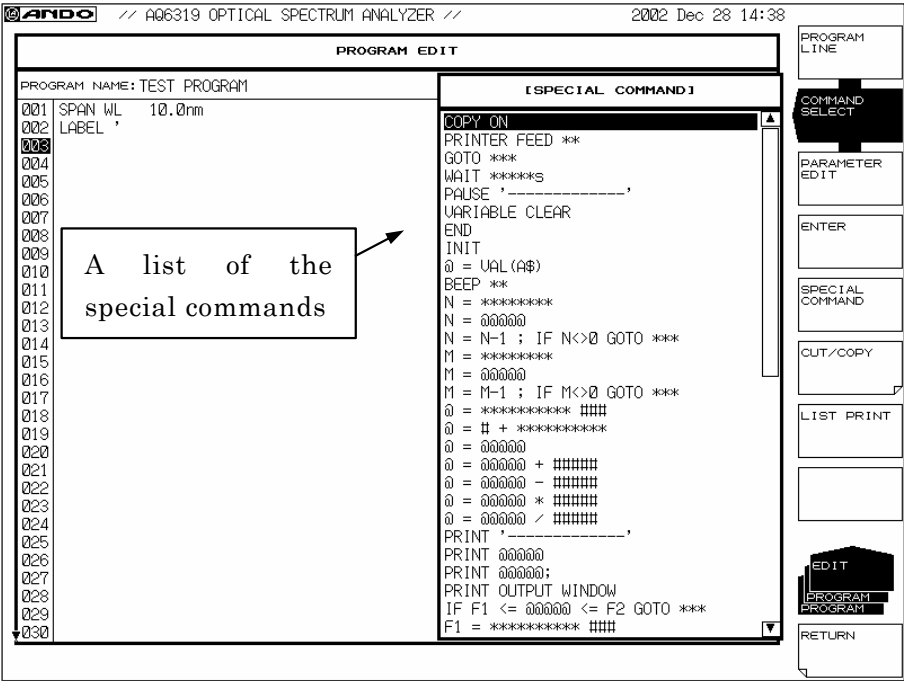


Figure 1-6: Screen Displaying a List of Special Commands

1.3.2 Setting a Parameter Command's Parameter

When a command with a parameter is entered in program input, the current set value or initial value defined for each command will be input. In this case, the parameter part of the command will be inverse video displayed.

Use the rotary knob or the numeric keys to modify the parameter and press the <ENTER> key to accept it.

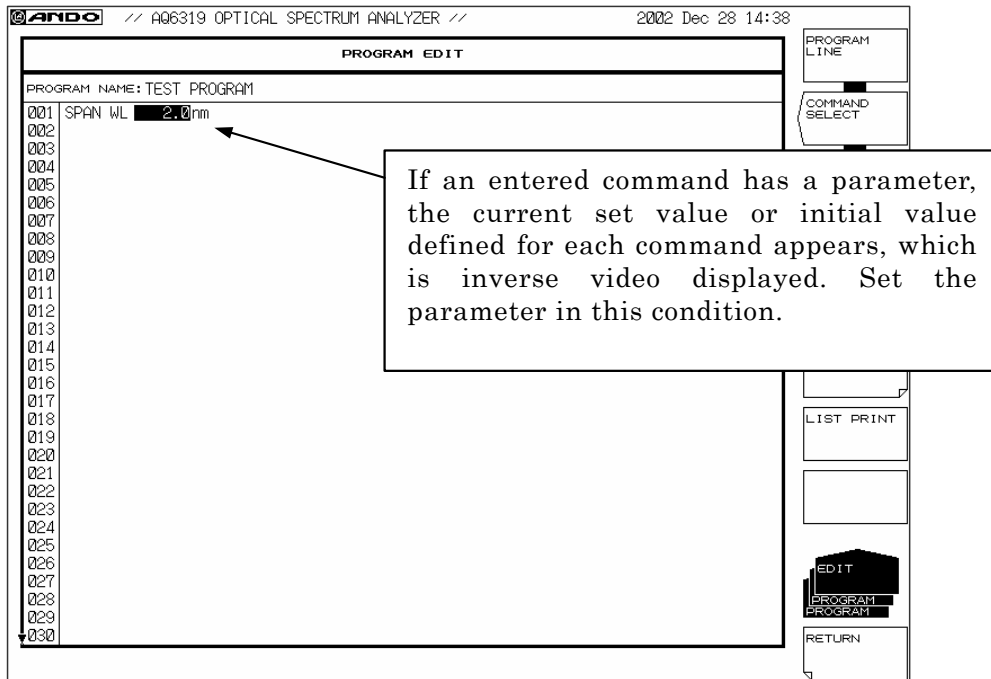


Figure 1-7: Entering a Command with a Parameter

1.3.3 Modifying the Parameter Part of a Registered Command

You can modify only the parameter part of a registered command in the program display area. The modifying procedure is as follows:

- ① On the PROGRAM EDIT screen as shown in Figure 1-4, use the rotary knob or step keys to select the line number of the command you wish to modify from among the already input commands with parameters.
- ② Press the <PARAMETER EDIT> key.
This causes the parameter part of the command to be inverse video displayed, allowing you to modify the parameter.

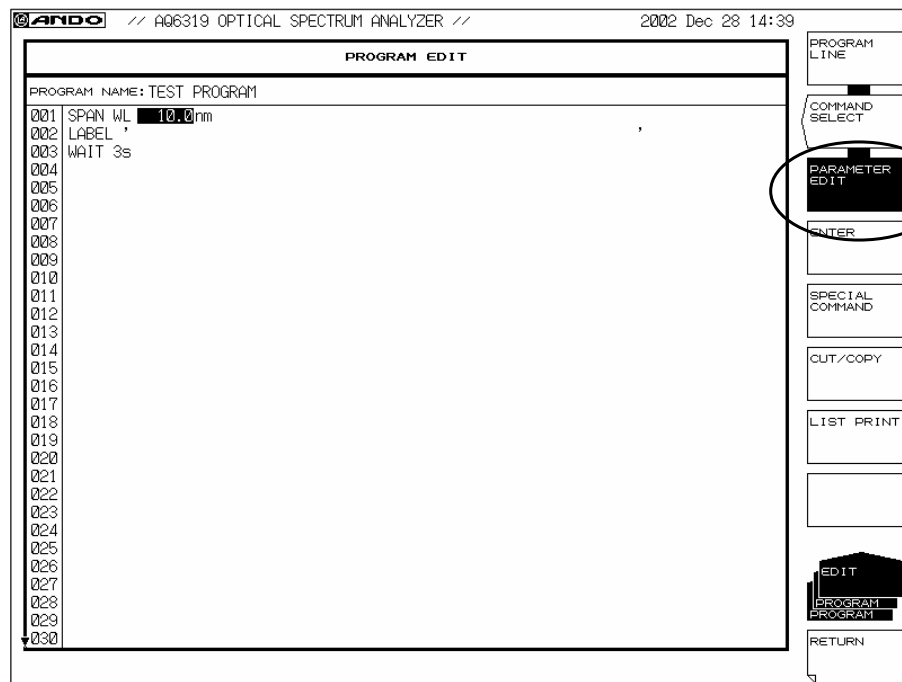


Figure 1-8: Modifying the Set Value of a Command with a Parameter

- ③ In the same way as for program input, use the rotary knob or step keys to change the parameter and press the <ENTER> key to accept it.

1.3.4 Setting a Character Input Command

When you select a command requiring character input (label input command or file-related function command), the character input window and the menu of the corresponding soft keys will be displayed in the same way as the label input. Enter a file name using the same procedure as that of label input and press the <DONE> key.

(☛ Section 6.21: General Purpose LABEL INPUT, in the Instruction Manual)

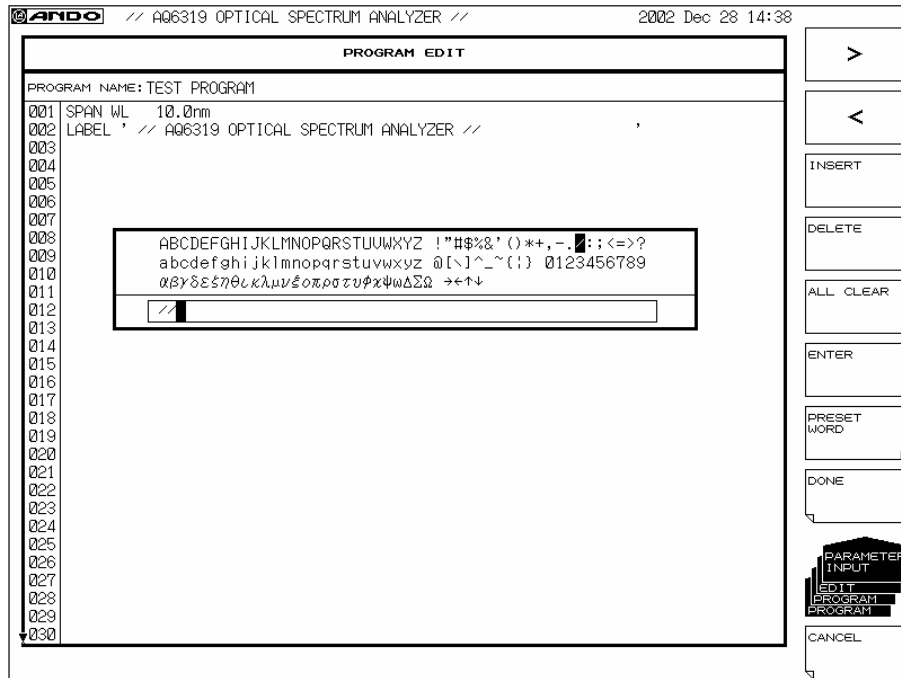


Figure 1-9: Setting a Character Input Command

1.3.5 Inserting, Deleting, or Copying a Line

You can copy, delete, or insert line(s) within the program area in the PROGRAM EDIT screen.

The following describes each operating procedure:

- Inserting a Line
 - ① On the PROGRAM EDIT screen as shown in Figure 1-4, press the <CUT/COPY> key. This will move you to the screen shown in Figure 1-10.
 - This screen allows edits such as copying or deleting a line.
 - ② Move the cursor to the location where you wish to insert a line and press the <LINE INSERT> key. This causes a blank line to be inserted into the position where the cursor is.

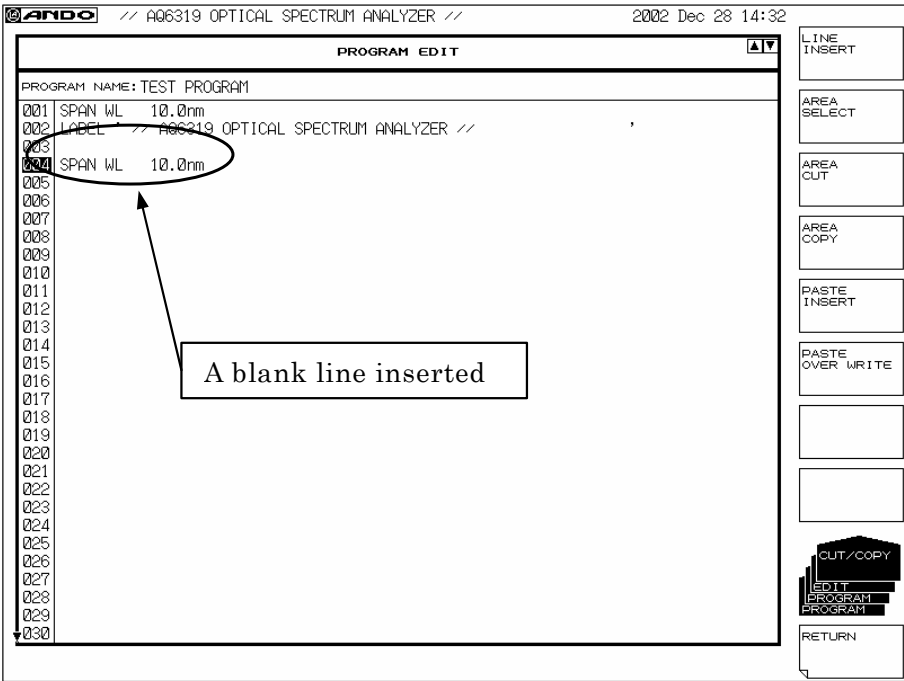


Figure1-10: Screen for Inserting a Line

CAUTION

- If programs have been registered in all 200 lines, no line can be inserted.

● Deleting line(s)

There are two methods to delete a line(s): deleting only one line and deleting a specified area of lines.

✓ Deleting a single line

- ① On the PROGRAM EDIT screen as shown in Figure 1-4, press the <CUT/COPY> key. This will move you to the screen shown in Figure 1-10.
- ② Move the cursor to the location where you wish to delete a line and press the <AREA CUT> key. This deletes the line inverse-video displayed and causes the successive programs to be shifted up by one line each.

✓ Deleting a specified area

- ① On the PROGRAM EDIT screen as shown in Figure 1-4, press the <CUT/COPY> key. This will move you to the screen shown in Figure 1-10.
- ② Use the rotary knob or step keys to select and display the line number of the starting point for the area specification in inverse video and press the <AREA SELECT> key.
- ③ Use the rotary knob or step keys to specify an area up to the line number of the ending point for the area specification to display the area in inverse video.
- ④ Press the <AREA CUT> key. This deletes the area specified in step ③ entirely and causes the remaining programs to be shifted up to the first line of the setting range.

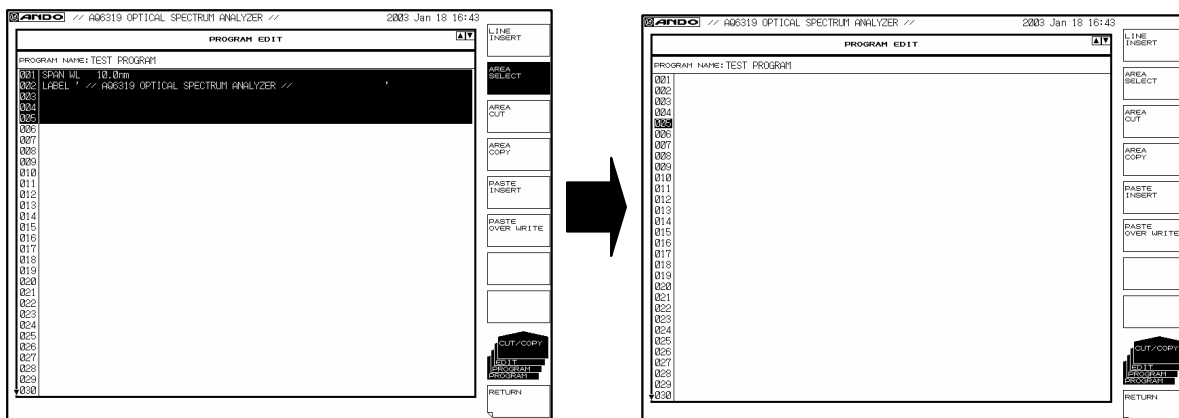


Figure 1-11: Deleting a Specified Area of Lines

● Copying line(s)

There are two methods to copy line(s): inserting a copied line to another position or overwriting the existing line of that position with the line being copied.

✓ To insert a copied line to another position:

- ① On the PROGRAM EDIT screen as shown in Figure 1-4, press the <CUT/COPY> key. This will move you to the screen shown in Figure 1-10.
- ② Use the rotary knob or step keys to select and display the line number of a line to copy in inverse video and press the <AREA COPY> key.
- ③ Use the rotary knob or step keys to move the inverse video display to a line where you wish to insert the copied line and press the <PASTE INSERT> key.

This causes the contents of the line you wish to copy to be inserted into that line, causing the program contents at and after the inverse video displayed line to be shifted down by one line each.

✓ To overwrite another location with the contents of a copied line:

- ① On the PROGRAM EDIT screen as shown in Figure 1-4, press the <CUT/COPY> key. This will move you to the screen shown in Figure 1-10.
- ② Use the rotary knob or step keys to select and display the line number of a line to copy in inverse video and press the <AREA COPY> key.
- ③ Use the rotary knob or step keys to move the inverse video display to a line where you wish to overwrite the copied line and press the <PASTE OVER WRITE> key.

This overwrites the noted line with the contents of the copied line.

NOTE

- A line copy can be accomplished by specifying an area of lines in the same way as line deletion and then inserting or overwriting their contents into or over another position.
Before pressing the <AREA COPY> key, use the <AREA SELECT> key to specify an area to be copied.

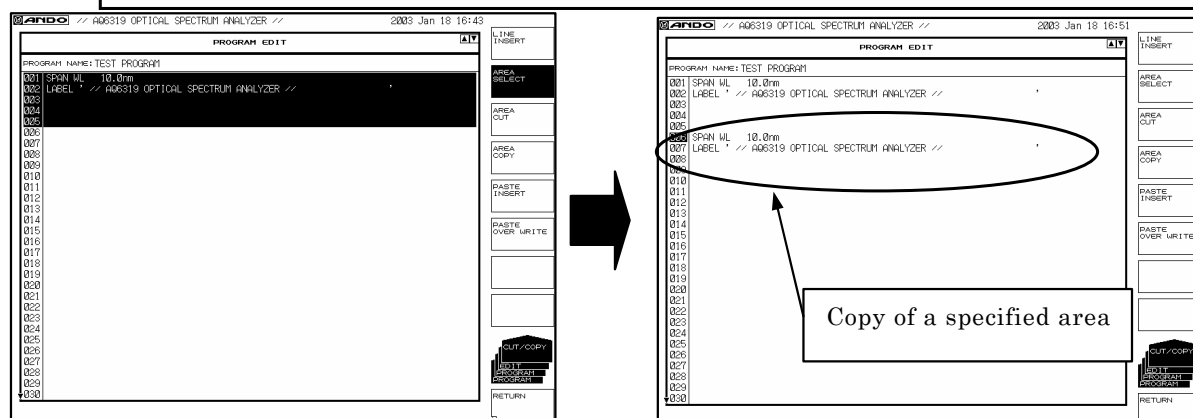


Figure 1-12: Copying a Specified Area of Lines

1.3.6 Printing out a Program List

Pressing the <LIST PRINT> key on the PROGRAM EDIT screen shown in Figure 1-4 causes a list of the first to the final lines of the program being edited to be output to the internal printer.

1.3.7 Copying or Deleting a Program

On the screen listing programs (Figure 1-1), you can delete a registered program or copy it to another program number.

● Copying a Program

The procedure for copying a registered program to another program number is as follows:

- ① Press the [PROGRAM] switch.
- ② Press the <PROGRAM EDIT> key.
- ③ Move the cursor to the program number of the program you wish to copy and press the <COPY> key.
This causes the program number list at the copy source to be underlined.
- ④ Select the program number of the copy destination and press the <OVER WRITE EXECUTE> key or the <MERGE EXECUTE> key.
This causes the contents of the program at the copy source to be copied to the line of the selected program number.
To cancel a copy, press the <CANCEL> key.

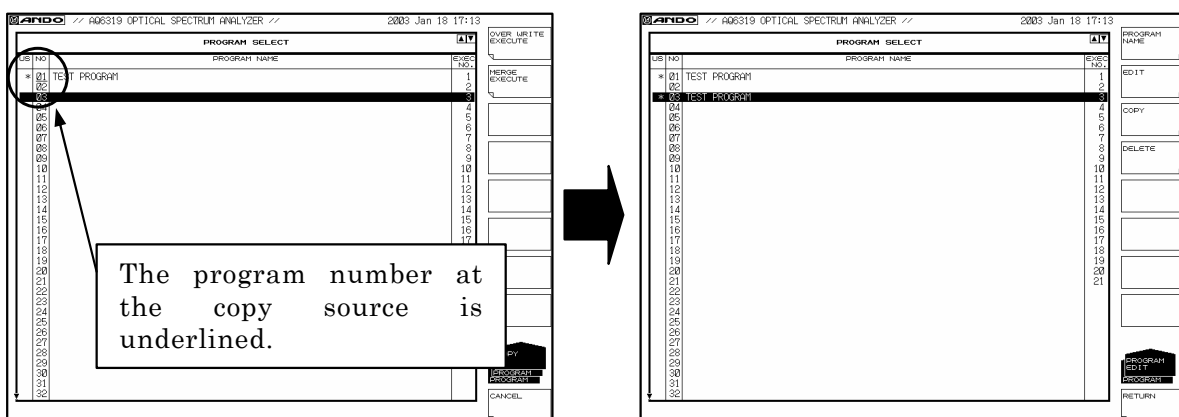


Figure 1-13: Copying a Program

If a program has already been registered in the program number at the copy destination, you can either overwrite the copy destination program with the copy source program or add the copy source program to a location following the registered program.

✓ To overwrite:

Press the <OVER WRITE EXECUTE> key.

This overwrites the contents at the program number selected in program copy step ④ with the contents of the program whose number has been selected in step ③.

✓ To add the program to a location following the registered program:

Press the <MERGE EXECUTE> key.

This causes the contents of the program whose number has been selected in program copy step ③ to be added to a location following the program at the number selected in step ④.

● Deleting a Program

The procedure of deleting a registered program is as follows:

① Press the [PROGRAM] switch.

② Press the <PROGRAM EDIT> key.

③ Move the cursor to the number of a program you wish to delete and press the <DELETE> key.

When you press this key, the confirmation message appears whether to perform deletion or not.

(☛ Figure 1-14: Screen Prompting Whether or not to Delete a Program)

To execute a deletion, press the <YES> key; to cancel a deletion, press the <NO> key.

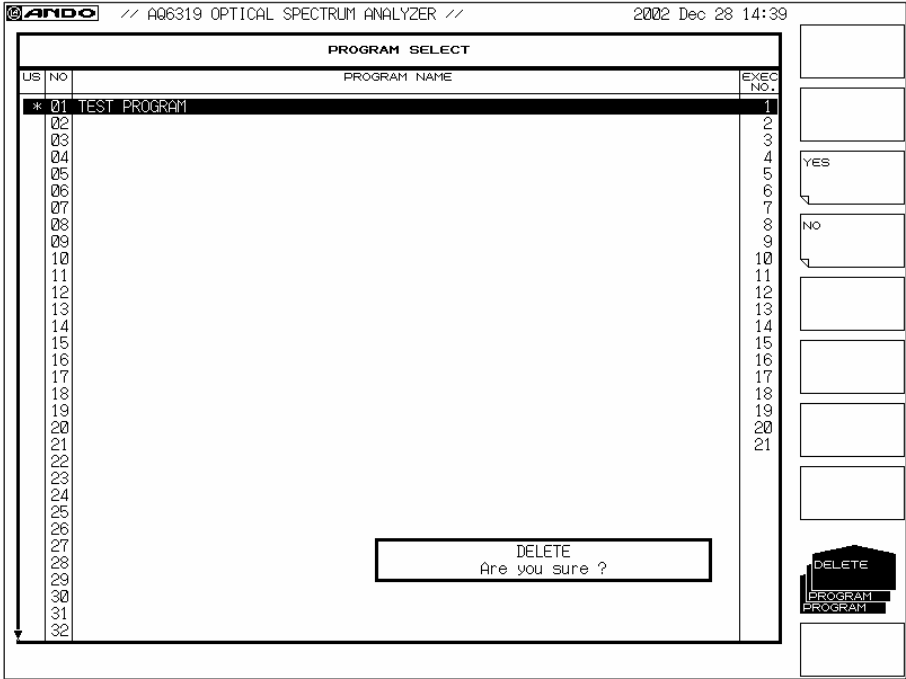


Figure 1-14: Screen Prompting Whether or not to Delete a Program

1.3.8 Registering Programs to the <EXECUTE1> to <EXECUTE21> Keys

You can register a created program to the <EXECUTE1> to <EXECUTE21> keys.

When created programs have been registered to these keys, simply pressing an <EXECUTE@> key allows a target program to be executed easily.

The Program Functions allow a maximum of 21 types of programs to be registered to the <EXECUTE@> keys.

When shipped from the factory, program numbers 01 to 21 have been respectively assigned to the <EXECUTE 1> to <EXECUTE 21> keys.

The procedure for registering a program to an <EXECUTE@@> key is as follows:

- ① Press the [PROGRAM] switch
- ② Press the <PROGRAM EXECUTE> key.
- ③ Move the cursor to the number of the program you wish to register to an <EXECUTE @@> key and then press the <EXECUTE KEY DEFINE> key. This causes an interrupt display area to appear. Use the rotary knob, step keys, or numeric keys to enter the key number you wish to register.
- ④ Press the <ENTER> key.

This causes the key number entered in step ③ to be set to the location of the cursor-indicated program number in the EXECUTE NO. list.

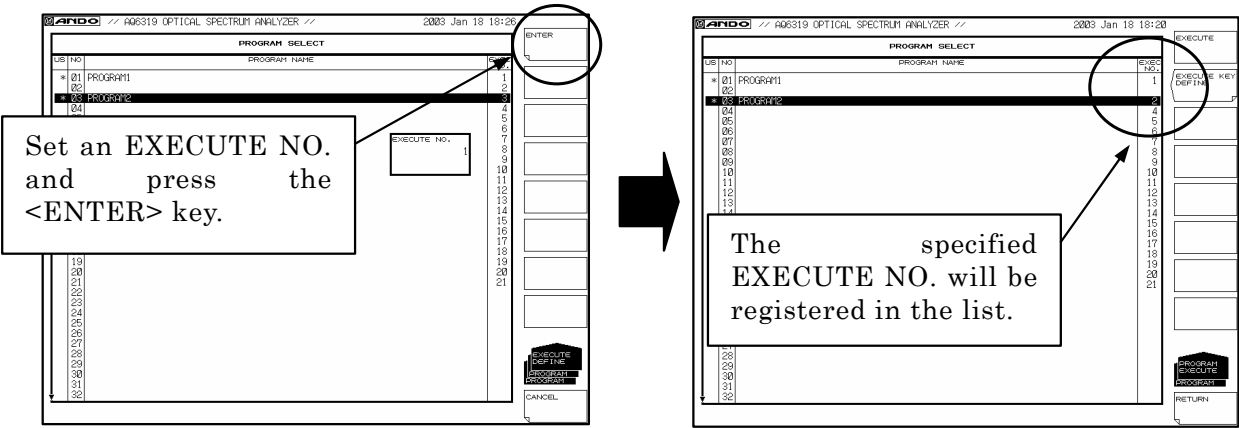


Figure 1-15: Screen for Registering EXECUTE Number

1.4 Executing a Program

There are two methods for executing a created program:

- Press the <PROGRAM EXECUTE> key of the [PROGRAM] switch to select the program to be executed from the program list and then press the <EXECUTE> key.
- On the soft-key menu displayed when pressing the [PROGRAM] switch, press one of the <EXECUTE1> to <EXECUTE21> keys.
Note that the program needs to have been pre-registered to the <EXECUTE@@> keys.
(☛ Subsection 1.3.8: Registering Programs to the <EXECUTE1> to <EXECUTE21> Keys)

1.4.1 Executing a Program Using the <PROGRAM EXECUTE> Key

The procedure for running a program is as follows:

- ① Press the [PROGRAM] switch.
- ② Press the <PROGRAM EXECUTE> key.
- ③ Move the cursor to the program you wish to run and press the <EXECUTE> key.

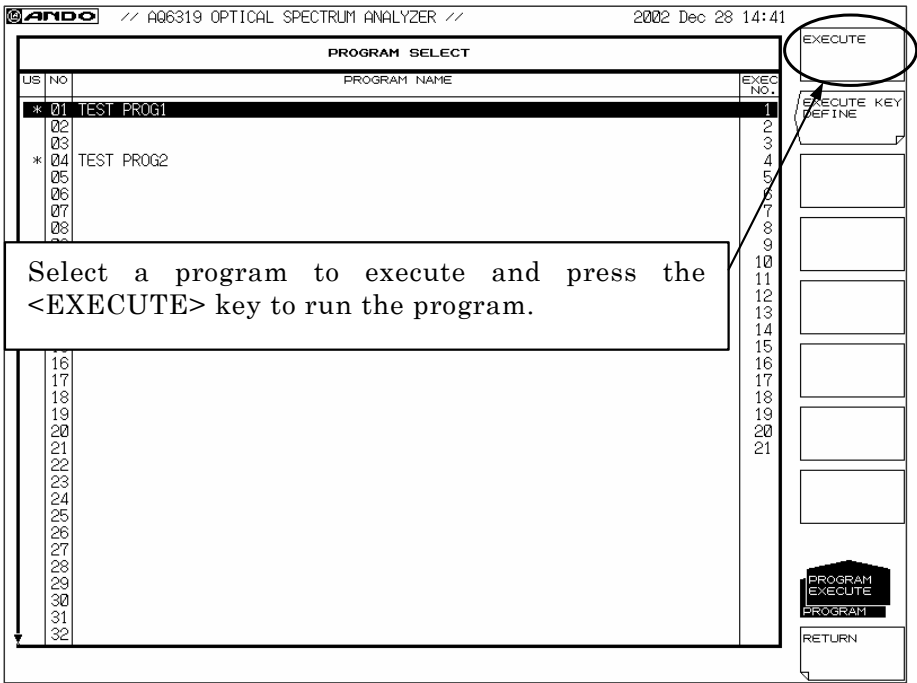


Figure 1-16: Screen Displayed When Pressing the <PROGRAM EXECUTE> Key

NOTE

- To stop the program during running , press the <PROGRAM EXIT> key.

1.4.2 Executing a Program Using the <EXECUTE1> to <EXECUTE21> Keys

The procedure for executing a program is as follows:

- ① Press the [PROGRAM] switch.
- ② Locate the program you wish to run in the <EXECUTE1> to <EXECUTE21> soft keys and press the corresponding soft key.

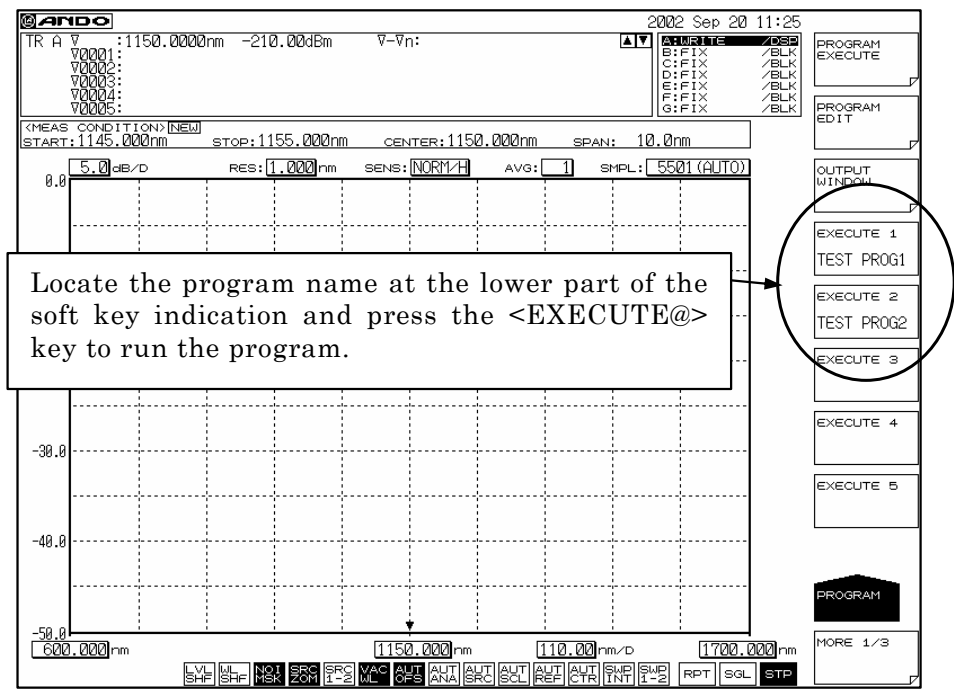


Figure 1-17: Executing a Program Using an <EXECUTE@> Key

NOTE

- To stop the program during running, press the <PROGRAM EXIT> key.
- When a program name has been entered in the programs registered in the <EXECUTE 1> to <EXECUTE 21> keys, the program name (of 11 characters) will also be displayed in the soft keys.

1.4.3 Pause Using the “PAUSE ‘-----’ ” Program Command

If a program containing the “PAUSE ‘-----’ ” special command is run, the program will pause when this command is executed.

When the “PAUSE ‘-----’ ” command is executed, the screen status becomes as shown in Figure 1-18, and a character string entered by “PAUSE ‘-----’ ” during editing of the program appears together with a message. The program then pauses.

To resume running of the program, press the <CONTINUE> soft key.

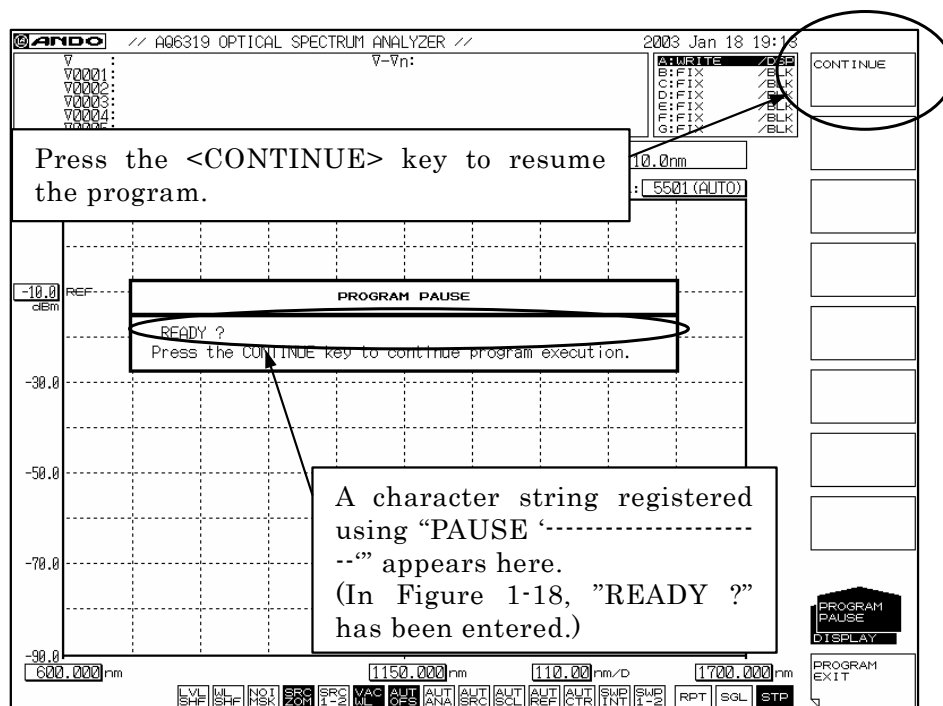


Figure 1-18: Screen Showing a Program Pause

NOTE

- If a program is executed in remote control, the “PAUSE ‘-----’ ” command will be overridden.

1.4.4 Entering a Parameter Using the “DATA INPUT -----‘;@’” Program Command

When a program containing the “DATA INPUT ----- ‘;@’” special command is run, a window prompting the entry of data will be displayed during execution of the program.

In this case, one of two types of windows will appear depending on the @ variable.

- When @ is a string variable:
Enter a file name in the same way as inputting a label and press the <DONE> key.
(☛ Section 6.21: General Purpose LABEL INPUT, in the Instruction Manual)

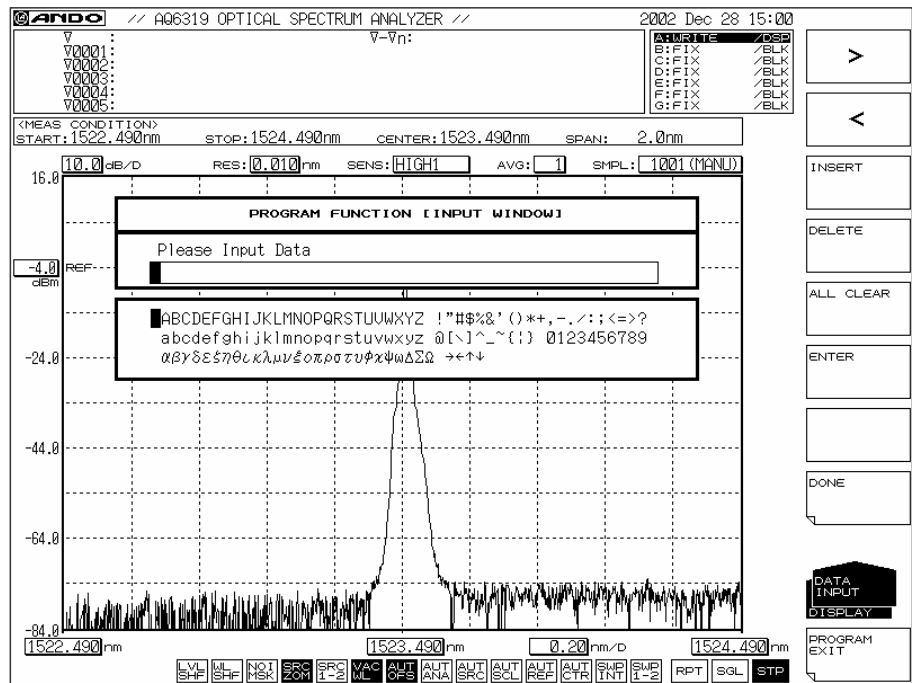


Figure 1-19: Screen for Executing “DATA INPUT ----- ‘;@’” when @ is a String Variable

NOTE

- If a program is executed in remote control, the “DATA INPUT ----- ‘;@’” command will be overridden.

- When @ is a numeric variable:
The INPUT WINDOW appears. Use the rotary knob, step keys, or numeric keys to set any numerical value.

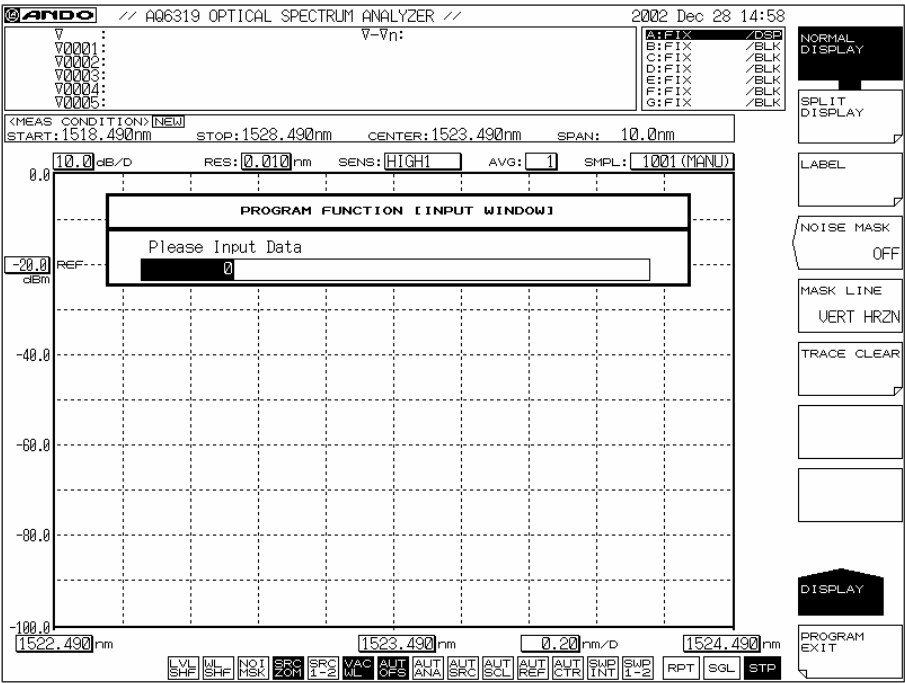


Figure 1-20: Screen for Executing “DATA INPUT ----- “@” when @ is a Numeric Variable

NOTE

- If a program is executed in remote control, the “DATA INPUT ----- “@” command will be overridden.

1.4.5 Outputting Data Using the “DATA OUTPUT @@@@” Program Command

If a program containing the “DATA OUTPUT @@@@” or “DATA OUTPUT @@@@,” special command is executed, an OUTPUT WINDOW displaying output data appears during execution of the program.

In this case, the contents of the variables specified by “@@@@” appear in the OUTPUT WINDOW.

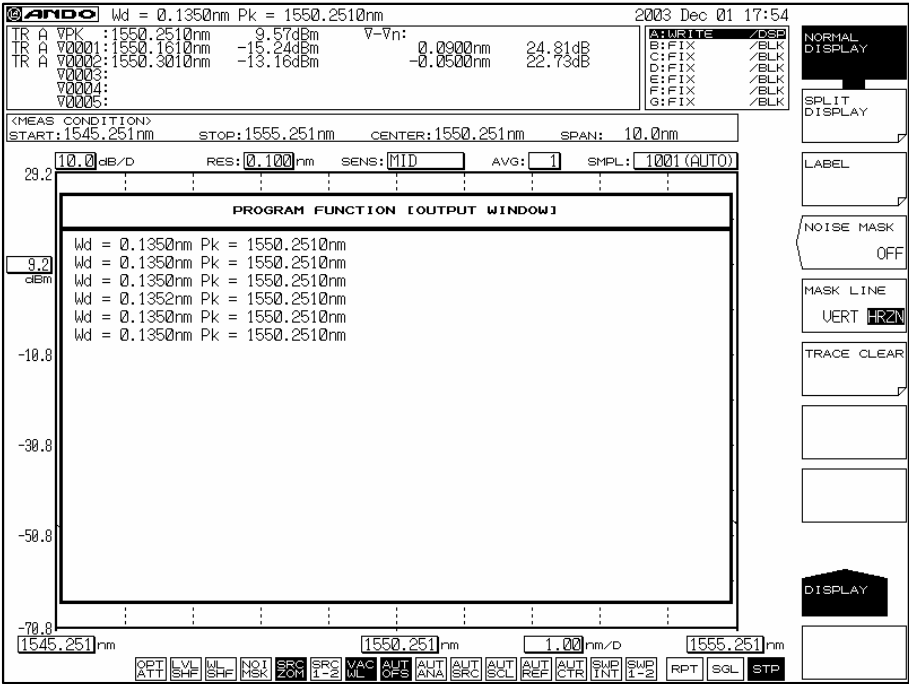


Figure 1-21: Screen Displaying the OUTPUT WINDOW

The Output Window has the following features.

- ✓ Capable of outputting a maximum of 200 lines of data.
The screen can display a maximum of 20 lines in the OUTPUT WINDOW. If data of more than 20 lines has been output, you can use the rotary knob or step keys to scroll through the OUTPUT WINDOW display.
- ✓ The OUTPUT WINDOW can be displayed during execution of a program.
To do so, use the “OUTPUT WINDOW ###” special command.
Note that the OUTPUT WINDOW disappears if the program ends.

NOTE

- To check the contents of the OUTPUT WINDOW after the program has ended, press the <OUTPUT WINDOW> key of the [PROGRAM] switch.

- ✓ The contents of the OUTPUT WINDOW can be displayed or printed out using the <OUTPUT WINDOW> key of the [PROGRAM] switch even if a program is not running.

The operating procedure for this is as follows:

- ① Press the [PROGRAM] switch.
- ② Press the <OUTPUT WINDOW> key.

This causes the OUTPUT WINDOW to appear on the screen.

To close the OUTPUT WINDOW, press the <CLOSE WINDOW> key.

To output the contents of the OUTPUT WINDOW to the internal printer, press the <PRINT OUTPUT WIN> key.

NOTE

- If no data is output to the OUTPUT WINDOW, the <OUTPUT WINDOW> key will be overridden.
- Even if the <HARD COPY DEVICE> key in the [SYSTEM] switch is set to other than INTERNAL, printing is done using an internal printer.

- ✓ Even when the program ends, the contents of the OUTPUT WINDOW will continue to be displayed unless the “OUTPUT WINDOW CLEAR” special command is executed.

- ✓ The contents of the OUTPUT WINDOW can be stored in a file.

The procedure for storing the data is as follows:

- ① Press the [FILE] switch.
- ② Select the <DATA> key using the <ITEM SELECT> key and then press the <WRITE> key.
- ③ Press the <OUTPUT ITEM SETTING> key and then set the <OUTPUT DISPLAY> key to ON.

(☛ Section 7-5: Data Writing/Reading Function, in the Operation Manual)

NOTE

- If data exceeding 200 lines is displayed in the OUTPUT WINDOW, data will be erased beginning from the first line in turn.
- Turning off the Power switch of this unit erases data in the OUTPUT WINDOW.

1.4.6 Error Encountered-upon Execution of a Program

If an error occurs during the running of a program, an error number indicating the details of the error is displayed in a window, and execution of the program is stopped.

A list of error numbers that may be indicated is shown in Table 1-1.

NOTE

- If an error occurs during execution of a program, an error number indicating the type of the error and a message are displayed in the same way as ordinary operations.

● Classification of the error numbers

The error numbers are classified as follows:

- 300 to 307 Errors caused by attempting a setting in manual operation which is disabled.
- 320 to 326 Special command-related errors
- 340 to 347 Input/output-related errors
- 360 to 369 Floppy disk-related errors
- 380, 381 Other errors

The noted numbers can be read out using the SYSTem:ERRor[:NEXT]? GPIB command. (☛ Section 2-4: GPIB Commands)

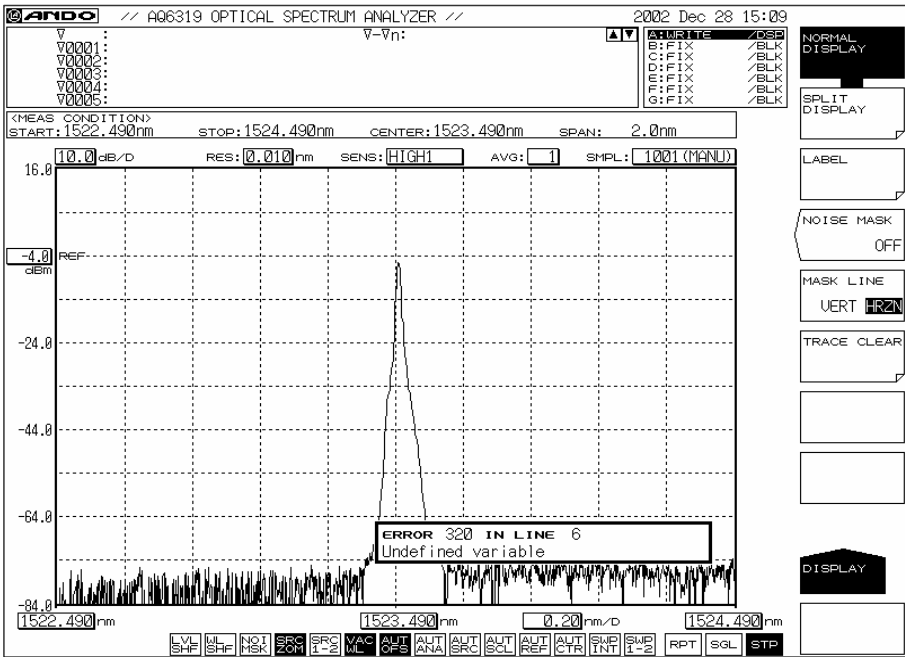


Figure 1-22: Indication of an Error During Execution of a Program

Table 1-1: List of Errors Possibly Encountered during Execution of a Program

No.	Message	Possible Cause of Error
300	Parameter out of range	A variable value is out of range or is not defined for a command that sets a parameter using variables.
302	Scale unit mismatch	There is a difference between the Y-axis scale of the active trace and the unit of a parameter in the "LINE MKR 3 or 4" command.
303	No data in active trace	Setting of the moving marker, a peak (or bottom) search, or activation of the analysis function was made with no data in the active trace.
304	Marker value out of range	Specified wavelength was out of the sweep range in the moving marker or line wavelength marker setting command.
305	No data in traces A or B	No waveform data in traces A or B when executing the "EDFA NF" command
306	Invalid data	Trace had no data when attempting to save it to memory or to write it to FD/HDD.
307	Unsuitable Write item	All data items were OFF at execution of "WRITE DATA".
320	Undefined variable	A command containing an undefined variable was executed.
321	Variable unit mismatch	The unit of each variable does not agree within a command containing two or more variables.
322	Overflow	An overflow occurred in an arithmetic operation.
323	Undefined marker variable	A command containing a marker-value variable was executed when no marker had been displayed.
324	Invalid marker variable	A command containing the corresponding variable was executed at a time other than immediately after execution of a spectrum width search, peak search, etc.
325	Undefined line number	GOTO command's jumping destination is a number other than 1 to 200.
326	F1 greater than F2	F1>F2 when the "IF F1 $\leq$ @@@@ $\leq$ F2" command was executed.
340	Printer paper empty	No printer paper
341	Printer head up	No print is made because the printer's head-up lever is raised.
345	Option does not respond	No response from an external device
346	Option is not connected	No external device is connected.
347	GP-IB2 not system controller	System controller connected to the GP-IB2 port has been set to an external computer.
360	Disk full	No file can be created due to insufficient free space on the floppy disk.
361	Disk not inserted	No floppy disk is in the FDD.
362	Disk is write protected	The floppy disk is write protected.
363	Disk not initialized	The floppy disk has not been initialized or has been formatted in a format not usable in this unit.
364	Directory full	Directory is full, therefore no file can be created.
365	File not found	Because no specified file is found, it cannot be read, or there is no file on the disk.
366	File is write protected	The file is specified to be read only, so that it cannot be rewritten or deleted.
367	No data	No data to store
368	File is not a trace file	A file cannot be read because it is not a trace file.
369	Illegal file name	A file cannot be saved due to an incorrect file name.
380	Undefined program	An attempt was made to run a program that is not defined.
381	Syntax error	Wrong command (a program has been rewritten for some reason)

1.5 Description of Commands

There are two types of commands used in a program: function commands which are input using a panel switch, and other special commands.

Table 1-4 shows a list of the function commands, while Table 1-5 shows a list of the special commands.

1.5.1 Description of Variables

Variables that can be used in a program are shown in Table 1-2.

Table 1-2: Variables Usable in a Program

Type	Variable Name	Description
Generalized variables	G	Contains a generalized numeric variable.
	H	Contains a generalized numeric variable.
	I	Contains a generalized numeric variable.
	J	Contains a generalized numeric variable.
	K	Contains a generalized numeric variable.
	P	Contains a generalized numeric variable.
	Q	Contains a generalized numeric variable.
	R	Contains a generalized numeric variable.
	X	Contains a generalized numeric variable.
	Y	Contains a generalized numeric variable.
	Z	Contains a generalized numeric variable.
	A\$	Contains a generalized string variable.
	B\$	Contains a generalized string variable.
	C\$	Contains a generalized string variable.
	D\$	Contains a generalized string variable.
GPIB variables	S	Performs serial polling and stores received status bytes. This variable is also used as a generalized numeric variable.
	A\$, B\$, C\$, D\$	Contains data received via the GP-IB2 port. This variable is also used as a generalized string variable.
Date & time/file name variables	FILE\$	Contains the name of a file to which the final access was made.
	TIME\$	Contains date and time data (e.g., 1994 Sep 08 20:45:37).
Marker variables	WM	Contains the wavelength value of the moving marker.
	W1	Contains the wavelength value of fixed marker 1.
	W2	Contains the wavelength value of fixed marker 2.
	W2-W1	Contains the wavelength difference between fixed markers 1 and 2.
	W(CH)	Contains the wavelength values of fixed markers (CH: 1 to 1024).
	LM	Contains the level value of the moving marker.
	L1	Contains the level value of fixed marker 1.
	L2	Contains the level value of fixed marker 2.
	L2-L1	Contains the level difference between fixed markers 1 and 2.
Analysis variables	L(CH)	Contains the level values of fixed markers (CH: 1 to 1024).
	SPWD	Contains spectrum width applied in making a spectrum width search.
	PKWL	Contains a peak (or bottom) wavelength value applied in making a peak (or bottom) search or spectrum width search.
	MEANWL	Contains center wavelength applied in making spectrum width search.
	PKLVL	Contains a peak (or bottom) level value applied in making a peak (or bottom) search or spectrum width search.

Type	Variable Name	Description
Analysis variables Analysis variables	MODN	Contains the number of modes applied in making a spectrum width search.
	SMSR	Contains the side mode suppression ratio (level difference) applied in making SMSR measurements.
	WDMCHN	Contains the number of channels detected in performing WDM analysis.
	WDMWL(CH)	Contains the center wavelength of channel CH used in performing WDM analysis.
	WDMLVL(CH)	Contains the level of channel CH used in performing WDM analysis.
	WDMSNR(CH)	Contains SNR of channel CH used in performing WDM analysis.
	NFCHN	Contains the number of channels detected in performing EDFA-NF analysis.
	NFWL(CH)	Contains the center wavelength of channel CH used in performing EDFA-NF analysis.
	NFLVLI(CH)	Contains the input signal level of channel CH used in performing EDFA-NF analysis.
	NFLVLO(CH)	Contains the output signal level of channel CH used in performing EDFA-NF analysis.
	NFASELV(CH)	Contains the ASE level of channel CH used in performing EDFA-NF analysis.
	NFGAIN(CH)	Contains the gain of channel CH used in performing EDFA-NF analysis.
	NFNF(CH)	Contains NF of channel CH used in performing EDFA-NF analysis.
	MKPWR	Contains power obtained in making between line-markers power measurements.
	PMD	Contains the PMD value obtained in PMD analysis.
Program control variables	M	Contains loop counter data.
	N	Contains loop counter data.
	F1	Contains a conditional judgment variable.
	F2	Contains a conditional judgment variable.
	CH	Contains an element number variable used in accessing an array variable. (1 to 1024)

1.5.2 Principles of Variable-based Arithmetic Operations

For assignment of units after arithmetic operations when a variable with a unit is used in the operation, see Table 1-3.

Table 1-3: Assignment of Units after Arithmetic Operation

Expression	Results
(With a unit) $\times$ (Without unit)	With a unit
(With a unit) $\div$ (Without unit)	With a unit
(Without unit) $+$ (Without unit)	Without unit
(Without unit) $-$ (Without unit)	Without unit
(Without unit) $\times$ (Without unit)	Without unit
(Without unit) $\div$ (Without unit)	Without unit
(nm) $+$ (nm)	(nm)
(nm) $-$ (nm)	(nm)
(nm) $\div$ (nm)	Without unit
(dB) $+$ (dB)	(dB)
(dB) $-$ (dB)	(dB)
(dBm) $+$ (dB)	(dBm)
(dBm) $-$ (dB)	(dBm)
(dBm) $-$ (dBm)	(dB)
(#W) $+$ (#W)	(#W)
(#W) $-$ (#W)	(#W)
(#W) $\div$ (#W)	Without unit

- For the units of dBm/nm, W/nm, dB/km, and %, dBm, W, dB, and without unit apply respectively in terms of variables.
- Arithmetic operations are made as noted above according to the unit of a variable, and the unit is appended to the result obtained after operation.
- If an arithmetic operation is made in any combination other than the above (addition, subtraction, multiplication, or division of variables with different units), the result of the operation has no units.
- The units of #W for the above are treated as follows:
 - 1 mW = 1
 - 1 μ W = 0.001
 - 1 nW = 0.000001
 - 1 pW = 0.000000001

1.5.3 Specifications of “@=VAL(@\$)” Command

- A character string other than the numerics located before a value (starting with a sign or number) in @\$ character string will be ignored, and numerics up to “,” or the next character string or delimiter are converted.
- If no numeric exists in @\$ character string, “0” is substituted for variable @.

1.5.4 List of Function Commands

Table 1-4: List of Function Commands

Program Command	Description	Parameter Ranges and Available Variables
SWEEP		
AUTO	Auto sweep	
REPEAT	Repeat sweep	
SINGLE	Single sweep	
STOP	Stops a sweep.	
SEGMENT MEASURE	Makes measurements only by a specified number of points starting at the position of the wavelength being stopped.	
SEGMENT POINT *****	Sets the number of points to measure with the <SEGMENT MEASURE> key.	1 to 50001 (1 step)
SWEEP MKR L1-L2 ###	Selects ON/OFF of sweep function.	###: ON or OFF
SWP INTVL *****sec	Sets interval time applied to repeat sweeps.	MINIMUM, 1 to 99999sec (1step) (If "0" is set, MINIMUM applies.)
CENTER		
CENTER WL ****.***nm	Sets measurement center wavelength.	600.000 to 1700.000nm (0.1 step)
CENTER WL @@@@	Sets the value of variable @@@@ to a measurement center wavelength.	@@@@: G, H, I, J, K, P, Q, R, S, X, Y, Z, WM, W1, W2, W (CH), PKWL, MEANWL, WDMWL (CH), NFWL (CH)
CENTER FREQ ***.***THz	Sets measurement center frequency.	176.5000 to 500.0000THz (0.01 step)
CENTER FREQ @@@@	Sets the value of variable @@@@ to measurement center frequency.	@@@@: G, H, I, J, K, P, Q, R, S, X, Y, Z, WM, W1, W2, W (CH), PKWL, MEANWL, WDMWL (CH), NFWL (CH)
START WL ****.***nm	Sets measurement-starting wavelength.	50.000 to 1700.000nm (0.1 step)
START FREQ ***.***THz	Sets measurement-starting frequency.	11.50000 to 500.0000THz (0.01 step)
STOP WL ****.***nm	Sets measurement-ending wavelength.	600.000 to 2250.000nm (0.1 step)
STOP FREQ ***.***THz	Sets measurement-ending frequency.	176.5000 to 665.0000 (0.01 step)
PEAK->CENTER	Sets the peak wavelength of the waveform on an active trace to measurement center wavelength.	
MEAN WL->CENTER	Makes a spectrum width search on an active trace and sets the resultant central wavelength to measurement center wavelength.	
AUTO CENTER ###	Selects ON/OFF of the <PEAK→CENTER> function, activated each time a sweep ends.	###: ON or OFF
VIEW SCALE->MEAS SCALE	Sets the current displayed conditions for measuring conditions.	

Program Command	Description	Parameter Ranges and Available Variables
SPAN		
SPAN WL ****.*nm	Sets the measuring span.	0, 0.1 to 1100.0nm (1 step)
SPAN WL @@@@	Sets the value of variable @@@@ to the measuring span.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,W2-W1,SPWD
SPAN FREQ ***.**THz	Sets the measuring span.	0, 0.10 to 330.000THz (0.1 step)
SPAN FREQ @@@@	Sets the value of variable @@@@ to the measuring span.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,W2-W1,SPWD
START WL ****.*nm	Sets measurement-starting wavelength.	50.000 to 1700.000nm (0.1 step)
START FREQ ***.**THz	Sets measurement-starting frequency.	11.50000 to 500.0000THz (0.01 step)
STOP WL ****.*nm	Sets measurement-ending wavelength.	600.000 to 2250.000nm (0.1 step)
STOP FREQ ***.**THz	Sets measurement-ending frequency.	176.5000 to 665.0000 (0.01 step)
$\Delta \lambda \rightarrow$ SPAN	Makes a spectrum width search on an active trace and sets the result to the measuring span.	
0nm SWEEP TIME **sec	Sets sweep time used when a sweep is made in a 0 nm measuring span.	MINIMUM, 1 to 50 (1 step) (If "0" is set, MINIMUM applies.)
VIEW SCALE->MEAS SCALE	Sets the current display conditions to measuring conditions.	

Program Command	Description	Parameter Ranges and Available Variables
LEVEL		
REFERENCE LEVEL ***.*dBm	Sets the reference level value used for LOG scaling.	-90.0 to 30.0 dBm (0.1 step)
REFERENCE LEVEL ***.*##	Sets the reference level value used for LIN scaling.	1.00 pW to 1000 mW (1.00 to 9.99 [pW, nW, μ W, mW]: 0.01 step 10.0 to 99.9 (100) [pW, nW, μ W, (mW)]: 0.1 step 100 to 999 [pW, nW, μ W, mW]: 1 step) For ##, select one of pW, nW, μ W, mW.
REFERENCE LEVEL @@@@@	Sets the value of variable @@@@@ to the reference level value.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,LM,L1,L2,L(CH),PKLVL,WDMLVL(CH),NFLVI(CH),NFLVO(CH),NFASLV(CH),MKPWR
LEVEL SCALE **.*dB/D	Sets a level scale value.	LINEAR, 0.1 to 10.0 dB/DIV (0.1 step) (If "0" is set, LINEAR applies.)
LEVEL SCALE @@@@@	Sets the value of variable @@@@@ to the level scale.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,L2-L1,SMSR,WDMSNR(CH),NFF(CH)
BASE LEVEL ****	Sets the bottom value used for linear scaling. This uses the unit set in REF LEVEL. If 90% of the top end value of the scale is exceeded, an error occurs during execution.	0.0 to 900.0 (0.1 step)
PEAK->REF LEVEL	Sets the peak level of the waveform on an active trace to the reference level value.	
AUTO REF LEVEL ###	Selects ON/OFF of the <PEAK->REF LEVEL> function, activated each time a sweep ends.	###: ON/OFF
LEVEL UNIT #####	Sets the unit of a level scale.	###: dBm, dBm/nm
Y SCALE DIVISION ##DIV	Sets the number of divisions for a level scale.	##: 8, 10, 12
REF LEVEL POSITION **DIV	Sets the reference level position on the level scale.	0 to 12 (1 step)
SUB LOG **.*dB/D	Sets a sub-scale value used for LOG scaling.	0.1 to 10.0dB/DIV (0.1 step)
SUB LIN *.*./D	Sets a sub-scale value used for LIN scaling.	0.005 to 1.250 (0.005 step)
SUB SCALE **.*dB/km	Sets a sub-scale value used for dB/km scaling.	0.1 to 10.0 (0.1 step)
SUB SCALE **.*%/D	Sets a sub-scale value used for %/D scaling.	0.5 to 125.0 (0.1 step)
OFFSET LEVEL **.*dB	Sets a sub-scale offset value used for LOG scaling.	-99.99 to 99.99 (0.1 step)
OFFSET LEVEL ***.*dB/km	Sets a sub-scale offset value used for dB/km scaling.	-99.99 to 99.99 (0.1 step)
SCALE MINIMUM **.*	Sets a sub-scale bottom value used for linear scaling.	0.00 to 12.50 (0.01 step)
SCALE MINIMUM ***.*%	Sets a sub-scale bottom value used for %/D scaling.	0.0 to 125.0 (0.1 step)
LENGTH **.*km	Sets fiber length.	0.001 to 99.999 (0.001 step)
AUTO SUB SCALE ###	Sets a sub-scale based on an operation trace waveform automatically.	###: ON/OFF
SUB REF LEVEL POSITION **DIV	Sets the reference level position on the sub-level scale.	0 to 12 (1 step)

Program Command	Description	Parameter Ranges and Available Variables
SETUP		
RESOLUTION WL *.****nm	Sets resolution.	0.010 to 1.000 (1-2-5 step)
RESOLUTION WL @@@@	Sets the value of variable @@@@ to resolution.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,W2-W1,SPWD
SENS NORMAL/HOLD	Sets measuring sensitivity to NORMAL/HOLD.	
SENS NORMAL/AUTO	Sets measuring sensitivity to NORMAL/AUTO.	
SENS MID	Sets measuring sensitivity to MID.	
SENS HIGH1	Sets measuring sensitivity to HIGH1.	
SENS HIGH2	Sets measuring sensitivity to HIGH2.	
SENS HIGH3	Sets measuring sensitivity to HIGH3.	
CHOPPER ###	Switches chopper mode.	###: OFF/CHOP/SWITCH
AVERAGE TIMES ***	Sets the number of averaging times.	1 to 999 (1 step)
AVERAGE TIMES @	Sets the number of averaging times to the value of variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,M,N
SAMPLING POINT AUTO ###	Sets sampling points per sweep automatically.	###: ON/OFF
SAMPLING POINT *****	Sets sampling points per sweep.	101 to 50001 (1 step)
SAMPLING POINT @	Sets the number of samples to the value of variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,M,N
SAMPLING INTERVAL *.****nm	Sets sampling intervals for measurements per sweep.	0.0010 to SPAN/50000 (0.0001 step)
SAMPLING INTERVAL @	Sets sampling intervals to the value of variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,M,N,W2-W1,SPWD
MEASURE WAVELENGTH AIR	Sets measurement wavelength to an air wavelength.	
MEASURE WAVELENGTH VACUUM	Sets measurement wavelength to a vacuum wavelength.	
X SCALE UNIT WAVELENGTH	Sets axis X to wavelength display mode.	
X SCALE UNIT FREQUENCY	Sets axis X to frequency display mode.	
EXTERNAL TRIGGER MODE ###	Sets external trigger mode.	###: ON/OFF
EXTERNAL TRIGGER EDGE RISE	Sets external trigger edge to RISE.	
EXTERNAL TRIGGER EDGE FALL	Sets external trigger edge to FALL.	
EXTERNAL TRIGGER DELAY ****.*μs	Sets external trigger delay time.	0.0 to 1000.0 (0.1 step)
OPTICAL ATTENUATOR ###	Sets insertion of internal attenuator.	###: ON/OFF
TLS SYNC SWEEP ###	Select the synchronous sweep function with the Tunable Laser Source	###: ON/OFF

Program Command	Description	Parameter Ranges and Available Variables
TRACE 1/2		
ACTIVE TRACE #	Sets trace # to active trace.	#: A to G
DISPLAY #	Sets trace # to display mode.	#: A to G
BLANK #	Sets trace # to invisible mode.	#: A to G
WRITE #	Sets trace # to write mode.	#: A to G
FIX #	Sets trace # to data-fixing mode.	#: A to G
MAX HOLD #	Sets trace # to max. value detection mode.	#: A to G
MIN HOLD #	Sets trace # to min. value detection mode.	#: A to G
ROLL AVG # ***	Sets trace # to sequential addition averaging mode.	#: A to G 2 to 100 (1 step)
C=A·B(LOG)	Sets trace C to TRACE A·B computation mode (LOG).	
C=B·A(LOG)	Sets trace C to TRACE B·A computation mode (LOG).	
C=A+B(LOG)	Sets trace C to TRACE A+B computation mode (LOG).	
C=A+B(LIN)	Sets trace C to TRACE A+B computation mode (LIN).	
C=A·B(LIN)	Sets trace C to TRACE A·B computation mode (LIN).	
C=B·A(LIN)	Sets trace C to TRACE B·A computation mode (LIN).	
C=1·k(A/B) k=*,****	Sets trace C to 1·k (TRACE A/B) computation mode.	1.0000 to 2000.0000 (0.0001 step)
C=1·k(B/A) k=*,****	Sets trace C to 1·k (TRACE B/A) computation mode.	1.0000 to 2000.0000 (0.0001 step)
F=C·D(LOG)	Sets trace F to TRACE C·D computation mode (LOG).	
F=D·C(LOG)	Sets trace F to TRACE D·C computation mode (LOG).	
F=C+D(LOG)	Sets trace F to TRACE C+D computation mode (LOG).	
F=D·E(LOG)	Sets trace F to TRACE D·E computation mode (LOG).	
F=E·D(LOG)	Sets trace F to TRACE E·D computation mode (LOG).	
F=D+E(LOG)	Sets trace F to TRACE D+E computation mode (LOG).	
F=C+D(LIN)	Sets trace F to TRACE C+D computation mode (LIN).	
F=C·D(LIN)	Sets trace F to TRACE C·D computation mode (LIN).	
F=D·C(LIN)	Sets trace F to TRACE D·C computation mode (LIN).	
F=D+E(LIN)	Sets trace F to TRACE D+E computation mode (LIN).	
F=D·E(LIN)	Sets trace F to TRACE D·E computation mode (LIN).	
F=E·D(LIN)	Sets trace F to TRACE E·D computation mode (LIN).	
G=C·F(LOG)	Sets trace G to TRACE C·F computation mode (LOG).	
G=F·C(LOG)	Sets trace G to TRACE F·C computation mode (LOG).	
G=C+F(LOG)	Sets trace G to TRACE C+F computation mode (LOG).	
G=E·F(LOG)	Sets trace G to TRACE E·F computation mode (LOG).	

Program Command	Description	Parameter Ranges and Available Variables
TRACE 2/2		
G=F·E(LOG)	Sets trace G to TRACE F·E computation mode (LOG).	
G=E+F(LOG)	Sets trace G to TRACE E+F computation mode (LOG).	
G=C+F(LIN)	Sets trace G to TRACE C+F computation mode (LIN).	
G=C·F(LIN)	Sets trace G to TRACE C·F computation mode (LIN).	
G=F·C(LIN)	Sets trace G to TRACE F·C computation mode (LIN).	
G=E+F(LIN)	Sets trace G to TRACE E+F computation mode (LIN).	
G=E·F(LIN)	Sets trace G to TRACE E·F computation mode (LIN).	
G=F·E(LIN)	Sets trace G to TRACE F·E computation mode (LIN).	
G=NORM A	Sets the normalized data of trace A to be displayed on trace G.	
G=NORM B	Sets the normalized data of trace B to be displayed on trace G.	
G=NORM C	Sets the normalized data of trace C to be displayed on trace G.	
G=CURVE FIT A **dB	Sets the curve-fit processing data of trace A to be displayed on trace G.	0 to 99 (1 step)
G=CURVE FIT B **dB	Sets the curve-fit processing data of trace B to be displayed on trace G.	0 to 99 (1 step)
G=CURVE FIT C **dB	Sets the curve-fit processing data of trace C to be displayed on trace G.	0 to 99 (1 step)
G=CURVE FIT PEAK A **dB	Sets the peak curve-fit processing data of trace A to be displayed on trace G.	0 to 99 (1 step)
G=CURVE FIT PEAK B **dB	Sets the peak curve-fit processing data of trace B to be displayed on trace G.	0 to 99 (1 step)
G=CURVE FIT PEAK C **dB	Sets the peak curve-fit processing data of trace C to be displayed on trace G.	0 to 99 (1 step)
G=MARKER FIT **dB	The current measurement scale performs a curve fit to the delta markers currently set.	0 to 99 (1 step)
CVFIT OPERATION AREA #####	Set the range of data used as applicale trace for calculation.	####: ALL / IN L1-L2 / OUT L1-L2
CURVE FIT/CURVE FIT PEAL ALGO #####	Set the fitting algorithm for obtaining curve-fits.	####: GAUSS / LORENZ / 3RD POLY / 4TH OPLY / 5TH POLY
TRACE #->#	Copies data from the trace of variable @ to that of variable @.	#: A to G
TRACE # CLEAR	Clears trace # data.	#: A to G
ALL TRACE CLEAR	Clears all trace data.	

Program Command	Description	Parameter Ranges and Available Variables
ZOOM		
ZOOM CENTER WL ****.***nm	Sets the display scale's center wavelength.	600.000 to 1700.000 (0.1 step)
ZOOM CENTER WL @@@@@	Sets the value of variable @@@@@ to the display scale's center wavelength.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W(CH), PKWL,MEANWL,WDMWL(CH),NFWL(CH)
ZOOM CENTER FREQ ***.****THz	Sets the display scale's center frequency.	176.5000 to 500.0000THz (0.01 step)
ZOOM CENTER FREQ @@@@@	Sets the value of variable @@@@@ to the display scale's center frequency.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W(CH), PKWL,MEANWL,WDMWL(CH),NFWL(CH)
ZOOM SPAN WL ****.*nm	Sets the display scale's span.	0.1 to 1100.0nm (0.1 step)
ZOOM SPAN WL @@@@@	Sets the value of variable @@@@@ to the display scale's span.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,W2-W1,SPWD
ZOOM SPAN FREQ ***.**THz	Sets the display scale's span.	0.01 to 330.00THz (0.1 step)
ZOOM SPAN FREQ @@@@@	Sets the value of variable @@@@@ to the display scale's span.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,W2-W1,SPWD
ZOOM START WL ****.***nm	Sets the starting wavelength of the display scale.	50.000 to 1699.950nm (0.1 step)
ZOOM START WL ***.****THz	Sets the starting frequency of the display scale.	11.5000 to 499.9950THz (0.01 step)
ZOOM STOP WL ****.***nm	Sets the ending wavelength of the display scale.	600.050 to 2250.000nm (0.1 step)
ZOOM STOP FREQ ***.****THz	Sets the ending frequency of the display scale.	176.5050 to 665.000 (0.01 step)
PEAK->ZOOM CENTER	Sets the peak wavelength of the waveform on an active trace to the display scale's center wavelength.	
OVERVIEW DISPLAY OFF	Sets OVERVIEW display during ZOOM to OFF.	
OVERVIEW DISPLAY LEFT	Sets OVERVIEW display during ZOOM to the left pane of the waveform screen.	
OVERVIEW DISPLAY RIGHT	Sets OVERVIEW display during ZOOM to the right pane of the waveform screen.	
OVERVIEW SIZE LARGE	Sets OVERVIEW display during ZOOM to a larger display.	
OVERVIEW SIZE SMALL	Sets OVERVIEW display during ZOOM to a smaller display.	
ZOOM INITIALIZE	Resets the display scale to the initial state.	

Program Command	Description	Parameter Ranges and Available Variables
DISPLAY		
NORMAL DISPLAY	Sets the screen into normal display mode.	
SPLIT DISPLAY	Sets the screen into split display mode.	
TRACE # UPPER	Sets trace # to the top screen of the split display.	#: A to G
TRACE # LOWER	Sets trace # to the bottom screen of the split display.	#: A to G
UPPER HOLD ###	Holds the top screen of the split display.	###: ON/OFF
LOWER HOLD ###	Holds the bottom screen of the split display.	###: ON/OFF
LABEL '---56 characters ---'	Displays a label comment in the label area. If a semicolon ";" is appended to the end of the comment, a comment (variable value) specified by the next LABEL command will be displayed successively.	
LABEL @@@@	Displays the contents of variable @@@@ in the label area.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W(CH), W2W1,LM,L1,L2,L2L1,L(CH),SPWD,MEAN WL,PKWL,PKLVL,MODN,GONO,MSR,WD MCHN,WDMWL(CH),WDMVL(CH),WDM NR(CH),FNCHN,NFWL(CH),NFLVI(CH),NF LVO(CH),NFASLV(CH),NFGAIN(CH),NFN F(CH), MKPWR,PMD,M,N,CH,A\$,B\$,C\$,D\$,FILE\$, TIME\$
LABEL @@@@;	Displays the contents of variable @@@@ in the label area. Comment (variable value) specified by the next LABEL command is displayed successively.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W(CH), W2W1,LM,L1,L2,L2L1,L(CH),SPWD,MEAN WL,PKWL,PKLVL,MODN,GONO,MSR,WD MCHN,WDMWL(CH),WDMVL(CH),WDM NR(CH),FNCHN,NFWL(CH),NFLVI(CH),NF LVO(CH),NFASLV(CH),NFGAIN(CH),NFN F(CH), MKPWR,PMD,M,N,CH,A\$,B\$,C\$,D\$,FILE\$, TIME\$
LABEL CLEAR	Clears the LABEL command in the label area.	
NOISE MASK ***dB	Displays waveform data with the data at or below the set level masked.	OFF (-999), -100 to 0 (1 step)
MASK LINE VERTICAL	Sets data at the mask value or less to zero in the noise masking function.	
MASK LINE HORIZONTAL	Sets data at the mask value or less to the mask value in the noise masking function.	
TRACE # CLEAR	Clears trace # data.	#: A to G
ALL TRACE CLEAR	Clears all trace data.	

Program Command	Description	Parameter Ranges and Available Variables
MARKER 1/2		
MARKER ****.***nm	Sets a marker to the specified wavelength position of the waveform on an active trace (according to a wavelength value).	600.000 to 1700.000 (0.1 step)
MARKER ***.****THz	Sets a marker to the specified wavelength position of the waveform on an active trace (according to a frequency value).	176.500 to 500.0000 (0.01 step)
MARKER @@@@	Sets a marker to the wavelength position of variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W(CH), MEANWL,PKWL,WDMWL(CH),NFWL(CH)
SET MARKER ****	Sets fixed marker **** to the moving marker position.	1 to 1024 (1 step)
SET MARKER @	Sets the fixed marker of variable @ to the moving marker position.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,S,N,M
CLEAR MARKER ****	Clears fixed marker ****.	1 to 1024 (1 step)
CLEAR MARKER@	Clears the fixed marker of variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,S,N,M
MARKER->CENTER	Sets the wavelength value of a marker to the measurement center wavelength.	
MARKER->ZOOM CENTER	Sets the wavelength value of a marker to the display scale's center wavelength.	
MARKER->REF LEVEL	Sets the marker level value to the reference level.	
ALL MARKER CLEAR	Clears markers from the screen.	
LINE MARKER1 ****.***nm	Sets line marker 1 to a specified wavelength position (according to a wavelength value).	600.000 to 1700.000 (0.1 step)
LINE MARKER1 ***.****THz	Sets line marker 1 to a specified frequency position (according to a frequency value).	10.0000 to 665.0000 (0.01 step)
LINE MARKER1 @@@@	Sets line marker 1 to the wavelength position of variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,S,WM,W1,W2,W(C H),MEANWL,PKWL,WDMWL(CH),NFWL(CH)
LINE MARKER2 ****.***nm	Sets line marker 2 to a specified wavelength position (according to a wavelength value).	600.000 to 1700.000 (0.1 step)
LINE MARKER2 ***.****THz	Sets line marker 2 to a specified frequency position (according to a frequency value).	10.0000 to 665.0000 (0.01 step)
LINE MARKER2 @@@@	Sets line marker 2 to the wavelength position of variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,S,WM,W1,W2,W(C H),W2-W1,PKWL,WDMWL(CH),NFWL(CH)
LINE MARKER3 ****.***dB	Sets line marker 3 to a specified level.	-139.90 to 159.90 (0.01 step)
LINE MARKER3 ****.***dBm	Sets line marker 3 to a specified level.	-210.00 to 50.00 (0.01 step)
LINE MARKER3 *.***##	Sets line marker 3 to a specified level.	0.0000 pW to 1.000E+05 mW (1.00 to 9.99 [pW, nW, μW, mW]: 0.01 step 10.0 to 99.9 (100) [pW, nW, μW, (mW)]: 0.1 step 100 to 999 [pW, nW, μW, mW]: 1 step) For ##, select one of pW, nW, μW, and mW.

Program Command	Description	Parameter Ranges and Available Variables
MARKER 2/2		
LINE MARKER3 **.***	Sets line marker 3 to a specified level.	0.00 to 25.00 (0.01 step)
LINE MARKER3 @@@@	Sets line marker 3 to the level position of variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,LM,L1,L2,L(CH),P KLVL,WDMLVL(CH),NFLVI(CH),NFLVO(C H),NFASELV(CH),MKPWR
LINE MARKER4 ****.***dB	Sets line marker 4 to a specified level.	-139.90 to 159.90 (0.01 step)
LINE MARKER4 ****.***dBm	Sets line marker 4 to a specified level.	-210.00 to 50.00 (0.01 step)
LINE MARKER4 *.***##	Sets line marker 4 to a specified level.	0.0000 pW to 1.000E+05 mW (1.00 to 9.99 [pW, nW, μW, mW]: 0.01 step 10.0 to 99.9 (100) [pW, nW, μW, (mW)]: 0.1 step 100 to 999 [pW, nW, μW, mW]: 1 step) For ##, select one of pW, nW, μW, and mW.
LINE MARKER4 **.***	Sets line marker 4 to a specified level.	0.00 to 25.00 (0.01 step)
LINE MARKER4 @@@@	Sets line marker 4 to the level position of variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,LM,L1,L2,L(CH),P KLVL,WDMLVL(CH),NFLVI(CH),NFLVO(C H),NFASELV(CH),MKPWR
MARKER L1-L2->SPAN	Sets an area enclosed by line markers 1 and 2 to a measuring span.	
MARKER L1-L2->ZOOM SPAN	Sets an area enclosed by line markers 1 and 2 to the display scale's span.	
LINE MARKER CLEAR	Clears line markers on the screen.	
MARKER OFFSET LIST	Displays a difference to the moving marker.	
MARKER SPACING LIST	Displays a difference to a neighboring marker.	
MARKER AUTO UPDATE ###	Makes the level position of a fixed marker following an active trace waveform.	###: ON/OFF
MARKER UNIT nm	Sets a wavelength marker value to the wavelength display.	
MARKER UNIT THz	Sets a wavelength marker value to the frequency display.	
SEARCH/ANA L1-L2 ###	Selects ON/OFF of the analysis function in an area enclosed by line markers 1 and 2.	###: ON/OFF
SEARCH/ANA ZOOM AREA ###	Selects ON/OFF of the analysis function in the display scale range.	###: ON/OFF
MARKER LIST PRINT	Prints out multiple marker values.	

Program Command	Description	Parameter Ranges and Available Variables
PEAK SEARCH		
PEAK SEARCH	Performs a peak search across waveforms on an active trace.	
BOTTOM SEARCH	Performs a bottom search across waveforms on an active trace.	
NEXT SRCH	Searches for a peak/bottom level next to the waveform peak/bottom level on an active trace.	
NEXT SRCH RIGHT	Searches for a peak/bottom level right to the waveform peak/bottom marker on an active trace.	
NEXT SRCH LEFT	Searches for a peak/bottom level left to the waveform peak/bottom marker on an active trace.	
SET MARKER ****	Sets fixed marker **** to the moving marker position.	1 to 1024 (1 step)
SET MARKER @	Sets the fixed marker of variable @ to the moving marker position.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,S,N,M
CLEAR MARKER ****	Clears fixed marker ****.	1 to 1024 (1 step)
CLEAR MARKER @	Clears the fixed marker of variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,S,N,M
ALL MARKER CLEAR	Clears all markers from the screen.	
AUTO SEARCH ###	Selects ON/OFF of the peak/bottom search function, activated each time a sweep is made.	###: ON/OFF
MODE DIFF **. **dB	Sets the level difference of the mode judgment criteria used for peak search or waveform analysis.	0.01 to 50.00 (0.01 step)
SEARCH/ANA L1-L2 ###	Selects ON/OFF of the analysis function in an area enclosed by line markers 1 and 2.	###: ON/OFF
SEARCH/ANA ZOOM AREA ###	Selects ON/OFF of the analysis function in the display scale range.	###: ON/OFF

Program Command	Description	Parameter Ranges and Available Variables
ANALYSIS 1/4		
SPEC WD THRESH **.**dB	Performs a THRESH method-based spectrum width search according to a specified threshold.	0.01 to 50.00 (0.01 step)
PARAM THRESH K **.**	Sets the magnification of a THRESH method-based spectrum width search.	1.00 to 10.00 (0.01 step)
PARAM THRESH MODE FIT ###	Sets ON/OFF of the function of setting a marker to a mode peak during execution of a THRESH method-based spectrum width search.	###: ON/OFF
SPEC WD ENV **.**dB	Performs an envelope method-based spectrum width search according to a specified threshold.	0.01 to 50.00 (0.01 step)
PARAM ENV TH2 **.**dB	Sets a lower-limit truncation value of an envelope method-based spectrum width search.	0.01 to 50.00 (0.01 step)
PARAM ENV K **.**dB	Sets the magnification of an envelope method-based spectrum width search.	1.00 to 10.00 (0.01 step)
SPEC WD RMS **.**dB	Performs an RMS method-based spectrum width search according to a specified threshold.	
PARAM RMS K .**.	Sets the magnification of an RMS method-based spectrum width search.	
SPEC WD PEAK RMS **.**dB	Performs a PEAK RMS method-based spectrum width search according to a specified threshold.	0.01 to 50.00 (0.01 step)
PARAM PEAK RMS K **.	Sets the magnification of a spectrum width search based on the PEAK RMS method.	1.00 to 10.00 (0.01 step)
SPEC WD NOTCH **.**dB	Makes notch width measurements according to a specified threshold.	0.01 to 50.00 (0.01 step)
PARAM NOTCH K **.**	Sets a magnification based on notch width measurement.	1.00 to 10.00 (0.01 step)
NOTCH FROM #####	Sets the reference for making notch width measurements.	#####: PEAK/BOTTOM
SMSR *	Sets the execution mode applied in SMSR measurements.	1 to 2 (1 step)
SMSR MASK ± **.**nm	Sets a mask range close to the peak during SMSR1 measurements.	0.00 to 99.99 (0.01 step)
POWER	Performs power analysis.	
POWER OFFSET **.**dB	Sets a correction value in power measurements.	-10.00 to 10.000 (0.01step)
DFB-LD ANALYSIS	Performs analysis necessary for DFB-LD.	
FP-LD ANALYSIS	Performs analysis necessary for FP-LD.	
LED ANALYSIS	Performs analysis necessary for LED.	
PMD ANALYSIS	Performs analysis necessary for PMD.	
PMD THRESH **.**dB	Sets a threshold value for PMD analysis.	0.01 to 50.00 (0.01step)
WDM ANALYSIS	Performs analysis necessary for WDM measurements.	
WDM THRESH **.**dB	Sets a threshold value for WDM analysis.	0.1 to 99.99 (0.01 step)
WDM MODE DIFF **.**dB	Sets the minimum peak/bottom difference for channel detection during WDM analysis.	0.01 to 50.00 (0.01 step)
WDM DISPLAY MASK OFF	Cancels setting of the level threshold used in masking channel display.	
WDM DISPLAY MASK ****.**dB	Sets the level threshold used in masking channel display.	-100.00 to 0.00 (0.1 step)
WDM NOISE ALGO AUTO FIX	Sets noise level measuring algorithm to AUTO FIX.	
WDM NOISE ALGO MANUAL FIX	Sets noise level measuring algorithm to MANUAL FIX.	
WDM NOISE ALGO AUTO CTR	Sets noise level measuring algorithm to AUTO CTR.	
WDM NOISE ALGO MANUAL CTR	Sets noise level measuring algorithm to MANUAL CTR.	

Program Command	Description	Parameter Ranges and Available Variables
ANALYSIS 2/4		
WDM NOISE ALGO PIT	Sets noise level measuring algorithm to PIT.	
WDM NOISE AREA **.**nm	Sets an area used for noise level analysis in a range centered on channel wavelength.	0.01 to 10.00 (0.01 step)
WDM NOISE AREA @	Sets an area used for noise level analysis to the area of variable @, centered on channel wavelength.	@: G,H,I,J,K,P,R,S,X,Y,Z
WDM MASK AREA **.**nm	Sets the spectral range of the signal light to be masked, centered on channel wavelength.	0.01 to 10.00 (0.01 step)
WDM MASK AREA @	Sets the spectral range of the signal light to be masked to the area of variable @, centering on channel wavelength.	@: G,H,I,J,K,P,R,S,X,Y,Z
WDM FITTING ALGO LINEAR	Places a fitting algorithm obtaining a noise level in linear interpolation mode.	
WDM FITTING ALGO GAUSS	Places a fitting algorithm obtaining a noise level in the Gauss distribution curve mode.	
WDM FITTING ALGO LORENZ	Places a fitting algorithm obtaining a noise level in the Lorenz curve mode.	
WDM FITTING ALGO 3RD POLY	Places a fitting algorithm obtaining a noise level in the 3 rd polynomial mode.	
WDM FITTING ALGO 4TH POLY	Places a fitting algorithm obtaining a noise level in the 4 th polynomial mode.	
WDM FITTING ALGO 5TH POLY	Places a fitting algorithm obtaining a noise level in the 5 th polynomial mode.	
WDM NOISE BANDWIDTH *.**nm	Sets bandwidth applied in measuring noise.	0.01 to 1.00 (0.01 step)
WDM DUAL TRACE ###	Makes setting so that both TRACEs A and B are used in analyzing WDM.	###: ON/OFF
WDM DISPLAY ABSOLUTE	Sets the display of WDM analysis results to absolute value display.	
WDM DISPLAY RELATIVE	Sets the display of WDM analysis results to relative value display.	
WDM DISPLAY DRIFT MEAS	Sets the display of WDM analysis results to drift value display (on the basis of the past measurement wavelength).	
WDM DISPLAY DIRFT GRID	Sets the display of WDM analysis results to drift value display (on the basis of grid wavelength).	
WDM CH RELATION #####	Sets the display format of an inter-channel level absolute value when WDM analysis display is in absolute value display.	#####: OFFSET/SPACING
WDM REF CHANNEL HIGHEST	Sets the reference channel used when CH RELATION setting is OFFSET to a channel of the highest level.	
WDM REF CHANNEL NO.****	Sets the reference channel used when CH RELATION setting is OFFSET.	1 to 1024 (1 step)
WDM MAX/MIN RESET	Resets MAX/MIN data obtained during drift measurements (MEAS, GRID).	
WDM OUTPUT SLOPE ###	Displays the least square approximation line of a channel peak.	###: ON/OFF
WDM POINT DISPLAY ###	Displays a data range used for fitting on the waveform screen.	###: ON/OFF
EDFA ANALYSIS	Performs analysis necessary for EDFA-NF measurements.	
EDFA NF THRESH **.**dB	Sets an EDFA-NF analysis threshold.	0.01 to 50.00 (0.01 step)

Program Command	Description	Parameter Ranges and Available Variables
ANALYSIS 3/4		
WDM MAX/MIN RESET	Resets MAX/MIN data obtained during drift measurements (MEAS, GRID).	
WDM OUTPUT SLOPE ###	Displays the least square approximation line of a channel peak.	###: ON/OFF
WDM POINT DISPLAY ###	Displays a data range used for fitting on the waveform screen.	###: ON/OFF
EDFA ANALYSIS	Performs analysis necessary for EDFA-NF measurements.	
EDFA NF THRESH **.**.dB	Sets an EDFA-NF analysis threshold.	0.01 to 50.00 (0.01 step)
EDFA NF MODE DIFF **.**.dB	Sets the minimum peak/bottom difference for channel detection during EDFA-NF analysis.	0.01 to 50.00 (0.01 step)
EDFA NF TRACE A OFST (IN) ***.**.dB	Sets the offset value of the signal light used in calculating NF and gain.	-99.99 to 99.99 (0.01 step)
EDFA NF TRACE A OFST (IN) @@@@	Sets the offset value of the signal light used in calculating NF and gain to variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z
EDFA NF TRACE B OFST (OUT) ***.**.dB	Sets the offset value of the output light used in calculating NF and gain.	-99.99 to 99.99 (0.01 step)
EDFA NF TRACE B OFST (OUT) @@@@	Sets the offset value of the output light used in calculating NF and gain to variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z
EDFA NF ASE ALGO AUTO FIX	Sets an ASE level algorithm to AUTO FIX.	
EDFA NF ASE ALGO MANUAL FIX	Sets an ASE level measuring algorithm to MANUAL FIX.	
EDFA NF ASE ALGO AUTO CTR	Sets an ASE level measuring algorithm to AUTO CTR.	
EDFA NF ASE ALGO MANUAL CTR	Sets an ASE level measuring algorithm to MANUAL CTR.	
EDFA NF ASE AREA **.**.nm	Sets an area used for ASE level analysis to a range centered on channel wavelength.	0.01 to 10.00 (0.01 step)
EDFA NF ASE AREA @	Sets an area used for ASE level analysis to a range centered on variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z
EDFA NF MASK AREA **.**.nm	Sets the spectral range of the signal light to be masked, centered on channel wavelength.	0.01 to 10.00 (0.01 step)
EDFA NF MASK AREA @	Sets the spectrum range of the signal light to be masked, centered on variable @@@@.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z
EDFA NF FITTING ALGO LINEAR	Places a fitting algorithm obtaining an ASE level in the linear interpolation mode.	
EDFA NF FITTING ALGO GAUSS	Places a fitting algorithm obtaining an ASE level in the Gauss distribution curve mode.	
EDFA NF FITTING ALGO LORENZ	Places a fitting algorithm obtaining an ASE level in the Lorenz curve mode.	
EDFA NF FITTING ALGO 3 RD POLY	Places a fitting algorithm obtaining an ASE level in the 3 rd polynomial mode.	
EDFA NF FITTING ALGO 4 TH POLY	Places a fitting algorithm obtaining an ASE level in the 4 th polynomial mode.	
EDFA NF FITTING ALGO 5 TH POLY	Places a fitting algorithm obtaining an ASE level in the 5 th polynomial mode.	
EDFA NF POINT DISPLAY ###	Displays a data range used for fitting on the waveform screen.	###: ON/OFF

Program Command	Description	Parameter Ranges and Available Variables
ANALYSIS 4/4		
FILTER (PEAK) ANALYSIS	Performs optical filter (PEAK) analysis.	
FILTER (BOTTOM) ANALYSIS	Performs optical filter (BOTTOM) analysis.	
WDM FILTER (PEAK) ANALYSIS	Performs multichannel-type optical filter (PEAK) analysis.	
WDM FILTER (BOTTOM) ANALYSIS	Performs multichannel-type optical filter (BOTTOM) analysis.	
SWITCH DISPLAY TO TRACE&TABLE	Displays both waveforms and tables in the display of analytical results.	
SWITCH DISPLAY TO TABLE	Displays only tables in the display of analytical results.	
SWITCH DISPLAY TO TRACE	Displays only traces in the display of analytical results.	
AUTO ANALYSIS ###	Selects ON/OFF of the waveform analysis function activated each time a sweep is made.	###: ON/OFF
ANALYSIS RESULT PRINT	Prints out analytical results.	
ANALYSIS RESULT WRITE HDD:'#####.DT5'	Saves analytical results to the hard disk with a file name specified by the user.	'#####.DT5' : File name
ANALYSIS RESULT WRITE FDD:'#####.DT5'	Saves analytical results to a floppy disk with a file name specified by the user.	'#####.DT5' : File name
ANALYSIS RESULT WRITE HDD	Saves analytical results to the hard disk. A file name will automatically be given.	
ANALYSIS RESULT WRITE FDD	Saves analytical results to a floppy disk. A file name will automatically be given.	
ANALYSIS RESULT WRITE HDD @@	Saves analytical results to the hard disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
ANALYSIS RESULT WRITE FDD @@	Saves analytical results to a floppy disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
SEARCH/ANA L1-L2 ###	Selects ON/OFF of the analysis function in an area enclosed by line markers 1 and 2.	###: ON/OFF
SEARCH/ANA ZOOM AREA ###	Selects ON/OFF of the analysis function in the display scale range.	###: ON/OFF
MEMORY		
SAVE #->MEMORY **	Writes the contents of a selected trace to the memory of a specified number.	00 to 63 (1 step) #: A,B,C,D,E,F,G
SAVE #->MEMORY @	Writes the contents of a selected trace to the memory of a variable-@ number.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z #: A,B,C,D,E,F,G
RECALL MEMORY **->#	Reads the contents of a selected trace from memory of a specified number.	00 to 63 (1 step) #: A,B,C,D,E,F,G
RECALL MEMORY @->#	Reads the contents of a selected trace from memory of a variable-@ number.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z #: A,B,C,D,E,F,G
CLEAR MEMORY **	Clears trace data in the memory.	00 to 63 (1 step)
CLEAR MEMORY @	Clears trace data in the memory specified by variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z

Program Command	Description	Parameter Ranges and Available Variables
FILE 1/3		
WRITE TRACE # HDD:'#####.***'	Saves specified trace data to the hard disk with a file name specified by the user.	#: A, B, C, D, E, F, G #####.***': File name
WRITE TRACE # FDD:'#####.***'	Saves specified trace data to a floppy disk with a file name specified by the user.	#: A, B, C, D, E, F, G #####.***': File name
WRITE TRACE # HDD	Saves specified trace data to the hard disk. A file name will automatically be given.	#: A,B,C,D,E,F,G
WRITE TRACE # FDD	Saves specified trace data to a floppy disk. A file name will automatically be given.	#: A,B,C,D,E,F,G
WRITE TRACE # HDD @@	Saves specified trace data to the hard disk under a file name specified by variable @@.	#: A,B,C,D,E,F,G @@: A\$,B\$,C\$,D\$
WRITE TRACE # FDD @@	Saves specified trace data to a floppy disk under a file name specified by variable @@.	#: A,B,C,D,E,F,G @@: A\$,B\$,C\$,D\$
TRACE WRITE: BINARY	Sets the data storage format to BINARY.	
TRACE WRITE: CSV	Sets the data storage format to CSV.	
WRITE MEMORY ** HDD:'#####.***'	Saves memory data to the hard disk with a file name specified by the user.	**: 00 to 63 (1 step) #####.***': File name
WRITE MEMORY ** FDD:'#####.***'	Saves memory data to a floppy disk with a file name specified by the user.	**: 00 to 63 (1 step) #####.***': File name
WRITE MEMORY ** HDD	Saves memory data to the hard disk. A file name will automatically be given.	**: 00 to 63 (1 step)
WRITE MEMORY ** FDD	Saves memory data to a floppy disk. A file name will automatically be given.	**: 00 to 63 (1 step)
WRITE MEMORY ** HDD @@	Saves memory data to the hard disk under a file name specified by variable @@.	**: 00 to 63 (1 step) @@: A\$,B\$,C\$,D\$
WRITE MEMORY ** FDD @@	Saves memory data to a floppy disk under a file name specified by variable @@.	**: 00 to 63 (1 step) @@: A\$,B\$,C\$,D\$
WRITE GRAPH HDD:'#####.***'	Saves graphic data to the hard disk with a file name specified by the user.	#####.***': File name
WRITE GRAPH FDD:'#####.***'	Saves graphic data to a floppy disk with a file name specified by the user.	#####.***': File name
WRITE GRAPH HDD	Saves graphic data to the hard disk. A file name will automatically be given.	
WRITE GRAPH FDD	Saves graphic data to a floppy disk. A file name will automatically be given.	
WRITE GRAPH HDD @@	Saves graphic data to the hard disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
WRITE GRAPH FDD @@	Saves graphic data to a floppy disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
GRAPH COLOR MODE: B&W	Sets the graphic color mode to black & white.	
GRAPH COLOR MODE: COLOR	Sets the graphic color mode to screen color mode.	
GRAPH TYPE: BMP	Sets the graphic file format to BMP.	
GRAPH TYPE: TIFF	Sets the graphic file format to TIFF.	
WRITE SETTING HDD:'#####.ST5'	Saves setting data to the hard disk with a file name specified by the user.	#####.ST5': File name
WRITE SETTING FDD:'#####.ST5'	Saves setting data to a floppy disk with a file name specified by the user.	#####.ST5': File name
WRITE SETTING HDD	Saves setting data to in the hard disk. A file name will automatically be given.	

Program Command	Description	Parameter Range and Available Variable
FILE 2/3		
WRITE SETTING FDD	Saves setting data to a floppy disk. A file name will automatically be given.	
WRITE SETTING HDD @@	Saves setting data to the hard disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
WRITE SETTING FDD @@	Saves setting data to a floppy disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
DATA:ADD WRITE	Specifies a data file for insert.	
DATA:OVER WRITE	Specifies a data file for overwrite.	
DATA WRITE CSV	Set storage data format to CSV	
DATA WRITE DT5	Set storage data format to DT5	
WRITE DATA HDD:'#####.DT5'	Saves data to the hard disk with a file name specified by the user.	'#####.DT5': File name
WRITE DATA FDD:'#####.DT5'	Saves data to a floppy disk with a file name specified by the user.	'#####.DT5': File name
WRITE DATA HDD	Saves data to the hard disk. A file name will automatically be given.	
WRITE DATA FDD	Saves data to a floppy disk. A file name will automatically be given.	
WRITE DATA HDD @@	Saves data to the hard disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
WRITE DATA FDD @@	Saves data to a floppy disk under a file name specified by variable @@.	@@: A\$,B\$,C\$,D\$
DATA DATE&TIME ###	Selects ON/OFF of date and time output.	###: ON/OFF
DATA LABEL ###	Selects ON/OFF of label output.	###: ON/OFF
DATA DATA AREA ###	Selects ON/OFF of data area output.	###: ON/OFF
DATA CONDITION ###	Selects ON/OFF of measuring conditions output.	###: ON/OFF
DATA TRACE DATA ###	Selects ON/OFF of waveform data output.	###: ON/OFF
DATA OUTPUT WINDOW ###	Selects ON/OFF of output of the Program Functions' OUTPUT WINDOW contents.	###: ON/OFF
READ TRACE # HDD:'#####.\$\$\$'	Reads specified trace data from the hard disk with a file name specified by the user.	#####.\$\$\$': File name #: A,B,C,D,E,F,G
READ TRACE # FDD:'#####.\$\$\$'	Reads specified trace data from a floppy disk with a file name specified by the user.	#####.\$\$\$': File name #: A,B,C,D,E,F,G
READ TRACE # HDD @@	Reads trace data of a file name specified by variable @@, from the hard disk.	#: A,B,C,D,E,F,G @@: A\$,B\$,C\$,D\$
READ TRACE # FDD @@	Reads trace data of a file name specified by variable @@, from a floppy disk.	#: A,B,C,D,E,F,G @@: A\$,B\$,C\$,D\$
READ MEMORY ** HDD:'#####.\$\$\$'	Reads memory data from the hard disk with a file name specified by the user.	#####.\$\$\$': File name 00 to 63 (1 step)
READ MEMORY ** FDD:'#####.\$\$\$'	Reads memory data from a floppy disk with a file name specified by the user.	#####.\$\$\$': File name 00 to 63 (1 step)
READ MEMORY ** HDD @@	Reads memory data of a file name specified by variable @@, from the hard disk	**: 00 to 63 (1 step) @@: A\$,B\$,C\$,D\$
READ MEMORY ** FDD @@	Reads memory data of a file name specified by variable @@, from a floppy disk	**: 00 to 63 (1 step) @@: A\$,B\$,C\$,D\$
READ SETTING HDD:'#####.\$\$\$'	Reads setting data from the hard disk with a file name specified by the user.	'#####.\$\$\$': File name
READ SETTING FDD:'#####.\$\$\$'	Reads setting data from a floppy disk with a file name specified by the user.	'#####.\$\$\$': File name

Program Command	Description	Parameter Ranges and Available Variables
FILE 3/3		
READ SETTING HDD @@	Reads setting data of a file name specified by variable @@, from the hard disk.	@@: A\$,B\$,C\$,D\$
READ SETTING FDD @@	Reads setting data of a file name specified by variable @@, from a floppy disk.	@@: A\$,B\$,C\$,D\$
READ DATA HDD:'#####.\$\$\$'	Reads data from the hard disk with a file name specified by the user.	'#####.\$\$\$': File name
READ DATA FDD:'#####.\$\$\$'	Reads data from a floppy disk with a file name specified by the user.	'#####.\$\$\$': File name
READ DATA HDD @@	Reads data of a file name specified by variable @@, from the hard disk.	@@: A\$,B\$,C\$,D\$
READ DATA FDD @@	Reads data of a file name specified by variable @@, from a floppy disk.	@@: A\$,B\$,C\$,D\$
READ TEMPLATE FDD:'#####.\$\$\$'	Reads template data from a floppy disk with a file name specified by the user.	'#####.\$\$\$': File name
READ TARGET LINE FDD:'#####.\$\$\$'	Reads target line data from a floppy disk with a file name specified by the user.	'#####.\$\$\$': File name
DELETE HDD:'#####.\$\$\$'	Deletes a file in the hard disk.	'#####.\$\$\$': File name
DELETE FDD:'#####.\$\$\$'	Deletes a file in a floppy disk.	'#####.\$\$\$': File name
DELETE HDD @@	Deletes a variable @@-specified file in the hard disk.	@@: A\$,B\$,C\$,D\$
DELETE FDD @@	Deletes a variable @@-specified file in a floppy disk.	@@: A\$,B\$,C\$,D\$
RENAME HDD:## @@	Renames the name of a file specified by variable ## to a file name specified by variable @@ in the hard disk.	##, @@: A\$,B\$,C\$,D\$
RENAME FDD:## @@	Renames the name of a file specified by variable ## to a file name specified by variable @@ in a floppy disk.	##, @@: A\$,B\$,C\$,D\$

Program Command	Description	Parameter Ranges and Available Variables
ADVANCE		
TEMPLATE GO/NO GO ##	Sets if GO/NO-GO judgment is made.	###: ON/OFF
TEMPLATE DISPLAY ###	Sets ON/OFF of template display. ON:UPPER LINE=ON LOWER LINE=ON TARGET LINE=ON	###: ON/OFF
TEMPLATE DISPLAY UPPER ###	Sets ON/OFF of upper line display.	###: ON/OFF
TEMPLATE DISPLAY LOWER ###	Sets ON/OFF of lower line display.	###: ON/OFF
TEMPLATE DISPLAY TARGET ###	Sets ON/OFF of target line display.	###: ON/OFF
TMPLATE TEST TYPE UPPER	Sets if GO/NO-GO judgment at the upper line is made.	
TMPLATE TEST TYPE LOWER	Sets if GO/NO-GO judgment at the lower line is made.	
TMPLATE TEST TYPE UP & LOW	Sets if GO/NO-GO judgment at the upper and lower lines is made.	
TMPLATE WL SHIFT ****.***nm	Sets the amount of wavelength shift of the template.	-999.999 to 999.999 (0.001 step)
TEMPLATE LEVEL SHIFT ***.**dB	Sets the amount of level shift of the template.	-99.99 to 99.99 (0.01 step)

Program Command	Description	Parameter Ranges and Available Variables
SYSTEM 1/2		
OPTICAL ALIGNMENT	Aligns the optical axis of a monochromator optical system.	
SELF WL CALIBRATION	Sets the light source to be wavelength calibrated for the internal light source.	
EXT WL CALIBRATION ****.***nm	Sets the light source to be wavelength calibrated for an external light source (laser type).	600.000 to 1700.000 (0.001 step)
EXT-GAS WL CALIBRATION ****.***nm	Sets the light source to be wavelength calibrated for an external light source (gas cell type).	600.000 to 1700.000 (0.001 step)
WL SHIFT **.***nm	Sets the amount of wavelength shift.	-5.000 to 5.000 (0.001 step)
LEVEL SHIFT ***.***dB	Sets the amount of level shift.	-60.000 to 60.000 (0.001 step)
SYSTEM GRID 200GHz	Sets system grid to a 200 GHz spacing grid table.	
SYSTEM GRID 100GHz	Sets system grid to a 100 GHz spacing grid table.	
SYSTEM GRID 50GHz	Sets system grid to a 50 GHz spacing grid table.	
SYSTEM GRID 25GHz	Sets system grid to a 25 GHz spacing grid table.	
SYSTEM GRID 12.5GHz	Sets system grid to a 12.5 GHz spacing grid table.	
CUSTOM GRID START WL ****.***nm	Inputs the starting wavelength of a user grid table.	1000.0000 to 1700.000 (0.0001 step)
CUSTOM GRID START FREQ ***.***THz	Inputs the starting frequency of a user grid table.	176.34851to 299.7924 (0.0001 step)
CUSTOM GRID STOP WL ****.***nm	Inputs the stop wavelength of a user grid table.	1000.0000 to 1700.000 (0.0001 step)
CUSTOM GRID STOP FREQ ***.***THz	Inputs the stop frequency of a user grid table.	176.34851to 299.7924 (0.0001 step)
CUSTOM GRID SPACING ***.***GHz	Inputs the grid spacing of a user grid table.	0.1 to 999.9 (0.1 step)
GRID REFERENCE WL ****.***nm	Inputs the reference wavelength of a user grid table.	1000.0000 to 1700.000 (0.0001 step)
GRID REFERENCE FREQ ***.***THz	Inputs the reference frequency of a user grid table.	176.34851to 299.79245 (0.0001 step)
REMOTE INTERFACE:GP-IB	Sets remote interface to GP-IB.	
REMOTE INTERFACE:RS-232C	Sets remote interface to RS-232C.	
REMOTE INTERFACE:ETHERNET	Sets remote interface to ETHERNET.	
TLS ADDRESS **	Sets the address of the Tunable Laser Source.	0 to 30 (1 step)
SELECT COLOR *	Selects the display color of the screen.	1 to 5 (1 step)
UNCAL WARNING DISPLAY ###	Displays UNCAL and warning.	
BUZZER CLICK ###	Sets whether the buzzer clicks when a key is pressed.	###: ON/OFF
BUZZER WARNING ###	Sets whether a buzzer sounds in the event of a warning error.	###: ON/OFF
LEVEL DISPLAY DIGIT *	Sets the number of display digits (decimal places) of level data displayed for an area marker and the analysis results.	1 to 3 (1 step)

Program Command	Description	Parameter Ranges and Available Variables
SYSTEM 2/2		
WINDOW TRANSPARENT ###	Selects ON/OFF of the function of displaying interrupt display and overview display windows in a semi-transparent display.	###: ON/OFF
AUTO OFFSET ###	Selects ON/OFF of the auto offset function.	###: ON/OFF

1.5.5 Lists of Special Commands

Table 1-5 Lists of Special Commands

Program Command	Description	Parameter Ranges and Available Variables
General Commands		
COPY ON	Produces a hard copy of the screen on a printer.	
PRINTER FEED **	Feeds printer paper.	1 to 10 (1 step)
GOTO ***	Makes a jump to line ***.	1 to 200 (1 step)
GOTO PROGRAM **	Makes a jump to program ** to run it from the first line. After completing running of program **, control returns to the original program. However, if there is an END command in program **, return to the jump source is not performed and the program ends. Note that when a program is executed using this command, variables are not initialized.	
WAIT *****S	Makes a wait of ***** seconds.	1 to 99999 (1 step)
PAUSE '---56 characters ---'	Pauses execution of a program and causes a message window to appear. This window displays a message in ' ' and an explanation of the <CONTINUE> key. Pressing the <CONTINUE> soft key closes the window and resumes the program. Note if a program is started via GPIB, no pause is made.	
VARIABLE CLEAR	Initializes all variables used in a program.	
END	Ends a program.	
INIT	Initializes all parameters, but does not clear variables.	
@=VAL(@\$)	Converts character variables @\$ to numerics and substitutes them into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
BEEP **	Sounds the buzzer for ** × 100 msec.	1 to 10 (1 step)

Program Command	Description	Parameter Ranges and Available Variables
Loop Control		
N=*****	Substitutes a value into variable N.	1 to 99999999 (1 step)
N=@@@@@@	Copies the contents of variable @@@@@@ to variable N.	@@@@@: MODN,WDMCHN,NFCHN,M
N·N-1;IF N<>0 GOTO ***	Subtracts "1" from variable N and, if the result is not "0," makes a jump to line ***.	1 to 200 (1 step)
M=*****	Substitutes a value into variable M.	1 to 99999999 (1 step)
M=@@@@@@	Copies the contents of variable @@@@@@ to variable M.	@@@@@: MODN,WDMCHN,NFCHN,N
M·M-1;IF M<>0 GOTO ***	Subtracts "1" from variable M and, if the result is not "0," makes a jump to line ***.	1 to 200 (1 step)

Program Command	Description	Parameter Ranges and Available Variables
Variable Computation		
@ = ***** ###	Substitutes a value into variable @. For ***** , a real number of 10 or fewer digits can be specified, including a sign and the decimal point.	-999999999 to 999999999 (1 step) @: G,H,I,J,K,P,Q,R,S,X,Y,Z,CH ###: nm, dB, dBm, pW, nW, μW, mW, W, ' '(without units)
@ = # + *****	Adds value ***** to variable # and substitutes the result into variable @. For ***** , a real number of 10 or fewer digits can be specified, including a sign and the decimal point. Setting a negative value causes a subtraction to be made from variable #.	-999999999 to 999999999 (1 step) @, #: G,H,I,J,K,P,Q,R,S,X,Y,Z,CH
@ = @@@@	Copies the contents of variable @@@@ to variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,CH @@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1, W(CH),LM,L1,L2,L2L1,L(CH),SPWD, MEANWL,PKWL,PKLVL,MODN,GONO, SMSR,WDMCHN,WDMWL(CH),WDMVL L(CH), WDMSNR(CH),FNCHN,NFWL(CH), NFLVL(CH),NFLVI(CH),NFLVO(CH), NFASLV(CH),NFGAIN(CH),NFNF(CH), MKPWR,PMD,M,N,CH
@ = @@@@ + #####	Performs addition, subtraction, multiplication, and/or division between variables.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z,CH @@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1, W(CH),LM,L1,L2,L2L1,L(CH),SPWD, MEANWL,PKWL,PKLVL,MODN,GONO, SMSR,WDMCHN,WDMWL(CH),WDMVL L(CH), WDMSNR(CH),FNCHN,NFWL(CH), NFLVL(CH),NFLVI(CH),NFLVO(CH), NFASLV(CH),NFGAIN(CH),NFNF(CH), MKPWR,PMD,M,N,CH
@ = @@@@ - #####		
@ = @@@@ * #####		
@ = @@@@ / #####		
@\$ = @\$	Copies the contents of string variable @\$ to string variable @\$.	@\$: A\$,B\$,C\$,D\$
@\$ = MID(@\$,@,@)	Substitutes form numeric variable @ separated from head of string variable @\$ to string for numeric variable @ into string variable @\$.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z @\$: A\$,B\$,C\$,D\$
@\$ = '---56 characters ---'	Substitutes a character string into string variable @\$.	@\$: A\$,B\$,C\$,D\$

Program Command	Description	Parameter Ranges and Available Variables
Printer Output		
PRINT '---56 characters ---'	Prints out a character string in ' '. If a semicolon ";" is appended to the end of the string, no line feed is made after printing, but a character string or the variable values specified by the next PRINT command are printed out successively.	
PRINT @@@@	Prints out the value of variable @@@@ with units appended.	@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1, W(CH),LM,L1,L2,L2L1,L(CH),SPWD, MEANWL,PKWL,PKLVL,MODN,GONO,SMS R,WDMCHN,WDMWL(CH),WDMLVL(CH), WDMSNR(CH),FNCHN,NFWL(CH), NFLVL(CH),NFLVI(CH),NFLVO(CH), NFASELV(CH),NFGAIN(CH),NFNF(CH), MKPWR,PMD,M,N,CH,A\$,B\$,C\$,D\$,FILE\$
PRINT @@@@@;	Prints out the value of variable @@@@@ with units appended. After printing, no line feed is made, but a character string or the variable values specified by the next PRINT command are printed out successively.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1, W(CH),LM,L1,L2,L2L1,L(CH),SPWD, MEANWL,PKWL,PKLVL,MODN,GONO,SMS R,WDMCHN,WDMWL(CH),WDMLVL(CH), WDMSNR(CH),FNCHN,NFWL(CH), NFLVL(CH),NFLVI(CH),NFLVO(CH), NFASELV(CH),NFGAIN(CH),NFNF(CH), MKPWR,PMD,M,N,CH,A\$,B\$,C\$,D\$,FILE\$
PRINT DATA AREA	Prints out the contents of the data area.	
PRINT OUTPUT WINDOW	Prints out the contents of OUTPUT WINDOW.	

Program Command	Description	Parameter Ranges and Available Variables
Condition Judgment		
IF F1 <= @@@@ <= F2 GOTO ***	If the value of variable @@@@ falls between F1 and F2, makes a jump to line ***.	1 to 200 (1 step) @@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1,W(CH),LM,L1,L2,L2L1,L(CH),SPWD,MEANWL,PKWL,PKLVL,MODN,GONO,SMSR,WDMCHN,WDMWL(CH),WDMLVL(CH),WDMSNR(CH),FNCHN,NFWL(CH),NFLVL(CH),NFLVI(CH),NFLVO(CH),NFASELV(CH),NFGAIN(CH),NFNF(CH),MKPWR,PMD,M,N,CH,A\$,B\$,C\$,D\$,FILE\$
F1 = ***** ###	Substitutes a value into variable F1. For ***** , a real number of 10 or fewer digits can be specified, including a sign and decimal point.	-999999999 to 999999999 (1 step) ###: nm, dB, dBm, pW, nW, μW, mW, W, ' ' (without units)
F2 = ***** ###	Substitutes a value into variable F2. For ***** , a real number of 10 or fewer digits can be specified, including a sign and decimal point.	-999999999 to 999999999 (1 step) ###: nm, dB, dBm, pW, nW, μW, mW, W, ' ' (without units)
F1 = @@@@@	Copies the contents of variable @@@@@ to variable F1.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1,W(CH),LM,L1,L2,L2L1,L(CH),SPWD,MEANWL,PKWL,PKLVL,MODN,GONO,SMSR,WDMCHN,WDMWL(CH),WDMLVL(CH),WDMSNR(CH),FNCHN,NFWL(CH),NFLVL(CH),NFLVI(CH),NFLVO(CH),NFASELV(CH),NFGAIN(CH),NFNF(CH),MKPWR,PMD,M,N,CH
F2 = @@@@@	Copies the contents of variable @@@@@ to variable F2.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1,W(CH),LM,L1,L2,L2L1,L(CH),SPWD,MEANWL,PKWL,PKLVL,MODN,GONO,SMSR,WDMCHN,WDMWL(CH),WDMLVL(CH),WDMSNR(CH),FNCHN,NFWL(CH),NFLVL(CH),NFLVI(CH),NFLVO(CH),NFASELV(CH),NFGAIN(CH),NFNF(CH),MKPWR,PMD,M,N,CH
@ = LEVEL(****.***nm)	Substitutes the level of the point of wavelength ****.*** nm on an active trace into variable @.	600.000 to 1700.000 (0.1 step) @: G,H,I,J,K,P,Q,R,S,X,Y,Z
@ = LEVEL(@@@@@)	Substitutes the level of the point of wavelength @@@@@ (variable) on an active trace into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z @@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W(CH),MEANWL,PKWL,WDMLVL(CH),WDMWL(CH),NFWL(CH)
IF @@@@@ < @@@@@ GOTO ***	Compares the large and small relationship of two variables and if the conditions are met, makes a jump to line ***.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1,W(CH),LM,L1,L2,L2L1,L(CH),SPWD,MEANWL,PKWL,PKLVL,MODN,GONO,SMSR,WDMCHN,WDMWL(CH),WDMLVL(CH),WDMSNR(CH),FNCHN,NFWL(CH),NFLVL(CH),NFLVI(CH),NFLVO(CH),NFASELV(CH),NFGAIN(CH),NFNF(CH),MKPWR,PMD,M,N,CH
IF @@@@@ <= @@@@@ GOTO ***		
IF @@@@@ = @@@@@ GOTO ***		
IF @@@@@ > @@@@@ GOTO ***		

Program Command	Description	Parameter Ranges and Available Variables
External Control 1/2		
SEND **---56 characters --- '	Specifies the external device of address ** connected to the [GP-IB2] connector as a listener and sends a command in ' ' to it. The delimiter is CR/LF.	0 to 30 (1 step)
SEND **---56 characters --- '@	Specifies the external device of address ** connected to the [GP-IB2] connector as a listener and sends the command in ' ' and then the value of variable @. The delimiter is CR/LF.	0 to 30 (1 step) @: G,H,I,J,K,P,Q,R,S,X,Y,Z
SEND **---20 characters --- '; @;'---20 characters ---'	Specifies the external device of address ** connected to the [GP-IB2] connector as a listener and sends the command in ' ', the value of variable @, and then the command in ' '. The delimiter is CR/LF.	0 to 30 (1 step) @: G,H,I,J,K,P,Q,R,S,X,Y,Z
SEND RS232C '---56 characters---'	The external device connected to the [RS-232C] connector as a listener and sends the command in ' ' to it. The delimiter depend on the setting of SET DELIMITER.	
SEND RS232C '---56 characters---'@	The external device connected to the [RS-232C] connector as a listener and sends the command in ' ' and then the value of variable @. The delimiter depend on the setting of SET DELIMITER.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
SEND RS232C '---20 characters---'@; '---20 characters---'	The external device connected to the [RS-232C] connector as a listener and sends the command in ' ', the value of variable @, and then the command in ' '. The delimiter depend on the setting of SET DELIMITER.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
SEND LAN @\$, '*****' ; '---56 characters ---' ※@\$: Computer Name or IP address **** : Port Number	Specified the external device of "Computer Name" or "IP address" and "Port Number" connected to the [LAN] connector as a listener and sends the command in ' ' to it. The delimiter depend on the setting of SET DELIMITER.	Port Number : 1024~65535 @ \$: A\$,B\$,C\$,D\$
SEND LAN @\$, '***** ' ; '---56 characters ---'@; ※@\$: Computer Name or IP address **** : Port Number	Specified the external device of "Computer Name" or "IP address" and "Port Number" connected to the [LAN] connector as a listener and sends the command in ' ' and then the value of variable @. The delimiter depend on the setting of SET DELIMITER.	Port Number : 1024~65535 @:G,H,I,J,K,P,Q,R,S,X,Y,Z @ \$: A\$,B\$,C\$,D\$
SEND LAN @\$, '***** ' , '--- 20 characters ---' '@; '---20 characters ---' ※@\$: Computer Name or IP address **** : Port Number	Specified the external device of "Computer Name" or "IP address" and "Port Number" connected to the [LAN] connector as a listener and sends the command in ' ', the value of variable @, and then the command in ' '. The delimiter depend on the setting of SET DELIMITER.	Port Number : 1024~65535 @:G,H,I,J,K,P,Q,R,S,X,Y,Z @ \$: A\$,B\$,C\$,D\$
RECEIVE **;@\$	Specifies the external device of address ** connected to the [GP-IB2] connector as a talker to receive a message and substitutes the received message into character variable @\$. Up to 512 characters can be received. The delimiter is CR/LF.	0 to 30 (1 step)

Program Command	Description	Parameter Ranges and Available Variables
External Control 2/2		
SEND RS-232C ‘—56characters—’;@\$	The external device connected to the [RS-232C] connector to sends the query command in ' ' and receive a message and substitutes the received message into character variable @\$. Up to 512 characters can be received. The delimiter depend on the setting of SET DELIMITER.	@ \$: A\$,B\$,C\$,D\$
SEND LAN @\$, ‘*****’,—56characters—’; @\$ ※@\$: Computer Name or IP address ***** : Port Number	Specified the external device of "Computer Name" or "IP address" and "Port Number" connected to the [LAN] connector to sends the query command in ' ' and receive a message and substitutes the received message into character variable @\$. Up to 512 characters can be received. The delimiter depend on the setting of SET DELIMITER.	Port Numeber : 1024~65535 @ \$: A\$,B\$,C\$,D\$
RESET OPTION	Performs remote cancellation, device clearing, and interface clearing with respect to the external device connected to the [GP-IB2] connector.	
SOLL **;S	Serial-polls the external device of address** connected to the [GP-IB2] connector and substitutes a status byte into variable S.	0 to 30 (1 step)
SET DELIMITER ###	Set the delimiter which the unit sending / Receiving remote commands for the external device connected to the [RS-232C] and [LAN] connector.	###:CR,LF,CR+LF
Substitution of Measuring Condition		
@ = CENTER	Substitutes the current measurement center wavelength into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
@ = SPAN	Substitutes the current sweep width into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
@ = REF LEVEL	Substitutes the current reference level into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
@ = RESOLUTION	Substitutes the current measuring resolution into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
@ = SAMPLING POINT	Substitutes the current sampling number into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
@ = ZOOM CENTER	Substitutes the current center display wavelength into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z
@ = ZOOM SPAN	Substitutes the current display width into variable @.	@: G,H,I,J,K,P,Q,R,S,X,Y,Z

Program Command	Description	Parameter Ranges and Available Variables
User Inputs/Outputs		
DATA INPUT '---56 characters ---';@	Pauses execution of a program and prompts the user to enter a numeric or a character string into variable @. On the screen the Input Window appears, displaying a character string in '. The window is in a numeric input state when variable @ is a numeric, or is in string input state when the variable is a character string.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,A\$,B\$,C\$,D\$
DATA OUTPUT '---56 characters ---'	Outputs a character string in ' ' to OUTPUT WINDOW. If a semicolon ";" is appended to the end of the string, no line feed is made after output of the string, but a character string or the variable values specified by the next DATA OUTPUT command are output successively.	
DATA OUTPUT @@@@@	Outputs the value of variable @@@@@ to OUTPUT WINDOW with units assigned.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1,W(CH),LM,L1,L2,L2L1,L(CH),SPWD,MEANWL,PKWL,PKLVL,MODN,GONO,SMSR,WDMCHN,WDMWL(CH),WDMVL(CH),WDMSNR(CH),FNCHN,NFWL(CH),NFLVL(CH),NFLVI(CH),NFLVO(CH),NFASLV(CH),NFGAIN(CH),NFNF(CH),MKPWR,PMD,M,N,CH,A\$,B\$,C\$,D\$,FILE\$
DATA OUTPUT @@@@@;	Outputs the value of variable @@@@@ to OUTPUT WINDOW with units assigned. After output of the string, no line feed is made, but a character string or the variable values specified by the next DATA OUTPUT command are output successively.	@@@@@: G,H,I,J,K,P,Q,R,S,X,Y,Z,WM,W1,W2,W2W1,W(CH),LM,L1,L2,L2L1,L(CH),SPWD,MEANWL,PKWL,PKLVL,MODN,GONO,SMSR,WDMCHN,WDMWL(CH),WDMVL(CH),WDMSNR(CH),FNCHN,NFWL(CH),NFLVL(CH),NFLVI(CH),NFLVO(CH),NFASLV(CH),NFGAIN(CH),NFNF(CH),MKPWR,PMD,M,N,CH,A\$,B\$,C\$,D\$,FILE\$
DATA OUTPUT DATA AREA	Outputs the contents of the data area to OUTPUT WINDOW.	
OUTPUT WINDOW CLEAR	Clears the contents of OUTPUT WINDOW.	
OUTPUT WINDOW ###	Sets whether to display OUTPUT WINDOW on the screen.	###: ON or OFF

1.6 Remote control with program function.

Using the program function, the unit can remote control the external device which is connected various interfaces. In addition, it is possible to remote control the multiple external device by one program source.

1.6.1 Remote control using [GP-IB] port

Using the program function, the unit can send remote commands and receive talker data and serial polling for the external device which connected [GP-IB] port.

Also it is possible to remote control which specifies the external device of address ** connected to the [GP-IB2] connector.

GP-IB address does not same to setting of <GP-IB2 PORT ADDRESS>key.

In such case, the unit cannot remote control the external device.

○ Sending command

SEND **1) 'control command / query command(56 characters)'

SEND **1) 'control command / query command (56 characters)';@

SEND **1) 'control command / query command (20 characters)';@;

control command / query command(20 characters)'

○ Receiving command

RECEIVE **1);@\$

1) ** = GP-IB command



Caution

- A controller such as a PC that is connected to the GP-IB2 port cannot remotely control the AQ6319.
- Even if an external device to be controlled by the AQ6319 using program functions or a Tunable Laser Source is connected to the GP-IB1 port, it cannot remote control the AQ6319.
- The two GP-IB ports are independent of each other. Thus, a controller connected to the GP-IB1 port cannot directly send a message to an external device connected to the GP-IB2 port.
- Also, with a controller connected to the GP-IB1 port, connecting the GP-IB1 port and the GP-IB2 port results in improper operations.
- Do not connect a cable between these ports, or press [SYSTEM] and then <GP-IB SETTING> to set <SYSTEM CONTROLLER> to OFF. The default is ON.

(☛ Section 6.16, SYSTEM in the Instruction Manual)

1.6.2 Remote control using [RS-232C] port

Using the program function , the unit can send remote commands and receive talker data and serial polling for the external device which connected [RS-232C] port.

In this case , please connect a cross cable to the RS-232C interface at the back side of the unit.

The parameters of the RS-232C port can be set by pressing the [SYSTEM] switch and then the <RS-232C SETTING> key.

(☛ Section 6-16 SYSTEM in the Instruction Manual)

If you want to receive query data for the external device , use the sending / receiving command. A query data is stored string variable @\$ which specified.

○ Sending command

SEND RS-232C 'control command(56 characters)'

SEND RS-232C 'control command (56 characters)';@

SEND RS-232C 'control command (20 characters)';@;

control command(20 characters)'

○ Sending / Receiving command

SENDRCV RS-232C 'query command(56 characters)';@\$

NOTE

- The unit should be change the setting of delimiter by the external device.
- If the setting for the delimiter is changed , use the SET SEND DELIMITER command in special commands.
(Default : CR+LF)

1.6.3 Remote control using [LAN] port

Using the program function , specified the external device of "Computer Name" or "IP address" and "Port Number" connected to the [LAN] connector to perform remote control.

"Computer Name" or " IP address" must be entered it in the character variable @\$ of program command.

If you want to recieve query data for the external device , use the sending / recieving command. A query data is stored string variable @\$ which specified.

○ Sending command

SEND LAN @\$¹⁾ ****²⁾ 'control command(56 characters)'

SEND LAN @\$¹⁾ ****²⁾ 'control command(56 characters)';@

SEND LAN @\$¹⁾ ****²⁾ 'control command(20 characters)';@;
control command(20 characters)'

○ Sending / Receiving command

SENDRCV LAN @\$¹⁾ ****²⁾ 'quert command(56characters)'

1) @\$ = Computer Name or IPaddress

2) **** = Port Number

注意 (CAUTION)

- It is necessary to set up the IP address of the unit correctly so that the remote control with [LAN] port.
- If a DHCP server is provided on the network to which this unit is connected, the IP address given to the unit is automatically set. Thus, set the item IP ADDRESS SETTING in the [SYSTEM] →<NETWORK SETTING>→<TCP/IP SETTING> to "☑AUTO (DHCP)".
- Please ask the network administrator of your company about the details of a network.

NOTE

- The unit should be change the setting of delimiter by the external device.
- If the setting for the delimiter is changed , use the SET SEND DELIMITER command in special commands.
(Default : CR+LF)

1.7 Sample Program

- (1) After the measuring conditions have been set, the program performs a single sweep. Then it searches for a spectrum width and peak wavelength to output the results to the label area and OUTPUT WINDOW. It repeats these operations 10 times with a wait of 3 sec. between repetitions.

001	CENTER WL 1550.00nm	}	Set the measuring conditions.
002	SPAN 10.0nm		
003	REFERENCE LEVEL -10.0dBm		
004	RESOLUTION 0.1nm		
005	AVERAGE TIMES 1		
006	SENS NORMAL/HOLD	}	Clear data in OUTPUT WINDOW. Display OUTPUT WINDOW.
007	OUTPUT WINDOW CLEAR		
008	OUTPUT WINDOW ON		
009	N=10		Set "10" to loop counter "N."
010	SINGLE		Perform a single sweep.
011	SPEC WD THRESH 20.0dB		Search for spectrum width.
012	DATA OUTPUT 'Wd = ;	}	Output spectrum width to OUTPUT WINDOW and the label area.
013	LABEL 'Wd = ;		
014	DATA OUTPUT SPWD;		
015	LABEL SPWD ;	}	Search for peak wavelength.
016	PEAK SEARCH		
017	DATA OUTPUT 'Pk = ;		
018	LABEL 'Pk = ;	}	Output the peak wavelength value to OUTPUT WINDOW and the label area.
019	DATA OUTPUT PKWL		
020	LABEL PKWL		
021	WAIT 3S		Cause a wait of 3 sec.
022	N=N-1 ; IF N <> 0 GOTO 10		Subtract "1" from loop counter "N" and if the result is not "0," make a jump to line 010.
023	END		Exit the program.

When the above program is executed, measured spectrum width and peak wavelength values are output to the OUTPUT WINDOW and level area as shown in Figure 1-23.

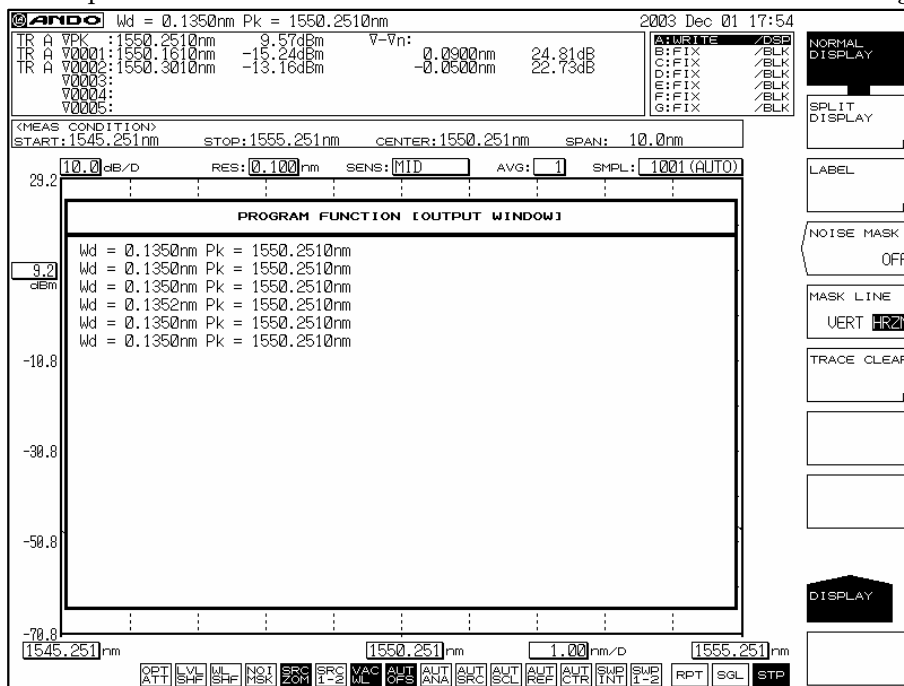


Figure 1-23: Screen for Running the Sample Program (1)

- (2) The following program specifies an external device connected to the [GP-IB 2] connector as a listener to send a device message and then specifies the device as a talker to receive data from it.

Received data is displayed on the OUTPUT WINDOW. The program repeats these operations 10 times.

001	OUTPUT WINDOW CLEAR	Clear OUTPUT WINDOW data.
002	OUTPUT WINDOW ON	Display OUTPUT WINDOW.
003	N=10	Set "10" to loop counter "N."
004	SEND 1'B,C1,E1,H1,S	Specify the external device of address 1 as a listener to send the contents of '' to it.
005	WAIT 1S	Cause a wait of 1 sec.
006	RECEIVE 1;A\$	Specify the device of address 1 as a talker to receive data from it and substitute received data into character variable A\$.
007	DATA OUTPUT @@@@	Output the contents of A\$ to OUTPUT WINDOW.
008	N=N-1;IF N<>0 GOTO 4	Subtract "1" from loop counter "N" and if the result is not "0," make a jump to line 004.
009	END	Exit the program.

When the noted program is run, data received from the external device is output to the OUTPUT WINDOW as shown in Figure 1-24.

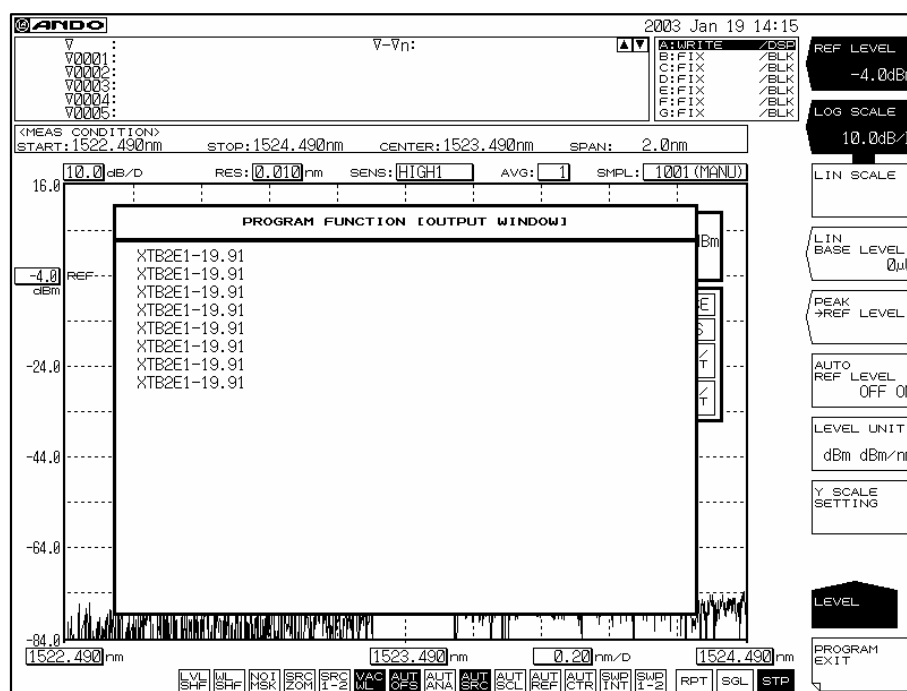


Figure 1-24: Screen for Running the Sample Program (2)

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2.1 General

The AQ6319 has two GPIB ports and one RS-232C port, allowing remote control using an external PC.

Moreover, this unit itself may become a controller to remote control other GPIB devices.

● GP-IB1 (IEEE 488.2)

This port is used to connect a controller such as a PC for remote controlling this unit. Connect a controller or another device controlled by the controller to this port.

(☛ Section 2.3, GP-IB1 Port)

This unit is controlled using remote commands.

Two types of remote commands are provided: this unit's native commands complying with SCPI (Standard Commands for Programmable Instruments) and commands compatible with the conventional model AQ6317.

(☛ Section 2.7, AQ6317 Compatible Mode)

● GP-IB2 (IEEE 488.1)

This port is used when this unit acts as a controller to remote control an external device. Connect the external device controlled by this unit's program functions to this port.

(☛ Section 2.8, GP-IB2 Port)

● RS-232C

This port is used to connect a controller such as a PC to remote control this unit.

(☛ Section 2.9, RS-232C Port)

● ETHERNET

This port is used to connect a controller such as a PC to remote control this unit.

(☛ Section 2.10, LAN Port)



Caution

- A controller such as a PC that is connected to the GP-IB2 port cannot remote control this unit.
- Even if an external device to be controlled by this unit using a Tunable Laser Source or program functions is connected to the GP-IB1 port, it cannot remote control this unit.
- The two GP-IB ports are independent of each other. Thus, a controller connected to the GP-IB1 port cannot directly send a message to an external device connected to the GP-IB2 port.
- Also, with a controller connected to the GP-IB1 port, connecting the GP-IB1 port and the GP-IB2 port results in improper operations.
- Do not connect a cable between these ports or press [SYSTEM] and then <GP-IB SETTING> to set <SYSTEM CONTROLLER> to OFF. The default is ON. (☛ Section 6.16, SYSTEM in the Instruction Manual)

2.2 Selecting the Remote Interface

To remote control the AQ6319 using an external PC, etc., it is necessary to pre-select the remote interface to use.

The interface to be used for controlling this unit can be selected from the following two types:

GP-IB1 port
RS-232C port
ETHERNET port

To select the interface, follow the procedure below:

- ① Press the [SYSTEM] switch.
- ② Press the <MORE 1/4> key and then the <REMOTE INTERFACE> key.
- ③ Use the soft key to select the interface to use for remote control.

GP-IB	The GP-IB1 port is used.
RS-232C	The RS-232C port is used.
ETHERNET	The LAN port is used.

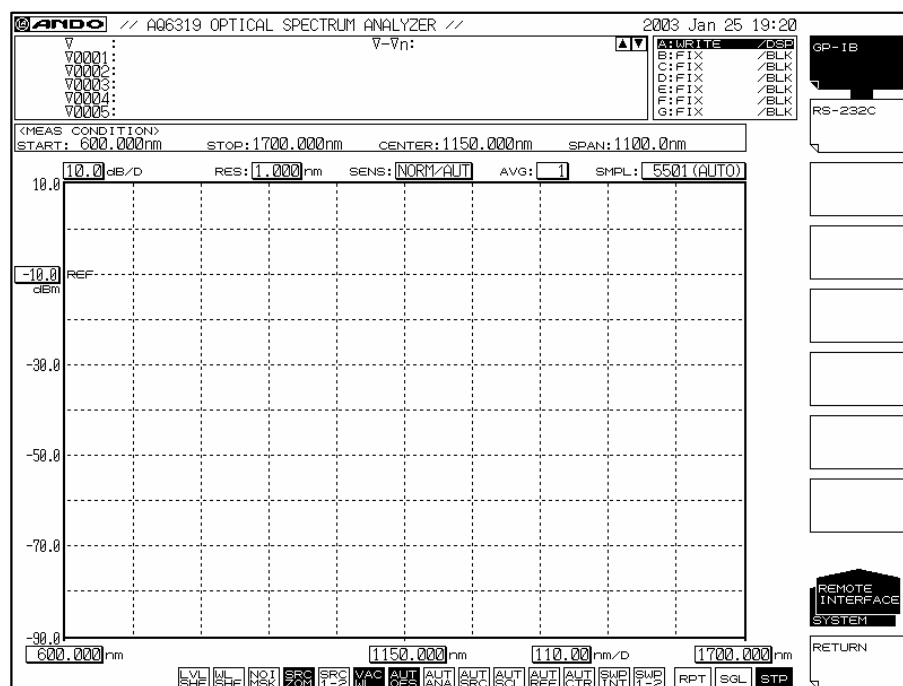


Figure 2-1: Screen for Selecting the Remote Interface

2.3 GP-IB1 Port

2.3.1 GP-IB1 Port's Interface Functions

Table 2-1 shows the interface functions of the GP-IB1 port.

Table 2-1: GP-IB1 Port's Interface Functions

Code	Interface Function
SH1	All capabilities of send function
AH1	All capabilities of receive function
T6	Basic talker talk and serial polling functions
L4	Basic listener listen function
SR1	Full service request functions
RL1	Full local lockout functions
PP0	Parallel polling function not provided
DC1	Full device clear functions
	Output buffer clear
	Input buffer clear (clearing of an unexecuted command)
	Error buffer clear
	Clearing of STB's and clear ESR
DT1	Device trigger function
C0	Controller function not provided

2.3.2 Connecting to GPIB System

Turn OFF all the power switches of this unit and devices (such as the controller).

Using the GPIB cable, connect the GP-IB1 connector on the rear of this unit to another device. Make sure to tighten the connector-fixing screws firmly.

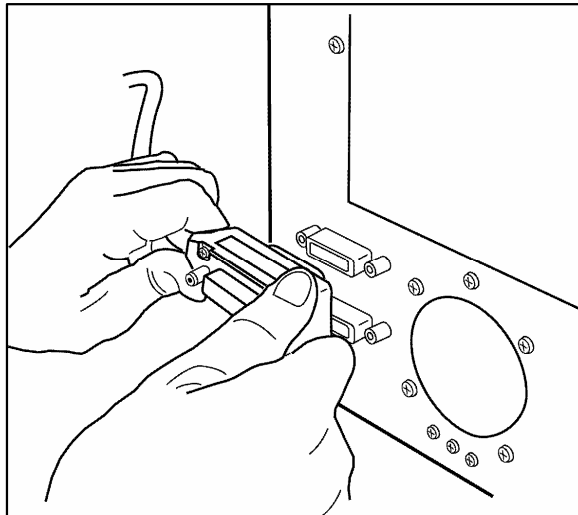


Figure 2-2: Connecting the GPIB Connector

⚠ Caution

- Connecting/disconnecting the GPIB cable with the power switch turned ON may cause the AQ6319 to malfunction or fail.

2.3.3 Setting the GPIB Address

To set the GP-IB1 port address of the AQ6319, press the [SYSTEM] switch and then the <MY ADDRESS> key. The default of port address is 1

When you press the soft key, the current address appears in the Interrupt Display Area on the screen. Use the rotary knob, step keys, numeric keypad, mouse, keyboard, or other interfaces to set the address in the range of 1 to 30. A setting value is saved at the built-in HDD of this unit.

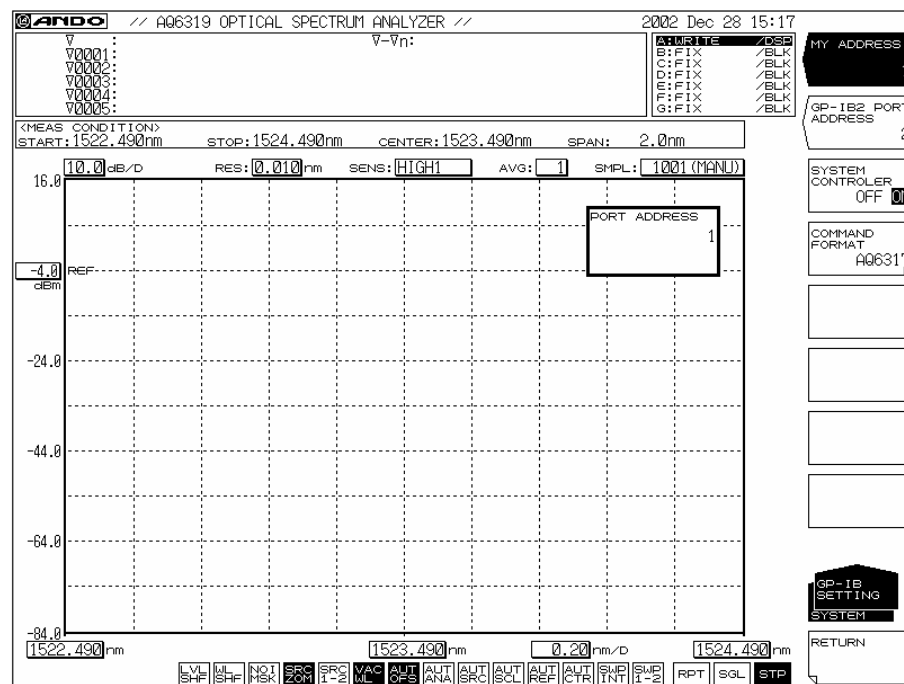


Figure 2-3: Setting the GPIB Address

2.3.4 Remote/Local Function

Local → Remote

When a listen address is sent from the controller with REN, ATN set to "True," this unit enters remote status.

In remote status, all panel switches and soft keys other than the <LOCAL> key are overridden. Also, the [REMOTE] lamp lights up, indicating that this unit is in remote status.

When an LLO (Local Lock Out) message is sent from the controller, this unit enters local lockout status. In this status, the <LOCAL> key is disabled and does not return this unit to the local status even when pressed.

To cancel the local lockout status, set REN to "False" from the controller.

Remote → Local

When the <LOCAL> key is pressed, this unit returns to the local status, allowing input of the switches on the panel. Also, the [REMOTE] lamp goes off. To return this unit to local status, send a GTL (Go To Local) message or set REN to "False" from the controller.

2.3.5 Sending/Receiving a Remote Command(s)

● Buffers

Command input buffer

The AQ6319's command input buffer is of a single stage and has a buffer size of 64 Kbytes.

When receiving the query data which buffer size exceeded, query data after buffer size is deleted. The remote command after the last command separate among the data of 64KBytes is deleted.

Talker output buffer

The unit's command output buffer is of a single stage and has a buffer sizeof 1Mbytes.

This unit is stores only the latest data.

(When this unit receives a talker command while there is still data in the buffer, it clears old data in preparation for receiving new data.)

When thie unit is performed an output statement contains multiple remote commands and rereceiving the query data which buffer size exceeded, use the following action.

- A query error bit(QYE) of standard event status register is set "1".query error bit.
 - This unit's talker output buffer is cleared.
 - Already recieved command is performed after this unit recieved the query data which buffer size exceeded.
- Note, however, that talker data by talker commands is not stored at the output buffer.

Error buffer

This unit's error buffer is of a single stage and stores only the latest error information.

● Message Terminators

This unit allows the following message terminators to be used.

Program Message Terminators

- Assertion of EOI (End-Of-Identify) signal
- LF (line feed) character
- LF+EOI

Here, LF is a line feed (0Ah) in ASCII. For CR + LF, because CR (0Dh) is recognized as "wsp," CR + LF can consequently also be used as a message terminator.

Also, for waveform binary transfer, only EOI acts as a message terminator.

Response message terminator

The response message terminator uses LF+EOI.

● Receipt of Remote Commands

- When completing receipt of a remote command, this unit releases the GPIB bus.
- When receiving the next command while a command action is being executed, this unit captures that command to store it in the receive buffer. Then, it releases the GPIB bus.
- When there is a remote command in the receive buffer, this unit does not capture a successive command even if there are commands on the GPIB bus.
- When the action of the preceding command is complete, this unit executes the command stored in the receive buffer and clears the buffer. Then it captures the next command into the receive buffer if there is one on the bus.
- When an output statement contains multiple remote commands, this unit captures them all and services them in the order they were written. In this case, unless the last command in the statement has started to be executed, this unit cannot capture the next command.

● Data Inquiry

- Inquiry of data by the external controller is made using a query command or a data output request from the controller.
- A query command has the form in which "?" is appended to the end of the command.
- For query commands with an argument, the argument is specified in the form of <wsp> + <argument> at the end of "?".
- When receiving a query command, this unit prepares a reply to the query command for the output buffer at that time.
- Data in the output buffer will be retained until this unit receives an input statement or a new query command from the controller.
- If multiple query commands are specified and written in succession using a semicolon ";", this unit prepares replies to all of them for the output buffer. In this case, this unit will collectively output all the prepared data when receiving the next data output request.

● Device Trigger Function

- When GET (Group Execute Trigger) is received, the unit will perform a single sweep.

2.4 Status Registers

The status registers that the AQ6319 has are shown in Table 2-2, List of the Status Registers. Also, an overall diagram of the status registers is shown in Figure 2-4, Overall Diagram of the Status Registers.

This unit has the following status registers defined by IEEE 488-2 and SCPI:

- Status byte register
- Standard event status register
- Operation status register
- Questionable status register

Moreover, this unit has an operation status bit (OPS) and a questionable status bit (QUS), each of which contains the summary information of each piece of register information, as the extension bits of the status byte register.

Table 2-2: List of the Status Registers

Register Name	Description
○ Status byte register	Register defined by IEEE 488.2
STB: Status Byte Register	Same as the above
SRE: Service Request Enable Register	Same as the above
○ Standard event status register	Register defined by IEEE 488.2
ESR: Standard Event Status Register	Same as the above
ESE: Standard Event Status Enable Register	Same as the above
○ Operation status register	Provides execution information of operation (such as being swept, being copied, and under calibration)
Operation Event Register	A register indicating the presence/absence of an event. An event will be latched.
Operation Event Enable Register	A condition mask register used when the summary bit (OPS) is created
○ Questionable status register	Not assigned yet
Questionable Event Register	A register indicating the presence/absence of an event. An event will be latched.
Questionable Event Enable Register	A condition mask register used when summary bit (QUS) is created

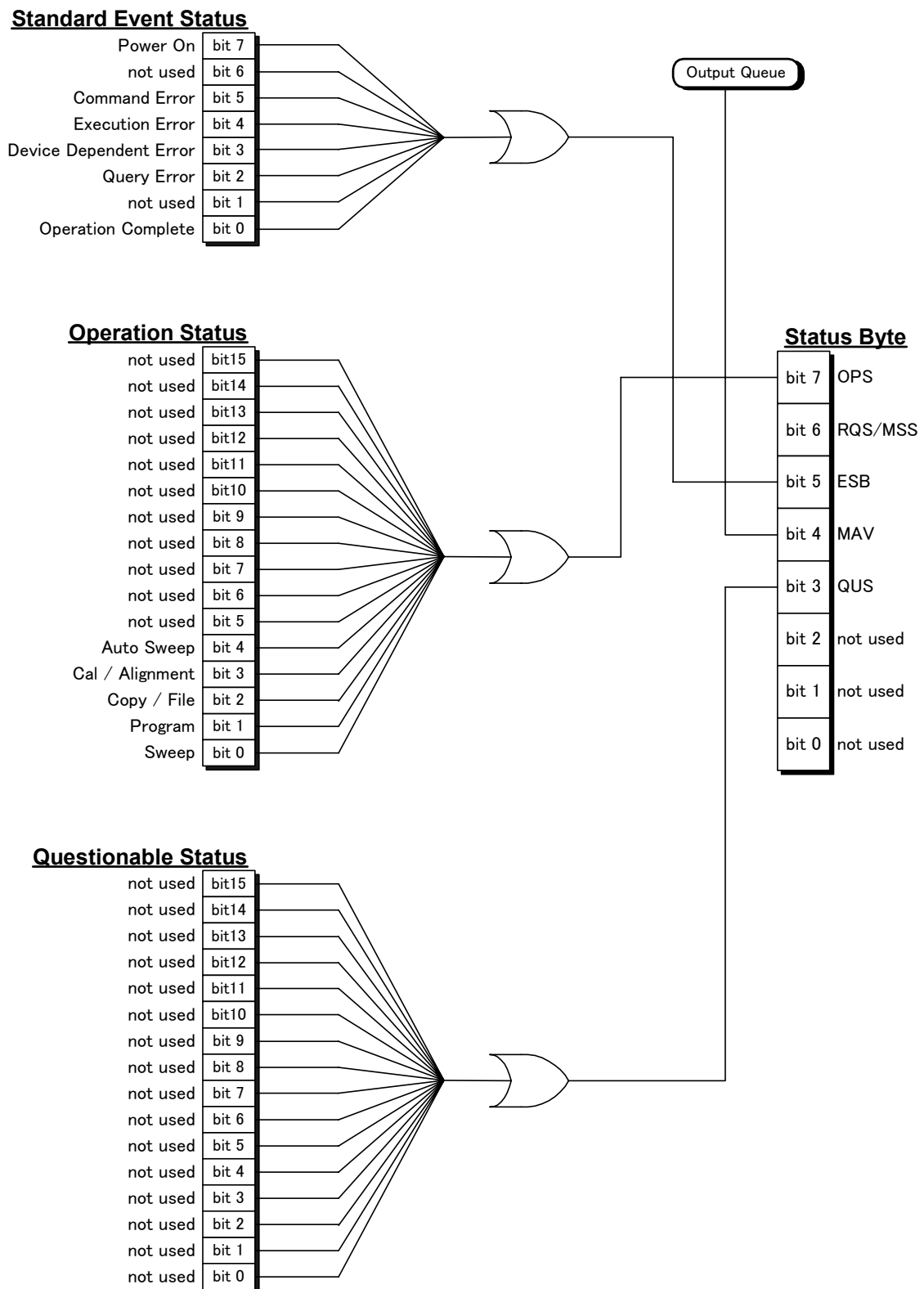


Figure 2-4: Overall Diagram of the Status Registers

2.4.1 Status Byte Register

• Configuration

The configuration of the status byte register is shown in Figure 2-5, Status Byte Register. The details and action of this register comply with the IEEE 488.2 standard.

This AQ6319 also provides the extended OPS and QUS bits to the status byte register.

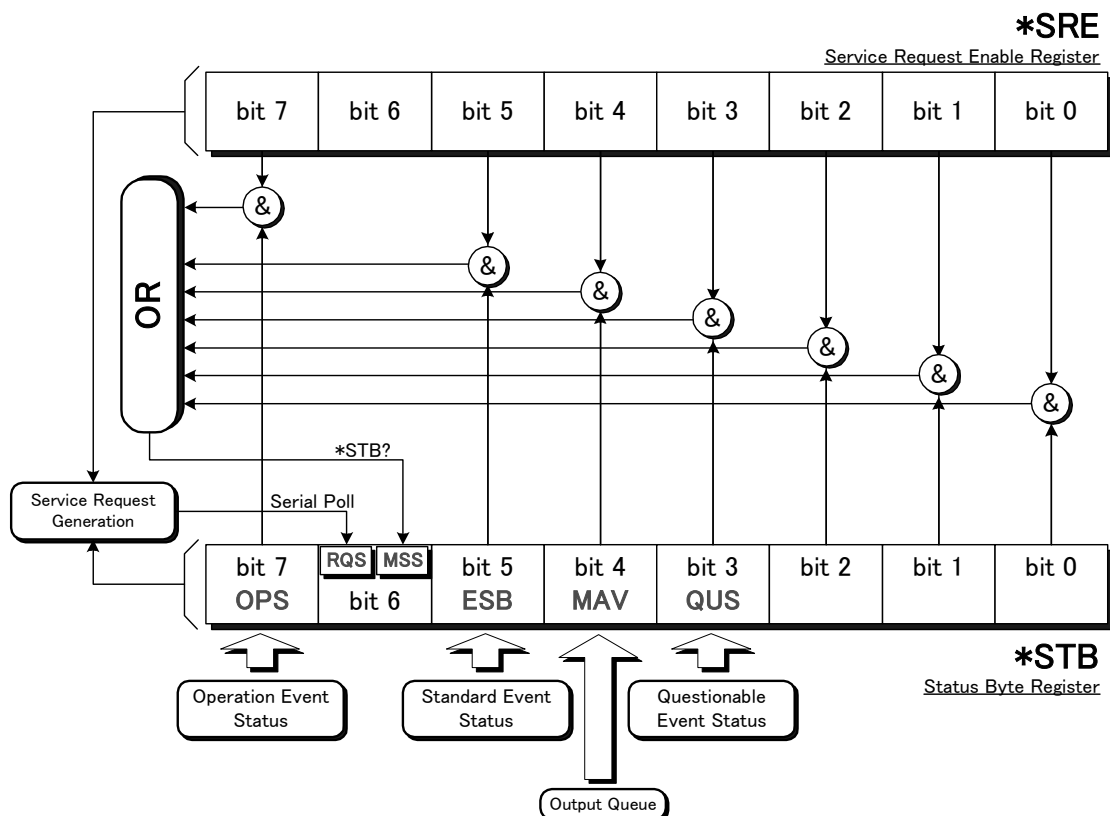


Figure 2-5: Status Byte Register

Table 2-3: Details of the Status Byte Register

Bit	Event Name	Description	Decimal value
Bit 7	OPS	Summary bit of operation status	128
Bit 6	RQS, MSS	"1" if there is more than one service request	64
Bit 5	ESB	Summary bit of standard event status register	32
Bit 4	MAV	"1" if the output buffer contains data	16
Bit 3	QUS	Summary bit of questionable event status	8
Bit 2	None	Not used (always 0)	0
Bit 1	None	Not used (always 0)	0
Bit 0	None	Not used (always 0)	0

• Status Byte Register

<Read>

This register can be read by a serial poll or the common *STB? query. Note that the information of bit 6 changes with a different reading method.

- Read by serial polling

An RQS message is read as bit 6 information.

After reading, the RQS message will be cleared.

- Read by the common *STB? query

An MSS summary message is read as bit 6 information.

Even after reading, the MSS message will be held.

Note that the statuses of any bits other than bit 6 do not change, even when the register is read.

Action for a read complies with the IEEE 488.2 standard.

<Write>

The contents of the register will be rewritten only when the status of an assigned status data structure has been changed. Action for a write complies with the IEEE 488.2 standard.

<Clear>

All event registers and queues, not including the output queues and MAV bit, will be cleared by the common *CLS command.

Action for a clear complies with the IEEE 488.2 standard.

• Service Request Enable Register

<Read>

This register can be read by the common *SRE? query.

The value of bit 6, an unassigned bit, is always "0." The contents of the register are not cleared even when read. Action for a read complies with the IEEE 488.2 standard.

<Write>

This register can be written by the common *SRE command.

The set value of bit 6, an unassigned bit, is always ignored. Action for a write complies with the IEEE 488.2 standard.

<Clear>

This register is cleared under any of the following conditions:

- Data "0" is set using the common *SRE command.
- Power ON Also, the register will not be cleared in the following cases:
 - Receipt of the *RST command
 - Receipt of the *CLS command
 - Device clear (DCL, SDC)

Action for a clear complies with the IEEE 488.2 standard.

2.4.2 Standard Event Status Register

• Configuration

The configuration of the standard event status register is shown in Table 2-4, Contents of the Standard Event Status Register. The details and action of this register comply with the IEEE 488.2 standards.

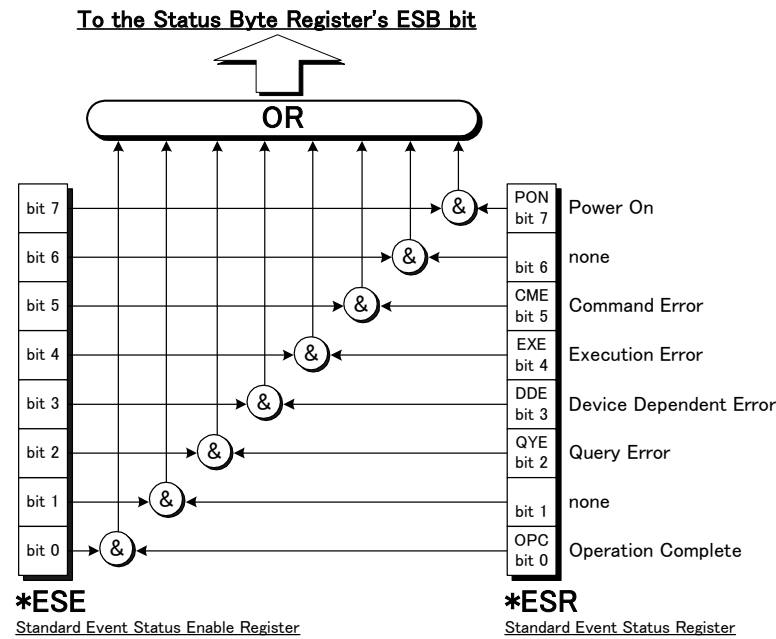


Figure 2-6: Standard Event Status Register

Table 2-4: Contents of the Standard Event Status Register

Bit	Event Name	Description	Decimal value
Bit 7	PON (Power ON)	Power is turned ON. Set to "1" at startup.	128
Bit 6	None	Not used (always "0")	0
Bit 5	CME (Command error)	· A syntax error or unrecognizable command is detected. · GET is encountered between the 1 st byte of a program message and the program message terminator.	32
Bit 4	EXE (Execution error)	· Program data following the program header is out of the effective range. · Receipt of a program message contradictory to device state.	16
Bit 3	DDE (Device-specific error)	Error caused by an event other than CME, EXE, or QYE	8
Bit 2	QYE (Query error)	· Access to an output queue was made with no output existing. · Output queue data was lost.	4
Bit 1	None	Not used (always 0)	0
Bit 0	OPC (Operation complete)	Completion of command action: * Enabled only when OPC is received * Disabled if OPC? is received	1

- **Standard Event Status Register**

- <Read>

- This register can be read by the common *ESR? query.

- Its contents will be cleared after reading.

- Action for a read complies with the IEEE 488.2 standard.

- <Write>

- The user cannot write to the register, but can only clear it.

- <Clear>

- This register is cleared by:

- Common *CLS command
 - Common *ESR? query

- Action for a clear complies with the IEEE 488.2 standard.

- **Standard Event Status Enable Register**

- <Read>

- This register can be read by the common *ESE? query.

- Action for a read complies with the IEEE 488.2 standard.

- <Write>

- This register can be written by the common *ESE command.

- Action for a write complies with the IEEE 488.2 standard

- <Clear>

- This register will be cleared under any of the following conditions:

- Data "0" is set using the common *ESE command.
 - Power ON It is not cleared in the following cases:
 - Receipt of the *RST command
 - Receipt of the *CLS command
 - Device clear (DCL, SDC)

- Action for a clear complies with the IEEE 488.2 standard.

2.4.3 Operation Status Register

This register reports the operation status of the AQ6319.

The operation condition register indicates this unit's condition.

A change in the operation condition register is latched into the operation event register. The user can refer to the operation event register to view changes in the operation status of this unit. After that change has been latched, the summary information of this unit event register is set to the OPS bit of the status byte register. In this case, only statuses corresponding to bits specified as "1" in the operation enable register are included in the summary information.

• Configuration

The configuration of the operation status register is shown in Figure 2-7, Structure of the Operation Status Register. The details of the operation status are given in Table 2-5, Operation Status.

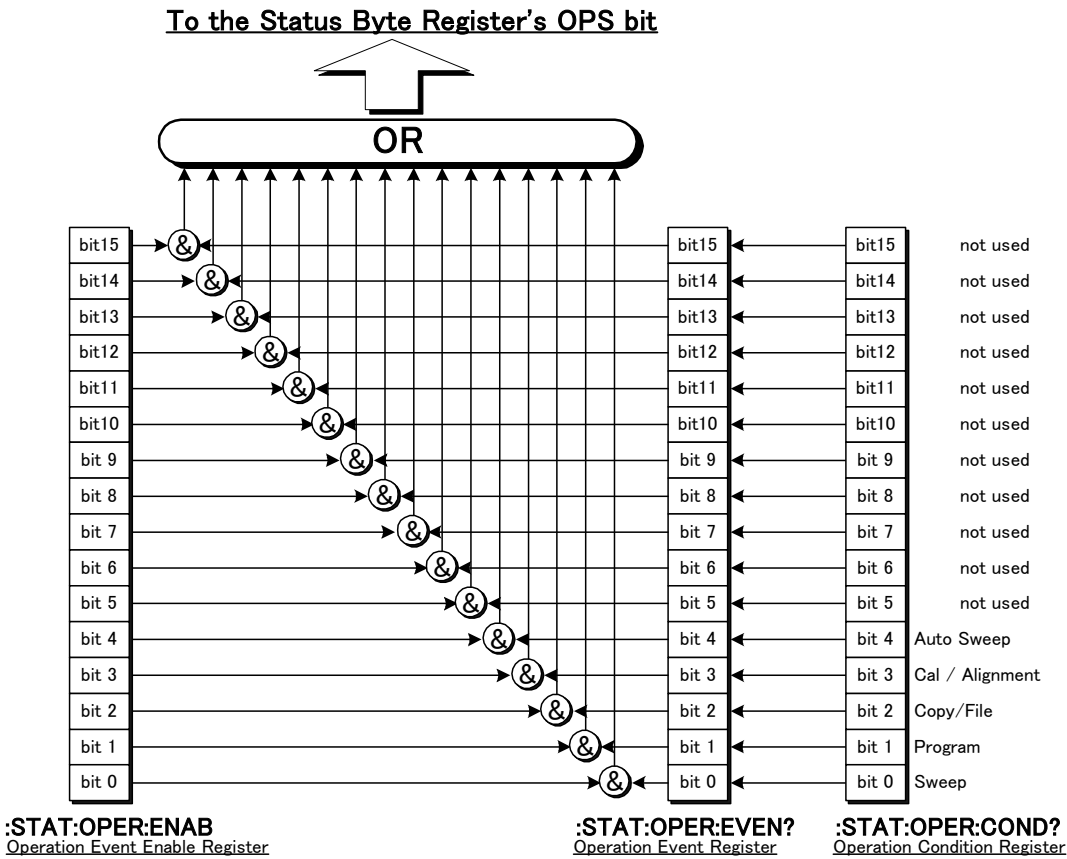


Figure 2-7: Structure of the Operation Status Register

Table 2-5: Operation Status

Bit	Event Name	Description	Decimal value
Bit 15	Not used	Spare (always 0)	0
Bit 14	Not used	Spare (always 0)	0
Bit 13	Not used	Spare (always 0)	0
Bit 12	Not used	Spare (always 0)	0
Bit 11	Not used	Spare (always 0)	0
Bit 10	Not used	Spare (always 0)	0
Bit 9	Not used	Spare (always 0)	0
Bit 8	Not used	Spare (always 0)	0
Bit 7	Not used	Spare (always 0)	0
Bit 6	Not used	Spare (always 0)	0
Bit 5	Not used	Spare (always 0)	0
Bit 4	Auto Sweep	Completion of auto sweep running action	16
Bit 3	Cal / Alignment	Completion of wavelength calibration or alignment	8
Bit 2	Copy / File	Completion of printout or file operation	4
Bit 1	Program	Completion of execution of the program functions	2
Bit 0	Sweep	Completion of a sweep	1

• Operation Condition Register

< Read >

This register can be read by the :STATus:OPERation:CONDition? query command. Its contents will not be cleared even when read.

< Write >

This register sets or resets a bit corresponding to a change in the status of this unit only when that change occurs. The user cannot write to the register.

< Clear >

The user cannot clear the register.

• Operation Event Register

< Read >

This register can be read by the :STATus:OPERation[:EVENT?] query command. Its contents are cleared when read.

< Write >

The user cannot write to the register, but can only clear it.

< Clear >

The register will be cleared by:

- A read using the :STATus:OPERation[:EVENT?] query command
- Initialization by the :STATus:PRESet command
- Common *CLS command
- Power ON

- **Operation Event Enable Register**

- <Read>

- This register can be read by the :STATus:OPERation:ENABle? query command.

- <Write>

- The register can be written by the :STATus:OPERation:ENABle command.

- <Clear>

- This register will be cleared under any of the following conditions.

- Data "0" is set by the :STATus:OPERation:ENABle command.
 - Power ON It is not cleared in the following cases:
 - Receipt of the *RST command
 - Receipt of the *CLS command
 - Device clear (DCL, SDC)

2.4.4 Questionable Status Register

This register reports the questionable status of the AQ6319.

As of this point, however, all bits of the register have not yet been assigned, but it can properly handle user operations such as read and write.

Moreover, the summary information of the questionable event register will be set to the QUS bit of the status byte register.

• Configuration

The configuration of the questionable status register is shown in Figure 2-8, Structure of the Questionable Status Register. The details of the questionable status are given in Table 2-6, Questionable Status.

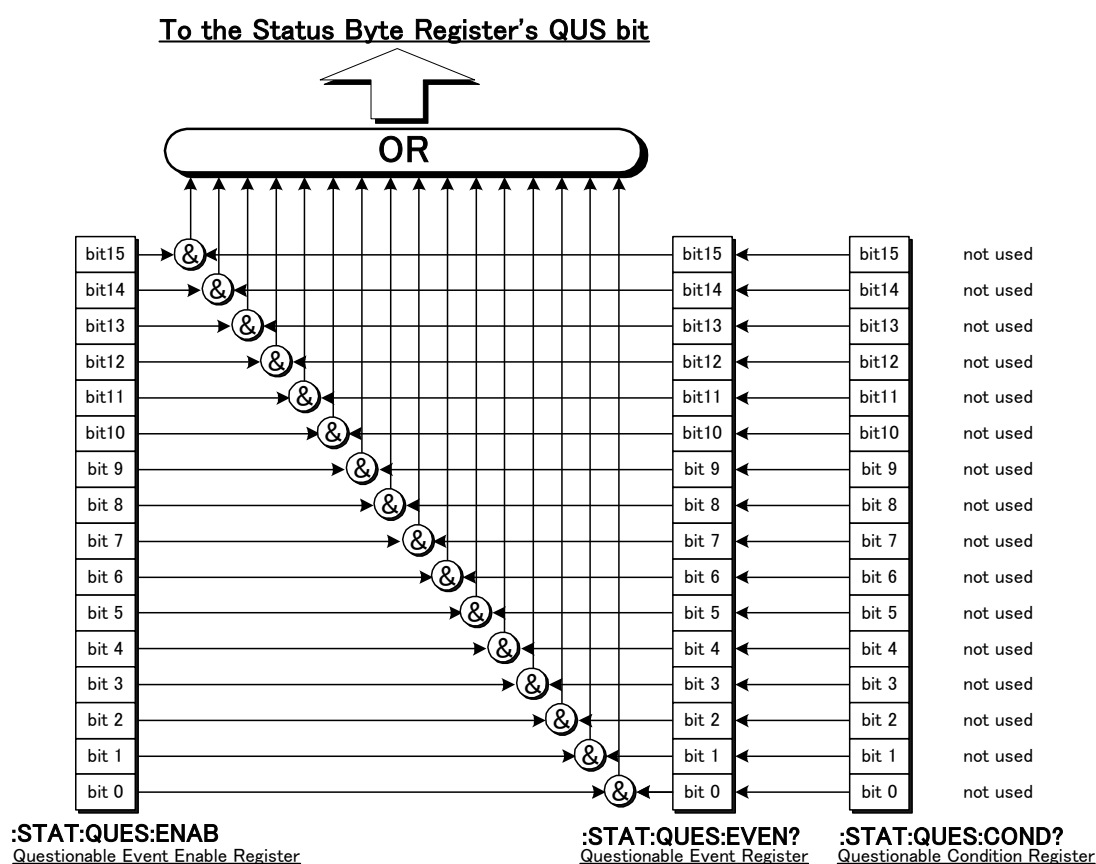


Figure 2-8: Structure of the Questionable Status Register

Table 2-6: Questionable Status

Bit	Event Name	Description	Decimal value
Bit 0~15	Not used	Spare (always 0)	0

- **Questionable Condition Register**

- <Read>

- This register can be read by the :STATus:QUEStionable:CONDition? query command. Its contents will not be cleared even when read.

- <Write>

- The register sets or resets a bit corresponding to a change in the status of this unit only when that change occurs. The user cannot write to this register.

- <Clear>

- The user cannot clear the register.

- **Questionable Event Register**

- <Read>

- This register can be read by the :STATus:QUEStionable[:EVENT?] query command. Its contents will be cleared when read.

- <Write>

- The user cannot write to the register, but can only clear it.

- <Clear>

- The register can be cleared by:

- A read using the :STATus:QUEStionable[:EVENT?] query command
 - Initialization by the :STATus:PRESet command
 - Common *CLS command
 - Power ON

Questionable Event Enable Register

<Read>

This register can be read by the :STATus:QUEStionable:ENABle? query command.

<Write>

The register can be written by the :STATus:QUEStionable:ENABle command.

<Clear>

This register will be cleared under any of the following conditions.

- Data "0" is set using the :STATus:QUEStionable:ENABle command.
- Power ON It is not cleared in the following cases:
 - Receipt of the *RST command
 - Receipt of the *CLS command
 - Device clear (DCL, SDC)

2.5 Overview of the Remote Commands

2.5.1 Command Mode

The AQ6319 supports the following two types of remote commands.

① AQ6319 commands

These are commands complying with SCPI (Standard Commands for Programmable Instruments).

Information contained in this manual is basically information about the AQ6319 commands.

For the following AQ6317-compatible commands, see 2.7, AQ6317-compatible Mode.

② AQ6317-compatible commands

These are commands compatible with the GPIB commands for AQ6317.

When remote-controlling this unit, which command is to be used can be set using the [SYSTEM] switch. (☛ Section 2.7, AQ6317-compatible Mode)

2.5.2 Rules of Syntax Description

The following information is intended for the common commands and instrument-specific commands contained in this manual.

Measured values and parameters are all sent and received in ASCII characters, not including special commands.

Table 2-7: Rules of Syntax Description

Rules	Description
	Indicates that one of the elements in a list should be selected. e.g., A B C = one of A, B, and C is used
[]	An item in square brackets is specified as desired.
{ }	An item in curly brackets can be specified multiple times within a command.
<wsp> *1	Space
<integer>	Integer value
<NRf>	Exponent-indicating value
<"file name">	A file name can be a maximum of 56 characters, including extensions, excluding the directory part. Enclose a character string using double quotations ("").
<trace name>	Trace name (TRA TRB TRC TRD TRE TRF TRG)
<marker>	Marker number (0: moving marker, 1 to 1024: fixed markers)
<"string">	Character string Enclose a character string using double quotations ("").

*1: About white space (<wsp>)

White space is defined as a character corresponding to 00h to 20h (not including 0Ah (LF)) of the ASCII character sets.

White space can be specified as desired, except when inserting it

between a command and parameters when specifying the parameters, and the case of using it as space in a character string such as a file name in a parameter. White space can be used to make a program legible.

2.5.3 Types of Commands

This unit's commands can be classified into the following three types:

(1) Sequential Commands

- These commands are the most general commands.
- The action of another command is not performed until the running of a sequential command is complete.
- Another action is not started until the running of the other command is complete.

(2) Command subject to overlapping

- A command subject to overlapping allows execution of an overlap command while it is being run.

E.g. of this command: :INITiate Makes a sweep.

(3) Overlap commands

- An overlap command can be executed while a command subject to overlapping is being run.
- These commands cannot be executed while a sequential command is being executed or if it has not yet been processed.

E.g. of command: :ABORt Stops measurement or calibration action.

*STB? Reads status byte.

2.5.4 Collective Transmission of Multiple Commands

Using the commands described in this manual, create a command string and send it to this unit.

If multiple commands are written in a single output statement by using a semicolon ";" to delimit each command, the commands will be executed in the order of which they have been written.

2.5.5 Format of a Remote Command

● Short and Long Forms

This unit's GP-IB commands cope with both short and long forms.
For the commands contained in this manual, the part written in capital letters is the short form of the command concerned.
E.g., the short form of the INITiate command is INIT.

● Upper- and Lower-Case Letters

This unit is not case-sensitive.
Return values are all written in upper-case letters.

● Grouping of SCPI Commands Using a Subsystem

This unit copes with the subsystem-based grouping of the SCPI commands.
Commands belonging to the same sub-system and existing at the same tree of the hierarchical structure of the subsystem can be sent in combination. In this case, each command should be delimited by a semicolon.

List of GPIB commands used in examples

:SENSe	:SETTing :ATTenuator :WAVelength :STOP :STARt
--------	---

- SENSe:WAVelength:STARt 1500NM;STOP 1600NM

(Possible)
- SENSe:WAVelength:STARt 1500NM;ATTenuator ON

(Not possible)

(Reason: They are not in the same hierarchy.)
- SENSe:WAVelength:STARt 1500NM;;STOP 1600NM

(Not possible)

(Reason: A colon ":" is unnecessary after a semicolon ";".)

● Numerics

- This unit copes with multiple notation methods when receiving a numeric(s).
- This unit uses only the basic units when transmitting a numeric(s).

The number of digits for a real part is fixed to one digit for the integer part (with a sign) and eight digits for decimal places.

The number of digits for an exponential part is fixed to 3.

E.g.: Receivable numerics (in case of 1550 nm)

1550 nm, 1.55 um, 1550E-9, 1.55E-6, and others

E.g.: Transmittable numerics (in case of 1550 nm)

+1.55000000E-006 only

- If a received numeric has a precision higher than the range of numerics handled inside this unit, lower decimal places will be rounded off rather than being discarded.
- This unit can handle the following multiplier suffixes:

Table 2-8: Multiplier Suffixes

Multiplier	Mnemonic	Multiplier	Mnemonic
1E18	EX (exa)	1E-3	M (milli)
1E15	PE (peta)	1E-6	U (micro)
1E12	T (tera)	1E-9	N (nano)
1E9	G (giga)	1E-12	P (pico)
1E6	MA (mega)	1E-15	F (femto)
1E3	K (kilo)	1E-18	A (atto)

● Specification of Parameters in a Command

To use parameters in a command, a space character must be placed between the command and parameters. Each parameter is delimited by a comma ",". A space may also be placed before and after a comma to make the command legible.

2.6 Detailed Description of the Remote Commands

2.6.1 Correspondence between the Soft Keys and Remote Commands

The following table shows the GPIB remote commands corresponding to the function keys used in the unit.

Table 2-9: Correspondence between the Soft Keys and the Remote Commands

Function	Control Command
SWEEP	
AUTO	:INITiate:SMODE<wsp>AUTO 3 ; INITiate
REPEAT	:INITiate:SMODE<wsp>REPeat 2 ; INITiate
SINGLE	:INITiate:SMODE<wsp>SINGLE 1 ; INITiate
STOP	:ABORT
SEGMENT MEASURE	:INITiate:SMODE<wsp>SEGment 4 ; INITiate
SEGMENT POINT****	:SENSe:SWEep:SEGment:POINts<wsp><integer>
SWEEP MKR L1-L2 ON/OFF	:SENSe:WAVelength:SRANge<wsp>OFF ON 0 1
SWEEP INTVL *****sec	:SENSe:SWEep:TIME:INTerval<wsp><integer> [SEC]
CENTER	
CENTER WL ****.***nm	:SENSe:WAVelength:CENTER<wsp><NRf>[M]
CENTER FREQ ***.****THz	:SENSe:WAVelength:CENTER<wsp><NRf>[HZ]
START WL ****.***nm	:SENSe:WAVelength:START<wsp><NRf>[M]
START FREQ ***.****THz	:SENSe:WAVelength:START<wsp><NRf>[HZ]
STOP WL ****.***nm	:SENSe:WAVelength:STOP<wsp><NRf>[M]
STOP FREQ ***.****THz	:SENSe:WAVelength:STOP<wsp><NRf>[HZ]
PEAK →CENTER	:CALCulate:MARKer:SCENter
AUTO CENTER ON/OFF	:CALCulate:MARKer:MAXimum:SCENter:AUTO<wsp>OFF ON 0 1
VIEW→MEAS	:DISPlay[:WINDow]:TRACe:X[:SCALe]:SMSCale
SPAN	
SPAN****.*nm	:SENSe:WAVelength:SPAN<wsp><NRf>[M]
START WL****.***nm	:SENSe:WAVelength:START<wsp><NRf>[M]
START FREQ***.****THz	:SENSe:WAVelength:START<wsp><NRf>[HZ]
STOP WL****.***nm	:SENSe:WAVelength:STOP<wsp><NRf>[M]
STOP FREQ***.****THz	:SENSe:WAVelength:STOP<wsp><NRf>[HZ]
0nm SWEEP TIME**sec	:SENSe:SWEep:TIME:0NM<wsp><integer>[SEC]
VIEW→MEAS	:DISPlay[:WINDow]:TRACe:X[:SCALe]:SMSCale

Function		Control Command
LEVEL		
REF LEVEL	LOG	:DISPlay[:WINDow]:Y1[:SCALe]:RLEVel<wsp><NRf>[DBM]
	LIEAR	:DISPlay[:WINDow]:Y1[:SCALe]:RLEVel<wsp><NRf>[NW UM MW]
LOG SCALE**. *dB/D		:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:PDIVision<wsp><NRf>[DB]
LIN SCALE		:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing<wsp>LINear 1
LIN BASE LEVEL**. *mW		:DISPlay[:WINDow]:Y1[:SCALe]:BLEVel<wsp><NRf>[MW]
PEAK→REF LEVEL		:CALCulate:MARKer:MAXimum:SRLevel
AUTO REF LEVEL ON/OFF		:CALCulate:MARKer:MAXimum:SRLevel:AUTO
LEVEL UNIT dBm / dBm/nm		:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:UNIT<wsp>DBM DBM/NM
Y SCALE SETTING	Y SCALE DIVISION 8/10/12	:DISPlay[:WINDow]:TRACe:Y[:SCALe]:DNUMBER<wsp>8 10 12
	REF LEVEL POSITION **DIV	:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RPOSition<wsp><integer>[DIV]
SUB LOG**. *dB/D		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<wsp><NRf>[DB]
SUB LIN*. ***/D		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<wsp><NRf>
SUB SCALE**. *dB/km		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<wsp><NRf>[DB/KM]
SUB SCALE**. **%/D		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<wsp><NRf>[%]
OFST LVL or SCALE MIN **. *dB		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:OLEVel<wsp><NRf>[DB]
LENGTH**. ***km		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:LENGth<wsp><NRf>[KM]
AUTO SUB SCALE ON/OFF		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:AUTO<wsp>OFF ON 0 1
SUB REF LVL POSITION **DIV		:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:RPOSition<wsp><integer>[DIV]

Function		Control Command
SETUP		
RESOLUTION *.***nm		:SENSe:BANDwidth:BWIDth[:RESolution]<wsp><NRf>[M]
SENS/MODE @@@@@@@	NORM/HOLD	:SENSe:SENSe<wsp>NHLd 0
	NORM/AUTO	:SENSe:SENSe<wsp>NAUT 1
	MID	:SENSe:SENSe<wsp>MID 2
	HIGH1	:SENSe:SENSe<wsp>HIGH1 3
	HIGH2	:SENSe:SENSe<wsp>HIGH2 4
	HIGH3	:SENSe:SENSe<wsp>HIGH3 5
CHOP MODE @@@@@ Ⓢ	OFF	:SENSe:CHOPPer<wsp>OFF 0
	CHOP	:SENSe:CHOPPer<wsp>ON 1 / :SENSe:CHOPPer<wsp>CHOP 1
	SWITCH	:SENSe:CHOPPer<wsp>SWITCh 2
AVG TIMES ***		:SENSe:AVERage:COUnT<wsp><integer>
SAMPLINT POINT AUTO		:SENSe:SWEEp:POINts:AUTO<wsp>OFF ON 0 1
SAMPLING POINT *****		:SENSe:SWEEp:POINts<wsp><integer>
SAMPLING INTVL *.****nm		:SENSe:SWEEp:STEP<wsp><NRf> [M]
MEAS WL AIR/VAC		:SENSe:CORRection:RVELocity:MEDium<wsp>AIR VACuum 0 1
HORZN SCALE nm/THz		:UNIT:X<wsp>WAVelength FREQuency 0 1
EXT TRIGGER MODE ON/OFF		:TRIGger[:SEQuence]:STATe<wsp>OFF ON 0 1
EXT TRIGGER SETTING	EDGE RISE/FALL	:TRIGger[:SEQuence]:SLOPe<wsp>RISE FALL 0 1
	DELAY ****.* μs	:TRIGger[:SEQuence]:DELay<wsp><NRf>[S]
OPT ATT ON/OFF		:SENSe:SETTing:ATTenuator<wsp>OFF ON 0 1
TLS SYNC SWEEP		:SENSe:SWEEp:TLSSync<wsp>OFF ON 0 1
ZOOM 1/2		
ZOOM CENTER WL ****.*nm		:DISPlay[:WINDow]:TRACe:X[:SCALe]:CENTer<wsp><NRf>[M]
ZOOM CENTER FREQ ***.*THz		:DISPlay[:WINDow]:TRACe:X[:SCALe]:CENTer<wsp><NRf>[HZ]
ZOOM SPAN ****.*nm		:DISPlay[:WINDow]:TRACe:X[:SCALe]:SPAN<wsp><NRf>[M]
ZOOM SPAN ***.*THz		:DISPlay[:WINDow]:TRACe:X[:SCALe]:SPAN<wsp><NRf>[HZ]
ZOOM START WL ****.*nm		:DISPlay[:WINDow]:TRACe:X[:SCALe]:STARt<wsp><NRf>[M]
ZOOM START FREQ ***.*THz		:DISPlay[:WINDow]:TRACe:X[:SCALe]:STARt<wsp><NRf>[HZ]
ZOOM STOP WL ****.*nm		:DISPlay[:WINDow]:TRACe:X[:SCALe]:STOP<wsp><NRf>[M]
ZOOM STOP FREQ ***.*THz		:DISPlay[:WINDow]:TRACe:X[:SCALe]:STOP<wsp><NRf>[HZ]
PEAK→ZOOM CTR		:CALCulate:MARKer:MAXimum:SZCEnter
OVERVIEW DISPLAY OFF/L/R		:DISPlay[:WINDow]:OVIew:POSition<wsp>OFF LEFT RIGHT 0 1 2
OVERVIEW SIZE LARGE/SMALL		:DISPlay[:WINDow]:OVIew:SIZE<wsp>LARGe SMALL 0 1

Function		Control Command
ZOOM 2/2		
INITIAL		:DISPlay[:WINDow]:TRACe:X[:SCALe]:INITialize
DISPLAY		
NORMAL DISPLAY		:DISPlay[:WINDow]:SPLit<wsp>OFF 0
SPLIT DISPLAY		:DISPlay[:WINDow]:SPLit<wsp>ON 1
SPLIT DISPLAY	TRACE A UP/LOW	:DISPlay[:WINDow]:SPLit:POSition<wsp>TRA,UP LOW 0 1
	TRACE B UP/LOW	:DISPlay[:WINDow]:SPLit:POSition<wsp>TRB,UP LOW 0 1
	TRACE C UP/LOW	:DISPlay[:WINDow]:SPLit:POSition<wsp>TRC,UP LOW 0 1
	TRACE D UP/LOW	:DISPlay[:WINDow]:SPLit:POSition<wsp>TRD,UP LOW 0 1
	TRACE E UP/LOW	:DISPlay[:WINDow]:SPLit:POSition<wsp>TRE,UP LOW 0 1
	TRACE F UP/LOW	:DISPlay[:WINDow]:SPLit:POSition<wsp>TRF,UP LOW 0 1
	TRACE G UP/LOW	:DISPlay[:WINDow]:SPLit:POSition<wsp>TRG,UP LOW 0 1
	HOLD	
	UPPER HOLD ON/OFF	:DISPlay[:WINDow]:SPLit:HOLD:UPPer<wsp>OFF ON 0 1
	LOWER HOLD ON/OFF	:DISPlay[:WINDow]:SPLit:HOLD:LOWer<wsp>OFF ON 0 1
LABEL		:DISPlay[:WINDow]:TEXT:DATA<wsp><string>
NOISE MASK ***dB		:DISPlay[:WINDow]:TRACe:Y:NMASt<wsp><NRf>[DB]
MASK LINE VERT / HRZN		:DISPlay[:WINDow]:TRACe:Y:NMASt:TYPE<wsp>VERTical HORIzontal 0 1
TRACE CLEAR	ALL TRACE	:DISPlay[:WINDow]:TEXT:CLEar

Function			Control Command
TRACE 1/2			
ACTIVE TRACE	A	:TRACe:ACTive<wsp>TRA	
	B	:TRACe:ACTive<wsp>TRB	
	C	:TRACe:ACTive<wsp>TRC	
	D	:TRACe:ACTive<wsp>TRD	
	E	:TRACe:ACTive<wsp>TRE	
	F	:TRACe:ACTive<wsp>TRF	
	G	:TRACe:ACTive<wsp>TRG	
VIEW @ DISP/BLANK		:TRACe:STATe:<:TRACe name><wsp>ON OFF 1 0	
WRITE @		:TRACe:ATTRibute:<:TRACe name><wsp>WRITe 0	
FIX @		:TRACe:STATe:<tarce name><wsp>FIX 1	
HOLD @	MAX HOLD	:TRACe:ATTRibute:<:TRACe name><wsp>MAX 2	
	MIN HOLD	:TRACe:ATTRibute:<:TRACe name><wsp>MIN 3	
ROLL AVG @ ***		:TRACe:ATTRibute:RAVG:<:TRACe name><wsp><integer>	
CALCULATE C @@@@@	LOG MATH @@@@	C=A·B(LOG)	:CALCulate:MATH:TRC<wsp>A·B(LOG)
		C=B·A(LOG)	:CALCulate:MATH:TRC<wsp>B·A(LOG)
		C=A+B(LOG)	:CALCulate:MATH:TRC<wsp>A+B(LOG)
	LIN MATH @@@@@	C=A+B(LIN)	:CALCulate:MATH:TRC<wsp>A+B(LIN)
		C=A·B(LIN)	:CALCulate:MATH:TRC<wsp>A·B(LIN)
		C=B·A(LIN)	:CALCulate:MATH:TRC<wsp>B·A(LIN)
		C=1·k(A/B) k: *.****	:CALCulate:MATH:TRC:K<wsp><NRf>; :CALCulate:MATH:TRC<wsp>1·K(A/B)
		C=1·k(B/A) k: *.****	:CALCulate:MATH:TRC:K<wsp><NRf>; :CALCulate:MATH:TRC<wsp>1·K(B/A)
CALCULATE F @@@@@	LOG MATH @@@@	F=C·D(LOG)	:CALCulate:MATH:TRF<wsp>C·D(LOG)
		F=D·C(LOG)	:CALCulate:MATH:TRF<wsp>D·C(LOG)
		F=C+D(LOG)	:CALCulate:MATH:TRF<wsp>C+D(LOG)
		F=D·E(LOG)	:CALCulate:MATH:TRF<wsp>D·E(LOG)
		F=E·D(LOG)	:CALCulate:MATH:TRF<wsp>E·D(LOG)
		F=D+E(LOG)	:CALCulate:MATH:TRF<wsp>D+E(LOG)
CALCulate F @@@@@	LIN MATH @@@@@	F=C+D(LIN)	:CALCulate:MATH:TRF<wsp>C+D(LIN)
		F=C·D(LIN)	:CALCulate:MATH:TRF<wsp>C·D(LIN)
		F=D·C(LIN)	:CALCulate:MATH:TRF<wsp>D·C(LIN)
		F=D+E(LIN)	:CALCulate:MATH:TRF<wsp>D+E(LIN)
		F=D·E(LIN)	:CALCulate:MATH:TRF<wsp>D·E(LIN)
		F=E·D(LIN)	:CALCulate:MATH:TRF<wsp>E·D(LIN)

機能		コントロールコマンド	
TRACE 2/2			
CALCulate G @@@@@	LOG MATH @@@@	G=C·F(LOG)	:CALCulate:MATH:TRG<wsp>C·F(LOG)
		G=F·C(LOG)	:CALCulate:MATH:TRG<wsp>F·C(LOG)
		G=C+F(LOG)	:CALCulate:MATH:TRG<wsp>C+F(LOG)
		G=E·F(LOG)	:CALCulate:MATH:TRG<wsp>E·F(LOG)
		G=F·E(LOG)	:CALCulate:MATH:TRG<wsp>F·E(LOG)
		G=E+F(LOG)	:CALCulate:MATH:TRG<wsp>E+F(LOG)
	LIN MATH @@@@	G=C+F(LIN)	:CALCulate:MATH:TRG<wsp>C+F(LIN)
		G=C·F(LIN)	:CALCulate:MATH:TRG<wsp>C·F(LIN)
		G=F·C(LIN)	:CALCulate:MATH:TRG<wsp>F·C(LIN)
		G=E+F(LIN)	:CALCulate:MATH:TRG<wsp>E+F(LIN)
		G=E·F(LIN)	:CALCulate:MATH:TRG<wsp>E·F(LIN)
		G=F·E(LIN)	:CALCulate:MATH:TRG<wsp>F·E(LIN)
	NORMALIZE @@@@	G=NORM A	:CALCulate:MATH:TRG<wsp>NORMA
		G=NORM B	:CALCulate:MATH:TRG<wsp>NORMB
		G=NORM C	:CALCulate:MATH:TRG<wsp>NORMC
	CURVE FIT @@@@	G=CVFIT A	:CALCulate:MATH:TRG<wsp>CVFTA
		G=CVFIT B	:CALCulate:MATH:TRG<wsp>CVFTB
		G=CVFIT C	:CALCulate:MATH:TRG<wsp>CVFTC
		G=MKR FIT	:CALCulate:MATH:TRG<wsp>MKRFT
		THRESH **dB	:CALCulate:MATH:TRG:CVFT:THResh<wsp><NRf>[DB]
		OPERATION AREA	:CALCulate:MATH:TRG:CVFT:OPARea<wsp>ALL INL1-L2 OUTL1-L2 0 1 2
		FITTING ALGO	:CALCulate:MATH:TRG:CVFT:FALGo<wsp>GAUSs LORENz 3RD 4TH 5TH 0 1 2 3 4
	CURVE FIT PK @@@@	G=PKCVFIT A	:CALCulate:MATH:TRG<wsp>PKCVFTA
		G=PKCVFIT B	:CALCulate:MATH:TRG<wsp>PKCVFTB
		G=PKCVFIT C	:CALCulate:MATH:TRG<wsp>PKCVFTC
		THRESH **dB	:CALCulate:MATH:TRG:PCVFt:THResh<wsp><NRf>[DB]
		OPERATION AREA	:CALCulate:MATH:TRG:CVFT:OPARea<wsp>ALL INL1-L2 OUTL1-L2 0 1 2
		FITTING ALGO	:CALCulate:MATH:TRG:CVFT:FALGo<wsp>GAUSs LORENz 3RD 4TH 5TH 0 1 2 3 4
TRACE LIST		-	
TRACE COPY		:TRACe:COpy<wsp><source :TRACe name>,<destination :TRACe name>	
TRACE CLEAR		:TRACe:DELeTe<wsp><:TRACe name>	

Function		Control Command
MARKER		
MKR ACTIVE ON/OFF , SET MARKER		:CALCulate:MARKer[:STATe]<wsp><marker>, ON 1 :CALCulate:MARKer:X<wsp><marker>,<NRf>[M HZ]
CLEAR MARKER		:CALCulate:MARKer[:STATe]<wsp><marker>,OFF 0
MARKER→CENTER		:CALCulate:MARKer:SCENter
MARKER →ZOOM CTR		:CALCulate:MARKer:SZCenter
MARKER →REF LEVEL		:CALCulate:MARKer:SRLevel
ALL MARKER CLEAR		:CALCulate:MARKer:AOff
LINE MKR 1 ON/OFF		:CALCulate:LMARker:X<wsp>1,<NRf>[M]
LINE MKR 2 ON/OFF		:CALCulate:LMARker:X<wsp>2,<NRf>[M]
LINE MKR 3 ON/OFF		:CALCulate:LMARker:Y<wsp>3,<NRf>[DBM]
LINE MKR 4 ON/OFF		:CALCulate:LMARker:Y<wsp>4,<NRf>[DBM]
MKR L1-L2→SPAN		:CALCulate:LMARker:SSPan
MKR L1-L2 →ZOOM SPAN		:CALCulate:LMARker:SZSPan
LINE MARKER ALL CLEAR		:CALCulate:LMARker:AOff
MAKER DISPLAY	OFFSET	:CALCulate:MARKer:FUNCTion:FORMat<wsp>OFFSet 0
	SPACING	:CALCulate:MARKer:FUNCTion:FORMat<wsp>SPACing 1
MARKER AUTO UPDATE ON/OFF		:CALCulate:MARKer:FUNCTion:UPDate<wsp>OFF ON 0 1
MARKER UNIT nm THz		:CALCulate:MARKer:UNIT<wsp>WAVelength FREQuency 0 1
SEARCH/ANA L1-L2 ON/OFF		:CALCulate:LMARker:SRANge<wsp>OFF ON 0 1
SEARCH/ANA ZOOM AREA ON/OFF		:DISPlay[:WINDow]:TRACe:X[:SCALe]:SRANge<wsp>OFF ON 0 1
MARKER LIST PRINT		:HCOPY[:IMMediate]:FUNCTion:MARKer:LIST
SEARCH 1/2		
PEAK SEARCH		:CALCulate:MARKer:MAXimum
BOTTOM SEARCH		:CALCulate:MARKer:MINimum
NEXT LEVEL SEARCH		:CALCulate:MARKer:MAXimum:NEXT or :CALCulate:MARKer:MINimum:NEXT
NEXT SEARCH RIGHT		:CALCulate:MARKer:MAXimum:RIGHT or :CALCulate:MARKer:MINimum:RIGHT
NEXT SEARCH LEFT		:CALCulate:MARKer:MAXimum:LEFT or :CALCulate:MARKer:MINimum:LEFT
SET MARKER		:CALCulate:MARKer[:STATe]<wsp><marker>, ON 1
CLEAR MARKER		:CALCulate:MARKer[:STATe]<wsp><marker>,OFF 0
ALL MARKER CLEAR		:CALCulate:MARKer:AOff
AUTO SEARCH ON/OFF		:CALCulate:MARKer:AUTO<wsp>OFF ON 0 1
MODE DIFF **.***dB		:CALCulate:PARAmeter:COMMon:MDIFF<wsp><NRf>[DB]
SEARCH/ANA L1-L2 ON/OFF		:CALCulate:LMARker:SRANge<wsp>OFF ON 0 1

Function		Control Command
SEARCH 2/2		
SEARCH/ANA ZOOM AREA ON/OFF		:DISPlay[:WINDow]:TRACe:X[:SCALe]:SRANge<wsp>OFF ON 0 1
ANALYSIS		
SPEC WIDTH @@@@@	THRESH	:CALCulate:CATegory<wsp>SWThresh 0
	ENVELOPE	:CALCulate:CATegory<wsp>SWENvelope 1
	RMS	:CALCulate:CATegory<wsp>SWRMs 2
	PEAK RMS	:CALCulate:CATegory<wsp>SWPKrms 3
	NOTCH	:CALCulate:CATegory<wsp>NOTCh 4
ANALYSIS1 @@@@@	DFB-LD	:CALCulate:CATegory<wsp>DFBLd 5
	FP-LD	:CALCulate:CATegory<wsp>FPLD 6
	LED	:CALCulate:CATegory<wsp>LED 7
	SMSR	:CALCulate:CATegory<wsp>SMSR 8
	POWER	:CALCulate:CATegory<wsp>POWer 9
	PMD	:CALCulate:CATegory<wsp>PMD 10
ANALYSIS2 @@@@@	WDM	:CALCulate:CATegory<wsp>WDM 11
	EDFA-NF	:CALCulate:CATegory<wsp>NF 12
	FILTER-PK	:CALCulate:CATegory<wsp>FILPk 13
	FILTER-BTM	:CALCulate:CATegory<wsp>FILBtm 14
	WDM FIL-PK	:CALCulate:CATegory<wsp>WFPeak 15
	WDM FIL-BTM	:CALCulate:CATegory<wsp>WFBtm 16
ANALYSIS EXECUTE (@@@@)		:CALCulate[:IMMediate]
SPEC WIDTH THRESH **,.*dB		:CALCulate:PARAmeter[:CATegory]:SWThresh:TH<wsp><NRf>[DB]
AUTO ANALYSIS ON/OFF		:CALCulate[:IMMediate]:AUTO<wsp>OFF ON 0 1
RESULT PRINT		:HCOPY[:IMMediate]:FUNCTion:CALCulate:LIST
RESULT SAVE		MMEMoRY:STORe:AREsult<wsp><"file name">[, INTERNAL FLOppy]
SEARCH/ANA L1-L2 ON/OFF		:CALCulate:LMARker:SRANge<wsp>OFF ON 0 1
SEARCH/ANA ZOOM AREA ON/OFF		:DISPlay[:WINDow]:TRACe:X[:SCALe]:SRANge<wsp>OFF ON 0 1

Function		Control Command
MEMORY		
SAVE	A TRACE →MEMORY	:MEMory:STORe<wsp><integer>,TRA
	B TRACE →MEMORY	:MEMory:STORe<wsp><integer>,TRB
	C TRACE →MEMORY	:MEMory:STORe<wsp><integer>,TRC
	D TRACE →MEMORY	:MEMory:STORe<wsp><integer>,TRD
	E TRACE →MEMORY	:MEMory:STORe<wsp><integer>,TRE
	F TRACE →MEMORY	:MEMory:STORe<wsp><integer>,TRF
	G TRACE →MEMORY	:MEMory:STORe<wsp><integer>,TRG
REC ALL	MEMORY →A TRACE	:MEMory:LOAD<wsp><integer>,TRA
	MEMORY →B TRACE	:MEMory:LOAD<wsp><integer>,TRB
	MEMORY →C TRACE	:MEMory:LOAD<wsp><integer>,TRC
	MEMORY →D TRACE	:MEMory:LOAD<wsp><integer>,TRD
	MEMORY →E TRACE	:MEMory:LOAD<wsp><integer>,TRE
	MEMORY →F TRACE	:MEMory:LOAD<wsp><integer>,TRF
	MEMORY →G TRACE	:MEMory:LOAD<wsp><integer>,TRG
MEMORY CLEAR		:MEMory:CLear<wsp><integer>

Function			Control Command	
FILE 1/2				
WRITE	DRIVE HDD/FDD		:MMEMory:CDRive<wsp>INTErnal FLOPpy	
	FILE NAME		:MMEMory:CDIREctory<wsp><directory name>	
	(TRACE)		:MMEMory:STORe:TRACe<wsp><trace name>,BIN CSV, <"file name">[,INTErnal FLOPpy]	
	(MEMORY)		:MMEMory:STORe:MEMory<wsp><integer>,BIN CSV, <"file name">[,INTErnal FLOPpy]	
	(GRAPHICS)		:MMEMory:STORe:GRAPhics<wsp>B&W COLor,BMP TIFF, <"file name">[,INTErnal FLOPpy]	
	(SETTING)		:MMEMory:STORe:SETTing<wsp><"file name">[,INTErnal FLOPpy]	
	(DATA)			:MMEMory:STORe:DATA<wsp><"file name">[,INTErnal FLOPpy]
		OUTPUT ITEM SETTING	DATE&TIME ON/OFF	:MMEMory:STORe:DATA:ITEM<wsp>DATE,OFF ON 0 1
			LABEL ON/OFF	:MMEMory:STORe:DATA:ITEM<wsp>LABel,OFF ON 0 1
			DATA AREA ON/OFF	:MMEMory:STORe:DATA:ITEM<wsp>DATA,OFF ON 0 1
			CONDITION ON/OFF	:MMEMory:STORe:DATA:ITEM<wsp>CONDition,OFF ON 0 1
			TRACE DATA ON/OFF	:MMEMory:STORe:DATA:ITEM<wsp>TRACe,OFF ON 0 1
		FILE TYPE CSV/DT5		:MMEMory:STORe:DATA:TYPE<wsp>CSV DT 0 1
	WRITE MODE ADD/OVER		:MMEMory:STORe:DATA:MODE<wsp>ADD OVER 0 1	
(PROGRAM)		:MMEMory:STORe:PROGram<wsp><integer>, <"file name">[,INTErnal FLOPpy]		
READ	DRIVE HDD/FDD		:MMEMory:CDRive<wsp>INTErnal FLOPpy	
	(TRACE)		:MMEMory:LOAD:TRACe<wsp><trace name>, <"file name">[,INTErnal FLOPpy]	
	(MEMORY)		:MMEMory:LOAD:MEMory<wsp><integer>, <"file name">[,INTErnal FLOPpy]	
	(SETTING)		:MMEMory:LOAD:SETTing<wsp><"file name">[,INTErnal FLOPpy]	
	(DATA)		:MMEMory:LOAD:DATA<wsp><"file name">[,INTErnal FLOPpy]	
	(PROGRAM)		:MMEMory:LOAD:PROGram<wsp><integer>, <"file name">[,INTErnal FLOPpy]	
	(TEMPLATE)		:MMEMory:LOAD:PROGram<wsp><"file name">[,INTErnal FLOPpy]	
FILE OPERATION	DRIVE HDD/FDD		:MMEMory:CDRive<wsp>INTErnal FLOPpy	
	DELETE		:MMEMory:DELeTe<wsp><"file name">[,INTErnal FLOPpy]	

Function		Control Command	
FILE 2/2			
FILE OPERATION	COPY	:MMEMory:REName<wsp><"source file name">[,INTernal FLOPpy] <"destination file name">[,INTernal FLOPpy]	
	RENAME	:MMEMory:REName<wsp><"new file name">, <"old file name">[,INTernal FLOPpy]	
	MAKE DIRECTORY	:MMEMory:MDIRectory<wsp><"directory name">[,INTernal FLOPpy]	
FD FORMAT (1.44MB)		:MMEMory:INITialize<wsp>[FLOPpy]	
PROGRAM			
PROGRAM EXECUTE		:PROGram:EXECute<wsp><integer>	
SYSTEM 1/2			
OPTICAL ALIGNMENT		:CALibration:ALIGn[:IMMediate]	
WL CALIBRATION	BUILT-IN SOURCE	:CALibration:WAVelength:INTernal[:IMMediate]	
	EXTERNAL LASER ****.***nm	:CALibration:WAVelength:EXTernal:SOURce<wsp>LASer 0; :CALibration:WAVelength:EXTernal:WAVelength<wsp><NRf>[M]	
	EXTERNAL GAS CELL ****.***nm	:CALibration:WAVelength:EXTernal:SOURce<wsp>GASCell 1; :CALibration:WAVelength:EXTernal:WAVelength<wsp><NRf>[M]	
WL SHIFT *.***nm		:SENSe:CORRection:WAVelength:SHIFt<wsp><NRf>[M]	
LVL SHIFT *.***dB		:SENSe:CORRection:LEVel:SHIFt<wsp><NRf>[DB]	
WL OFFSET TABLE		:CALibration:WAVelength:OFFSet:TABLE<wsp><integer>,<NRf> [DB]	
LVL OFFSET TABLE		:CALibration:POWER:OFFSet:TABLE<wsp><integer>,<NRf> [DB]	
GRID EDITOR	200GHz SPACING	:SYSTem:GRID<wsp>200GHZ 4	
	100GHz SPACING	:SYSTem:GRID<wsp>100GHZ 3	
	50GHz SPACING	:SYSTem:GRID<wsp>50GHZ 2	
	25GHz SPACING	:SYSTem:GRID<wsp>25GHZ 1	
	12.5GHz SPACING	:SYSTem:GRID<wsp>12.5GHZ 0	
	CUSTOM		:SYSTem:GRID<wsp>CUSTom 5
		START WL ****.****nm	:SYSTem:GRID:CUSTom:STARt<wsp><NRf>[M HZ]
		STOP WL ****.****nm	:SYSTem:GRID:CUSTom:STOP<wsp><NRf>[M HZ]
		SPACING *.**GHz	:SYSTem:GRID:CUSTom:SPACing<wsp><NRf>[GHZ]
		VALUE EDIT	
		INSERT	:SYSTem:GRID:CUSTom:INSert<wsp><NRf>[M HZ]
DELETE		:SYSTem:GRID:CUSTom:DELeTe<wsp><integer>	
REFERENCE WAVELENGTH ****.****nm	:SYSTem:GRID:REFerence<wsp><NRf>[HZ]		
GP-IB2 PORT ADDRESS **		:SYSTem:COMMunicate:GPIB2:ADDReSS<wsp><integer>	
TLS ADDRESS **		:SYSTem:COMMunicate:GPIB2:TLS:ADDReSS<wsp><integer>	

Function		Control Command
SYSTEM 2/2		
HARD COPY DEVOC E	INTERNAL	:HCOPY:DESTination<wsp>INTernal 0
	EXTERNAL	:HCOPY:DESTination<wsp>EXTernal 1
	FILE	:HCOPY:DESTination<wsp>FILE 2
EXTERNAL PRINTER@@@		:HCOPY:DEvice:LANGuage<wsp>
SET CLOCK		:SYSTem:DATE<wsp><year>,<month>,<day> :SYSTem:TIME<wsp><hour>,<minutes>,<seconds>
SELECT COLOR	COLOR 1	:DISPlay:COLor<wsp>1
	COLOR 2	:DISPlay:COLor<wsp>2
	COLOR 3	:DISPlay:COLor<wsp>3
	COLOR 4	:DISPlay:COLor<wsp>4
	COLOR 5	:DISPlay:COLor<wsp>5
	B&W	:DISPlay:COLor<wsp>0
UNCAL WARN DISPLAY ON/OFF		:SYSTem:DISPlay:UNCal<wsp>OFF ON 0 1
BUZZER SETTING	CLICK ON/OFF	:SYSTem:BUZZer:CLICk<wsp>OFF ON 0 1
	WARNING ON/OFF	:SYSTem:BUZZer:WARNing<wsp>OFF ON 0 1
LEVEL DISP	1DIG	:UNIT:POWer:DIGit<wsp>1
	2DIG	:UNIT:POWer:DIGit<wsp>2
	3DIG	:UNIT:POWer:DIGit<wsp>3
WINDOW TRANSPARENT ON/OFF		:SYSTem:DISPlay:TRANsparent<wsp>OFF ON 0 1
AUTO OFFSET ON/OFF		:CALibration:ZERO[:AUTO]<wsp>OFF ON 0 1 ONCE
PARAMETER INITIALIZE	ALL CLEAR	:SYSTem:INITialize
VERSION		-
COPY		
COPY		:HCOPY[:IMMediate]
FEED		
FEED		:HCOPY[:IMMediate]:FEED

2.6.2 ANALYSIS Setting Parameters

In setting [ANALYSIS] setting parameters, the analysis parameters differ with the analysis type. Thus, the <PARAMETER SETTING> key commands must be set independently of the regular key commands.

An analysis parameter setting command is shown in Table 2-10.

Table 2-10: List of ANALYSIS Parameter Setting Commands

ANALYSIS Parameter		Control Command
SPEC WIDTH		
THRESH	THRESH LEVEL **. **dB	:CALCulate:PARAmeter[:CATegory]:SWTHresh:TH<wsp><NRf>[DB]
	K **. **	:CALCulate:PARAmeter[:CATegory]:SWTHresh:K<wsp><NRf>
	MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]:SWTHresh:MFIT<wsp> OFF ON 0 1
ENVELOPE	THRESH LEVEL1 **. **dB	:CALCulate:PARAmeter[:CATegory]:SWENvelope:TH1<wsp><NRf>[DB]
	THRESH LEVEL2 **. **dB	:CALCulate:PARAmeter[:CATegory]:SWENvelope:TH2<wsp><NRf>[DB]
	K **. **	:CALCulate:PARAmeter[:CATegory]:SWENvelope:K
PEAK RMS	THRESH LEVEL **. **dB	:CALCulate:PARAmeter[:CATegory]:SWPKrms:TH<wsp><NRf>[DB]
	K **. **	:CALCulate:PARAmeter[:CATegory]:SWPKrms:K<wsp><NRf>[DB]
NOTCH TYPE	THRESH LEVEL **. **dB	:CALCulate:PARAmeter[:CATegory]:NOTCh:TH<wsp><NRf>[DB]
	K **. **	:CALCulate:PARAmeter[:CATegory]:NOTCh:K<wsp><NRf>[DB]
	PEAK	:CALCulate:PARAmeter[:CATegory]:NOTCh:TYPE<wsp>PEAK 0
	BOTTOM	:CALCulate:PARAmeter[:CATegory]:NOTCh:TYPE<wsp>BOTTom 1
ANALYSIS 1 1/3		
DFB-LD -XdB WIDTH	ALGO	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SWIDth,ALGO,<data>
	THRESH **. **dB	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp> SWIDth,TH,<NRf>[DB]
	THRESH2 **. **dB	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SWIDth,TH2,<NRf>[DB]
	K	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SWIDth,K,<NRf>
	MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SWIDth,MFIT,OFF ON 0 1
	MODE DIFF *. **dB	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SWIDth,MDIFf,<NRf>[D B]
SMSR	SMSR MODE	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SMSR,SMODE,SMSR1 S MSR2
	SMSR MASK ± *. **nm	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SMSR,SMASk,<NRf>[M]
	MODE DIFF *. **dB	:CALCulate:PARAmeter[:CATegory]:DFBLd<wsp>SMSR,MDIFf,<NRf>[DB]
FP-LD SPECTRU M WIDTH	ALGO	:CALCulate:PARAmeter[:CATegory]:FPLD<wsp>SWIDth,ALGO,<data>
	THRESH **. **dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> SWIDth,TH,<NRf>[DB]
	THRESH2 **. **dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp>SWIDth,TH2,<NRf>[DB]
	K	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp>SWIDth,K,<NRf>

ANALYSIS Parameter			Control Command
ANALYSIS 1 2/3			
FP-LD	SPECTRUM WIDTH	MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp>SWIDth,MFIT,OFF ON 0 1
		MODE DIFF *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp>SWIDth,MDIFF,<NRf>[DB]
	MEAN WAVELENGTH	ALGO	:CALCulate:PARAmeter[:CATegory]:FPLD<wsp>MWAVelength,ALGO,<data>
		THRESH *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MWAVelength,TH,<NRf>[DB]
		THRESH2 *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MWAVelength,TH2,<NRf>[DB]
		K	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MWAVelength,K,<NRf>
		MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MWAVelength,MFIT,OFF ON 0 1
		MODE DIFF *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MWAVelength,MDIFF,<NRf>[DB]
	TOTAL POWER	OFFSET LEVEL *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> TPOwer,OFFSet,<NRf>[DB]
	MODE NO.	ALGO	CALCulate:PARAmeter[:CATegory]:FPLD<wsp>MNUMber,ALGO,<data>
		THRESH *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MNUMber,TH,<NRf>[DB]
		THRESH2 *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MNUMber,TH2,<NRf>[DB]
		K	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MNUMber,K,<NRf>
		MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MNUMber,MFIT,OFF ON 0 1
		MODE DIFF *.*dB	:CALCulate:PARAmeter[:CATegory]: FPLD <wsp> MNUMber,MDIFF,<NRf>[DB]
LED	SPECTRUM WIDTH	ALGO	:CALCulate:PARAmeter[:CATegory]:LED<wsp>SWIDth,ALGO,<data>
		THRESH *.*dB	:CALCulate:PARAmeter[:CATegory]: LED <wsp> SWIDth,TH,<NRf>[DB]
		THRESH2 *.*dB	:CALCulate:PARAmeter[:CATegory]: LED <wsp>SWIDth,TH2,<NRf>[DB]
		K	:CALCulate:PARAmeter[:CATegory]: LED <wsp>SWIDth,K,<NRf>
		MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]: LED <wsp>SWIDth,MFIT,OFF ON 0 1
		MODE DIFF *.*dB	:CALCulate:PARAmeter[:CATegory]: LED <wsp>SWIDth,MDIFF,<NRf>[DB]
	MEAN WAVELENGTH	ALGO	:CALCulate:PARAmeter[:CATegory]: LED <wsp>MWAVelength,ALGO,<data>
		THRESH *.*dB	:CALCulate:PARAmeter[:CATegory]: LED <wsp> MWAVelength,TH,<NRf>[DB]

ANALYSIS Parameter			Control Command	
ANALYSIS1 3/3				
LED	MEAN WAVELENGTH	THRESH2 **. **dB	:CALCulate:PARAmeter[:CATegory]: LED <wsp> MWAVelength,TH2,<NRf>[DB]	
		K	:CALCulate:PARAmeter[:CATegory]: LED <wsp> MWAVelength,K,<NRf>	
	MEAN WAVELENGTH	MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]: LED <wsp> MWAVelength,MFIT,OFF ON 0 1	
		MODE DIFF *. **dB	:CALCulate:PARAmeter[:CATegory]: LED <wsp> MWAVelength,MDIFf,<NRf>[DB]	
	TOTAL POWER	OFFSET LEVEL *. **dB	:CALCulate:PARAmeter[:CATegory]: LED <wsp> TPOWer,OFFSet,<NRf>[DB]	
SMSR	SMSR MODE		:CALCulate: PARAmeter[:CATegory]:SMSR:MODE<wsp>SMSR1 SMSR2	
	SMSR MASK ±*. **dB		:CALCulate: PARAmeter[:CATegory]:SMSR:MASK<wsp><NRf>[M]	
POWER	OFFSET LEVEL *. **dB		:CALCulate:PARAmeter[:CATegory]:POWer:OFFSet<wsp><NRf>[DB]	
PMD	THRESH LEVEL *. **dB		:CALCulate:PARAmeter[:CATegory]:PMD:TH<wsp><NRf>[DB]	
ANALYSIS 2 1/7				
WDM	CHANNEL DETECTION SETTING	THRESH LEVEL **. **dB	:CALCulate:PARAmeter[:CATegory]:WDM:TH<wsp><NRf>[DB]	
		MODE DIFF **. **dB	:CALCulate:PARAmeter[:CATegory]:WDM:MDIFf<wsp><NRf>[DB]	
		DISPLAY MASK OFF/ON *. **dB	:CALCulate:PARAmeter[:CATegory]:WDM:DMASk<wsp><NRf>[DB]	
	INTERPOLATION SETTING	NOISE ALGO	AUTO-FIX	:CALCulate:PARAmeter[:CATegory]:WDM::NALGo<wsp>AFIX 0
			MANUAL-FIX	:CALCulate:PARAmeter[:CATegory]:WDM::NALGo<wsp>MFIx 1
AUTO-CTR			:CALCulate:PARAmeter[:CATegory]:WDM::NALGo<wsp>ACENter 2	

ANALYSIS Parameter			Control Command
ANALYSIS 2 2/7			
WDM	INTERPOLATION SETTING	NOISE ALGO	MANUAL-CTR :CALCulate:PARAmeter[:CATegory]:WDM::NALGo<wsp>MCENter 3
		PIT	:CALCulate:PARAmeter[:CATegory]:WDM::NALGo<wsp>PIT 4
		FITTING AREA	:CALCulate:PARAmeter[:CATegory]:WDM:NARea<wsp><NRf>[M]
		MASK AREA	:CALCulate:PARAmeter[:CATegory]:WDM:MARea<wsp><NRf>[M]
		FITTING ALGO	LINEAR :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp>LINear 0
			GAUSS :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp>GAUSs 1
			LORENZ :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp>LORenZ 2
			3RD POLY :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp>3RD 3
			4TH POLY :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp>4TH 4
			5TH POLY :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp>5TH 5
		NOISE BW *.**nm	:CALCulate:PARAmeter[:CATegory]:WDM:NBW<wsp><NRf>[M]
		DUAL TRACE ON/OFF	:CALCulate:PARAmeter[:CATegory]:WDM:DUAL<wsp>OFF ON 0 1
	DISPLAY SETTING	DISPLAY TYPE	ABSOLUTE :CALCulate:PARAmeter[:CATegory]:WDM:DTYPe<wsp>ABSolute 0
			RELATIVE :CALCulate:PARAmeter[:CATegory]:WDM:DTYPe<wsp>RELatibe 1
			DRIFT(MEAS) :CALCulate:PARAmeter[:CATegory]:WDM:DTYPe<wsp>MDRift 2
			DRIFT(GRID) :CALCulate:PARAmeter[:CATegory]:WDM:DTYPe<wsp>GDRift 3
		CH RELATION	OFFSET :CALCulate:PARAmeter[:CATegory]:WDM:RELation<wsp>OFFSet 0
			SPACING :CALCulate:PARAmeter[:CATegory]:WDM:RELation<wsp>SPACing 1
		REF CH	:CALCulate:PARAmeter[:CATegory]:WDM:RCH<wsp><integer>
		MAX/MIN RESET	:CALCulate:PARAmeter[:CATegory]:WDM:MMReset
		OUTPUT SLOPE ON/OFF	:CALCulate:PARAmeter[:CATegory]:WDM:OSLope<wsp>OFF ON 0 1
		POINT DISPLAY ON/OFF	:CALCulate:PARAmeter[:CATegory]:WDM:PDISplay<wsp>OFF ON 0 1
EDFA-NF	CHANNEL DETECTION	THRESH LEVEL **.**dB	:CALCulate:PARAmeter[:CATegory]:NF:TH<wsp><NRf>[DB]
		MODE DIFF **.**dB	:CALCulate:PARAmeter[:CATegory]:NF:MDIFF<wsp><NRf>[DB]
	INTERPOLATION SETTING	OFFSET(IN) **.**dB	:CALCulate:PARAmeter[:CATegory]:NF:IOFFset<wsp><NRf>[DB]
		OFFSET(OUT) **.**dB	:CALCulate:PARAmeter[:CATegory]:NF:OOFFset<wsp><NRf>[DB]
		ASE ALGO AUTO-FIX	:CALCulate:PARAmeter[:CATegory]:NF:AALGo<wsp>AFIX 0

ANALYSIS Parameter				Control Command
ANALYSIS 2 3/7				
EDFA-NF	INTERPOLATION SETTING	ASE ALGO	MANUAL-FIX	:CALCulate:PARAmeter[:CATegory]:NF:AALGo<wsp>MFIx 1
			AUTO-CTR	:CALCulate:PARAmeter[:CATegory]:NF:AALGo<wsp>ACENter 2
EDFA-NF	INTERPOLATION SETTING	ASE ALGO	MANUAL-CTR	:CALCulate:PARAmeter[:CATegory]:NF:AALGo<wsp>MCENter 3
		FITTING AREA		:CALCulate:PARAmeter[:CATegory]:NF:FARea<wsp><NRf>[M]
		MASK AREA		:CALCulate:PARAmeter[:CATegory]:NF:MARea<wsp><NRf>[M]
		FITTING ALGO	LINEAR	:CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp>LINear 0
			GAUSS	:CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp>GAUSs 1
			LORENZ	:CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp>LORenZ 2
			3RD POLY	:CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp>3RD 3
			4TH POLY	:CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp>4TH 4
			5TH POLY	:CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp>5TH 5
		POINT DISPLAY ON/OFF		:CALCulate:PARAmeter[:CATegory]:NF:PDISplay<wsp>OFF ON 0 1
FILTER-PK	PEAK LEVEL	SW ON/OFF		:CALCulate:PARAmeter[:CATegory]:FILPk<wsp>PLEVel,SW,OFF ON 0 1
	PEAK WAVELENGTH	SW ON/OFF		:CALCulate:PARAmeter[:CATegory]:FILPk<wsp>PWAVelength,SW,OFF ON 0 1
	WAVELENGTH	ALGO		:CALCulate:PARAmeter[:CATegory]:FILPk<wsp>MWAVelength,ALGO,<data>
		THRESH LEVEL *.**dB		:CALCulate:PARAmeter[:CATegory]:FILPk<wsp>MWAVelength,TH,<NRf>[DB]
		K		:CALCulate:PARAmeter[:CATegory]: FILPk<wsp>MWAVelength,K,<NRf>
		MODE FIT ON/OFF		:CALCulate:PARAmeter[:CATegory]: FILPk<wsp>MWAVelength,MFIT,OFF ON 0 1
CENTER WAVELENGTH	MODE DIFF *.**dB			:CALCulate:PARAmeter[:CATegory]: FILPk<wsp>MWAVelength,MDIFF,<NRf>[DB]

ANALYSIS Parameter		Control Command	
ANALYSIS 2 4/7			
FILTER-PK	SPECTRUM WIDTH	SW ON/OFF	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp>SWIDth,SW,OFF ON 0 1
		ALGO	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp> SWIDth,ALGO,<data>
		THRESH LEVEL **. **dB	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp> SWIDth,TH,<NRf> [DB]
		K	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> SWIDth,K,<NRf>
		MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> SWIDth,MFIT,OFF ON 0 1
		MODE DIFF *. **dB	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> SWIDth,MDIFf,<NRf>[DB]
FILTER-PK	CROSS TALK	SW ON/OFF	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp>XTALk,SW,OFF ON 0 1
		ALGO	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp> XTALk,ALGO,<data>
		THRESH LEVEL **. **dB	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp> XTALk,TH,<NRf> [DB]
		K	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> XTALk,K,<NRf>
		MODE FIT ON/OFF	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> XTALk,MFIT,OFF ON 0 1
		MODE DIFF *. **dB	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> XTALk,MDIFf,<NRf>[DB]
		CH SPACE ± *. **nm	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> XTALk,CSPace,<NRf>[M]
		SEARCH AREA ± *. **nm	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> XTALk,SARea,<NRf>[M]
	RIPPLE WIDTH	SW ON/OFF	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp>RWIDth,SW,OFF ON 0 1
		THRESH LEVEL **. **dB	:CALCulate:PARAmeter[:CATegory]:FILPk<wsp> RWIDth,TH,<NRf> [DB]
		MODE DIFF *. **dB	:CALCulate:PARAmeter[:CATegory]: FILPk<wsp> RWIDth,MDIFf,<NRf>[DB]
FILTER BOTTOM	BOTTOM LEVEL	SW ON/OFF	:CALCulate:PARAmeter[:CATegory]:FILBtm<wsp>BLEVel,SW,OFF ON 0 1
	BOTTOM WAVELENGTH	SW ON/OFF	:CALCulate:PARAmeter[:CATegory]: FILBtm<wsp>BWAVelength,SW,OFF ON 0 1

ANALYSIS Parameter		Control Command
ANALYSIS 2 5/7		
FILTER BOTTOM	CENTER WAVELENGTH	SW ON/OFF
		ALGO
		THRESH LEVEL **. **dB
	CENTER WAVELENGTH	MODE DIFF *. **dB
	NOTCH WIDTH	SW ON/OFF
		ALGO
		THRESH LEVEL **. **dB
		MODE DIFF *. **dB
	CROSS TALK	SW ON/OFF
		ALGO
		THRESH LEVEL **. **dB
		MODE DIFF *. **dB
		CH SPACE $\pm$ *. **nm
		SEARCH AREA $\pm$ *. **nm
WDM FIL-PK CHANNEL DETECTION/ NOMINAL WAVELENGTH		ALGO
		THRESH LEVEL **. **dB
		MODE DIFF *. **dB
		TEST BAND *. ***nm

ANALYSIS Parameter		Control Command
ANALYSIS 2 6/7		
WDM FIL-PK	PEAK WAVELENGTH TH/LEVEL	SW ON/OFF :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp>PWAVelength,SW,OFF ON 0 1
	XdB WIDTH / CENTER WAVELENGTH	SW ON/OFF :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp>CWAVelength,SW,OFF ON 0 1
	XdB STOP BAND	SW ON/OFF :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp>SBANd,SW,OFF ON 0 1
	XdB STOP BAND	THRESH LEVEL **.*dB :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp> SBANd,TH,<NRf> [DB]
	XdB PASS BAND	SW ON/OFF :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp>PBANd,SW,OFF ON 0 1
		THRESH LEVEL **.*dB :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp> PBANd,TH,<NRf>[DB]
		TEST BAND *.*nm :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp> PBANd,TBANd,<NRf>[DB]
	RIPPLE	SW ON/OFF :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp>RIPPlE,SW,OFF ON 0 1
		TEST BAND *.*nm :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp> RIPPlE,TBANd,<NRf>[DB]
	CROSS TALK	SW ON/OFF :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp>XTALk,SW,OFF ON 0 1
		SPACING *.*nm :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp> XTALk,SPACIng,<NRf>[M]
		TEST BAND *.*nm :CALCulate:PARAmeter[:CATegory]: WFPeak<wsp> XTALk,TBANd,<NRf>[DB]

ANALYSIS Parameter		Control Command
ANALYSIS 2 7/7		
WDM FIL-BTM	CHANNEL DETECTION/ NOMINAL WAVELENGTH	ALGO
		:CALCulate:PARAmeter[:CATegory]: WFBottom<wsp>NWAVelength,ALGO,<data>
		THRESH LEVEL **.***dB
		:CALCulate:PARAmeter[:CATegory]: WFPeak<wsp> WFBottom,TH,<NRf> [DB]
	MODE DIFF	*.***dB
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>NWAVelength,MDIFF,<NRf>[DB]
	TEST BAND	*.***nm
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>NWAVelength,TBAND<NRf>[DB]
	BOTTM WAVELENGTH/LEVEL	SW ON/OFF
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>BWAVelength,SW,OFF ON 0 1
	XdB NOTCH WIDTH/CE NTER	SW ON/OFF
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>CWAVelength,SW,OFF ON 0 1
	XdB STOP BAND	ALGO
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>SBAND,ALGO,<data>
	THRESH LEVEL	**.*dB
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp> SBAND,TH,<NRf>[DB]
	XdB ELIMINAT ION BAND	SW ON/OFF
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>EBAND,SW,OFF ON 0 1
	XdB ELIMINATIO N BAND	THRESH LEVEL **.*dB
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp> EBAND,TH,<NRf>[DB]
	TEST BAND	*.***nm
		:CALCulate:PARAmeter[:CATegory]: WFBottom<wsp> EBAND,TBAND,<NRf>[DB]
	RIPPLE	SW ON/OFF
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>RIPPLE,SW,OFF ON 0 1
	TEST BAND	*.***nm
		:CALCulate:PARAmeter[:CATegory]: WFBottom<wsp> RIPPLE,TBAND,<NRf>[DB]
	CROSS TALK	SW ON/OFF
		:CALCulate:PARAmeter[:CATegory]: WFBottom <wsp>XTALK,SW,OFF ON 0 1
		SPACING *.***nm
		:CALCulate:PARAmeter[:CATegory]: WFBottom<wsp> XTALK,SPACing,<NRf>[M]
	TEST BAND	*.***nm
		:CALCulate:PARAmeter[:CATegory]: WFBottom<wsp> XTALK,TBAND,<NRf>[DB]

2.6.3 Remote Command Tree

Command	Parameter	Page
COMMON command		
*CLS	None	2-53
*ESE	<integer>	2-53
*ESE?	None	2-54
*ESR?	None	2-54
*IDN?	None	2-55
*OPC	None	2-55
*OPC?	None	2-56
*RST	None	2-56
*SRE	<integer>	2-57
*SRE?	none	2-57
*STB?	none	2-58
*TRG	none	2-58
*TST?	none	2-58
*WAI	none	2-59
ABORt		2-60
CALCulate		
:CATegory	SWTHresh SWENvelope SWRMs SWPKrms NOTCh DFBLd FPLD LED SMSR POWer PMD WDM NF FILPk FILBtm WFPeak WFBtm	2-62
:DATA?	none	2-62
:CGAin?	none	2-63
:CNF?	none	2-63
:CPOWers?	none	2-64
:CSNR?	none	2-64
:CWAVelengths?	none	2-65
:NCHannels?	none	2-65
[:IMMediate]	none	2-66
:AUTO	OFF ON 0 1	2-66
:LMARker		
:AOFF	none	2-66
:SRANge	OFF ON 0 1	2-67
:SSPan	none	2-67
:SZSPan	none	2-68
:X	1 2, <NRf>[M HZ]	2-68
:Y	3 4, <NRf>[DBM/DB/%]	2-68
:MARKer		
:AOFF	none	2-68
:AUTO	OFF ON 0 1	2-69
:FUNCtion		
:FORMat	OFFSet SPACing 0 1	2-69
:UPDate	OFF ON 0 1	2-69
:MAXimum	none	2-70
:LEFT	none	2-70
:NEXT	none	2-70
:RIGHt	none	2-70
:SCENter	none	2-71
:AUTO	OFF ON 0 1	2-71
:SRLevel	none	2-71
:AUTO	OFF ON 0 1	2-72
:SZCenter	none	2-72
:MINimum	none	2-72
:LEFT	none	2-73
:NEXT	None	2-73
:RIGHt	none	2-73
:SCENter	none	2-73
:SRLevel	none	2-74

[:STATE]	<marker>,OFF ON 0 1	2-74
:SZCenter	None	2-74
:UNIT	WAVelength FREQuency	2-75
:X	<marker>,<NRf> [M HZ]	2-76
:Y?	<marker>	2-76
:MATH		
:TRC	A-B(LOG) B-A(LOG) A+B(LOG) A+B(LIN) A-B(LIN) B-A(LIN) 1-K(A/B) 1-K(B/A)	2-76
:K	<NRf>	2-77
:TRF	C-D(LOG) D-C(LOG) C+D(LOG) D-E(LOG) E-D(LOG) D+E(LOG) C+D(LIN) C-D(LIN) D-C(LIN) D+E(LIN) D-E(LIN) E-D(LIN)	2-77
:TRG	C-F(LOG) F-C(LOG) C+F(LOG) E-F(LOG) F-E(LOG) E+F(LOG) C+F(LIN) C-F(LIN) F-C(LIN) E+F(LIN) E-F(LIN) F-E(LIN) NORMA NORMB NORMC CVFTA CVFTB CVFTC MKRFT PKCVFTA PKCVFTB PKCVFTC	2-78
:CVFT		
:FLAGo	GAUSS LORENz 3RD 4 TH 5TH 0 1 2 3 4	2-78
:THResh	<integer>[DB]	2-79
:OPARea	ALL INL1-L2 OUTL1-L2 0 1 2	2-79
:PCVft:THResh	<integer>[DB]	2-80
:PARAmeter		
[:CATegory]		
:DFBLd	<item>,<paramater name>,<data>	2-80
:FILBtm	<item>,<paramater name>,<data>	2-81
:FILPk	<item>,<paramater name>,<data>	2-82
:FPLD	<item>,<paramater name>,<data>	2-83
:LED	<item>,<paramater name>,<data>	2-84
:NF		
:AALGo	AFIX MFIx ACENter MCENter 0 1 2 3	2-85
:FALGo	LINear GAUSs LORenz 3RD 4TH 5TH 0 1 2 3 4 5	2-85
:FARea	<NRf>[M]	2-86
:IOFFset	<NRf>[DB]	2-86
:MARea	<NRf>[M]	2-86
:MDIFf	<NRf>[DB]	2-87
:OOFfset	<NRf>[DB]	2-87
:PDISplay	OFF ON 0 1	2-87
:TH	<NRf>[DB]	2-87
:NOTCh		
:K	<NRf>	2-88
:TH	<NRf>[DB]	2-88
:TYPE	PEAK BOTToM 0 1	2-88
:PMD:TH	<NRf>[DB]	2-88
:POWer:OFFSet	<NRf>[DB]	2-89
:SMSR		
:MASK	<NRf>[M]	2-89
:MODE	SMSR1 SMSR2	2-89
:SWENvelope		
:K	<NRf>	2-89
:TH1	<NRf>[DB]	2-90
:TH2	<NRf>[DB]	2-90
:SWPKrms		
:K	<NRf>	2-90
:TH	<NRf>[DB]	2-91
:SWRMs		
:K	<NRf>	2-91
:TH	<NRf>[DB]	2-91
:SWTHresh		
:K	<NRf>	2-92
:MFIT	OFF ON 0 1	2-92
:TH	<NRf>[DB]	2-92

:WDM		
:DMASK	<NRf>[DB]	2-93
:DTYPE	ABSolute RELative MDRift GDRift 0 1 2 3	2-93
:DUAL	OFF ON 0 1	2-94
:FALGo	LINear GAUSs LORenz 3RD 4TH 5TH 0 1 2 3 4 5	2-94
:MARea	<NRf>[M]	2-95
:MDIFF	<NRf>[DB]	2-95
:MMReset	none	2-95
:NALGo	AFIX MFIx ACENter MCENter PIT 0 1 2 3 4	2-96
:NARea	<NRf>[M]	2-96
:NBW	<NRf>[M]	2-96
:OSLope	OFF ON 0 1	2-97
:PDISplay	OFF ON 0 1	2-97
:RCH	<integer>	2-97
:RELation	OFFSet SPACing 0 1	2-98
:TH	<NRf>[DB]	2-98
:WFBottom	<item>,<paramater name>,<data>	2-99
:WFPeak	<item>,<paramater name>,<data>	2-100
:COMMON		
:MDIFF	<NRf>[DB]	2-101
CALibration		
:ALIGN[:IMMediate]	none	2-102
:POWer		
:OFFSet:TABLE	<integer>,<NRf>[DB]	2-102
:WAVelength		
:EXTernal		
:[:IMMediate]	none	2-102
:SOURce	LASEr GASCell	2-103
:WAVelength	<NRf>M	2-103
:INTernal[:IMMediate]	none	2-103
:OFFSet:TABLE	<integer>,<NRf>	2-104
:ZERO[:AUTO]	OFF ON 0 1 ONCE	2-104
DISPlay		
:COLor	0 1 2 3 4 5	2-105
[:WINDow]		
:OVlew		
:POSition	OFF LEFT RIGHT 0 1 2	2-105
:SIZE	LARGE SMALL 0 1	2-105
:SPLIt	OFF ON 0 1	2-106
:HOLD		
:LOWer	OFF ON 0 1	2-106
:UPPer	OFF ON 0 1	2-106
:POSition	<trace name>,UP LOW 0 1	2-107
:TEXT		
:CLEar	none	2-107
:DATA	<"string">	2-107
:TRACe		
:X[:SCALe]		
:CENTer	<NRf> [M HZ]	2-108
:INITialize	none	2-108
:SMSCale	none	2-108
:SPAN	<NRf> [M HZ]	2-108
:SRANge	OFF ON 0 1	2-109
:STARt	<NRf> [M HZ]	2-109
:STOP	<NRf> [M HZ]	2-109
:Y		
:NMASt	<NRf>DB	2-109

:TYPE	VERTical HORizontal 0 1	2-110
[:SCALe]		
:DNUMber	8 10 12	2-110
:Y1		
[:SCALe]		
:BLEVel	<NRf>[W MW UW NW]	2-110
:PDIVision	<NRf>[DB]	2-111
:RLEVel	<NRf>[DBM W MW UW NW]	2-111
:RPOSition	<integer>[DIV]	2-111
:SPACing	LOGarithmic LINear 0 1	2-112
:UNIT	DBM W DBM/NM W/NM 0 1 2 3	2-112
:Y2		
[:SCALe]		
:AUTO	OFF ON 0 1	2-112
:LENGth	<NRf>[KM]	2-113
:OLEVel	<NRf>[DB DB/KM]	2-113
:PDIVision	<NRf>[DB DB/KM %]	2-114
:RPOSition	<integer>[DIV]	2-114
:SMINimum	<NRf>[%]	2-114
:UNIT	DB LINear DB/KM % 0 1 2 3	2-115
FORMat		
[:DATA]	REAL[,64 ,32] ASCii	2-116
HCOPy		
:DESTination	INTernal EXTernal FILE 0 1 2	2-118
:DEVice:COLor	OFF ON 0 1	2-118
[:IMMediate]	none	2-118
:FEED	[<integer>]	2-119
:FUNCTION		
:CALCulate:LIST	none	2-119
:MARKer:LIST	none	2-119
INITiate		
[:IMMediate]	none	2-120
:SMODE	SINGle REPeat AUTO SEGment 1 2 3 4	2-120
MEMory		
:CLEar	<integer>	2-121
:EMPTy?	<integer>	2-121
:LOAD	<integer>,<trace name>	2-121
:STORE	<integer>,<trace name>	2-122
MMEMory		
:CATalog?	[INTernal FLOPPy]	2-123
:CDIRectory	<"directory name">	2-123
:CDRive	INTernal FLOPPy	2-124
:COPY	<"source file name">,[INTernal FLOPPy], <"destination file name">,[INTernal FLOPPy]	2-124
:DELeTe	<"file name">,[INTernal FLOPPy]	2-124
:INITialize	[FLOPPy]	2-124
:LOAD		
:MEMory	<integer>,<"filename"> [,INTernal FLOPPy]	2-125
:PROGram	<integer>,<"filename"> [,INTernal FLOPPy]	2-125
:SETTing	<"filename"> [,INTernal FLOPPy]	2-125
:TEMPlate	<template>,<"filename"> [,INTernal FLOPPy]	2-125
:TRACe	<trace name>,<"filename"> [,INTernal FLOPPy]	2-126
:MDIRectory	<"directory name">,[INTernal FLOPPy]	2-126
:REName	<"new file name">,<"old file name">,[INTernal FLOPPy]	2-126
:STORE		
:ARESult	<"filename"> [,INTernal FLOPPy]	2-126
:DATA	<"filename"> [,INTernal FLOPPy]	2-127
:ITEM	DATE LABel DATA CONDition TRACe,OFF ON 0	2-127

:MODE	1 ADD OVER	2-127
:TYPE	CSV DT 0 1	2-128
:GRAPHics	B&W COLor, BMP TIFF,<"filename"> [,INTernal FLOPPy]	2-128
:MEMory	<integer>,BIN CSV,<"filename"> [,INTernal FLOPPy]	2-128
:PROGram	<integer>,<"filename"> [,INTernal FLOPPy]	2-129
:SETTing	<"filename"> [,INTernal FLOPPy]	2-129
:TEMPlate	<template>,<"filename"> [,INTernal FLOPPy]	2-129
:TRACe	<trace name>,BIN CSV,<"filename"> [,INTernal FLOPPy]	2-129
PROGram		
:EXECute	<integer>	2-130
SENSe		
:AVERage:COUNT	<integer>	2-131
:BANDwidth[:BWIDTH[:RESolution]	<NRf>[M]	2-131
:CHOPper	OFF ON(CHOP) SWITCh 0 1 2	2-131
:CORRection		
:LEVel:SHIFt	<NRf>[DB]	2-131
:RVELocity:MEDium	AIR VACuum 0 1	2-132
:WAVelength:SHIFt	<NRf>[M]	2-132
:SENSe	NHLD NAUT MID HIGH1 HIGH2 HIGH3 0 1 2 3 4 5	2-133
:SETTing		
:ATTenuator	OFF ON 0 1	2-133
:SWEep		
:POINts	<integer>	2-134
:AUTO	OFF ON 0 1	2-135
:SEGMENT:POINts	<integer>	2-135
:STEP	<NRf>[M]	2-135
:TIME		
:ONM	<integer> [SEC]	2-136
:INTERval	<integer> [SEC]	2-136
:TLS Sync	OFF ON 0 1	2-136
:WAVelength		
:CENTer	<NRf>[M HZ]	2-136
:SPAN	<NRf>[M HZ]	2-137
:SRANge	OFF ON 0 1	2-137
:STARt	<NRf>[M HZ]	2-137
:STOP	<NRf>[M HZ]	2-137
STATus		
:OPERation		
:CONDition?	none	2-138
:ENABle	<integer>	2-138
[:EVENT?]	none	2-138
:PRESet		2-139
:QUESTionable		
:CONDition?	none	2-139
:ENABle	<integer>	2-139
[:EVENT?]	none	2-140
SYSTem		
:BUZZer		
:CLICk	OFF ON 0 1	2-141
:WARNing	OFF ON 0 1	2-141
:COMMunicate		
:CFORmat	AQ6317 AQ6319 0 1	2-142
:GPIB2:ADDRess	<integer>	2-142

:GPIB2:TLS:ADDRess	<integer>	2-143
:DATE	yyyy,mm,dd	2-143
:DISPlay		
:TRANSPARENT	OFF ON 0 1	2-143
:UNCAL	OFF ON 0 1	2-143
:ERRor		
[:NEXT]?	none	2-144
:GRID	12.5GHZ 50GHZ 100GHZ 200GHZ CUSTom 0 1 2 3 4 5	2-144
:CUSTom		
:CLEAr:ALL	none	2-144
:DELeTe	<grid number>	2-144
:INSert	<NRf>[M HZ]	2-145
:SPACing	<NRf>[GHZ]	2-145
:STARt	<NRf>[M HZ]	2-145
:STOP	<NRf>[M HZ]	2-145
:REFerence	<NRf>[M HZ]	2-146
:PRESet	none	2-146
:TIME	hh,mm,ss	2-146
:VERSion?		2-146
TRACe		
:ACTive	<trace name>	2-147
:ATTRibute[:<trace name>]	WRITE FIX MAX MIN RAVG CALC	2-147
:RAVG[:<trace name>]	<integer>	2-148
:COPY	<source trace>,<distination trace>	2-148
[:DATA]		
:SNUMber?	<trace name>	2-148
:X?	<trace name>[,<start point>,<stop point>]	2-149
:Y?	<trace name>[,<start point>,<stop point>]	2-150
:DELeTe	<trace name>	2-151
:ALL		2-151
:STATe[:<trace name>]	OFF ON 0 1	2-151
:TEMPlate		
:DATA	<template>,<wavelength>,<level>	2-152
:ADELeTe	<template>	2-152
:ETYPe	<template>,NONE A B 0 1 2	2-152
:MODE	<template>,ABSolute RELative 0 1	2-153
:DISPlay	<template>,OFF ON 0 1	2-153
:GONogo	OFF ON 0 1	2-153
:LEVel:SHIFt	<NRf>[DB]	2-154
:RESult?		2-154
:TTYPe	UPPer LOWer U&L 0 1 2	2-154
:WAVelength:SHIFt	<NRf>[M]	2-154
TRIGger		
[:SEQUence]		
:DELay	<NRf>[S MS US]	2-155
:SLOPe	RISE FALL 0 1	2-155
:STATe	OFF ON 0 1	2-155
UNIT		
:POWER:DIGit	1 2 3	2-156
:X	WAVelength FREQuency 0 1	2-156

2.6.4 Common Commands

This AQ6319 supports the common commands indicated as "Required" in the IEEE 488.2 Std. Column in Table 2-11, Correspondence between the IEEE 488.2 Standard and Common Commands.

Table 2-11: Correspondence between the IEEE 488.2 Standard and Common Commands

Command	Name	IEEE 488.2 Std.	AQ6319
*AAD	Accept Address Command	Option	
*CAL?	Calibration Query	Option	
*CLS	Clear Status Command	Required	○
*DDT	Define Device Trigger Command	Option if DT1	
*DDT?	Define Device Trigger Query	Option if DT1	
*DLF	Disable Listener Function Command	Option	
*DMC	Define Macro Command	Option	
*EMC	Enable Macro Command	Option	
*EMC?	Enable Macro Query	Option	
*ESE	Standard Event Status Enable	Required	○
*ESE?	Standard Event Status Enable Query	Required	○
*ESR?	Standard Event Status Register Query	Required	○
*GMC?	Get Macro Contents Query	Option	
*IDN?	Identification Query	Required	○
*IST?	Individual Status Query	Required for PP1	
*LMC?	Learn Macro Query	Option	
*LRN?	Learn Device Setup Query	Option	
*OPC	Operation Complete Command	Required	○
*OPC?	Operation Complete Query	Required	○
*OPT	Option Identification Query	Option	
*PCB	Pass Control Back Command	Required if not C0	
*PMC	Purge Macro Command	Option	
*PRE	Parallel Poll Register Enable Command	Required for PP1	
*PRE?	Parallel Poll Register Enable Query	Required for PP1	
*PSC	Power On Status Clear Command	Option	
*PSC?	Power On Status Clear Query	Option	
*PUD	Protected User Data Command	Option	
*PUD?	Protected User Data Query	Option	
*RCL	Recall Command	Option	
*RDT	Resource Description Transfer	Option	
*RDT?	Resource Description Transfer Query	Option	
*RST	Reset Command	Required	○
*SAV	Save Command	Option	
*SRE	Service Request Enable Command	Required	○
*SRE?	Service Request Enable Query	Required	○
*STB?	Read Status Byte Query	Required	○
*TRG	Trigger Command	Required if DT1	○
*TST?	Self-Test Query	Required	○
*WAI	Wait-to-Continue Command	Required	○

CLS (Clear Status Command)*<Operation>**

Clears all event status registers, the summary of which is reflected in the status byte register.

<Syntax>

*CLS

<Description>

- Clears all queues, with the exception of the output queue, and all event registers, with the exception of the MAV summary message.
- After executing this command, OCIS (Operation Complete Command Idle State) and OQIS (Operation Complete Query Idle State) are brought about.

<Query Response>

None

<Command Type>

Sequential command

ESE (Standard Event Status Enable Command)*<Operation>**

Sets the standard event enable register.

<Syntax>

*ESE<wsp><integer>

<Parameters>

<integer> : 0 to 255

<Description>

- An item having had its bit set becomes enabled.
- Presets to the default value in the following cases:
 - Power ON
 - When "0" is set
- The set value remains the same in the following cases:
 - *RST command
 - *CLS command
 - Device clear (DCL, SDC)

<Query Response>

<integer> 0 to 255

<Default>

0

<Command Type>

Sequential command

ESE? (Standard Event Status Enable Query)*<Operation>**

Reads the standard event enable register.

<Syntax>

*ESE?

<Description>

None

<Query Response>

<integer> 0 to 255

<Command Type>

Sequential command

ESR? (Standard Event Status Register Query)*<Operation>**

Reads and then clears the contents of the standard event status register.

<Syntax>

*ESR?

<Description>

- The return value of this query is not affected by ESE (Event Status Enable Register).

<Query Response>

<integer> 0 to 255

<Command Type>

Overlap command

***IDN? (Identification Query)**

<Operation>

Gets a character string indicating this unit type and firmware version.

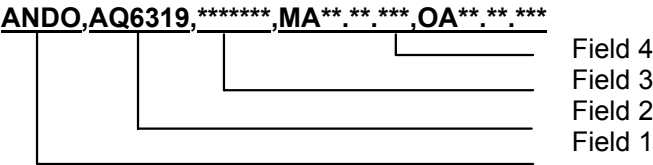
<Syntax>

*IDN?

<Description>

- Outputs 4 field data delimited by a comma.
Field 1: Manufacturer "ANDO"
Field 2: Model "AQ6319"
Field 3: Serial number Actual serial number (8-digit character string)
Field 4: Firmware version. Each version of main and sub software (delimited by a comma ",")

<Query Response>



<Command Type>

Sequential command

***OPC (Operation Complete Command)**

<Operation>

Sets bit 0 (OPC) of the standard event status register (ESR) if operations waiting to be processed have all been completed.

<Syntax>

*OPC

<Description>

- At the time this command is recognized, the command changes from OCIS (Operation Complete Command Idle State) to OCAS (Operation Complete Command Active State). When the no-operation pending flag is set to "True," it sets bit 0 (OCR) of ESR and returns to OCIS.
- If any of the following conditions are established, this command is disabled and is forced to return to OCIS.
 - (1) Power ON
 - (2) Device clear
 - (3) *CLS or *RST command

<Command Type>

Overlap command

OPC? (Operation Complete Query)*<Operation>**

Outputs "1" if operations waiting to be processed have been completed.

<Syntax>

*OPC?

<Description>

- At the time this command is recognized, the command changes from OCIS (Operation Complete Command Idle State) to OCAS (Operation Complete Command Active State). When the no-operation pending flag is set to "True," it sets "1" to the output queue and returns to OCIS.
- If any of the following conditions are established, this command is disabled and is forced to return to OCIS.
 - (1) Power ON
 - (2) Device clear
 - (3) *CLS or *RST command

<Query Response>

1

<Command Type>

Overlap command

RST (Reset Command)*<Operation>**

Executes a device clear to return this unit to the known status.

<Syntax>

*RST

<Description>

- Stops operation being processed and returns this unit to the known set value (default value) immediately.
- This unit's parameters are cleared.
- The following items will remain the same.
 - Status of the GPIB interface
 - GPIB address
 - Output queue
 - SRE
 - ESE
 - Calibration data affecting this unit's specifications

<Command Type>

Overlap command

SRE (Service Request Enable Command)*<Operation>**

Sets the service request enable register.

<Syntax>

*SRE<wsp><integer>

<Parameters>

<integer> : 0 to 255

<Description>

- An item having had its bit set becomes enabled.
- Resets to the default value in the following cases:
 - Power ON
 - When "0" is set
- The set value remains the same in the following cases:
 - *RST
 - *CLS
 - Device clear (DCL, SDC)

<Default>

0

<Command Type>

Sequential command

SRE? (Service Request Enable Query)*<Operation>**

Reads the service request enable register.

<Syntax>

*SRE?

<Description>

None

<Query Response>

<integer> 0 to 255

<Command Type>

Sequential command

STB? (Read Status Byte Query)*<Operation>**

Reads the current value of the status byte register.

<Syntax>

*STB?

<Description>

- STB will not be cleared even when the contents of the register are read.

<Query Response>

<integer> 0 to 255

<Command Type>

Overlap command

TRG (Trigger Command)*<Operation>**

Performs a <SINGLE> sweep under the sweep conditions established immediately before receiving the command.

<Syntax>

*TRG

<Description>

Performs a <SINGLE> sweep regardless of the setting condition of the :INITiate:CONTinuous command.

<Command Type>

Command subject to overlapping

TST? (Self Test Query)*<Operation>**

Performs this unit's self-test and returns the status.

<Syntax>

*TST?

<Description>

Of the initialization sequence to be run at startup, this command executes the following operations to output their results. During initialization, the screen maintains the waveform display.

- Motor's return to the origin
- AMP auto-offset

<Query Response>

<integer> 0: Normal, 1: Motor initialization error, 2: AMP offset error

<Command Type>

Sequential command

WAI (Wait to Continue Command)*<Operation>**

Prevents this unit from executing another command until the execution of the current command is complete.

This command is valid only when the successive command is an overlap command and is invalid if another type of command follows.

<Syntax>

*WAI

<Description>

- Becomes invalid by device clear.

<Command Type>

Sequential command

2.6.5 Instrument-Specific Commands

2

ABORt Sub System Command**ABORt**

Stops operations such as measurements and calibration.

Syntax: ABORt

Description: Objects to be stopped are as follows:
:CALibration:ALIGn[:IMMediate]
:CALibration:WAVelength
:INITiate
:PROGram:EXECute
:HCOPY[:INITiate]
:HCOPY[:INITiate]:FUNCTion:CALCulate:LIST
:HCOPY[:INITiate]:FUNCTion:MARKer:LIST

Parameters: None

Query Response: None

Example: abort

Command Type: Overlap command

CALCulate Sub System Command

<General>

- Command about following function is summarized in this Sub System.
 - Analysis function (Spectrum Width, ANALYSIS1 , ANALYSIS2)
 - Marker function(Δ marker , line marker)
 - Calculation function of trace

- The following procedure is performed in order to remote control carry out Analysis Function.
 - ① Selects the Analysis Algorithm(CALCulate:CATegory command)
 - ② Setting the Analysis Parameter(CALCulate:PARAmeter command)
 - ③ Execute the Analysis Function(CALCulate[:IMMediate] command)
 - ④ Gets the Analysis results(CALCulate:DATA? Command)

- The following command is used in order to remote control carry out Peak/Bottom search function.
 - CALCulate:MARKer:MAXimum | MINimum command

- The following command is used to in order to remote control carry out Marker function.
 - Δ marker: CALCulate:MARKer command
 - Line marker: CALCulate:LMARker command

- The following command is used to in order to remote control carry out Calculation function of tree.
 - CALCulate:MATH command

CALCulate:CATegory

Sets or acquires the type of analysis.

- Syntax:**
- :CALCulate:CATegory<wsp>SWThresh | SWENvelope | SWRMs | SWPKrms | NOTCh | DFBld | FPLD | LED | SMSR | POWer | PMD | WDM | NF FILPk | FILBtm | WFPeak | WFBtm | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16
 - :CALCulate:CATegory?

Description:

- Even when this command is executed, no analysis is performed unless the CALCulate[:IMMediate] command is executed.

Parameters: Specify the type of analysis using the following parameters.

SWThresh 0	Spectrum width analysis (THRESH)
SWENvelope 1	Spectrum width analysis (ENVELOPE)
SWRMs 2	Spectrum width analysis (RMS)
SWPKrms 3	Spectrum width analysis (PEAK-RMS)
NOTCh 4	Notch width analysis
DFBld 5	DFB-LD parameter analysis
FPLD 6	FP-LD parameter analysis
LED 7	LED parameter analysis
SMSR 8	SMSR analysis
POWer 9	Power analysis
PMD 10	PMD analysis
WDM 11	WDM analysis
NF 12	NF analysis
FILPk 13	Filter peak analysis
FILBtm 14	Filter bottom analysis
WFPeak 15	WDM FIL-PK analysis
WFBtm 16	WDM FIL-BTM analysis

Query Response: <integer> (0 to 16)

Example:

```
calc:cat swth
calc:cat? → 0
```

Command Type: Sequential command

CALCulate:DATA?

Acquires the analysis results.

- Syntax:**
- :CALCulate:DATA?

Description:

- The analysis result at the time of the last analysis execution is acquired.
- If the analysis function has not been executed, a query error occurs.

Parameters: None

Query Response: See section 2.6.4 "Output Format of the CALCulate :DATA? Query".

Example:

```
calc:data?
```

Command Type: Sequential command

CALCulate:DATA:CGain?

Acquires the gain value of the EDFA-NF analysis results.

- Syntax:** ▪ :CALCulate:DATA:CGain?
- Description:** ▪ If the analysis function has not been executed, a query error occurs.
- "0" is returned if there is no relevant return value (for example, if the analysis made is other than EDFA-NF analysis).
- The number of channels to be output can be acquired by the :CALCulate:DATA:NCHannels? command.
- The value is output as ASCII data or binary data according to the setting of FORMat[:DATA].
- Parameters:** None
- Query Response:** For ASCII data: <NRf>,<NRf>,...<NRf>
 For binary data: '#'<integer><byte num><data bytes>
- Example:** calc:data:cgain? → +1.00000000E+001,+1.00000000E+001
- Command Type:** Sequential command

CALCulate:DATA:CNF?

Acquires the NF value of the EDFA-NF analysis results.

- Syntax:** ▪ :CALCulate:DATA:CNF?
- Description:** ▪ If the analysis function has not been executed, a query error occurs.
- "0" is returned if there is no relevant return value (for example, if the analysis made is other than EDFA-NF analysis).
- The number of channels to be output can be acquired by the :CALCulate:DATA:NCHannels? command.
- The value is output as ASCII data or binary data according to the setting of FORMat[:DATA].
- Parameters:** None
- Query Response:** For ASCII data: <NRf>,<NRf>,...<NRf>
 For binary data: '#'<integer><byte num><data bytes>
- Example:** calc:data:cnf? → +1.00000000E+001,+1.00000000E+001
- Command Type:** Sequential command

CALCulate:DATA:CPOWers?

Acquires the level value of the WDM, EDFA-NF, WDM FIL-PK, or WDM FIL-BTM analysis results.

Syntax: ▪ :CALCulate:DATA:CPOWers?

Description:

- If the analysis function has not been executed, a query error occurs.
- "0" is returned if there is no relevant return value.
- The number of channels to be output can be acquired by the :CALCulate:DATA:NCHannels? command.
- The value to be output depends on the analysis performed.
 - WDM : LEVEL or MEAS LEVEL
 - EDFA-NF : INPUT LEVEL
 - WDM FIL-PK : PEAK LEVEL (output even if SW is OFF)
 - WDM FIL-BTM : PEAK LEVEL (output even if SW is OFF)
- The value is output as ASCII data or binary data according to the setting of FORMat[:DATA].

Parameters: None

Query Response: For ASCII data: <Nrf>,<Nrf>.....<Nrf>
For binary data: '#'<integer><byte num><data bytes>

Example: calc:data:cpowers?
+1.00000000E+001,+1.00000000E+001

Command Type: Sequential command

CALCulate:DATA:CSNR?

Acquires the SNR value of WDM analysis results.

Syntax: ▪ :CALCulate:DATA:CSNR?

Description:

- If the analysis function has not been executed, a query error occurs.
- "0" is returned if there is no relevant return value (for example, if analysis made is other than WDM analysis).
- The number of channels to be output can be acquired by the :CALCulate:DATA:NCHannels? command.
- The value is output as ASCII data or binary data according to the setting of FORMat[:DATA].

Parameters: None

Query Response: For ASCII data: <Nrf>,<Nrf>,<Nrf>
For binary data: '# '<integer><byte num><data bytes>

Example: calc:data:csnr? → +4.00000000E+001,+4.00000000E+001

Command Type: Sequential command

CALCulate:DATA:CWAVelengths?

Acquires the wavelength value of WDM, EDFA-NF, WDM FIL-PK, or WDM FIL-BTM analysis results.

Syntax:	• :CALCulate:DATA:CWAVelengths?
Description:	• If the analysis function has not been executed, a query error occurs. • "0" is returned if there is no relevant return value. • The number of channels to be output can be acquired by the :CALCulate:DATA:NCHannels? command. • The value to be output depends on the analysis performed. - WDM : WAVELENGTH or MEAS WL - EDFA-NF : WAVELENGTH - WDM FIL-PK : NOMINAL WAVELENGTH - WDM FIL-BTM : NOMINAL WAVELENGTH • The value is output as ASCII data or binary data according to the setting of FORMat[:DATA].
Parameters:	None
Query Response:	For ASCII data: <NRf>,<NRf>,...<NRf> For binary data: '#<integer><byte num><data bytes>'
Example:	calc:data:cpowers? → +1.55000000E-006,+1.56000000E-006
Command Type:	Sequential command

CALCulate:DATA :NCHannels?

Acquires the number of channels of the WDM, EDFA-NF, WDM FIL-PK, or WDM FIL-BTM analysis results.

Syntax:	• :CALCulate:DATA :NCHannels?
Description:	• If the analysis function has not been executed, a query error occurs. • "0" is returned if there is no relevant return value. • The value is output as ASCII data, regardless of the setting of FORMat[:DATA].
Parameters:	None
Query Response:	<integer>
Example:	calc:data:nchannels? → 16
Command Type:	Sequential command

CALCulate[:IMMediate]

Executes analysis.

Acquires the result of whether analysis has been performed.

Syntax:	• :CALCulate[:IMMediate] • :CALCulate[:IMMediate]?	
Description:	• Analysis is performed according to the latest analysis settings. • Analysis is performed on the following occasions: ① When CALCulate[:IMMediate] command is executed ② When parameter setting is modified by CALCulate[:IMMediate] command	
Parameters:	None	
Query Response:	0 1	0: Not performed, 1: Performed
Example:	<pre>calc calc? → 1</pre>	
Command Type:	Sequential command	

CALCulate[:IMMediate]:AUTO

Sets or acquires ON/OFF of the automatic analysis function.

Syntax:	• :CALCulate[:IMMediate]:AUTO<wsp>OFF ON 0 1 • :CALCulate[:IMMediate]:AUTO?	
Description:	• When the automatic analysis function is ON, the AQ6319 automatically activates an analysis function that is active after a sweep has ended.	
Parameters:	OFF ON 0 1	
Query Response:	0 1	0:OFF, 1:ON
Example:	<pre>calc:auto on calc:auto? → 1</pre>	
Command Type:	Sequential command	

CALCulate:LMARker:AOff

Clears all line markers.

Syntax:	• :CALCulate:LMARker:AOff
Description:	None
Parameters:	None
Query Response:	None
Example:	<pre>calc:lmar:aoff</pre>
Command Type:	Sequential command

CALCulate:LMARker:SRANge

Sets or acquires whether to limit an analytical range to the spacing between line markers L1 and L2.

Syntax:	▪ :CALCulate:LMARker:SRANge<wsp>OFF ON 0 1 ▪ :CALCulate:LMARker:SRANge?	
Description:	None	
Parameters:	OFF ON 0 1	
Query Response:	0 1	0:OFF, 1:ON
Example:	calc:lmar:sran on calc:lmar:sran? → 1	
Command Type:	Sequential command	

CALCulate:LMARker:SSPan

Sets spacing between line markers L1 and L2 for span.

Syntax:	▪ :CALCulate:LMARker:SSPan	
Description:	None	
Parameters:	None	
Query Response:	None	
Example:	calc:lmar:ssp	
Command Type:	Sequential command	

CALCulate:LMARker:SZSPan

Sets spacing between line markers L1 and L2 for zoom span.

Syntax:	• :CALCulate:LMARker:SZSPan
Description:	None
Parameters:	None
Query Response:	None
Example:	calc:lmar:szsp
Command Type:	Sequential command

CALCulate:LMARker:X

Places line markers L1 and L2 to specified positions.

Acquires the positions of line markers L1 and L2.

Syntax:	• :CALCulate:LMARker:X<wsp>1 2,<NRf>[M HZ] • :CALCulate:LMARker:X?<wsp>1 2
Description:	• A query error occurs in the following case: A specified line marker has not been placed.
Parameters:	1 2 Line marker number <NRf> Position of a line marker
Query Response:	<NRf> Position of a line marker
Example:	calc:lmar:x 1,1550.000nm calc:lmar:x? 1 → +1.55000000E-006
Command Type:	Sequential command

CALCulate:LMARker:Y

Places line markers L3 and L4 in specified positions.

Acquires the positions of line markers L3 and L4.

Syntax:	• :CALCulate:LMARker:Y<wsp>3 4,<NRf>[DBM DB %] • :CALCulate:LMARker:Y?<wsp>3 4
Description:	• A query error occurs in the following cases. A specified line marker has not been placed.
Parameters:	3 4 Line marker number <NRf> Position of a line marker
Query Response:	<NRf> Position of a line marker
Example:	calc:lmar:y 3,-10dBm calc:lmar:y? 3 → -1.00000000E+001
Command Type:	Sequential command

CALCulate:MARKer:AOff

Clears all markers.

Syntax:	• :CALCulate:MARKer:AOff
Description:	None
Parameters:	None
Query Response:	None
Example:	calc:mark:aoff
Command Type:	Sequential command

CALCulate:MARKer:AUTO

Sets or acquires ON/OFF of the auto search function.

Syntax:	• :CALCulate:MARKer:AUTO<wsp>OFF ON 0 1 • :CALCulate:MARKer:AUTO?
Description:	• When the auto search function is ON, this unit automatically performs a peak/bottom search through an active trace after a sweep has ended.
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	calc:mark:auto on calc:mark:auto? → 1
Command Type:	Sequential command

CALCulate:MARKer:FUNCTION:FORMAT

Sets the format of a difference value displayed in the area marker and acquires the format set.

Syntax:	• :CALCulate:MARKer:FUNCTION:FORMAT<wsp>OFFSet SPACing 0 1 • :CALCulate:MARKer:FUNCTION:FORMAT?
Description:	None
Parameters:	OFFSet 0 Displays the difference of each marker relative to the moving marker. SPACing 1 Displays the difference of each marker relative to a neighboring marker.
Query Response:	0 1 0:OFFSet, 1:SPACing
Example:	calc:mark:func:form spacing calc:mark:func:form? → 1
Command Type:	Sequential command

CALCulate:MARKer:FUNCTION:UPDATE

Sets or acquires ON/OFF of the automatic update function of fixed markers used when updating an active trace.

Syntax:	• :CALCulate:MARKer:FUNCTION:UPDATE<wsp>OFF ON 0 1 • :CALCulate:MARKer:FUNCTION:UPDATE?
Description:	• When the automatic update function is ON, the level positions of fixed markers automatically follow the waveform during updating of an active trace.
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	calc:mark:func:update on calc:mark:func:update? → 1
Command Type:	Sequential command

CALCulate:MARKer:MAXimum

Detects a peak and places the moving marker on that peak.

Syntax:	• :CALCulate:MARKer:MAXimum
Description:	None
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:max</code>
Command Type:	Sequential command

CALCulate:MARKer:MAXimum:LEFT

Detects the nearest peak existing on the left side of the current position of the moving marker and places the moving marker on that peak.

Syntax:	• :CALCulate:MARKer:MAXimum:LEFT
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:max:left</code>
Command Type:	Sequential command

CALCulate:MARKer:MAXimum:NEXT

Detects the highest peak that is below the level of the current position of the moving marker and places the moving marker on that peak.

Syntax:	• :CALCulate:MARKer:MAXimum:NEXT
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:max:next</code>
Command Type:	Sequential command

CALCulate:MARKer:MAXimum:RIGHT

Detects the nearest peak existing on the right side of the current position of the moving marker and places the moving marker on that peak.

Syntax:	• :CALCulate:MARKer:MAXimum:RIGHT
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:max:right</code>
Command Type:	Sequential command

CALCulate:MARKer:MAXimum:SCENter

Detects the peak wavelength and sets it as the measurement center waveform.

Syntax:	• :CALCulate:MARKer:MAXimum:SCENter
Description:	None
Parameters:	None
Query Response:	None
Example:	calc:mark:max:scenter
Command Type:	Sequential command

CALCulate:MARKer:MAXimum:SCENter:AUTO

Sets or acquires ON/OFF of the function to automatically detect the peak wavelength and set it as the measurement center wavelength.

Syntax:	• :CALCulate:MARKer:MAXimum:SCENter:AUTO<wsp>OFF ON 0 1 • :CALCulate:MARKer:MAXimum:SCENter:AUTO?
Description:	• When this function is ON, this unit automatically detects the peak wavelength of an active trace wavelength each time a sweep has ended, and sets it as the measurement center wavelength.
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	calc:mark:max:scenter:auto on calc:mark:max:scenter:auto? → 1
Command Type:	Sequential command

CALCulate:MARKer:MAXimum:SRLevel

Detects the peak level and sets it for the reference level.

Syntax:	• :CALCulate:MARKer:MAXimum:SRLevel
Description:	None
Parameters:	None
Query Response:	None
Example:	calc:mark:max:srlevel
Command Type:	Sequential command

CALCulate:MARKer:MAXimum:SRLevel:AUTO

Sets or acquires ON/OFF of the function to automatically detect the peak level and set it as the reference level.

Syntax:	▪ :CALCulate:MARKer:MAXimum:SRLevel:AUTO<wsp>OFF ON 0 1		
Description:	▪ When this function is ON, the unit automatically detects the level of an active trace waveform each time a sweep has ended, and sets it as the reference level.		
Parameters:	OFF ON 0 1		
Query Response:	0 1	0:OFF, 1:ON	
Example:	calc:mark:max:srlevel:auto on calc:mark:max:srlevel:auto? → 1		
Command Type:	Sequential command		

CALCulate:MARKer:MAXimum:SZCenter

Detects the peak wavelength and sets it as the display center wavelength.

Syntax:	• :CALCulate:MARKer:MAXimum:SZCenter		
Description:	None		
Parameters:	None		
Query Response:	None		
Example:	calc:mark:max:szcenter		
Command Type:	Sequential command		

CALCulate:MARKer:MINimum

Detects the bottom and places the moving marker on that bottom.

Syntax:	• :CALCulate:MARKer:MINimum		
Description:	None		
Parameters:	None		
Query Response:	None		
Example:	calc:mark:min		
Command Type:	Sequential command		

CALCulate:MARKer:MINimum:LEFT

Detects the nearest bottom existing on the left side of the current position of the moving marker and places the moving marker on that bottom.

Syntax:	• :CALCulate:MARKer:MINimum:LEFT
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:min:left</code>
Command Type:	Sequential command

CALCulate:MARKer:MINimum:NEXT

Detects the lowest bottom that is above the level of the current position of the moving marker and places the moving marker on that bottom.

Syntax:	• :CALCulate:MARKer:MINimum:NEXT
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:min:next</code>
Command Type:	Sequential command

CALCulate:MARKer:MINimum:RIGHT

Detects the nearest bottom existing on the right side of the current position of the moving marker and places the moving marker on that bottom.

Syntax:	• :CALCulate:MARKer:MINimum:RIGHT
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:min:right</code>
Command Type:	Sequential command

CALCulate:MARKer:SCENter

Sets the wavelength of the current moving marker as the measurement center waveform.

Syntax:	• :CALCulate:MARKer:SCENter
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	<code>calc:mark:scenter</code>
Command Type:	Sequential command

CALCulate:MARKer:SRLevel

Sets the current level of the moving marker for the reference level.

Syntax:	• :CALCulate:MARKer:SRLevel
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	calc:mark:srlevel
Command Type:	Sequential command

CALCulate:MARKer[:STATe]

Specified marker is arranged in the position of moving marker.

If moving marker is specified, it arranges in the center of measurment display.

Specified marker is deleted.

Syntax:	• :CALCulate:MARKer[:STATe]<wsp><marker>,OFF ON 0 1 • :CALCulate:MARKer[:STATe]<wsp><marker>
Description:	• An execution error occurs in the following case: When moving marker is not active and it's going to set fixed marker.
Parameters:	<marker> marker number (0; moving marker) OFF ON 0 1
Query Response:	0 1 0:OFF , 1:ON
Example:	calc:mark:stat 1,on calc:mark:stat 1 → 1
Command Type:	Sequential command

CALCulate:MARKer:SZCenter

Sets the current wavelength of the moving marker for the display center wavelength.

Syntax:	• :CALCulate:MARKer:SZCenter
Description:	• An execution error occurs in the following case: The moving marker is OFF.
Parameters:	None
Query Response:	None
Example:	calc:mark:szcenter
Command Type:	Sequential command

CALCulate:MARKer:UNIT

Sets and acquires the display unit of marker values.

Syntax:	• :CALCulate:MARKer:UNIT<wsp>WAVelength FREQuency 0 1 • :CALCulate:MARKer:UNIT?	
Description:	None	
Parameters:	WAVelength 0	Sets the display unit of marker values to that of wavelength.
	FREQuency 1	Sets the display unit of marker values to that of frequency.
Query Response:	0 1	0: WAVelength, 1: FREQuency
Example:	<pre>calc:mark:unit freq calc:mark:unit? → 1</pre>	
Command Type:	Sequential command	

CALCulate:MARKer:X

Places a specified marker in a specified position.

Acquires the X value of the specified marker.

Syntax:	• :CALCulate:MARKer:X<wsp><marker>,<NRf>[M HZ] • :CALCulate:MARKer:X?<wsp><marker>
Description:	• If an already located marker is specified, that marker will be moved to a specified position. • A query error occurs in the following case: A specified marker is not located.
Parameters:	<marker> Marker number (0: moving marker) <NRf> Marker position
Query Response:	<NRf> Marker position
Example:	<pre>calc:mark:x 0,1550.000nm calc:mark:x? 0 → +1.55000000E-006</pre>
Command Type:	Sequential command

CALCulate:MARKer:Y?

Acquires the Y value of a specified marker.

Syntax:	• :CALCulate:MARKer:Y?<wsp><marker>
Description:	• This unit of the marker level to be acquired is dependent on the Y-axis unit of the active trace. • A query error occurs in the following case: A specified marker is not located.
Parameters:	<marker> Marker number (0: moving marker)
Query Response:	<NRf> Marker level
Example:	<pre>calc:mark:y? 0 → -1.00000000E+001</pre>
Command Type:	Sequential command

CALCulate:MATH:TRC

Sets and acquires the calculation function of trace C.

Syntax:	• :CALCulate:MATH:TRC<wsp>A-B(LOG) B-A(LOG) A+B(LOG) A+B(LIN) A-B(LIN) B-A(LIN) 1-K(A/B) 1-K(B/A) • :CALCulate:MATH:TRC?
Description:	• When the calculation function of trace C is set using this command, the attribute of trace C automatically becomes attribute "CALC".
Parameters:	A-B(LOG) B-A(LOG) A+B(LOG) A+B(LIN) A-B(LIN) B-A(LIN) 1-K(A/B) 1-K(B/A)
Query Response:	Same as the above However, if trace C is not a calculation trace, "NONE" is returned.
Example:	<pre>calc:math:trc a-b(log) calc:math:trc? → A-B(LOG)</pre>
Command Type:	Sequential command

CALCulate:MATH:TRC:K

Sets and acquires parameter K of the trace-C calculation function.

Syntax:	• :CALCulate:MATH:TRC:K<wsp><NRf> • :CALCulate:MATH:TRC:K?
Description:	None
Parameters:	<NRf> Parameter K
Query Response:	Same as above
Example:	<pre>calc:math:trc:k 0.1</pre> <pre>calc:math:trc:k? → +1.00000000E-001</pre>
Command Type:	Sequential command

CALCulate:MATH:TRF

Sets and acquires the calculation function of trace F.

Syntax:	• :CALCulate:MATH:TRF<wsp>C-D(LOG) D-C(LOG) C+D(LOG) D-E(LOG) E-D(LOG) D+E(LOG) C+D(LIN) C-D(LIN) D-C(LIN) D+E(LIN) D-E(LIN) E-D(LIN) • :CALCulate:MATH:TRF?
Description:	• When the calculation function of trace F is set using this command, the attribute of trace F automatically becomes attribute "CALC".
Parameters:	C-D(LOG) D-C(LOG) C+D(LOG) D-E(LOG) E-D(LOG) D+E(LOG) C+D(LIN) C-D(LIN) D-C(LIN) D+E(LIN) D-E(LIN) E-D(LIN)
Query Response:	Same as the above However, if trace F is not a calculation trace, "NONE" is returned.
Example:	<pre>calc:math:trf c-d(log)</pre> <pre>calc:math:trf? → C-D(LOG)</pre>
Command Type:	Sequential command

CALCulate:MATH:TRG

Sets and acquires the calculation function of trace G.

Syntax: •:CALCulate:MATH:TRG<wsp>C-F(LOG) | F-C(LOG) | C+F(LOG) |
E-F(LOG) | F-E(LOG) | E+F(LOG) | C+F(LIN) | C-F(LIN) | F-C(LIN) |
E+F(LIN) | E-F(LIN) | F-E(LIN) | NORMA| NORMB | NORMC | CVFTA
| CVFTB | CVFTC| MKRFT | PKCVFTA | PKCVFTB | PKCVFTC
•:CALCulate:MATH:TRG?

Description: • When the calculation function of trace G is set using this command,
the attribute of trace G automatically becomes attribute "CALC".

Parameters: C-F(LOG) | F-C(LOG) | C+F(LOG) | E-F(LOG) | F-E(LOG) |
E+F(LOG) | C+F(LIN) | C-F(LIN) | F-C(LIN) | E+F(LIN) | E-F(LIN) |
F-E(LIN) | NORMA| NORMB | NORMC | CVFTA | CVFTB | CVFTC|
MKRFT | PKCVFTA | PKCVFTB | PKCVFTC

Query Response: Same as the above
However, if trace G is not a calculation trace, "NONE" is returned.

Example: calc:math:trg c-f(log)
calc:math:trg? → C-F(LOG)

Command Type: Sequential command

CALCulate:MATH:TRG:CVFT:FALGo

Sets and acquires the fitting algorhytm during curve fit and peak curve fit.

Syntax: •:CALCulate:MATH:TRG:FALGo<wsp><algorhythm>
•:CALCulate:MATH:TRG:FALGo?

Description: Setting of fitting algo is common to curve fit and peak curve fit.

Parameters: <algorhythm> fitting algorhythm
GAUSS | 0 : GAUSS
LORENz | 1 : LORENZ
3RD | 2 : 3RD POLY
4TH | 3 : 4TH POLY
5TH | 4 : 5TH POLY

Query Response: <integer> fitting algorhythm
0 : GAUSS
1 : LORENZ
2 : 3RD POLY
3 : 4TH POLY
4 : 5TH POLY

Example: calc:math:trg:falگو gauss
calc:math:trg:falگو? → 1

Command Type: Sequential command

CALCulate:MATH:TRG:CVFT:THResh

Sets and acquires a threshold level applied during curve fit.

Syntax:	• :CALCulate:MATH:TRG:CVFT:THResh<wsp><integer>[DB] • :CALCulate:MATH:TRG:CVFT:THResh?
Description:	None
Parameters:	<NRf> Threshold level [dB]
Query Response:	Same as the above
Example:	calc:math:trg:cvft:thr 10db calc:math:trg:cvft:thr? → 10
Command Type:	Sequential command

CALCulate:MATH:TRG:CVFT:OPARea

Sets and acquires a calculation area during curve fit and peak curve fit.

Syntax:	• :CALCulate:MATH:TRG:CVFT:OPARea<wsp>ALL INL1-L2 OUTL1-L2 0 1 2 • :CALCulate:MATH:TRG:CVFT:OPARea?
Description:	Setting of calculation area is common to curve fit and peak curve fit.
Parameters:	ALL 0 All the data of calculation trace is made applicable for calculation. INL1-L2 1 The data of between line markers is made applicable for calculation. OUTL1-L2 2 The data of outside line markers is made applicable for calculation.
Query Response:	Same as the above
Example:	calc:math:trg:cvft:opar inl1-l2 calc:math:trg:cvft:opar? → 1
Command Type:	Sequential command

CALCulate:MATH:TRG:PCVfT:THResh

Sets and acquires a threshold level applied during peak curve fit.

Syntax: • :CALCulate:MATH:TRG:PCVfT:THResh<wsp><integer>[DB]
 • :CALCulate:MATH:TRG:PCVfT:THResh?

Description: None

Parameters: <NRf> Threshold level [dB]

Query Response: Same as the above

Example: calc:math:trg:pcvf:thr 10db
 calc:math:trg:pcvf:thr? → 10

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:DFBLd

Sets or acquires parameters for the DFB-LD analysis function.

Syntax: • :CALCulate:PARAmeter[:CATegory]:DFBLd<wsp><item>
 ,<parameter>,<data>

• :CALCulate:PARAmeter[:CATegory]:DFBLd?<wsp><Item>
,<parameter>

Description: • If a non-existing parameter is used for a combination, an execution error occurs. (For example, a combination of SWIDth and SMODE)

Parameters: <item> Analytical item that sets parameter(s)
 <parameter> Parameter to be set
 <data> Setting data

<item>	<parameter>	<data>
SWIDth	ALGO	ENVELOpe THResh RMS PKRMs
	TH	<NRf> [DB]
	TH2	<NRf> [DB]
	K	<NRf>
	MFIT	OFF ON 0 1
SMSR	MDIFF	<NRf> [DB]
	SMODE	SMSR1 SMSR2
	SMASK	<NRf> [M]
	MDIFF	<NRf> [DB]

Query Response: Same as the above

Example: calc:par:dfbld swidth,algo,thresh
 calc:par:dfbld? swidth,algo → THR

Example 2: calc:par:dfbld smsr,smask,0.5nm
 calc:par:dfbld? smsr,smask → +5.00000000E-010

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:FILBtm

Sets or acquires parameters of the FILTER-BTM analysis function.

Syntax

- :CALCulate:PARAmeter[:CATegory]:FILBtm<wsp><item>
,<parameter>,<data>
- :CALCulate:PARAmeter[:CATegory]:FILBtm?<wsp><item>
,<parameter>

Description:

- If a non-existing parameter is used for a combination, an execution error occurs. (For example, a combination of CWAVelength and SARea)

Parameters:

<item> Analytical item that sets parameter(s)
 <parameter> Parameter to be set
 <data> Setting data

<item>	<parameter>	<data>
BLEVel	SW	OFF ON 0 1
BWAVelength	SW	OFF ON 0 1
CWAVelength	SW	OFF ON 0 1
	ALGO	PEAK BOTTom
	TH	<NRf> [DB]
NWIDth	MDIFF	<NRf> [DB]
	SW	OFF ON 0 1
	ALGO	PEAK BOTTom
XTALk	TH	<NRf> [DB]
	MDIFF	<NRf> [DB]
	CSPace	<NRf> [M]
	SARea	<NRf> [M]

Query Response: Same as the above

Example:

```
calc:par:filbtm cwavlength,algo,bottom
calc:par:filbtm? cwavlength,algo → BOTTom
```

Example 2:

```
calc:par:filbtm xtalk,cspace,0.2nm
calc:par:filbtm? xtalk,cspace → +2.00000000E-010
```

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:FILPk

Sets or acquires parameters of the FILTER PEAK analysis function.

Syntax:

- :CALCulate:PARAmeter[:CATegory]:FILPk<wsp><item>
,<parameter>,<data>
- :CALCulate:PARAmeter[:CATegory]:FILPk?<wsp><item>
,<parameter>

Description:

- If a non-existing parameter is used for a combination, an execution error occurs. (For example, a combination of SWIDTH and CSPace)

Parameters:

<item>	Analytical item that sets parameter(s)
<parameter>	Parameter to be set
<data>	Setting data

<item>	<parameter>	<data>
PLEVel	SW	OFF ON 0 1
PWAVelength	SW	OFF ON 0 1
MWAVelength	SW	OFF ON 0 1
	ALGO	THResh RMS
	TH	<NRf> [DB]
	K	<NRf>
	MFIT	OFF ON 0 1
SWIDTH	MDIFF	<NRf> [DB]
	SW	OFF ON 0 1
	ALGO	THResh RMS
	TH	<NRf> [DB]
	K	<NRf>
XTALK	MFIT	OFF ON 0 1
	MDIFF	<NRf> [DB]
	CSPace	<NRf> [M]
	SARea	<NRf> [M]
	SW	OFF ON 0 1
RWIDTH	TH	<NRf> [DB]
	MDIFF	<NRf> [DB]

Query Response: Same as the above

Example:

```
calc:par:filpk swidth,algo,thresh
calc:par:filpk? swidth,algo → THR
```

Example 2:

```
calc:par:filpk xtalk,cspace,0.5nm
calc:par:filpk? xtalk,cspace → +5.00000000E-010
```

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:FPLD

Sets or acquires parameters of the FP-LD analysis function.

Syntax:

- :CALCulate:PARAmeter[:CATegory]:FPLD<wsp><item>
,<parameter r>,<data>
- :CALCulate:PARAmeter[:CATegory]:FPLD?<wsp><item>
,<parameter>

Description:

- If a non-existing parameter is used for combination, an execution error occurs. (For example, a combination of SWIDth and OFFSET)

Parameters:

<item> Analytical item that sets parameter(s)

<parameter> Parameter to be set

<data> Setting data

<item>	<parameter>	<data>
SWIDth	ALGO	ENVelope THResh RMS PKRM s
	TH	<NRf> [DB]
	TH2	<NRf> [DB]
	K	<NRf>
	MFIT	OFF ON 0 1
	MDIFf	<NRf> [DB]
MWAVelength	ALGO	ENVelope THResh RMS PKRM s
	TH	<NRf> [DB]
	TH2	<NRf> [DB]
	K	<NRf>
	MFIT	OFF ON 0 1
	MDIFf	<NRf> [DB]
TPOWer	OFFSet	<NRf> [DB]
MNUMber	ALGO	ENVelope THResh RMS PKRM s
	TH	<NRf> [DB]
	TH2	<NRf> [DB]
	K	<NRf>
	MFIT	OFF ON 0 1
	MDIFf	<NRf> [DB]

Query Response: Same as the above

Example:

calc:par:fpld swidth,algo,thresh

calc:par:fpld? swidth,algo → THR

Example 2:

calc:par:fpld tpower,offset,1.0db

calc:par:fpld? tpower,offset → +1.00000000E+000

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:LED

Sets or acquires parameters of the LED analysis function.

Syntax:

- :CALCulate:PARAmeter[:CATegory]:LED<wsp><item>,<parameter>,<data>
- :CALCulate:PARAmeter[:CATegory]:LED?<wsp><item>,<parameter>

Description:

- If a non-existing parameter is used for combination, an execution error occurs. (For example, a combination of SWIDth and OFFSet)

Parameters:

<item> Analytical item that sets parameter(s)

<parameter> Parameter to be set

<data> Setting data

<item>	<parameter>	<data>
SWIDth	ALGO	ENVELOpe THResh RMS PKRMs
	TH	<NRf> [DB]
	TH2	<NRf> [DB]
	K	<NRf>
	MFIT	OFF ON 0 1
	MDIFF	<NRf> [DB]
MWAVelength	ALGO	ENVELOpe THResh RMS PKRMs
	TH	<NRf> [DB]
	TH2	<NRf> [DB]
	K	<NRf>
	MFIT	OFF ON 0 1
	MDIFF	<NRf> [DB]
TPOWer	OFFSet	<NRf> [DB]

Query Response: Same as the above

Example:

```
calc:par:led swidth,algo,thresh
calc:par:led? swidth,algo → THR
```

Example 2:

```
calc:par:led tpower,offset,1.0db
calc:par:led? tpower,offset → +1.00000000E+000
```

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:NF:AALGo

Sets or acquires the measurement algorithm applied to ASE level measurements made by the NF analysis function.

- Syntax:**
- :CALCulate:PARAmeter[:CATegory]:NF:AALGo<wsp><algorhythm>
 - :CALCulate:PARAmeter[:CATegory]:NF:AALGo?
- Parameters:**
- <algorhythm> Measurement algorithm
- AFIX | 0: AUTO FIX
- MFIX | 1: MANUAL FIX
- ACENter | 2: AUTO CENTER
- MCENter | 3: MANUAL CENTER
- Query Response:**
- <integer> Measurement algorithm
- 0: AUTO FIX
- 1: MANUAL FIX
- 2: AUTO CENTER
- 3: MANUAL CENTER
- Example:**
- calc:par:nf:aalgo mfix
- calc:par:nf:aalgo? → 1
- Command Type:** Sequential command

CALCulate:PARAmeter[:CATegory]:NF:FALGo

Sets or acquires the fitting function used in ASE level measurements made by the NF analysis function.

- Syntax:**
- :CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp><algorhythm>
 - :CALCulate:PARAmeter[:CATegory]:NF:FALGo?
- Description:** None
- Parameters:**
- <algorhythm> Fitting function
- LINear | 0: LINEAR
- GAUSs | 1: GAUSS
- LORenZ | 2: LORENZ
- 3RD | 3: 3RD POLY
- 4TH | 4: 4YH POLY
- 5TH | 5: 5TH POLY
- Query Response:**
- <integer> Measurement algorithm
- 0: LINEAR
- 1: GAUSS
- 2: LORENZ
- 3: 3RD POLY
- 4: 4YH POLY
- 5: 5TH POLY
- Example:**
- calc:par:nf:falgo gauss
- calc:par:nf:falgo? → 1
- Command Type:** Sequential command

CALCulate:PARAmeter[:CATegory]:NF:FARea

Sets or acquires the fitting range used in ASE level measurements made by the NF analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:NF:FARea<wsp><NRf>[M] • :CALCulate:PARAmeter[:CATegory]:NF:FARea?
Description:	When the fitting range is set to "Between CH" (and ASE measurement algorithm is set to "AUTO-CTR" or "MANUAL-CTR"), then the command returns 0.
Parameters:	<NRf> Fitting range [m]
Query Response:	Same as the above
Example:	<pre>calc:par:nf:farea 0.80nm</pre> <pre>calc:par:nf:farea? → +8.00000000E-10</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:NF:IOFFset

Sets or acquires the level offset value (signal light) of the NF analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:NF:IOFFset<wsp><NRf>[DB] • :CALCulate:PARAmeter[:CATegory]:NF:IOFFset?
Description:	None
Parameters:	<NRf> Level offset value of signal light [dB]
Query Response:	Same as the above
Example:	<pre>calc:par:nf:ioffset 10.00</pre> <pre>calc:par:nf:ioffset? → +1.00000000E+001</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:NF:MARea

Sets or acquires the mask range used in ASE level measurements made by the NF analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:NF:MARea<wsp><NRf>[M] • :CALCulate:PARAmeter[:CATegory]:NF:MARea?
Description:	When the mask range is set to "-" (and ASE level measurement function is set to "LINEAR"), the command returns 0.
Parameters:	<NRf> Mask range [m]
Query Response:	Same as the above
Example:	<pre>calc:par:nf:marea 0.40nm</pre> <pre>calc:par:nf:marea? → +4.00000000E-10</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:NF:MDIFF

Sets or acquires the peak-bottom difference of channel detection used in the NF analysis function.

Syntax:

- :CALCulate:PARAmeter[:CATegory]:NF:MDIFF<wsp><NRf>[DB]
- :CALCulate:PARAmeter[:CATegory]:NF:MDIFF?

Description: None

Parameters: <NRf> Peak-bottom difference [dB]

Query Response: Same as the above

Example:

```
calc:par:nf:mdiff 3.00db
calc:par:nf:mdiff → +3.00000000E+000
```

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:NF:OOFFSET

Sets or acquires the level offset value (output light) of the NF analysis function.

Syntax:

- :CALCulate:PARAmeter[:CATegory]:NF:OOFFSET<wsp><NRf>[DB]
- :CALCulate:PARAmeter[:CATegory]:NF:OOFFSET?

Description: None

Parameters: <NRf> Level offset value of output light [dB]

Query Response: Same as the above

Example:

```
calc:par:nf:offset 10.00
calc:par:nf:offset? → +1.00000000E+001
```

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:NF:PDISplay

Sets or acquires whether to display data used for fitting of the NF analysis function on the waveform screen.

Syntax:

- :CALCulate:PARAmeter[:CATegory]:NF:PDISplay<wsp>OFF | ON | 0 | 1
- :CALCulate:PARAmeter[:CATegory]:NF:PDISplay?

Description:

- When this set value is 1 (ON), this unit displays data used for fitting on the waveform screen.

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
calc:par:nf:pdisplay on
calc:par:nf:pdisplay? → 1
```

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:NF:TH

Sets or acquires the threshold level of channel detection used in the NF analysis function.

Syntax:

- :CALCulate:PARAmeter[:CATegory]:NF:TH<wsp><NRf>[DB]
- :CALCulate:PARAmeter[:CATegory]:NF:TH?

Description: None

Parameters: <NRf> Threshold level [dB]

Query Response: Same as the above

Example:

```
calc:par:nf:th 20.00db
calc:par:nf:th → +2.00000000E+001
```

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:POWER:OFFSet

Sets or acquires the offset value of the power analysis function.

Syntax: ▪ :CALCulate:PARAmeter[:CATegory]:POWER:OFFSet<wsp><NRf>
[DB]
 ▪ :CALCulate:PARAmeter[:CATegory]:POWER:OFFSet?

Description: None

Parameters: <NRf> Offset value [dB]

Query Response: Same as the above

Example: calc:par:pow:offset 1.00db
 calc:par:pow:offset? → +1.00000000E+000

CALCulate:PARAmeter[:CATegory]:SMSR:MASK

Sets or acquires the mask value of the SMSR analysis function.

Syntax: ▪ :CALCulate:PARAmeter[:CATegory]:SMSR:MASK<wsp><NRf>[M]
 ▪ :CALCulate:PARAmeter[:CATegory]:SMSR:MASK?

Description: None

Parameters: <NRf> Mask value [m]

Query Response: Same as the above

Example: calc:par:smsr:mask 2.0nm
 calc:par:smsr:mask? → +2.00000000E-009

CALCulate:PARAmeter[:CATegory]:SMSR:MODE

Sets or acquires the analytical mode of the SMSR analysis function.

Syntax: ▪ :CALCulate:PARAmeter[:CATegory]:SMSR:MODE<wsp>SMSR1 |
SMSR2
 ▪ :CALCulate:PARAmeter[:CATegory]:SMSR:MODE?

Description: None

Parameters: SMSR1 | SMSR2 SMSR analytical mode

Query Response: Same as the above

Example: calc:par:smsr:mode smsr1
 calc:par:smsr:mode? → SMSR1

CALCulate:PARAmeter[:CATegory]:SWENvelope:K

Sets or acquires the magnification of the envelope method-based spectrum width analysis function.

Syntax: ▪ :CALCulate:PARAmeter[:CATegory]:SWENvelope:K
 ▪ :CALCulate:PARAmeter[:CATegory]:SWENvelope:K

Description: None

Parameters: <NRf> Magnification

Query Response: Same as the above

Example: calc:par:swen:k 2.00
 calc:par:swen:k? → +2.00000000E+000

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:SWENvelope:TH1

Sets or acquires the threshold level of a search made by the envelope method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWENvelope:TH1<wsp><NRf> [DB] • :CALCulate:PARAmeter[:CATegory]:SWENvelope:TH1?
Description:	None
Parameters:	<NRf> Search's threshold level [dB]
Query Response:	Same as the above
Example:	<pre>calc:par:swen:th1 3.00 calc:par:swen:th1? → +3.00000000E+000</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWENvelope:TH2

Sets or acquires the limit value of the envelope method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWENvelope:TH2<wsp><NRf> [DB] • :CALCulate:PARAmeter[:CATegory]:SWENvelope:TH2?
Description:	None
Parameters:	<NRf> limit value [dB]
Query Response:	Same as the above
Example:	<pre>calc:par:swen:th2 10.00db calc:par:swen:th2? → +1.00000000E+001</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWPKrms:K

Sets or acquires the magnification of the PEAK RMS method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWPKrms:K<wsp><NRf>[DB] • :CALCulate:PARAmeter[:CATegory]:SWPKrms:K?
Description:	None
Parameters:	<NRf> Magnification
Query Response:	Same as the above
Example:	<pre>calc:par:swpk:k 2.00 calc:par:swpk:k? → +2.00000000E+000</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWPKrms:TH

Sets or acquires the threshold level of the PEAK RMS method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWPKrms:TH<wsp><NRf>[DB] • :CALCulate:PARAmeter[:CATegory]:SWPKrms:TH?
Description:	None
Parameters:	<NRf> Threshold level [dB]
Query Response:	Same as the above
Example:	<pre>calc:par:swpk:th 3.00db calc:par:swpk:th? → +3.00000000E+000</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWRMS:K

Sets or acquires the magnification of the RMS method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWRMS:K<wsp><NRf>[DB] • :CALCulate:PARAmeter[:CATegory]:SWRMS:K?
Description:	None
Parameters:	<NRf> Magnification
Query Response:	Same as the above
Example:	<pre>calc:par:swrms:k 2.00 calc:par:swrms:k? → +2.00000000E+000</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWRMS:TH

Sets or acquires the threshold level of the RMS method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWRMS:TH<wsp><NRf>[DB] • :CALCulate:PARAmeter[:CATegory]:SWRMS:TH?
Description:	None
Parameters:	<NRf> Threshold level [dB]
Query Response:	Same as the above
Example:	<pre>calc:par:swrms:th 3.00db calc:par:swrms:th? → +3.00000000E+000</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWTHresh:K

Sets or acquires the magnification of the THRESH method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWTHresh:K<wsp><NRf> • :CALCulate:PARAmeter[:CATegory]:SWTHresh:K?
Description:	None
Parameters:	<NRf> Magnification
Query Response:	Same as the above
Example:	<pre>calc:par:swth:k 2.00 calc:par:swth:k? → +2.00000000E+000</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWTHresh:MFIT

Sets or acquires whether to enable the mode fit of the THRESH method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWTHresh:MFIT<wsp>OFF ON 0 1 • :CALCulate:PARAmeter[:CATegory]:SWTHresh:MFIT?
Description:	None
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	<pre>calc:par:swth:mfit on calc:par:swth:mfit? → 1</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:SWTHresh:TH

Sets or acquires the threshold level of the THRESH method-based spectrum width analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:SWTHresh:TH<wsp><NRf>[DB] • :CALCulate:PARAmeter[:CATegory]:SWTHresh:TH?
Description:	None
Parameters:	<NRf> Threshold level [dB]
Query Response:	Same as the above
Example:	<pre>calc:par:swth:th 3.00db calc:par:swth:th? → +3.00000000E+000</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:DMASk

Sets or acquires the threshold level of the channel mask used in the WDM analysis function.

- Syntax:**
- :CALCulate:PARAmeter[:CATegory]:WDM:DMASk<wsp><NRf>[dB]
 - :CALCulate:PARAmeter[:CATegory]:WDM:DMASk?
- Description:** Channels the level of which is below this parameter will not be detected as a channel.
- To turn off the channel mask function, set the threshold level to -999.
- Parameters:** <NRf> Threshold level [dB] (-999 : channel mask OFF)
- Query Response:** Same as the above
- Example:**
- ```
calc:par:wdm:dmask -999
calc:par:wdm:dmask? → -9.99000000E+002
```
- Command Type:** Sequential command

**CALCulate:PARAmeter[:CATegory]:WDM:DTYPE**

Sets or acquires the display type of the analysis results of the WDM analysis function.

- Syntax:**
- :CALCulate:PARAmeter[:CATegory]:WDM:DTYPE<wsp><display type>
  - :CALCulate:PARAmeter[:CATegory]:WDM:DTYPE?
- Description:** None
- Parameters:**
- |                |                                                              |
|----------------|--------------------------------------------------------------|
| <display type> | Display type of analysis result                              |
| ABSolute   0:  | Absolute value display                                       |
| RELative   1:  | Relative value display                                       |
| MDRift   2:    | Drift value display based on the past measurement wavelength |
| GDRift   3:    | Drift value display based on the grid wavelength             |
- Query Response:** <integer>
- |    |                                                              |
|----|--------------------------------------------------------------|
| 0: | Absolute value display                                       |
| 1: | Relative value display                                       |
| 2: | Drift value display based on the past measurement wavelength |
| 3: | Drift value display based on the grid wavelength             |
- Example:**
- ```
calc:par:wdm:dtype absolute
calc:par:wdm:dtype? → 0
```
- Command Type:** Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:DUAL

Sets or acquires the SNR calculation mode of the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:DUAL<wsp>OFF ON 0 1 • :CALCulate:PARAmeter[:CATegory]:WDM:DUAL?
Description:	• When this set value is 1 (ON), SNR calculation uses both traces A and B data. • When this set value is 0 (OFF), SNR calculation uses active trace data.
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	<pre>calc:par:wdm:dual on calc:par:wdm:dual? → 1</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:FALGo

Sets or acquires the fitting function used in noise level measurements made by the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp><algorithm> • :CALCulate:PARAmeter[:CATegory]:WDM:FALGo?
Description:	None
Parameters:	<algorithm> Fitting function LINEar 0: LINEAR GAUSs 1: GAUSS LOReNZ 2: LORENZ 3RD 3: 3RD POLY 4TH 4: 4YH POLY 5TH 5: 5TH POLY
Query Response:	<integer> Measurement algorithm 0: LINEAR 1: GAUSS 2: LORENZ 3: 3RD POLY 4: 4YH POLY 5: 5TH POLY
Example:	<pre>calc:par:wdm:falگو gauss calc:par:wdm:falگو? → 1</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:MARea

Sets or acquires a mask range used in noise level measurements made by the WDM analysis function.

Syntax: • :CALCulate:PARAmeter[:CATegory]:WDM:MARea<wsp><NRf>[M]
 • :CALCulate:PARAmeter[:CATegory]:WDM:MARea?

Description: None

Parameters: <NRf> Mask area [m]

Query Response: Same as the above

Example: calc:par:wdm:marea 0.40nm
 calc:par:wdm:marea? → +4.00000000E-10

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:MDIFF

Sets or acquires the peak-bottom difference of channel detection used in the WDM analysis function.

Syntax: • :CALCulate:PARAmeter[:CATegory]:WDM:MDIFF<wsp><NRf>[DB]
 • :CALCulate:PARAmeter[:CATegory]:WDM:MDIFF?

Description: None

Parameters: <NRf> Peak-bottom difference [dB]

Query Response: Same as the above

Example: calc:par:wdm:mdiff 3.00db
 calc:par:wdm:mdiff → +3.00000000E+000

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:MMReset

Resets the maximum and minimum of the drift values of the WDM analysis function.

Syntax: • :CALCulate:PARAmeter[:CATegory]:WDM:MMReset

Description: When "DISPLAY TYPE" (set
 by :CALCulate:PARAmeter[:CATegory]:WDM:DTYPE command is set
 to other than "DRIFT", an execution error occurs.

Parameters: None

Query Response: None

Example: calc:par:wdm:mmreset

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:NALGo

Sets or acquires a measurement algorithm applied to noise level measurements made by the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:NALGo<wsp><algorithm> • :CALCulate:PARAmeter[:CATegory]:WDM:NALGo?
Description:	None
Parameters:	<algorithm> Measurement algorithm AFIX 0 : AUTO FIX MFIX 1 : MANUAL FIX ACENter 2 : AUTO CENTER MCENter 3 : MANUAL CENTER PIT 4 : PIT
Query Response:	<integer> Measurement algorithm 0 : AUTO FIX 1 : MANUAL FIX 2 : AUTO CENTER 3 : MANUAL CENTER 4 : PIT
Example:	calc:par:wdm:nalgo acenter calc:par:wdm:nalgo? → 2
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:NARea

Sets or acquires a measurement range used in noise level measurements made by the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:NARea<wsp><NRf>[M] • :CALCulate:PARAmeter[:CATegory]:WDM:NARea?
Description:	None
Parameters:	<NRf> Noise area [m]
Query Response:	Same as the above
Example:	calc:par:wdm:narea 0.80nm calc:par:wdm:narea? → +8.00000000E-10
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:NBW

Sets or acquires the noise bandwidth of the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:NBW<wsp><NRf>[M] • :CALCulate:PARAmeter[:CATegory]:WDM:NBW?
Description:	None
Parameters:	<NRf> Noise bandwidth [m]
Query Response:	Same as the above
Example:	calc:par:wdm:nbw 0.10nm calc:par:wdm:nbw? → +1.00000000E-010
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:OSLOpe

Sets or acquires whether to enable the function of obtaining the least square approximation line in the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:OSLOpe<wsp>OFF ON 0 1 • :CALCulate:PARAmeter[:CATegory]:WDM:OSLOpe?
Description:	• When this set value is 1 (ON), this unit calculates the least square approximation line of the peak of each channel and draws it on the waveform screen.
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	<pre>calc:par:wdm:oslope on calc:par:wdm:oslope? → 1</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:PDISplay

Sets or acquires whether to display data used for fitting of the WDM analysis function on the waveform screen.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:PDISplay<wsp>OFF ON 0 1 • :CALCulate:PARAmeter[:CATegory]:WDM:PDISplay?
Description:	• When this set value is 1 (ON), data used for fitting is displayed on the waveform screen.
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	<pre>calc:par:wdm:pdisplay on calc:par:wdm:pdisplay? → 1</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:RCH

Sets or acquires the reference channel used in calculating the offset wavelength/level of the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:RCH<wsp><integer> • :CALCulate:PARAmeter[:CATegory]:WDM:RCH?
Description:	• When this set value is "0," the channel with the highest level is regarded as the reference channel.
Parameters:	<integer> Reference channel number (0: channel with the highest level)
Query Response:	Same as the above
Example:	<pre>calc:par:wdm:rch 10 calc:par:wdm:rch? → 10</pre>
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:RELation

Sets or acquires the display type of wavelength/level relative values of the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:RELation<wsp>OFFSet SPACing 0 1 • :CALCulate:PARAmeter[:CATegory]:WDM:RELation?
Description:	When "DISPLAY TYPE" (set by :CALCulate:PARAmeter[:CATegory]:WDM:DTYPE command is set to other than "ABSOLUTE", an execution error occurs.
Parameters:	OFFSet 0 Displays an offset value based on any channel. SPACing 1 Displays an offset value relative to a neighboring channel.
Query Response:	0 1 0 : OFFSET, 1 : SPACING
Example:	calc:par:wdm:relation spacing calc:par:wdm:relation? → 1
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WDM:TH

Sets or acquires the threshold level of channel detection used in the WDM analysis function.

Syntax:	• :CALCulate:PARAmeter[:CATegory]:WDM:TH<wsp><NRf>[dB] • :CALCulate:PARAmeter[:CATegory]:WDM:TH?
Description:	None
Parameters:	<NRf> Threshold level [dB]
Query Response:	Same as the above
Example:	calc:par:wdm:th 20.00db calc:par:wdm:th → +2.000000000E+001
Command Type:	Sequential command

CALCulate:PARAmeter[:CATegory]:WFBOTTOM

Sets or acquires parameters of the WDM FILTER-BTM analysis function.

- Syntax:**
- :CALCulate:PARAmeter[:CATegory]:WFBOTTOM<wsp><item>,<parameter>,<data>
 - :CALCulate:PARAmeter[:CATegory]:WFBOTTOM?<wsp><item>,<parameter>

Description:

- If a non-existing parameter is used for a combination, an execution error occurs. (For example, a combination of NWAVelength and SPACing)

Parameters:

<item> Analytical item that sets parameter(s)

<parameter> Parameter to be set

<data> Setting data

<item>	<parameter>	<data>
NWAVelength	ALGO	BOTTom NPEak NBOTtom GFIT GRID
	MDIFF	<NRf> [DB]
	TH	<NRf> [DB]
	TBAND	<NRf> [M]
BWAVelength	SW	OFF ON 0 1
CWAVelength	SW	OFF ON 0 1
	ALGO	NPEak NBOTtom
SBAND	TH	<NRf> [DB]
	TH	OFF ON 0 1
EBAND	TH	<NRf> [DB]
	TBAND	<NRf> [M]
	SW	OFF ON 0 1
RIPPlE	TBAND	<NRf> [M]
	SW	OFF ON 0 1
XTALk	SPACing	<NRf> [M]
	TBAND	<NRf> [M]
	SW	OFF ON 0 1

Query Response: Same as the above

Example: calc:par:wfbottom nwav,algo,npeak

calc:par:wfbottom? nwav,algo → NPE

Example 2: calc:par:wfbottom bwavelength,sw,off

calc:par:wfbottom? bwavelength,sw → 0

Command Type: Sequential command

CALCulate:PARAmeter[:CATegory]:WFPeak

Sets or acquires parameters of the WDM filter-peak analysis function.

- Syntax:**
- :CALCulate:PARAmeter[:CATegory]:WFPeak<wsp><item>,<parameter>,<data>
 - :CALCulate:PARAmeter[:CATegory]:WFPeak?<wsp><item>,<parameter>

Description:

- If a non-existing parameter is used for a combination, an execution error occurs. (For example, a combination of NWAVelength and SPACing)

Parameters:

<item> Analytical item that sets parameter(s)

<parameter> Parameter to be set

<data> Setting data

<item>	<parameter>	<data>
NWAVelength	ALGO	PEAK MEAN GFIT GRID
	MDIFF	<NRf> [DB]
	TH	<NRf> [DB]
	TBAND	<NRf> [M]
PWAVelength	SW	OFF ON 0 1
CWAVelength	SW	OFF ON 0 1
	TH	<NRf> [DB]
SBAND	SW	OFF ON 0 1
	TH	<NRf> [DB]
PBAND	SW	OFF ON 0 1
	TH	<NRf> [DB]
	TBAND	<NRf> [M]
RIPPlE	SW	OFF ON 0 1
	TBAND	<NRf> [M]
XTALk	SW	OFF ON 0 1
	SPACing	<NRf> [M]
	TBAND	<NRf> [M]

Query Response: Same as the above

Example:

```
calc:par:wfpeak nwav,algo,peak
calc:par:wfpeak? nwav,algo → PEAK
```

Example 2:

```
calc:par:wfpeak pwavelength,sw,off
calc:par:wfpeak? pwavelength,sw → 0
```

Command Type: Sequential command

CALCulate:PARAmeter:COMMon:MDIFf

Sets or acquires the peak-bottom difference parameter of channel detection used in the WDM analysis function.

Syntax:

- :CALCulate:PARAmeter:COMMon:MDIFf<wsp><NRf>[DB]
- :CALCulate:PARAmeter:COMMon:MDIFf?

Description: None

Parameters: <NRf> Peak-bottom difference [dB]

Query Response: Same as the above

Example:

```
calc:par:common:mdiff 3.00db  
calc:par:common:mdiff → +3.00000000E+000
```

Command Type: Sequential command

CALibration Sub System Command**CALibration:ALIGn[:IMMediate]**

Adjusts the optical axis of the internal monochrometer.

Syntax: • :CALibration:ALIGn[:IMMediate]
Description: None
Parameters: None
Query Response: None
Example: cal:align
Command Type: Command subject to overlapping

CALibration:POWer:OFFSet:TABLE

Sets or acquires the level offset table.

Syntax: • :CALibration:POWer:OFFSet:TABLE<wsp><integer>,<NRf> [dB]
 • :CALibration:POWer:OFFSet:TABLE?<wsp><integer>
Description: • Of the level offset table, the command sets or acquires the offset value of a wavelength specified by <integer>.
Parameters: <integer> Wavelength [nm]
 <NRf> Level offset value [dB]
Query Response: <NRf> Level offset value [dB]
Example: cal:pow:offset:table 1550,-0.1db
 cal:pow:offset:table? 1550 → -1.00000000E-001
Command Type: Sequential command

CALibration:WAVelength:EXTernal[:IMMediate]

Performs wavelength calibration using an external reference light source.

Syntax: • :CALibration:WAVelength:EXTernal[:IMMediate]
Description: • The type of the external reference light source to be used for calibration is set using the CALibration:WAVelength:EXTernal:SOURce command.
 • The wavelength of the external reference light source to be used for calibration is set using the CALibration:WAVelength:EXTernal:WAVelenght command.
Parameters: None
Query Response: None
Example: cal:wav:external
Command Type: Command subject to overlapping

CALibration:WAVelength:EXTeRnal:SOURce

Sets or acquires the type of the light source used for external reference light source-based wavelength calibration.

Syntax:	• :CALibration:WAVelength:EXTeRnal:SOURce<wsp>LASer GASCell 0 1 • :CALibration:WAVelength:EXTeRnal:SOURce?
Description:	• Of the level offset table, the command sets or acquires the offset value of a wavelength specified by <integer>.
Parameters:	LASer 0 A laser is used as the external reference light source. GASCell 1 A gas cell is used as the external reference light source.
Query Response:	0: Laser 1: Gas cell
Example:	<pre>cal:wav:ext:source laser cal:wav:ext:source? → 0</pre>
Command Type:	Sequential command

CALibration:WAVelength:EXTeRnal:WAVelength

Sets or acquires the wavelength of the light source used for external reference light source-based wavelength calibration.

Syntax:	• :CALibration:WAVelength:EXTeRnal:WAVelength<wsp><NRf>[M] • :CALibration:WAVelength:EXTeRnal:WAVelength?
Description:	None
Parameters:	<NRf> Wavelength of the external reference light source [nm]
Query Response:	Same as the above
Example:	<pre>cal:wav:ext:wavelength 1550.000nm cal:pow:offset:table? → +1.55000000E-006</pre>
Command Type:	Sequential command

CALibration:WAVelength:INTeRnal[:IMMeDiate]

Performs wavelength calibration using the internal reference light source.

Syntax:	• :CALibration:WAVelength:INTeRnal[:IMMeDiate]
Description:	None
Parameters:	None
Query Response:	None
Example:	<pre>cal:wav:internal</pre>
Command Type:	Command subject to overlapping

CALibration:WAVelength:OFFSet:TABLE

Sets or acquires the wavelength offset table.

Syntax:	• :CALibration:WaAVelength:OFFSet:TABLE<wsp><integer>,<NRf> • :CALibration WaAVelength:POWer:OFFSet:TABLE?<wsp><integer>
Description:	• Of the wavelength offset table, the command sets or acquires the offset value of a wavelength specified by <integer>.
Parameters:	<integer> Wavelength (specified in nm) <NRf> Wavelength offset value (specified in nm)
Query Response:	<NRf> Wavelength offset value (specified in nm)
Example:	<pre>cal:wav:offset:table 1550, -0.1</pre> <pre>cal:wav:offset:table? 1550 → -1.00000000E-001</pre>
Command Type:	Sequential command

CALibration:ZERO[:AUTO]

Sets or acquires whether to enable the auto offset function of the level.

Syntax:	• :CALibration:ZERO[:AUTO]<wsp>OFF ON 0 1 ONCE • :CALibration:ZERO[:AUTO]?
Description:	• If "ONCE" is selected in the parameter, offset adjustment is carried out once. In this case, ON/OFF of this setting does not change.
Parameters:	OFF ON 0 1 ONCE
Query Response:	0 1 0:OFF, 1:ON
Example:	<pre>cal:zero once</pre> <pre>cal:zero? → 1</pre>
Command Type:	Sequential command

DISPlay Sub System Command

DISPlay:COLor

Sets or acquires the screen color mode.

Syntax:	• :DISPlay:COLor<wsp><mode> • :DISPlay:COLor?
Description:	None
Parameters:	<mode> Screen color mode (0 to 5) 0: Black and white mode 1 to 5: Modes 1 to 5
Query Response:	Same as the above
Example:	disp:col 1 disp:col? → 1
Command Type:	Sequential command

DISPlay[:WINDow]:OVlew:POSition

Sets or acquires the ON/OFF and position of the OVERVIEW display shown during zoom operation.

Syntax:	• :DISPlay[:WINDow]:OVlew:POSition<wsp>OFF LEFT RIGHT 0 1 2 • :DISPlay[:WINDow]:OVlew:POSition?
Description:	None
Parameters:	OFF 0 Display is OFF. LEFT 1 The overview display appears on the left of the screen. RIGHT 2 The overview display appears on the right of the screen.
Query Response:	0: Display is OFF. 1: The overview display is on the left of the screen. 2: The overview display is on the right of the screen.
Example:	disp:oview:pos right disp:oview:pos? → 2
Command Type:	Sequential command

DISPlay[:WINDow]: OVlew:SIZE

Sets or acquires the size of the OVERVIEW display shown during zoom operation.

Syntax:	• :DISPlay[:WINDow]:OVlew:SIZE<wsp>LARGE SMALL 0 1 • :DISPlay[:WINDow]:OVlew:SIZE?
Description:	None
Parameters:	LARGE 0: Larger OVERVIEW size SMALL 1: Smaller OVERVIEW size
Query Response:	0: Larger OVERVIEW size 1: Smaller OVERVIEW size
Example:	disp:oview:size large disp:oview:size? → 0
Command Type:	Sequential command

DISPlay[:WINDow]:SPLit

Sets or acquires whether to split the screen display into two parts.

Syntax:

- :DISPlay[:WINDow]:SPLit<wsp>OFF | ON | 0 | 1
- :DISPlay[:WINDow]:SPLit?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
disp:split on
disp:split? → 1
```

Command Type: Sequential command

DISPlay[:WINDow]:SPLit:HOLD:LOWer

Sets or acquires whether to fix a trace assigned to the lower area when the screen is in the upper/lower 2-split display mode.

Syntax:

- :DISPlay[:WINDow]:SPLit:HOLD:LOWer<wsp>OFF | ON | 0 | 1
- :DISPlay[:WINDow]:SPLit:HOLD:LOWer?

Description:

- An execution error occurs in the following case:
Screen display is not in the upper/lower 2-split display mode.

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
disp:split:hold:lower on
disp:split:hold:lower? → 1
```

Command Type: Sequential command

DISPlay[:WINDow]:SPLit:HOLD:UPPer

Sets or acquires whether to fix a trace assigned to the upper screen area when the screen is in the upper/lower 2-split display mode.

Syntax:

- :DISPlay[:WINDow]:SPLit:HOLD:UPPer<wsp>OFF | ON | 0 | 1
- :DISPlay[:WINDow]:SPLit:HOLD:UPPer?

Description:

- An execution error occurs in the following case:
Screen display is not in the upper/lower 2-split display mode.

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
disp:split:hold:upper on
disp:split:hold:upper? → 1
```

Command Type: Sequential command

DISPlay[:WINDow] :SPLit:POSition

Sets or acquires whichever display area, upper or lower, is used to display a trace when the screen is in the upper/lower 2-split display mode.

Syntax:	• :DISPlay[:WINDow]:SPLit:POSition<wsp><trace name>,UP LOW 0 1 • :DISPlay[:WINDow]:SPLit:POSition?<wsp><trace name>	
Description:	None	
Parameters:	<trace name>	A trace name (TRA, TRB, TRC, TRD, TRE, TRF, TRG)
	UP 0	Displays a trace on the upper area.
	LOW 1	Displays a trace on the lower area.
Query Response:	0:	Trace is displayed on the upper area.
	1:	Trace is displayed on the lower area.
Example:	<pre>disp:split:pos tra,up disp:split:pos? tra → 0</pre>	
Command Type:	Sequential command	

DISPlay[:WINDow]:TEXT:CLEAr

Clears labels.

Syntax:	• :DISPlay[:WINDow]:TEXT:CLEAr
Description:	None
Parameters:	None
Query Response:	None
Example:	<pre>disp:text:clear</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TEXT:DATA

Sets or acquires a label.

Syntax:	• :DISPlay[:WINDow]:TEXT:DATA<wsp><string> • :DISPlay[:WINDow]:TEXT:DATA?	
Description:	• A label character string has a maximum length of 56 characters. If a label of more than 56 characters is specified, characters from and exceeding the 57th will be ignored. • If there is no label, one space character is returned.	
Parameters:	<string>	Label character string (56 characters max.)
Query Response:	Same as the above	
Example:	<pre>disp:text:data "Optical Spectrum Analyzer" disp:text:data? → Optical Spectrum Analyzer</pre>	
Command Type:	Sequential command	

DISPlay[:WINDow]:TRACe:X[:SCALe]:CENTer

Sets or acquires the center wavelength of the X-axis of the display scale.

Syntax:	• :DISPlay[:WINDow]:TRACe:X[:SCALe]:CENTer<wsp><NRf>[M] • :DISPlay[:WINDow]:TRACe:X[:SCALe]:CENTer?
Description:	None
Parameters:	<NRf> Center wavelength [m]
Query Response:	Same as the above
Example:	<pre>disp:trace:x:center 1550.000nm disp:trace:x:center? → +1.550000000E-006</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TRACe:X[:SCALe]:INITialize

Initializes the X-axis parameters of the display scale.

Syntax:	• :DISPlay[:WINDow]:TRACe:X[:SCALe]:INITialize
Description:	• The following parameters are initialized based on the measurement scale after this command has been executed. ZOOM CENTER, ZOOM SPAN, ZOOM START, ZOOM STOP
Parameters:	None
Query Response:	None
Example:	<pre>disp:trace:x:init</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TRACe:X[:SCALe]:SMSCale

Sets parameters of the current display scale to the measurement scale.

Syntax:	• :DISPlay[:WINDow]:TRACe:X[:SCALe]:SMSCale
Description:	• The following parameters are initialized based on the display scale after this command has been executed. CENTER, SPAN, START, STOP
Parameters:	None
Query Response:	None
Example:	<pre>disp:trace:x:smyscale</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TRACe:X[:SCALe]:SPAN

Sets or acquires the X-axis span of the display scale.

Syntax:	• :DISPlay[:WINDow]:TRACe:X[:SCALe]:SPAN<wsp><NRf>[M] • :DISPlay[:WINDow]:TRACe:X[:SCALe]:SPAN?
Description:	None
Parameters:	<NRf> Span [m]
Query Response:	Same as the above
Example:	<pre>disp:trace:x:span 20.0nm disp:trace:x:span? → +2.000000000E-008</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TRACe:X[:SCALe]:SRANge

Sets or acquires whether to limit an analytical range to the display scale range.

Syntax: • :DISPlay[:WINDow]:TRACe:X[:SCALe]:SRANge<wsp>OFF | ON | 0
 | 1
 • :DISPlay[:WINDow]:TRACe:X[:SCALe]:SRANge?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example: disp:trace:x:srange on
 disp:trace:x:srange? → 1

Command Type: Sequential command

DISPlay[:WINDow]:TRACe:X[:SCALe]:STARt

Sets or acquires the starting wavelength of the X-axis of the display scale.

Syntax: • :DISPlay[:WINDow]:TRACe:X[:SCALe]:STARt<wsp><NRf>[M]
 • :DISPlay[:WINDow]:TRACe:X[:SCALe]:STARt?

Description: None

Parameters: <NRf> Starting wavelength [m]

Query Response: Same as the above

Example: disp:trace:x:start 1540.000nm
 disp:trace:x:start? → +1.54000000E-006

Command Type: Sequential command

DISPlay[:WINDow]:TRACe:X[:SCALe]:STOP

Sets or acquires the stop waveform of the X-axis of the display scale.

Syntax: • :DISPlay[:WINDow]:TRACe:X[:SCALe]:STOP<wsp><NRf>[M]
 • :DISPlay[:WINDow]:TRACe:X[:SCALe]:STOP?

Description: None

Parameters: <NRf> Ending wavelength [m]

Query Response: Same as the above

Example: disp:trace:x:stop 1560.000nm
 disp:trace:x:stop? → +1.56000000E-006

Command Type: Sequential command

DISPlay[:WINDow]:TRACe:Y:NMASt

Sets whether to mask the display of waveforms the level of which is at or below a set threshold level or acquires the condition of whether the relevant waveform display is masked.

Syntax: • :DISPlay[:WINDow]:TRACe:Y:NMASt<wsp><NRf>[DB]
 • :DISPlay[:WINDow]:TRACe:Y:NMASt?

Description: • The display of waveforms the level of which is at or below this parameter will be masked.
 To deactivate the masking function, set the threshold level to -999.

Parameters: <NRf> Threshold level [dB] (-999: Masking function OFF)

Example: disp:trace:y:mask -999
 disp:trace:y:mask? → -9.99000000E+002

Command Type: Sequential command

DISPlay[:WINDow]:TRACe:Y:NMASk:TYPE

Sets or acquires the display method applied when masking the waveform display the level of which is at or below a threshold level.

Syntax:	• :DISPlay[:WINDow]:TRACe:Y:NMASk:TYPE<wsp>VERTical HORizontal 0 1 • :DISPlay[:WINDow]:TRACe:Y:NMASk:TYPE?
Description:	None
Parameters:	VERTical 0: Waveforms are displayed, with waveforms the level of which is at or below the mask value made to be zero. HORizontal 1: Waveforms are displayed, with waveforms the level of which is at or below the mask value made to be the mask value.
Query Response:	0: Waveforms are displayed, with waveforms the level of which is at or below the mask value made to be zero. 1: Waveforms are displayed, with waveforms the level of which is at or below the mask value made to be the mask value.
Example:	<pre>disp:trace:y:nmask:type vertical disp:trace:y:nmask:type? → 0</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TRACe:Y[:SCALe]:DNUMber

Sets or acquires the number of display divisions of the level axis.

Syntax:	• :DISPlay[:WINDow]:TRACe:Y[:SCALe]:DNUMber<wsp>8 10 12 • :DISPlay[:WINDow]:TRACe:Y[:SCALe]:DNUMber?
Description:	None
Parameters:	8 10 12 : Number of display divisions
Query Response:	Same as the above
Example:	<pre>disp:trace:y:dnumber 10 disp:trace:y:dnumber? → 10</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TRACe:Y1[:SCALe]:BLEVel

Sets or acquires the base level applied when the main scale of the level axis is linear.

Syntax:	• :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:BLEVel<wsp><NRf>[W] • :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:BLEVel?
Description:	• If a unit other than W is specified, an execution error occurs.
Parameters:	<NRf> Base level value [W]
Query Response:	Same as the above
Example:	<pre>disp:trace:y1:blevel 1.0mw disp:trace:y1:blevel? → +1.00000000E-003</pre>
Command Type:	Sequential command

DISPlay[:WINDow]:TRACe:Y1[:SCALe]:PDIVision

Sets or acquires the main scale of the level axis.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:PDIVision<wsp><NRf>[DB]
 - :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:PDIVision?
- Description:**
- If a unit other than dB is specified, an execution error occurs.
- Parameters:** <NRf> Level scale [dB]
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y1:pdiv 5.0db
disp:trace:y1:pdiv? → +5.00000000E+000
```
- Command Type:** Sequential command

**DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RLEVel**

Sets or acquires the reference level of the main scale of the level axis.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RLEVel<wsp><NRf>[DBM | W]
  - :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RLEVel?
- Description:**
- When the unit is omitted in the parameter, the reference level is set in dBm if the main scale of the level axis is in the LOG mode or is set in W if it is in the linear mode.
  - If the setting condition of the LOG/linear mode of the level axis' main scale does not agree with the unit specified in the parameter of the command, the parameter of this command is translated matching the LOG/linear mode of the main scale. (The unit of the scale will remain the same.)
- For example, when the main scale is in the LOG mode, if the reference level is set to 1 mW using this command, the reference level will be set to 0 dBm.
- Parameters:** <NRf> Reference level [dBm | W]
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y1:rlev -30dbm
disp:trace:y1:rlev? → -3.00000000E+001
```
- Command Type:** Sequential command

DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RPOSition

Sets or acquires the position of the reference level for the main scale of the level axis.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RPOSition<wsp><integer>[DIV]
 - :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RPOSition?
- Description:**
- If a value greater than the number of display divisions of the level axis is specified for the position of the reference level, the position of this level is treated as the top of the scale.
- Parameters:** <integer> Position of the reference level
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y1:rpos 10div
disp:trace:y1:rpos? → 10
```
- Command Type:** Sequential command

**DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing**

Sets or acquires the main scale mode of the level axis.

|                        |                                                                                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing&lt;wsp&gt;LOGarithmic   LINear   0   1</li> <li>• :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing?</li> </ul> |
| <b>Description:</b>    | None                                                                                                                                                                                         |
| <b>Parameters:</b>     | LOGarithmic   0: LOG scale<br>LINear   1: Linear scale                                                                                                                                       |
| <b>Query Response:</b> | 0: LOG scale<br>1: Linear scale                                                                                                                                                              |
| <b>Example:</b>        | disp:trace:y1:spacing linear<br>disp:trace:y1:spacing? → 1                                                                                                                                   |
| <b>Command Type:</b>   | Sequential command                                                                                                                                                                           |

**DISPlay[:WINDow]:TRACe:Y1[:SCALe]:UNIT**

Sets or acquires the units of the main scale of the level axis.

|                        |                                                                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:UNIT&lt;wsp&gt;&lt;unit&gt;</li> <li>• :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:UNIT?</li> </ul> |
| <b>Description:</b>    | None                                                                                                                                                                   |
| <b>Parameters:</b>     | <unit> Unit<br>DBM   0: dBm<br>W   1: W<br>DBM/NM   2: dBm/nm or dBm/THz<br>W/NM   3: W/nm or W/THz                                                                    |
| <b>Query Response:</b> | 0: dBm<br>1: W<br>2: dBm/nm or dBm/THz<br>3: W/nm or W/THz                                                                                                             |
| <b>Example:</b>        | disp:trace:y1:unit dbm/nm<br>disp:trace:y1:unit? → 2                                                                                                                   |
| <b>Command Type:</b>   | Sequential command                                                                                                                                                     |

**DISPlay[:WINDow]:TRACe:Y2[:SCALe]:AUTO**

Sets or acquires the automatic setting function of the subscale of the level axis.

|                        |                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:AUTO&lt;wsp&gt;OFF   ON   0   1</li> <li>• :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:AUTO?</li> </ul> |
| <b>Description:</b>    | None                                                                                                                                                                       |
| <b>Parameters:</b>     | OFF   ON   0   1                                                                                                                                                           |
| <b>Query Response:</b> | 0   1                      0:OFF, 1:ON                                                                                                                                     |
| <b>Example:</b>        | disp:trace:y2:auto on<br>disp:trace:y2:auto? → 1                                                                                                                           |
| <b>Command Type:</b>   | Sequential command                                                                                                                                                         |

**DISPlay[:WINDow]:TRACe:Y2[:SCALe]:LENGth**

Sets or acquires the parameter of the optical fiber length used when the unit of the subscale of the level axis is dB/km.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:LENGth<wsp><NRf>[KM]
  - :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:LENGth?
- Description:** When the unit of the subscale is set to other than "dB/km", an execution error occurs.
- Parameters:** <NRf> Length of optical fiber [km]
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y2:length 99.999km
disp:trace:y2:length? → +9.99990000E+001
```
- Command Type:** Sequential command

DISPlay[:WINDow]:TRACe:Y2[:SCALe]:OLEVel

Sets or acquires the offset level of the subscale of the level axis.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:OLEVel<wsp><NRf>[DB | DB/KM]
 - :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:OLEVel?
- Description:**
- When the unit of the subscale is set to other than "dB" or "dB/km", an execution error occurs. If the unit is not specified in the parameter, dB is set if the subscale of the level axis is in the dB mode or dB/km is set if it is in the dB/km mode.
 - If a unit different from the current set unit (:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT) of the subscale is specified, an execution error occurs.
- Parameters:** <NRf> Offset level [dB | dB/km]
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y2:olevel 10db/km
disp:trace:y2:olevel? → +1.00000000E+001
```
- Command Type:** Sequential command

**DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision**

Sets or acquires the subscale of the level axis.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<wsp><NRf>[DB | DB/KM | %]
  - :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision?
- Description:**
- If the unit is not specified in the parameter, the set unit of the subscale of the level axis is used as the set unit of this parameter.
  - If a unit different from the current set unit (:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT) of the subscale is specified, an execution error occurs.
- Parameters:** <NRf> Level scale [dB | dB/km | %]
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y2:pdiv 5.0%
disp:trace:y2:pdiv? → +5.00000000E+000
```
- Command Type:** Sequential command

DISPlay[:WINDow]:TRACe:Y2[:SCALe]:RPOSition

Sets or acquires the position of the reference level of the level axis subscale.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:RPOSition<wsp><integer> [DIV]
 - :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:RPOSition?
- Description:**
- If a value greater than the number of display divisions of the level axis is specified for the position of the reference level, the position of this level is treated as the top of the scale.
- Parameters:** <integer> Position of the reference level
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y2:rpos 10div
disp:trace:y2:rpos? → 10
```
- Command Type:** Sequential command

**DISPlay[:WINDow]:TRACe:Y2[:SCALe]:SMINimum**

Sets or acquires the value of the bottom of the scale applied when the subscale of the level axis is set to the linear or % mode.

- Syntax:**
- :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:SMINimum<wsp><NRf>[%]
  - :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:SMINimum?
- Description:**
- If the unit is not specified in the parameter, the set unit of the subscale of the level axis is used as the set unit of this parameter.
  - If a unit different from the current set unit (:DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT) of the subscale is specified, an execution error occurs.
- Parameters:** <NRf> Value of the bottom of the scale [%]
- Query Response:** Same as the above
- Example:**
- ```
disp:trace:y2:smin 0%
disp:trace:y2:smin? → 0
```
- Command Type:** Sequential command

DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT

Sets or acquires the unit of the subscale of the level axis.

Syntax:	• :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT<wsp><unit> • :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT?										
Description:	None										
Parameters:	<table><tr><th><unit></th><th>Unit</th></tr><tr><td>DB 0:</td><td>dB display</td></tr><tr><td>LINear 1:</td><td>Linear display</td></tr><tr><td>DB/KM 2:</td><td>dB/km display</td></tr><tr><td>% 3:</td><td>% display</td></tr></table>	<unit>	Unit	DB 0:	dB display	LINear 1:	Linear display	DB/KM 2:	dB/km display	% 3:	% display
<unit>	Unit										
DB 0:	dB display										
LINear 1:	Linear display										
DB/KM 2:	dB/km display										
% 3:	% display										
Query Response:	<table><tr><td>0:</td><td>dB display</td></tr><tr><td>1:</td><td>Linear display</td></tr><tr><td>2:</td><td>dB/km display</td></tr><tr><td>3:</td><td>% display</td></tr></table>	0:	dB display	1:	Linear display	2:	dB/km display	3:	% display		
0:	dB display										
1:	Linear display										
2:	dB/km display										
3:	% display										
Example:	<pre>disp:trace:y2:unit db/km disp:trace:y2:unit? → 2</pre>										
Command Type:	Sequential command										

FORMat Sub System Command**2****FORMat[:DATA]**

Sets or acquires the format used for data transfer via GPIB.

Syntax:

- :FORMat[:DATA]<wsp>REAL[,64 | ,32 | ASCii]
- :FORMat[:DATA]?

Description:

- When the format is set to REAL (binary) using this command, the output data of the following commands are produced in the REAL format.

```
:CALCulate:DATA:CGain?
:CALCulate:DATA:CNF?
:CALCulate:DATA:CPOWers?
:CALCulate:DATA:CSNR?
:CALCulate:DATA:CWAVelengths?
:TRACe[:DATA]:X?
:TRACe[:DATA]:Y?
```

- The default is the ASCII mode.
- When the *RST command is executed, the format is reset to the ASCII mode.

- The ASCII format outputs a list of numerics each of which is delimited by a comma (.).

E.g.: 12345,12345,....

- By default, the REAL format outputs data in fixed length blocks of 64 bits, floating-point binary numerics.
- If "REAL,32" is specified in the parameter, data is output in the 32-bit, floating-point binary form.

- The fixed length block is defined by IEEE 488.2 and consists of "#" (ASCII), one numeric (ASCII) indicating the number of bytes that specifies the length after #, length designation (ASCII), and binary data of a specified length in this order. Binary data consists of a floating-point data string of 8 bytes (64 bits) or 4 bytes (32 bits). Floating-point data consists of lower-order bytes to higher-order bytes.

E.g.: #18 [eight <byte data>]
 #280 [eighty <byte data>]
 #48008 [8008 <byte data>]

- For data output in the 32-bit floating-point binary form, cancellation of significant digits is more likely to occur in comparison with transfer of data in the 64-bit, floating-point binary form.

Parameters:

ASCii	ASCII format (default)
REAL[,64]	REAL format (64 bits)
REAL,32	REAL format (32 bits)

Query Response:	Same as the above
Example:	format:data real format:data? → REAL,64
Example2:	format:data real,32 format:data? → REAL,32
Example3:	format:data asc format:data? → ASCII
Command Type:	Sequential command

HCOPY Sub System Command**HCOPY:DESTination**

Sets or acquires the printing output destination.

Syntax:

- :HCOPY:DESTination<wsp>INTernal | EXTernal | FILE | 0 | 1 | 2
- :HCOPY:DESTination?

Description: None

Parameters: INTernal | 0: Internal printer
EXTernal | 1: External printer
FILE | 2: File

Query Response: 0: Internal printer
1: External printer
2: File

Example: hcopy:dest file
hcopy:dest? → 2

Command Type: Sequential command

HCOPY:DEvice:COLor

Sets or acquires the printing mode to the color mode or the black and white mode or acquires the printing mode set.

Syntax:

- :HCOPY:DEvice:COLor<wsp>OFF | ON | 0 | 1
- :HCOPY:DEvice:COLor?

Description:

- Setting of this command is initialized to OFF if the *RST command is executed.
- Setting of this command is applied when output to an external printer.

Parameters: OFF | 0: Black and white mode
ON | 1: Color mode

Query Response: Same as the above

Example: hcopy:device:color on
hcopy:device:color? → 1

Command Type: Sequential command

HCOPY[:IMMediate]

Makes a hard copy of the screen display.

Syntax:

- :HCOPY[:IMMediate]

Description: None

Parameters: None

Query Response: None

Example: hcopy

Command Type: Command subject to overlapping

HCOPY[:IMMediate]:FEED

Feeds printer paper to the internal printer.

- Syntax:** ▪ :HCOPY[:IMMediate]:FEED<wsp>[<integer>]
Description: ▪ If <integer> is not specified, printer paper is fed by approx. 5 mm.
Parameters: <integer> Specify the amount of feed in 1 to 10 (unit: ×5 mm).
Query Response: None
Example: hcopy:feed
Command Type: Sequential command

HCOPY[:IMMediate]:FUNCTION:CALCulate:LIST

Prints the results of the execution of an analysis function.

- Syntax:** ▪ :HCOPY[:IMMediate]:FUNCTION:CALCulate:LIST
Description: ▪ An execution error occurs in the following case:
 The analysis function has not been performed.
Parameters: None
Query Response: None
Example: hcopy:func:calc:list
Command Type: Command subject to overlapping

HCOPY[:IMMediate]:FUNCTION:MARKer:LIST

Prints a marker list.

- Syntax:** ▪ :HCOPY[:IMMediate]:FUNCTION:MARKer:LIST
Description: ▪ No execution error occurs even if a marker does not exist.
Parameters: None
Query Response: None
Example: hcopy:func:mark:list
Command Type: Command subject to overlapping

INITiate Sub System Command**INITiate[:IMMediate]**

Makes a sweep.

Syntax: • :INITiate[:IMMediate]

Description:

- A sweep can be stopped using the :ABORT command.
- The sweep mode (AUTO, SINGLE, REPEAT, or SEGMENT MEASURE) is set using the :INITiate:SMODE command.
- If this command is executed while the sweep mode is in REPEAT (:INITiate:SMODE REPeat), the operation of the command is complete at the instant a sweep starts. In this case, this command is regarded as a sequential command.
- If this command is executed while the sweep mode is one of AUTO, SINGLE, and SEGMENT MEASURE, the operation of the command is complete at the instant a sweep ends. In this case, this command is regarded as a command subject to overlapping.

Parameters: None

Query Response: None

Example: init

Command Type:

- When the sweep mode is in REPEAT:
Sequential command
- If the sweep mode is any one of AUTO, SINGLE, and SEGMENT MEASURE:
Command subject to overlapping

INITiate:SMODE

Sets or acquires the sweep mode.

Syntax:

- :INITiate:SMODE<wsp><sweep mode>
- :INITiate:SMODE?

Description: None

Parameters:

<sweep mode>	Sweep mode
SINGLE 1:	SINGLE sweep mode
REPeat 2:	REPEAT sweep mode
AUTO 3:	AUTO sweep mode
SEGment 4:	SEGMENT MEASURE sweep mode

Query Response:

1:	SINGLE sweep mode
2:	REPEAT sweep mode
3:	AUTO sweep mode
4:	SEGMENT MEASURE sweep mode

Example:

```
init:smode repeat
init:smode? → 2
```

Command Type: Sequential command

MEMory Sub System Command

MEMory:CLEar

Clears the contents of a specified waveform memory.

- Syntax:** • :MEMory:CLEar<wsp><integer>
- Description:** • No execution error occurs even if a specified waveform memory has already been cleared.
- Parameters:** <integer> Memory number
- Query Response:** None
- Example:** mem:clear 10
- Command Type:** Sequential command

MEMory:EMPTy?

Acquires the condition of whether a waveform has been specified in a specified waveform memory.

- Syntax:** • :MEMory:EMPTy?<wsp><integer>
- Description:** None
- Parameters:** <integer> Memory number
- Query Response:** 0: Not registered yet
 1: Registered
- Example:** mem:empty? 10 → 1
- Command Type:** Sequential command

MEMory:LOAD

Loads a waveform from a specified waveform memory into a specified trace.

- Syntax:** • :MEMory:LOAD<wsp><integer>,<trace name>
- Description:** • When a waveform is not registered in the specified waveform memory, a warning message appears.
 (Warning 107: Unsuitable memory number)
- Parameters:** <integer> Memory number
 <trace name> Trace (TRA, TRB, TRC, TRD, TRE, TRF, TRG)
- Query Response:** None
- Example:** mem:load 10,TRA
- Command Type:** Sequential command

MEMory:STORe

Stores the waveform of a specified trace into a specified waveform memory.

Syntax: • :MEMory:STORe<wsp><integer>,<trace name>

Description: • When waveform data do not exist in the specified trace, a warning message appears.

(Warning 107: Unsuitable memory number)

Parameters: <integer> Memory number

<trace name> Trace (TRA, TRB, TRC, TRD, TRE, TRF, TRG)

Query Response: None

Example: mem:store 10,TRA

Command Type: Sequential command

MMEMory Sub System Command

<Common Items>

- To include a directory name in <"filename">, specify the path in the following manner.
 1. Specification of an absolute path
When the head of <"file name"> is character "\", specify the absolute path.
 2. Specification of a relative path
When the head of <"file name"> is any character other than "\", specify the relative path from the current directory.
- The current directory is specified using the :MMEMory:CDIRectory command.
- If INTERNAL | FLOPPy is not specified, access is made to the current drive. The current drive should be specified using the :MMEMory:CDRive command.
- If a file name extension is omitted when storing a file, an extension corresponding to the data type will be appended to the file name.
- It is not possible to omit the extension of a file name when loading a file.

MMEMory:CATalog

Acquires a list of all files in the current directory.

Syntax:	• :MMEMory:CATalog?<wsp>[INTERNAL FLOPPy][,<directory name>]
Description:	None
Parameters:	INTERNAL: Acquires a file list in the current directory of the internal HDD. FLOPPy: Acquires a file list in the current directory of FDD. directory name: Directory name
Query Response:	<free size>,<file number>,<file name>,<file name>, ... ,<file name> <free size> : <NRf> Disk's free size [KB] (1KB=1024 bytes) <file number> : <integer> Number of files <file name> : File name
Example:	mmem:catalog? internal, " \test\sample" → +1.91176800E+006,2,test0001.wv5,test0002.wv5
Command Type:	Sequential command

MMEMory:CDIRectory

Changes or acquires the current directory.

Syntax:	• :MMEMory:CDIRectory<wsp><directory name> • :MMEMory:CDIRectory?
Description:	None
Parameters:	<directory name>: Directory name to be changed
Query Response:	<directory name>: Current directory name (absolute path from the root directory)
Example:	mmem:cdir " \test\sample" mmem:cdir? → \test\sample
Command Type:	Sequential command

MMEMory:CDrive

Changes or acquires the current drive.

Syntax:	• :MMEMory:CDrive<wsp>INTernal FLOppy • :MMEMory:CDrive?
Description:	None
Parameters:	INTernal: Makes the current drive the internal HDD. FLOppy: Makes the current drive the FDD.
Query Response:	Same as the above
Example:	<pre>mmem:cdrive internal</pre> <pre>mmem:cdrive → INT</pre>
Command Type:	Sequential command

MMEMory:COPY

Copies a specified file.

Syntax:	• :MMEMory:COPY<wsp><"source file name">[,INTernal FLOppy],<"destination file name">[,INTernal FLOppy]
Description:	None
Parameters:	<"source file name">: File name at the copy source <"destination file name">: File name at the copy destination
Query Response:	None
Example:	<pre>mmem:copy "test001.wv5", , "test002.wv5"</pre>
Command Type:	Sequential command

MMEMory:DELeTe

Deletes a specified file.

Syntax:	• :MMEMory:DELeTe<wsp><"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<"file name"> Name of a file to be deleted
Query Response:	None
Example:	<pre>mmem:del "test002.wv5", internal</pre>
Command Type:	Sequential command

MMEMory:INITialize

Initializes a floppy disk in the FDD of this unit.

Syntax:	• :MMEMory:INITialize<wsp>[FLOppy]
Description:	None
Parameters:	FLOppy : Initializes an FD.
Query Response:	None
Example:	<pre>mmem:init floppy</pre>
Command Type:	Sequential command

MMEMory:LOAD:MEMory

Loads a specified waveform file into a specified memory.

Syntax:	· :MMEMory:LOAD:MEMory<wsp><integer>,<"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<integer>: Number of the memory into which a file is loaded <"file name">: Name of a file to be loaded INTernal FLOppy: Drive at the loading source
Query Response:	None
Example:	mmem:load:mem 1,"test001.wv5",internal
Command Type:	Sequential command

MMEMory:LOAD:PROGram

Loads a specified program file into a specified program number.

Syntax:	· :MMEMory:LOAD:PROGram<wsp><integer>,<"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<trace name>: Number of the program into which a file is loaded <"file name">: Name of a file to be loaded INTernal FLOppy: Drive at the loading source
Query Response:	None
Example:	mmem:load:prog 1,"test001.pg5",internal
Command Type:	Sequential command

MMEMory:LOAD:SETTing

Loads a specified setting file.

Syntax:	· :MMEMory:LOAD:SETTing<wsp><"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<"file name">: Name of the file to be loaded INTernal FLOppy: Drive at the loading source
Query Response:	None
Example:	mmem:load:setting "test001.st5",internal
Command Type:	Sequential command

MMEMory:LOAD:TEMPlate

Loads a specified template file.

Syntax:	· :MMEMory:LOAD:TEMPlate<wsp><template>,<"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<template> Template at the loading destination (UPPER LOWER TARGET) <"file name">: Name of a file to be loaded INTernal FLOppy: Drive at the loading source
Query Response:	None
Example:	mmem:load:template:setting upper,"test001.csv",internal
Command Type:	Sequential command

MMEMory:LOAD:TRACe

Loads a specified waveform file into a specified trace.

Syntax:	• :MMEMory:LOAD:TRACe<wsp><trace name>,<"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<trace name>: Trace into which a file is loaded <"file name">: Name of a file to be loaded INTernal FLOppy: Drive at the loading source
Query Response:	None
Example:	mmem:load:trace tra,"test001.wv5",internal
Command Type:	Sequential command

MMEMory:MDIRectory

Creates a new directory.

Syntax:	• :MMEMory:MDIRectory<wsp><"directory name">[,INTernal FLOppy]
Description:	None
Parameters:	<"directory name">: Name of a directory to be created INTernal FLOppy: Drive at the directory-creating destination
Query Response:	None
Example:	mmem:mdir "sample2",internal
Command Type:	Sequential command

MMEMory:REName

Renames a specified file.

Syntax:	• :MMEMory:REName<wsp><"new file name">,<"old file name">[,INTernal FLOppy]
Description:	None
Parameters:	<"new file name">: A new file name <"old file name">: An old file name
Query Response:	None
Example:	mmem:ren "test001.wv5","test002.wv5",internal
Command Type:	Sequential command

MMEMory:STORe:ARESt

Stores a variety of analysis results to a specified file.

Syntax:	• :MMEMory:STORe:ARESt<wsp><"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<"file name"> : Name of the file to which data is stored INTernal FLOppy : Drive at the storage destination
Query Response:	None
Example:	mmem:store:areult "test001",internal
Command Type:	Sequential command

MMEMory:STORe:DATA

Stores a variety of data to a specified file.

- Syntax:** ▪ :MMEMory:STORe:DATA<wsp><"file name">[,INTernal | FLOPpy]
- Description:** ▪ The type of data to be stored is specified using the
MMEMory:STORe:DATA:ITEM command.
- Whether to insert data into or overwrite the file with it when storing it
is specified using the MMEMory:STORe:DATA:MODE command.
- Parameters:** <"file name">: Name of the file to which data is stored
 INTernal | FLOPpy: Drive at the storage destination
- Query Response:** None
- Example:** mmem:store:data "test001",internal
- Command Type:** Sequential command

MMEMory:STORe:DATA:ITEM

Sets or acquires an item to be used when storing data.

- Syntax:** ▪ :MMEMory:STORe:DATA:ITEM<wsp><item>,OFF | ON | 0 | 1
 ▪ :MMEMory:STORe:DATA:ITEM?<wsp><item>
- Description:** None
- Parameters:** <item> Item
 DATE : Date and time at the time of storage
 LABel : Label
 DATA : Data in the DATA area
 OWINdow: Output window
 CONDition : Setting conditions
 TRACe : Waveform data
 OFF | ON | 0 | 1 Setting of whether to store data
- Query Response:** 0 | 1 0:OFF, 1:ON
- Example:** mmem:store:data:item trace,off
 mmem:store:data:item? trace → 0
- Command Type:** Sequential command

MMEMory:STORe:DATA:MODE

Sets whether to insert data into or overwrite an existing file with the data when storing it or acquires the condition of whether data is inserted or overwritten.

- Syntax:** ▪ :MMEMory:STORe:DATA:MODE<wsp>ADD | OVER | 0 | 1
 ▪ :MMEMory:STORe:DATA:MODE?
- Description:** None
- Parameters:** ADD | 0: Insert mode
 OVER | 1: Overwrite mode
- Query Response:** 0: Insert mode
 1: Overwrite mode
- Example:** mmem:store:data:mode over
 mmem:store:data:mode? → 1
- Command Type:** Sequential command

MMEMory:STORe:DATA:TYPE

Sets or acquires file format to be used when storing data.

Syntax:	• :MMEMory:STORe:DATA:TYPE<wsp>CSV DT 0 1 • :MMEMory:STORe:DATA:TYPE?
Description:	None
Parameters:	CSV 0 : CSV storage format DT 1 : DT5 storage format
Query Response:	0 : CSV storage format 1 : DT5 storage format
Example:	<pre>mmem:store:data:type dt5</pre> <pre>mmem:store:data:type? → 1</pre>
Command Type:	Sequential command

MMEMory:STORe:GRAPHics

Stores a waveform screen to a specified graphic file.

Syntax:	• :MMEMory:STORe:GRAPHics<wsp>B&W COLor,BMP TIFF, <"file name">[,INTernal FLOppy]
Description:	None
Parameters:	B&W COLor: Color mode applied when storing the screen B&W: Black and white mode, COLor: Color mode BMP TIFF: Storage format (BMP: BMP form, TIFF: TIFF form) <"file name">: Name of the file to which a waveform screen is stored INTernal FLOppy : Drive at the storage destination
Query Response:	None
Example:	<pre>mmem:store:graphics color,bmp,"test001",internal</pre>
Command Type:	Sequential command

MMEMory:STORe:MEMory

Stores a specified memory to a specified waveform file.

Syntax:	• :MMEMory:STORe:MEMory<wsp><integer>,BIN CSV, <"file name">[,INTernal FLOppy]
Description:	None
Parameters:	<integer>: Number of a memory the contents of which are stored BIN CSV: Storage format (BIN: binary form, CSV: text form) <"file name">: Name of a file to be stored INTernal FLOppy: Drive at the storage destination
Query Response:	None
Example:	<pre>mmem:store:mem 1, csv, "test001", internal</pre>
Command Type:	Sequential command

MMEMory:STORe:PROGram

Stores a specified program to a specified file.

Syntax:	<code>:MMEMory:STORe:PROGram<wsp><integer>,<"file name">[,INTernal FLOPpy]</code>
Description:	None
Parameters:	<code><integer></code>: Number of a program to be stored <code><"file name"></code>: Name of the file into which a program is stored <code>INTernal FLOPpy</code>: Drive at the storage destination
Query Response:	None
Example:	<code>mmem:store:prog 1,"test001",internal</code>
Command Type:	Sequential command

MMEMory:STORe:SETTing

Stores setting information to a specified file.

Syntax:	<code>:MMEMory:STORe:SETTing<wsp><"file name">[,INTernal FLOPpy]</code>
Description:	None
Parameters:	<code><"file name"></code>: Name of the file into which the settings are stored <code>INTernal FLOPpy</code>: Drive at the storage destination
Query Response:	None
Example:	<code>mmem:store:setting "test001",internal</code>
Command Type:	Sequential command

MMEMory:STORe:TEMPlate

Stores specified template data to a specified file

Syntax:	<code>:MMEMory:STORe:TEMPlate<wsp><template>,<"file name">[,INTernal FLOPpy]</code>
Description:	None
Parameters:	<code><template></code>: Template to be stored (UPPER LOWER TARGET) <code><"file name"></code>: Name of the file to which template data is stored <code>INTernal FLOPpy</code>: Drive at the storage destination
Query Response:	None
Example:	<code>mmem:store:temp upper,"test001",internal</code>
Command Type:	Sequential command

MMEMory:STORe:TRACe

Stores a specified trace in a specified waveform file.

Syntax:	<code>:MMEMory:STORe:TRACe<wsp><trace name>[,BIN CSV,<"file name">[,INTernal FLOPpy]</code>
Description:	None
Parameters:	<code><trace name></code>: Trace to be stored <code>BIN CSV</code>: Storage format (BIN: binary form, CSV: text form) <code><"file name"></code>: Name of a file to be stored <code>INTernal FLOPpy</code>: Drive at the storage destination
Query Response:	None
Example:	<code>mmem:store:trace tra,csv,"test001",internal</code>
Command Type:	Sequential command

PROGram Sub System Command**2****PROGram:EXECute**

Executes a specified program.

Syntax:	• :PROGram:EXECute<wsp><integer>
Description:	None
Parameters:	<integer> Number of a program to be executed
Query Response:	None
Example:	prog:exec 1
Command Type:	Command subject to overlapping

SENSe Sub System Command

SENSe:AVERage:COUNT

Sets or acquires the averaging times for each measurement point.

Syntax:

- :SENSe:AVERage:COUNT<wsp><integer>
- :SENSe:AVERage:COUNT?

Description: None

Parameters: <integer> Averaging times

Query Response: Same as the above

Example:

```
sens:aver:count 100
sens:aver:count? → 100
```

Command Type: Sequential command

SENSe:BANDwidth[:BWIDth]:RESolution

Sets or acquires the measurement resolution.

Syntax:

- :SENSe:BANDwidth[:BWIDth]:RESolution<wsp><NRf>[M]
- :SENSe:BANDwidth[:BWIDth]:RESolution?

Description: None

Parameters: <NRf> Measurement resolution [m]

Query Response: Same as the above

Example:

```
sens:bwid:res 10pm
sens:band? → +1.000000000E-012
```

Command Type: Sequential command

SENSe:CHOPper

Sets or acquires ON/OFF of the chopper.

Syntax:

- :SENSe:CHOPper<wsp>OFF | ON(CHOP) | SWITCh | 0 | 1 | 2
- :SENSe:CHOPper?

Description:

- Even when this parameter is ON, if the setting of measuring sensitivity (:SENSe:SENSe command) is NORMAL HOLD, NORMAL, or AUTO, the chopper does not function.
- If you want to set the CHOP, send to command whether ON or CHOP.

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
sens:chop on
sens:chop? → 1
```

Command Type: Sequential command

SENSe:CORRection:LEVel:SHIFt

Sets or acquires the level offset value.

Syntax:	• :SENSe:CORRection:LEVel:SHIFt<wsp><NRf>[DB] • :SENSe:CORRection:LEVel:SHIFt?
Description:	None
Parameters:	<NRf> Level offset value [dB]
Query Response:	Same as the above
Example:	<pre>sens:corr:lev:shift 0.2db sens:corr:lev:shift? → +2.00000000E-001</pre>
Command Type:	Sequential command

SENSe:CORRection:RVELocity:MEDium

Sets or acquires whether air or vacuum is used as the wavelength reference.

Syntax:	• :SENSe:CORRection:RVELocity:MEDium <wsp>AIR VACuum 0 1 • :SENSe:CORRection:RVELocity:MEDium?
Description:	None
Parameters:	AIR 0: Assumes air to be the reference. VACuum 1: Assumes vacuum to be the reference.
Query Response:	0: Air is assumed to be the reference. 1: Vacuum is assumed to be the reference.
Example:	<pre>sens:corr:rvel:medium vacuum sens:corr:rvel:medium? → 1</pre>
Command Type:	Sequential command

SENSe:CORRection:WAVelength:SHIFt

Sets or acquires the wavelength offset value.

Syntax:	• :SENSe:CORRection:WAVelength:SHIFt<wsp><NRf>[M] • :SENSe:CORRection:WAVelength:SHIFt?
Description:	None
Parameters:	<NRf> Wavelength offset value [m]
Query Response:	Same as the above
Example:	<pre>sens:corr:wav:shift 0.05nm sens:corr:wav:shift? → +5.00000000E-011</pre>
Command Type:	Sequential command

SENSe:SENSe

Sets or acquires the measuring sensitivity.

Syntax:

- SENSE:SENSe<wsp><sense>
- SENSE:SENSe?

Description: Chopper ON/OFF is set by SENSE:CHOPper command.

Parameters: <sense> Sensitivity setting parameters

NHLD | 0: NORMAL HOLD

NAUT | 1: NORMAL AUTO

MID | 2: MID

HIGH1 | 3: HIGH 1 or HIGH1/CHOP

HIGH2 | 4: HIGH 2 or HIGH2/CHOP

HIGH3 | 5: HIGH 3 or HIGH3/CHOP

Query Response:

0: NORMAL HOLD

1: NORMAL AUTO

2: MID

3: HIGH 1 or HIGH1/CHOP

4: HIGH 2 or HIGH2/CHOP

5: HIGH 3 or HIGH3/CHOP

Example:

sens:sense mid

sens:sense? → 2

Command Type: Sequential command

SENSe:SETTing:ATTenuator

Sets or acquires ON/OFF of the optical attenuator.

Syntax:

- SENSE:SETTing:ATTenuator<wsp>OFF | ON | 0 | 1
- SENSE:SETTing:ATTenuator?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

sens:setting:att on

sens:setting:att? → 1

Command Type: Sequential command

SENSe:SWEep:POINts

Sets or acquires the sampling number to be measured.

Syntax:

- :SENSe:SWEep:POINts<wsp><integer>
- :SENSe:SWEep:POINts?

Description:

- When the function of automatically setting the sampling number to be measured (SENSe:SWEep:POINts:AUTO command) is ON, the sampling number to be measured that has been set can be acquired.
- When the function of automatically setting the number of samples to be measured (SENSe:SWEep:POINts:AUTO command) is ON, this command will be automatically set to OFF.
- When the sampling number to be measured is set using this command, the sampling intervals for measurements (SENSe:SWEep:STEP) will be automatically set.

Parameters: <integer> The sampling number to be measured

Query Response: Same as the above

Example:

```
sens:sweep:points 20001  
sens:sweep:points? → 20001
```

Command Type: Sequential command

SENSe:SWEep:POINts:AUTO

Sets or acquires the function of automatically setting the sampling number to be measured.

Syntax:	• :SENSe:SWEep:POINts:AUTO<wsp>OFF ON 0 1 • :SENSe:SWEep:POINts:AUTO?
Description:	• When the capability to automatically set the sampling number to be measured is set to ON using this command, the sampling number to be measured and the sampling intervals for measurements (SENSe:SWEep:STEP) will be automatically set.
Parameters:	OFF ON 0 1
Query Response:	0 1 0:OFF, 1:ON
Example:	<pre>sens:sweep:points:auto on sens:sweep:points:auto? → 1</pre>
Command Type:	Sequential command

SENSe:SWEep:SEGment:POINts

Sets or acquires the number of sampling points to be measured at one time when performing SEGMENT MEASURE.

Syntax:	• SENSe:SWEep:SEGment:POINts<wsp><integer> • SENSe:SWEep:SEGment:POINts?
Description:	None
Parameters:	<integer> The number of sampling points for measurements
Query Response:	Same as the above
Example:	<pre>sens:sweep:segm:points 100 sens:sweep:segm:points? → 100</pre>
Command Type:	Sequential command

SENSe:SWEep:STEP

Sets or acquires the sampling intervals for measurements.

Syntax:	• :SENSe:SWEep:STEP<wsp><NRf> [M] • :SENSe:SWEep:STEP?
Description:	• When the function of automatically setting the sampling number to be measured (SENSe:SWEep:POINts:AUTO command) is ON, the sampling intervals for measurements that has been set can be acquired. • When the function of automatically setting the sampling number to be measured (SENSe:SWEep:POINts:AUTO command) is ON, this command will be automatically set to OFF. • If the sampling interval for measurements is set using this command, the sampling number to be measured (SENSe:SWEep:POINts) will be automatically set.
Parameters:	<NRf> The sampling number for measurement [m]
Query Response:	Same as the above
Example:	<pre>sens:sweep:step 1pm sens:sweep:step? → +1.00000000E-012</pre>
Command Type:	Sequential command

SENSe:SWEep:TIME:0NM

Sets or acquires the time taken from the start to the end of measurements when measurement is made in the 0-nm sweep mode.

Syntax:

- :SENSe:SWEep:TIME:0NM<wsp><integer>[SEC]
- :SENSe:SWEep:TIME:0NM?

Description: None

Parameters: <integer> Measurement time [sec] (0: Minimum)

Query Response: Same as the above

Example:

```
sens:sweep:time:0nm 10sec
sens:sweep:time:0nm? → 10
```

Command Type: Sequential command

SENSe:SWEep:TIME:INTerval

Sets or acquires the time taken from the start of a sweep to that of the next sweep when repeat sweeps are made.

Syntax:

- :SENSe:SWEep:TIME:INTerval<wsp><integer>[SEC]
- :SENSe:SWEep:TIME:INTerval?

Description: None

Parameters: <integer> Measurement time [sec] (0: Minimum)

Query Response: Same as the above

Example:

```
sens:sweep:time:interval 100sec
sens:sweep:time:interval? → 100
```

Command Type: Sequential command

SENSe:SWEep:TLSSync

Sets or acquires the function of synchronous sweep function with the Tunable Laser Source

Syntax:

- :SENSe:SWEep:TLSSync<wsp>OFF | ON | 0 | 1
- :SENSe:SWEep:TLSSync?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0 : OFF, 1: ON

Example:

```
sens:sweep:tlssync on
sens:sweep:tlssync? → 1
```

Command Type: Sequential command

SENSe:WAVelength:CENTer

Sets or acquires the center wavelength of the measurement conditions.

Syntax:

- :SENSe:WAVelength:CENTer<wsp><NRf>[M]
- :SENSe:WAVelength:CENTer?

Description: None

Parameters: <NRf> Measurement center wavelength [m]

Query Response: Same as the above

Example:

```
sens:wav:center 1550.000nm
sens:wav:cent? → +1.55000000E-006
```

Command Type: Sequential command

SENSe:WAVelength:SPAN

Sets or acquires the measurement span of the measurement conditions.

Syntax:

- :SENSe:WAVelength:SPAN<wsp><NRf>[M]
- :SENSe:WAVelength:SPAN?

Description: None

Parameters: <NRf> Measurement span [m]

Query Response: Same as the above

Example:

```
sens:wav:span 20.0nm
sens:wav:span? → +2.00000000E-008
```

Command Type: Sequential command

SENSe:WAVelength:SRANge

Sets or acquires whether to limit a sweep range to the spacing between line markers L1 and L2.

Syntax:

- :SENSe:WAVelength:SRANge<wsp>OFF | ON | 0 | 1
- :SENSe:WAVelength:SRANge?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
sens:wav:srange on
sens:wav:srange? → 1
```

Command Type: Sequential command

SENSe:WAVelength:STARt

Sets or acquires the measurement-starting wavelength of the measurement conditions.

Syntax:

- :SENSe:WAVelength:STARt<wsp><NRf>[M]
- :SENSe:WAVelength:STARt?

Description: None

Parameters: <NRf> Measurement-starting wavelength [m]

Query Response: Same as the above

Example:

```
sens:wav:start 1540.000nm
sens:wav:start? → +1.54000000E-006
```

Command Type: Sequential command

SENSe:WAVelength:STOP

Sets or acquires the measurement-ending wavelength of the measurement conditions.

Syntax:

- :SENSe:WAVelength:STOP<wsp><NRf>[M]
- :SENSe:WAVelength:STOP?

Description: None

Parameters: <NRf> Measurement-ending wavelength [m]

Query Response: Same as the above

Example:

```
sens:wav:stop 1560.000nm
sens:wav:stop? → +1.56000000E-006
```

Command Type: Sequential command

STATus Sub System Command**STATus:OPERation:CONDition?**

Reads the contents of the operation status condition register.

Syntax: • :STATus:OPERation:CONDition?

Description: None

Parameters: None

Query Response: <integer> Contents of the operation status condition register

Example: stat:operation:cond? → 1

Command Type: Sequential command

STATus:OPERation:ENABLE

Reads the contents of the operation status enable register or writes data to this register.

Syntax: • :STATus:OPERation:ENABLE<wsp><integer>
 • :STATus:OPERation:ENABLE?

Description: None

Parameters: <integer> Contents of the operation status enable register

Query Response: Same as the above

Example: stat:operation:enable 8
 stat:operation:enable? → 8

Command Type: Sequential command

STATus:OPERation[:EVENT]?

Reads the contents of the operation status event register.

Syntax: • :STATus:OPERation[:EVENT]?

Description: None

Parameters: None

Query Response: <integer> Contents of the operation status event register

Example: stat:operation? → 1

Command Type: Sequential command

STATus:PRESet

Clears the operation status event register and sets all bits of the operation status enable register to "0."

Syntax: `·:STATus:PRESet`

Description: When this command is executed, the following registers will be affected.

- The operation status event register is cleared to "0."
- All bits of the operation status enable register sets to "0."
- The questionable status event register is cleared to "0."
- All bits of the questionable status enable register sets to "0."

The following registers will not be affected by execution of this command.

- Standard event status register
- Standard event status enable register

Parameters: None

Query Response: None

Example: `stat:preset`

Command Type: Sequential command

STATusQUESTIONable:CONDition?

Reads the contents of the questionable status condition register.

Syntax: `·:STATus:QUESTIONable:CONDition?`

Description: None

Parameters: None

Query Response: `<integer>` Contents of the questionable status condition register

Example: `stat:ques:cond? → 1`

Command Type: Sequential command

STATus:QUESTIONable:ENABLE

Reads the contents of the questionable status enable register or writes data to this register.

Syntax: `·:STATus:QUESTIONable:ENABLE<wsp><integer>`
`·:STATus:QUESTIONable:ENABLE?`

Description: None

Parameters: `<integer>` Contents of the questionable status enable register

Query Response: Same as the above

Example: `stat:ques:enable 8`
`stat:ques:enable? → 8`

Command Type: Sequential command

STATus:QUESTionable[:EVENT]?**2**

Reads the contents of the questionable status event register.

Syntax: · :STATus:QUESTionable[:EVENT]?

Description: None

Parameters: None

Query Response: <integer> Contents of the questionable status event register

Example: stat: questionable? → 1

Command Type: Sequential command

SYSTem Sub System Command

SYSTem:BUZZer:CLICk

Sets or acquires whether to sound the buzzer when a click is made.

Syntax:

- :SYSTem:BUZZer:CLICk<wsp>OFF | ON | 0 | 1
- :SYSTem:BUZZer:CLICk?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
syst:buzzer:click on
syst:buzzer:click? → 1
```

Command Type: Sequential command

SYSTem:BUZZer:WARNing

Sets or acquires whether to sound the buzzer when generating an alarm.

Syntax:

- :SYSTem:BUZZer:WARNing<wsp>OFF | ON | 0 | 1
- :SYSTem:BUZZer:WARNing?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
syst:buzzer:warn on
syst:buzzer:warn? → 1
```

Command Type: Sequential command

SYSTem:COMMunicate:CFORmat

Sets or acquires the GPIB command format of this unit.

Syntax: • :SYSTem: COMMunicate:CFORmat<wsp><mode>

• :SYSTem: COMMunicate:CFORmat?

Description: • This command is valid only in the AQ6319 mode. If it is executed while this unit is in the AQ6317-compatible mode, a command error occurs.

• To set the GPIB command format while this unit is in the AQ6317-compatible mode, use the following commands.

Control command: CFORM* (*: 0 = AQ6317-compatible mode, 1 = AQ6319 mode)

Query command: CFORM? (return value: 0 = AQ6317-compatible mode, 1 = AQ6319 mode)

• To use a GPIB command to place this unit into the AQ6317-compatible mode, regardless of the status during execution of the command, execute the following command. Note that if this unit has already been in the AQ6317-compatible mode at the time of executing this command, a command error occurs, but you can ignore it.

:SYSTem:COMMunicate:CFORmat<wsp>AQ6317

• To use a GPIB command to place this unit into the AQ6319 mode, regardless of the status during execution of the command, execute the following command. Note that if this unit has already been in the AQ6319 mode at the time of executing the command, a command error occurs, but you can ignore it.

CFORM1

• For details on switching GPIB command mode, refer to section 2.7.3 "Switching the Command Mode Using a GPIB Command".

Parameters: <mode> GPIB command format

AQ6317 | 0 AQ6317-compatible mode

AQ6319 | 1 AQ6319 mode

Query Response: Same as the above

Example: syst:comm:cformat aq6319

syst:comm:cformat? → 1

Command Type: Sequential command

SYSTem:COMMunicate:GPIB2:ADDRess

Sets or acquires the GPIB address of the GP-IB2 port of this unit.

Syntax: • :SYSTem: COMMunicate:GPIB2:ADDRess<wsp><integer>

• :SYSTem: COMMunicate:GPIB2:ADDRess?

Description: None

Parameters: <integer> GPIB address of GP-IB2 port

Query Response: Same as the above

Example: syst:comm:gpi2:addr 2

syst:comm:gpi2:addr? → 2

Command Type: Sequential command

SYSTem:COMMunicate:GPIB2:TLs:ADDRess

Sets or acquires the GPIB address of the Tunable Laser Source connecting GP-IB2 port of this unit.

Syntax:

- :SYSTem: COMMunicate:GPIB2:TLs:ADDRess<wsp><integer>
- :SYSTem: COMMunicate:GPIB2:TLs:ADDRess?

Description: None

Parameters: <integer> Sets GP-IB address of Tunable Laser Source using synchronous sweep function.

Query Response: Same as the above

Example:

```
syst:comm:gpi2:tl:addr 20
syst:comm:gpi2:tl:addr? → 20
```

Command Type: Sequential command

SYSTem:DATE

Sets or acquires the system date of this unit.

Syntax:

- :SYSTem:DATE<wsp><year>,<month>,<day>
- :SYSTem:DATE?

Description: None

Parameters:

<year>	Year
<month>	Month
<day>	Day

Query Response: Same as the above

Example:

```
syst:date 2002,11,01
syst:date? → 2002,11,1
```

Command Type: Sequential command

SYSTem:DISPlay:TRANSPARENT

Sets or acquires whether to make the Interrupt Window and OVERVIEW Window of the measurement screen semi-transparent.

Syntax:

- :SYSTem:DISPlay:TRANSPARENT<wsp>OFF | ON | 0 | 1
- :SYSTem:DISPlay:TRANSPARENT?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
syst:disp:tran off
syst:disp:tran? → 0
```

Command Type: Sequential command

SYSTem:DISPlay:UNCAL

Sets or acquires whether to display an alarm message in the event of UNCAL.

Syntax:

- :SYSTem:DISPlay:UNCAL<wsp>OFF | ON | 0 | 1
- :SYSTem:DISPlay:UNCAL?

Description: None

Parameters: OFF | ON | 0 | 1

Query Response: 0 | 1 0:OFF, 1:ON

Example:

```
syst:disp:uncal off
syst:disp:uncal? → 0
```

Command Type: Sequential command

SYSTem:ERRor[:NEXT]?

Inquires about data in an error queue and deletes it from the queue.

Syntax: • :SYSTem:ERRor[:NEXT]?

Description: None

Parameters: None

Query Response: <integer> Error number

Example: syst:err? → 100

Command Type: Sequential command

SYSTem:GRID

Sets or acquires the grid setting of this unit.

Syntax: • :SYSTem:GRID<wsp><grid>
 • :SYSTem:GRID?

Description: None

Parameters: <grid> Grid setting

12.5GHZ 0:	12.5GHz spacing
25GHZ 1:	25GHz spacing
50GHZ 2:	50GHz spacing
100GHZ 3:	100GHz spacing
200GHZ 4:	200GHz spacing
CUSTom 5:	User-specified custom setting

Query Response: Same as the above
0: 12.5GHz spacing
1: 25GHz spacing
2: 50GHz spacing
3: 100GHz spacing
4: 200GHz spacing
5: User-specified custom setting

Example: syst:grid 50GHZ
 syst:grid? → 2

Command Type: Sequential command

SYSTem:GRID:CUSTom:CLEar:ALL

Clears the user-specified custom grid and returns it to the default value.

Syntax: • :SYSTem:GRID:CUSTom:CLEar:ALL

Description: None

Parameters: None

Query Response: None

Example: syst:grid:cust:clear:all

Command Type: Sequential command

SYSTem:GRID:CUSTom:CLEar:DELeTe

Deletes the specified grid of the custom grid.

Syntax: • :SYSTem:GRID:CUSTom:DELeTe<wsp><integer>

Description: None

Parameters: <integer> Number of a grid to be deleted

Query Response: None

Example: syst:grid:cust:del 10

Command Type: Sequential command

SYSTem:GRID:CUSTom:INSert

Inserts a new grid when the grid setting is in the custom grid.

Syntax: ▪ :SYSTem:GRID:CUSTom:INSert<wsp><NRf>[M | HZ]
Description: An execution error occurs in the following case:
 :SYSTem:GRID is in setting other than CUSTom.
Parameters: <NRf> Grid wavelength/frequency to be inserted [m | Hz]
Query Response: None
Example: syst:grid:cust:ins 1550.123nm
Command Type: Sequential command

SYSTem:GRID:CUSTom:SPACing

Sets or acquires the grid intervals of the custom grid.

Syntax: ▪ :SYSTem:GRID:CUSTom:SPACing<wsp><NRf>[GHZ]
Description: An execution error occurs in the following case:
 :SYSTem:GRID is in setting other than CUSTom.
Parameters: <NRf> Grid intervals [GHz]
Query Response: None
Example: syst:grid:cust:spac 12.5
 syst:grid:cust:spac? → +1.25000000E+001
Command Type: Sequential command

SYSTem:GRID:CUSTom:STARt

Sets or acquires the starting wavelength of the custom grid.

Syntax: ▪ :SYSTem:GRID:CUSTom:STARt<wsp><NRf>[M | HZ]
 ▪ :SYSTem:GRID:CUSTom:STARt?
Description: An execution error occurs in the following case:
 :SYSTem:GRID is in setting other than CUSTom.
Parameters: <NRf> Grid's starting wavelength [m | Hz]
Query Response: Same as the above
Example: syst:grid:cust:start 1550.000nm
 syst:grid:cust:start? → +1.55000000E-006
Command Type: Sequential command

SYSTem:GRID:CUSTom:STOP

Sets or acquires the ending wavelength of the custom grid.

Syntax: ▪ :SYSTem:GRID:CUSTom:STOP<wsp><NRf>[M | HZ]
 ▪ :SYSTem:GRID:CUSTom:STOP?
Description: An execution error occurs in the following case:
 :SYSTem:GRID is in setting other than CUSTom.
Parameters: <NRf> Grid's ending wavelength [m | Hz]
Query Response: Same as the above
Example: syst:grid:cust:stop 1560.000nm
 syst:grid:cust:stop? → +1.56000000E-006
Command Type: Sequential command

SYSTem:GRID:REFeRence

Sets or acquires the reference frequency of the grid setting of this unit.

Syntax: • :SYSTem:GRID:REFeRence<wsp><NRf>[HZ]
 • :SYSTem:GRID:REFeRence?

Description: None

Parameters: <NRf> Grid's reference frequency [Hz]

Query Response: Same as the above

Example: syst:grid:ref 193.1000thz
 syst:grid:ref? → +1.930000000E+014

Command Type: Sequential command

SYSTem:PRESet

Initializes the unit status.

Syntax: • :SYSTem:PRESet

Description: None

Parameters: None

Query Response: None

Example: syst:preset

Command Type: Sequential command

SYSTem:TIME

Sets or acquires the system time of this unit.

Syntax: • :SYSTem:TIME<wsp><hour>,<minutes>,<seconds>
 • :SYSTem:TIME?

Description: None

Parameters: <hour> Hour
 <minutes> Minute
 <seconds> Second

Query Response: Same as the above

Example: syst:time 22,10,01
 syst:time? → 22,10,1

Command Type: Sequential command

SYSTem:VERSion?

Acquires the SCPI compatibility version of this unit.

Syntax: • :SYSTem:VERSion?

Description: None

Parameters: None

Query Response: <strings> SCPI version

Example: syst:vers? → 1999.0

Command Type: Sequential command

TRACe Sub System Command

TRACe:ACTive

Sets or acquires an active trace.

Syntax:	▪ :TRACe:ACTive<wsp><trace name> ▪ :TRACe:ACTive?		
Description:	None		
Parameters:	<trace name>	Active trace (TRA TRB TRC TRD TRE TRF TRG)	
Query Response:	Same as the above		
Example:	trace:act tra trace:act? → TRA		
Command Type:	Sequential command		

TRACe:ATTRibute[:<trace name>]

Sets or acquires the attribute of a specified trace.

Syntax:	• :TRACe:ATTRibute[:<trace name>]<wsp><attribute> • :TRACe:ATTRibute[:<trace name>]?		
Description:	• If <trace name> is omitted, this command is executed with respect to the active trace. • When <trace name> is specified, the specified trace becomes the active trace after executing the command. • When the attribute is set to a CALC trace, the expression should be set using the :CALCulate:MATH command.		
Parameters:	<trace name>	Trace (TRA TRB TRC TRD TRE TRF TRG)	
	<attribute>	Attribute	
	WRITe 0:	WRITE	
	FIX 1:	FIX	
	MAX 2:	MAX HOLD	
	MIN 3:	MINIMUM HOLD	
	RAVG 4:	ROLL AVG	
	CALC 5:	CALC	
Query Response:	0:	WRITE	
	1:	FIX	
	2:	MAX HOLD	
	3:	MINIMUM HOLD	
	4:	ROLL AVG	
	5:	CALC	
Example:	trace:attr:tra write trace:attr:tra? → 0		
Command Type:	Sequential command		

TRACe:ATTRibute:RAVG[:<trace name>]

Sets or acquires the averaging time of a specified trace.

Syntax:	• :TRACe:ATTRibute:RAVG[:<trace name>]<wsp><integer> • :TRACe:ATTRibute:RAVG[:<trace name>]?
Description:	• When this command is executed, the attribute of the set trace goes to ROLL AVG. • If <trace name> is omitted, this command is executed with respect to the active trace. • When <trace name> is specified, the specified trace becomes the active trace after executing the command.
Parameters:	<trace name> Trace (TRA TRB TRC TRD TRE TRF TRG) <integer> Averaging time of ROLL AVG
Query Response:	<integer> Averaging time of ROLL AVG
Example:	<pre>trace:attr:ravg:tra 10 trace:attr:ravg:tra? → 10</pre>
Command Type:	Sequential command

TRACe:COPY

Copies the data of a specified trace to another trace.

Syntax:	• :TRACe:COPY<wsp><source trace name>,<destination trace name>
Description:	None
Parameters:	<source trace name> Trace at the copy source <destination trace name> Trace at the copy destination
Query Response:	None
Example:	<pre>trace:copy tra, trb</pre>
Command Type:	Sequential command

TRACe[:DATA]:SNUMber?

Acquires the sampling number of a specified trace.

Syntax:	• :TRACe[:DATA]:SNUMber?<wsp><trace name>
Description:	• If a specified trace has no data, "0" is returned.
Parameters:	<trace name> Trace from which data is desired to be acquired
Query Response:	<integer> The sampling number (101 to 50001)
Example:	<pre>trace:data:snumber? → 50001</pre>
Command Type:	Sequential command

TRACe[:DATA]:X?

Acquires the wavelength axis data of a specified trace.

Syntax: ▪ :TRACe[:DATA]:X?<wsp><trace name>[,<start point>,<stop point>]

Description:

- Data is output in the unit of wavelength value (m), regardless of whether this unit is in the wavelength mode or in the frequency mode.
- If the parameter <start point> or <stop point> is omitted, all sampling data of a specified trace will be output.
- The number of output data can be acquired by executing ":TRACe[:DATA]:SNUMber?".
- Data is output in either ASCII or binary form, depending on the setting of :FORMat[:DATA].

Parameters:	<trace name>	Trace from which data is transferred (TRA TRB TRC TRD TRE TRF TRG)
	<start point>	A sampling range to be transferred (starting point) (1 to 50001)
	<stop point>	A sampling range to be transferred (ending point) (1 to 50001)

Query Response: For ASCII data: <NRf>,<NRf>.....<NRf>
For binary data: '# '<integer><byte num><data byte>

```
Example:      trace:x? tra
                →
                +1.55000000E-006,+1.55001000E-006,+1.55002000E-006,
                .....

```

Command Type: Sequential command

TRACe[:DATA]:Y?

Acquires the level axis data of a specified trace.

Syntax: · :TRACe[:DATA]:Y?<wsp><trace name>[,<start point>,<stop point>]

Description:

- The data is output in order of its wavelength from the shortest level to the longest, irrespective of the wavelength/frequency mode.
- When the level scale is LOG, data is output in LOG values.
- When the level scale is linear, data is output in linear values (mW).
- If the parameter <start point> or <stop point> is omitted, all sampling data of a specified trace will be output. The number of output data can be acquired by executing ":TRACe[:DATA]:SNUMber?".
- Data is output in either ASCII or binary form, depending on the setting of :FORMat[:DATA].

Parameters:

<trace name>	Trace from which data is transferred (TRA TRB TRC TRD TRE TRF TRG)
<start point>	A range of samples to be transferred (starting point) (1 to 50001)
<stop point>	A range of samples to be transferred (ending point) (1 to 50001)

Query Response: For ASCII data: <NRf>,<NRf>,...<NRf>
For binary data: '#<integer><byte num><data byte>

Example: trace:y? tra
→
-1.00000000E+001,-1.00000000E+001,-1.00000000E+001,
....

Command Type: Sequential command

TRACe:DELeTe

Deletes the data of a specified trace.

Syntax:	• :TRACe:DELeTe<wsp><trace name>		
Description:	None		
Parameters:	<trace name>	Trace to be deleted (TRA TRB TRC TRD TRE TRF TRG)	
Query Response:	None		
Example:	trace:del tra		
Command Type:	Sequential command		

TRACe:DELeTe:ALL

Deletes the data of all traces.

Syntax:	• :TRACe:DELeTe:ALL		
Description:	None		
Parameters:	None		
Query Response:	None		
Example:	trace:del:all		
Command Type:	Sequential command		

TRACe:STATe[:<trace name>]

Sets or acquires the display status of a specified trace.

Syntax:	• :TRACe:STATe[:<trace name>]<wsp>OFF ON 0 1 • :TRACe:ACTive?		
Description:	• If <trace name> is omitted, the command is executed with respect to the active trace. • When <trace name> is specified, the specified trace becomes the active trace after executing the command.		
Parameters:	<trace name>Trace (TRA TRB TRC TRD TRE TRF TRG) OFF 0: Makes a trace invisible (BLANK). ON 1: Makes a trace visible (DISP).		
Query Response:	0 1 0:OFF, 1:ON		
Example:	trace:state off trace:state? → 0		
Command Type:	Sequential command		

TRACe:TEMPlate:DATA

Adds data to a specified template or acquires data from it.

- Syntax:**
- :TRACe:TEMPlate:DATA<wsp><template>,<wavelength>[M],<level>[dB]
 - :TRACe:TEMPlate:DATA?<wsp><template>
- Description:**
- Data is added to a specified template.
 - After data has been added, data will be rearranged in the order of the wavelength.
 - If an attempt is made to add data exceeding the maximum number of template data, an execution error occurs.
- Parameters:**
- <template> Template (UPPer | LOWer | TARGet)
- <wavelength> Wavelength of template data to be added [nm]
- <level> Level of template data to be added [dB]
- Query Response:**
- <integer>,<wavelength>,<level>,<wavelength>,<level>, ... ,<level>
- <integer> Number of data points
- <wavelength> Wavelength value [m]
- <level> Level value [dB]
- Example:**
- ```
trace:template:data target,1550nm,-10dbm
trace:template:data? target
→ 3,+1.54000000E-006,-1.00000000E+001,
 +1.54500000E-006,-5.00000000E+000,
 +1.55000000E-006,-1.00000000E+001
```
- Command Type:**    Sequential command

**TRACe:TEMPlate:DATA:ADELete**

Deletes all data of a specified template.

- Syntax:**
- :TRACe:TEMPlate:DATA:ADELete<wsp><template>
- Description:**
- All data of a specified template will be deleted.
- Parameters:**
- <template>    Template (UPPer | LOWer | TARGet)
- Query Response:**   None
- Example:**
- ```
trace:template:data:adel target
```
- Command Type:** Sequential command

TRACe:TEMPlate:DATA:ETYPe

Sets or acquires the extrapolation mode of a specified template.

- Syntax:**
- :TRACe:TEMPlate:DATA:ETYPe<wsp><template>,<type>
- Description:** None
- Parameters:**
- <template> Template (UPPer | LOWer | TARGet)
- <type> Extrapolation type
- A | 1 Extrapolation type A
- B | 2 Extrapolation type B
- NONE | 0 Without extrapolation
- Query Response:** Same as the above
- Example:**
- ```
trace:template:data:etype target,none
trace:template:data:etype? target → NONE
```
- Command Type:**    Sequential command

**TRACe:TEMPlate:DATA:MODE**

Sets the absolute value or relative value mode of a specified template or acquires the set mode of the template.

|                        |                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | • :TRACe:TEMPlate:DATA:MODE<wsp><template>, <mode>                                                                                            |
| <b>Description:</b>    | None                                                                                                                                          |
| <b>Parameters:</b>     | <template> Template (UPPer   LOWer   TARGet)<br><mode> Mode (ABSolute   RELative)<br>ABSolute   0 Absolute mode<br>RELative   1 Relative mode |
| <b>Query Response:</b> | Same as the above                                                                                                                             |
| <b>Example:</b>        | trace:template:data:mode target,relative<br>trace:template:data:mode? target → REL                                                            |
| <b>Command Type:</b>   | Sequential command                                                                                                                            |

**TRACe:TEMPlate:DISPlay**

Sets or acquires ON/OFF of the display of a specified template.

|                        |                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | • :TRACe:TEMPlate:DISPlay<wsp><template>, OFF   ON   0   1<br>• :TRACe:TEMPlate:DISPlay?<wsp><template> |
| <b>Description:</b>    | None                                                                                                    |
| <b>Parameters:</b>     | <template> Template (UPPer   LOWer   TARGet)<br>OFF   0 Display OFF<br>ON   1 Display ON                |
| <b>Query Response:</b> | Same as the above                                                                                       |
| <b>Example:</b>        | trace:template:display target,off<br>trace:template:display? tatget → 0                                 |
| <b>Command Type:</b>   | Sequential command                                                                                      |

**TRACe:TEMPlate:GONOGgo**

Sets or acquires ON/OFF of the go/no-go decision function of the template function.

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| <b>Syntax:</b>         | • :TRACe:TEMPlate:GONOGgo<wsp>OFF   ON   0   1<br>• :TRACe:TEMPlate:GONOGgo?   |
| <b>Description:</b>    | None                                                                           |
| <b>Parameters:</b>     | OFF   0 Go/no-go decision function OFF<br>ON   1 Go/no-go decision function ON |
| <b>Query Response:</b> | Same as the above                                                              |
| <b>Example:</b>        | trace:template:gono off<br>trace:template:gonogo? → 0                          |
| <b>Command Type:</b>   | Sequential command                                                             |

**TRACe:TEMPlate:LEVel:SHIFt**

Sets or acquires the amount of a level shift for a template.

|                        |                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :TRACe:TEMPlate:LEVel:SHIFt&lt;wsp&gt;&lt;NRf&gt;[DB]</li> <li>• :TRACe:TEMPlate:LEVel:SHIFt?</li> </ul> |
| <b>Description:</b>    | None                                                                                                                                              |
| <b>Parameters:</b>     | <NRf>      Amount of a level shift [dB]                                                                                                           |
| <b>Query Response:</b> | Same as the above                                                                                                                                 |
| <b>Example:</b>        | <pre>trace:template:level:shift -1db trace:template:level:shift? → -1.00000000E+000</pre>                                                         |
| <b>Command Type:</b>   | Sequential command                                                                                                                                |

**TRACe:TEMPlate:RESult?**

Acquires the results of go/no-go decision of the template function.

|                        |                                   |
|------------------------|-----------------------------------|
| <b>Syntax:</b>         | • :TRACe:TEMPlate:RESult?         |
| <b>Description:</b>    | None                              |
| <b>Parameters:</b>     | None                              |
| <b>Query Response:</b> | <integer>      1 : Go , 0 : No go |
| <b>Example:</b>        | trace:template:result? → 1        |
| <b>Command Type:</b>   | Sequential command                |

**TRACe:TEMPlate:TTYPe**

Sets or acquires the type of go/no-go decision of the template function.

|                        |                                                                                                                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :TRACe:TEMPlate:TTYPe&lt;wsp&gt;&lt;type&gt;</li> <li>• :TRACe:TEMPlate:TTYPe?</li> </ul>                                                                                      |
| <b>Description:</b>    | None                                                                                                                                                                                                                    |
| <b>Parameters:</b>     | <type>      Decision type<br>UPPER   0:    Makes a decision about the upper line only.<br>LOWER   1:    Makes a decision about the lower line only.<br>U&L   2:      Makes a decision about both upper and lower lines. |
| <b>Query Response:</b> | Same as the above                                                                                                                                                                                                       |
| <b>Example:</b>        | <pre>trace:template:ttype u&amp;l trace:template:ttype? → 2</pre>                                                                                                                                                       |
| <b>Command Type:</b>   | Sequential command                                                                                                                                                                                                      |

**TRACe:TEMPlate:WAVelength:SHIFt**

Sets or acquires the amount of a wavelength shift of a template.

|                        |                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :TRACe:TEMPlate:WAVelength:SHIFt&lt;wsp&gt;&lt;NRf&gt;[M]</li> <li>• :TRACe:TEMPlate:WAVelength:SHIFt?</li> </ul> |
| <b>Description:</b>    | None                                                                                                                                                       |
| <b>Parameters:</b>     | <NRf>      Amount of a wavelength shift [m]                                                                                                                |
| <b>Query Response:</b> | Same as the above                                                                                                                                          |
| <b>Example:</b>        | <pre>trace:template:wav:shift -5nm trace:template:wav:shift? → -5.00000000E-009</pre>                                                                      |
| <b>Command Type:</b>   | Sequential command                                                                                                                                         |

## TRIGger Sub System Command

### TRIGger[:SEQuence]:DELay

Sets or acquires a trigger delay.

|                        |                                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :TRIGger[:SEQuence]:DELay&lt;wsp&gt;&lt;NRf&gt;[S]</li> <li>• :TRIGger[:SEQuence]:DELay?</li> </ul> |
| <b>Description:</b>    | <ul style="list-style-type: none"> <li>• When this command is executed, the external trigger mode is enabled.</li> </ul>                     |
| <b>Parameters:</b>     | (TRIGger[:SEQuence]:STATe ON)<br><NRf> Delay [sec]                                                                                           |
| <b>Query Response:</b> | Same as the above                                                                                                                            |
| <b>Example:</b>        | <pre>trig:delay 100.0us trig:delay? → +1.00000000E-004</pre>                                                                                 |
| <b>Command Type:</b>   | Sequential command                                                                                                                           |

### TRIGger[:SEQuence]:SLOPe

Sets or acquires a trigger edge.

|                        |                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :TRIGger[:SEQuence]:SLOPe&lt;wsp&gt;RISE   FALL   0   1</li> <li>• :TRIGger[:SEQuence]:SLOPe?</li> </ul> |
| <b>Description:</b>    | <ul style="list-style-type: none"> <li>• When this command is executed, the external trigger mode becomes enabled.</li> </ul>                     |
| <b>Parameters:</b>     | (TRIGger[:SEQuence]:STATe ON)<br>RISE   0: RISE<br>FALL   1: FALL                                                                                 |
| <b>Query Response:</b> | 0: RISE<br>1: FALL                                                                                                                                |
| <b>Example:</b>        | <pre>trig:slope rise trig:slope? → 0</pre>                                                                                                        |
| <b>Command Type:</b>   | Sequential command                                                                                                                                |

### TRIGger[:SEQuence]:STATe

Sets or acquires the external trigger mode.

|                        |                                                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Syntax:</b>         | <ul style="list-style-type: none"> <li>• :TRIGger[:SEQuence]:STATe&lt;wsp&gt;OFF   ON   0   1</li> <li>• :TRIGger[:SEQuence]:STATe?</li> </ul> |
| <b>Description:</b>    | None                                                                                                                                           |
| <b>Parameters:</b>     | OFF   ON   0   1                                                                                                                               |
| <b>Query Response:</b> | 0   1                      0:OFF, 1:ON                                                                                                         |
| <b>Example:</b>        | <pre>trig:state on trig:state? → 1</pre>                                                                                                       |
| <b>Command Type:</b>   | Sequential command                                                                                                                             |



## UNIT Sub System Command

2

### UNIT:POWer:DIgit

Sets or acquires the number of decimal places of a level value to be displayed.

**Syntax:**

- :UNIT:POWer:DIgit<wsp>1 | 2 | 3
- :UNIT:POWer:DIgit?

**Description:** None

**Parameters:** 1 | 2 | 3: The number of decimal places to be displayed

**Query Response:** Same as the above

**Example:**

```
unit:pow:digit 3
unit:pow:digit? → 3
```

**Command Type:** Sequential command

### UNIT:X

Sets or acquires the unit of axis X.

**Syntax:**

- :UNIT:X<wsp>WAVelength | FREQuency | 0 | 1
- :UNIT:X?

**Description:** None

**Parameters:** WAVelength | 0: Wavelength  
FREQuency | 1: Frequency

**Query Response:** 0 | 1 0: Wavelength, 1: Frequency

**Example:**

```
unit:x freq
unit:x? → 1
```

**Command Type:** Sequential command

### 2.6.6 Output Format of the CALCulate:DATA? Query

#### (1) Output of the Analytical Results

The analytical results of analysis functions are collectively output using the CALCulate:DATA? command.

Note that if analysis functions have not been executed, a query error occurs.

#### (2) Output Data Format for Each Analysis Function

The output data format of each analysis function is as shown below.

For the information of abbreviations such as <center wl>, see Table 2-1, List of Abbreviations of Data Output using the CALCulate:DATA? Command.

○THRESH , ENVELOPE, PK-RMS

<center wl>,<spec wd>,<mode num>

○RMS

<center wl>,<spec wd>

○NOTCH

<center wl>,<notch wd>

○SMSR

<peak wl>,<peak lvl>,<2nd peak wl>,<2nd peak lvl>,<delta wl>,<delta lvl>

○POWER

<total pow>

○DFB-LD

<spec wd>,<peak wl>,<peak lvl>,<mode ofst>,<smsr>

○FP-LD

<spec wd>,<peak wl>,<peak lvl>,<center wl>,<total pow>,<mode num>

○LED

<spec wd>,<peak wl>,<peak lvl>,<center wl>,<total pow>

○PMD

<left mode peak>,<right mode peak>,<pmd>

## ○WDM

## •ABSOLUTE, CH RELATION=OFFSET

<ch num>,<center wl>,<peak lvl>,<offset wl>, <offset lvl>, <noise>,  
<snr>,...

## •ABSOLUTE, CH RELATION=SPACING

<ch num>,<center wl>,<peak lvl>,<spacing>,<lvl diff>,<noise>,<snr>,...

## •RELATIVE

<ch num>,<grid wl>,<center wl>,<rel wl>,<peak lvl>,<noise>,<snr>,...

## •DRIFT(MEAS)

<ch num>,<grid wl>,<center wl>,<wl diff max>,<wl diff min>,<ref lvl>,  
<peak lvl>, <lvl diff max>, <lvl diff min>,...

## •DRIFT(GRID)

<ch num>,<ref wl>,<center wl>,<wl diff max>,<wl diff min>,<ref lvl>,  
<peak lvl>, <lvl diff max>, <lvl diff min>,...

## ○EDFA-NF

<ch num>,<center wl>,<input lvl>,<output lvl>,<ase lvl>, <resoln>, <gain>,  
<nf>,...

## ○FILTER-PK

<peak wl>,<peak lvl>,<center wl>,<spec wd>,<l-xtalk>,<r-xtalk>, <ripple>

## ○FILTER-BTM

<btm wl>, <btm lvl>, <center wl>, <notch wd>, <l-xtalk>, <r-xtalk>

## ○WDM FIL-PK

<ch num>, <nominal wl>, <peak wl>, <peak lvl>, <xdb wd>, <center wl>,  
<xdb sb>, <xdb pb>, <ripple>, <l-xtalk>, <r-xtalk>,...

※ Items with SW set to OFF are also output.

## ○WDM FIL-BTM

<ch num>, <nominal wl>, <btm wl>, <btm lvl>, <xdb ntwd>, <center wl>,  
<xdb sb>, <xdb eb>, <ripple>, <l-xtalk>, <r-xtalk>,...

※ Items with SW set to OFF are also output.

Table 2-12: List of Abbreviations of Data Output  
Using the CALCulate:DATA? Command

| Abbreviation      | Description                     | Format    | Output Unit |
|-------------------|---------------------------------|-----------|-------------|
| <center wl>       | Center wavelength               | <NRf>     | M / Hz      |
| <spec wd>         | Spectrum width                  | <NRf>     | m / Hz      |
| <mode num>        | Mode number                     | <integer> |             |
| <notch wd>        | Notch width                     | <NRf>     | m / Hz      |
| <peak wl>         | Peak wavelength                 | <NRf>     | m / Hz      |
| <peak lvl>        | Peak level                      | <NRf>     | DBm         |
| <2nd peak wl>     | 2 <sup>nd</sup> peak wavelength | <NRf>     | m / Hz      |
| <2nd peak lvl>    | 2 <sup>nd</sup> peak level      | <NRf>     | DB          |
| <delta wl>        | Wavelength difference           | <NRf>     | m / Hz      |
| <delta lvl>       | Level difference                | <NRf>     | DB          |
| <mode ofst>       | Mode offset                     | <NRf>     | m / Hz      |
| <smsr>            | SMSR value                      | <NRf>     | DB          |
| <total pow>       | Total power value               | <NRf>     | dB / W      |
| <mode num>        | Mode number                     | <integer> |             |
| <left mode peak>  | Mode peak frequency (left)      | <NRf>     | Hz          |
| <right mode peak> | Mode peak frequency (right)     | <NRf>     | Hz          |
| <pmd>             | PMD value                       | <NRf>     | Ps          |
| <ch num>          | Channel number                  | <integer> |             |
| <offset wl>       | Offset wavelength               | <NRf>     | m / Hz      |
| <offset lvl>      | Offset level                    | <NRf>     | DB          |
| <noise>           | Noise level                     | <NRf>     | dBm/NBW     |
| <snr>             | SNR value                       | <NRf>     | dB          |
| <grid wl>         | Grid wavelength                 | <NRf>     | m / Hz      |
| <rel wl>          | Relative wavelength             | <NRf>     | m / Hz      |
| <wl diff max>     | Wavelength difference (max.)    | <NRf>     | m / Hz      |
| <wl diff min>     | Wavelength difference (min.)    | <NRf>     | m / Hz      |
| <ref lvl>         | Relative level                  | <NRf>     | dB          |
| <lvl diff max>    | Level difference (max.)         | <NRf>     | dB          |
| <lvl diff min>    | Level difference (min.)         | <NRf>     | dB          |
| <input lvl>       | Input level                     | <NRf>     | dBm         |
| <output lvl>      | Output level                    | <NRf>     | dBm         |
| <ase lvl>         | ASE level                       | <NRf>     | dBm/RES     |
| <resoln>          | Measurement resolution          | <NRf>     | m           |
| <gain>            | Gain                            | <NRf>     | dB          |
| <nf>              | NF value                        | <NRf>     | dB          |
| <l-xtalk>         | Crosstalk value (left)          | <NRf>     | dB          |
| <r-xtalk>         | Crosstalk value (right)         | <NRf>     | dB          |
| <ripple>          | Ripple width                    | <NRf>     | m / Hz      |
| <nominal wl>      | Reference wavelength            | <NRf>     | m / Hz      |
| <xdb wd>          | Xdb width                       | <NRf>     | m / Hz      |
| <xdb sb>          | XdB stop-band                   | <NRf>     | m / Hz      |
| <xdb pb>          | XdB passband                    | <NRf>     | m / Hz      |
| <xdb eb>          | XdB elimination band            | <NRf>     | m / Hz      |

## 2.7 AQ6317-compatible GPIB Commands

### 2.7.1 How to Switch the Command Mode

Because remote control via the GPIB interface of the AQ6319 complies with the IEEE 488.2 standard, it is not compatible with conventional model AQ6317 (complying with the IEEE 488.1 standard) as to the remote commands and internal actions.

However, using the [SYSTEM] switch and the <GP-IB SETTING> and <COMMAND FORMAT> keys to set the GPIB setting of this unit to "AQ6317-compatible mode" allows some of the AQ6317 commands to be used on the AQ6319. In this case, the operations of the status registers also become compatible with the AQ6317 commands.

Note that switching the command mode causes all the contents of the status registers and queues and receive buffer and talker output buffer to be initialized.

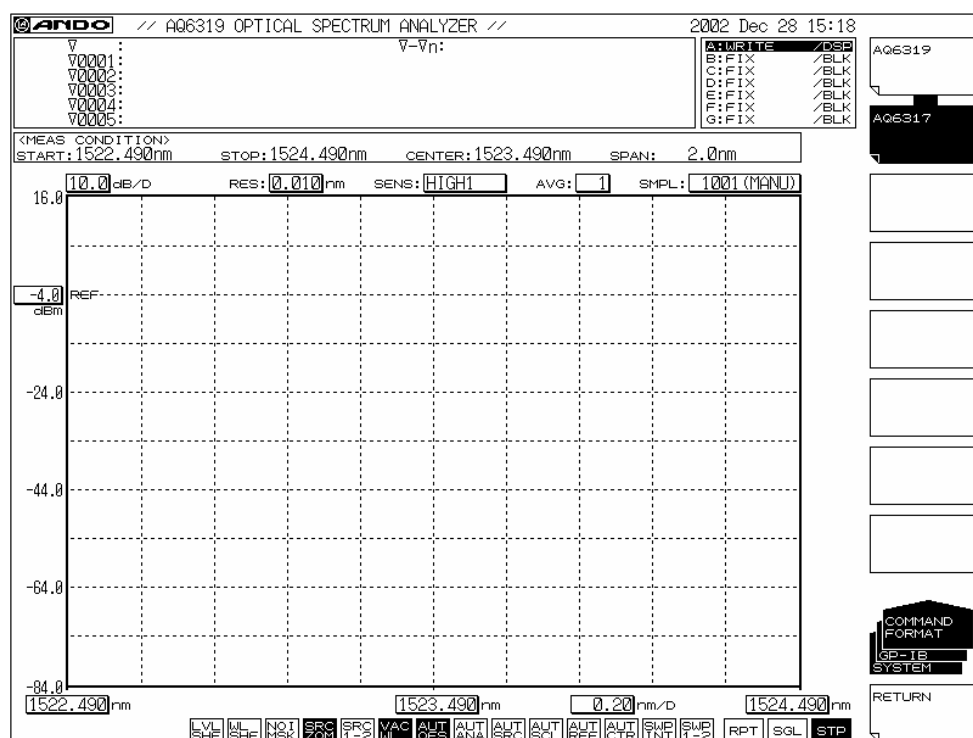


Figure 2-9: Switching the Command Mode

### 2.7.2 Operations in the AQ6317-compatible Mode

The AQ6319 operates as follows when it is remote controlled in the AQ6317-compatible mode.

- For the control commands and talker commands, the majority of AQ6317 commands are available.
- Talker data is output in the AQ6317-compatible format.
- To send multiple commands at one time, use a comma "," as a separator.
- If receiving multiple query commands in a single line, this unit outputs only data relative to the last query command.

### 2.7.3 Switching the Command Mode Using a GPIB Command

The GPIB command mode can also be switched using the following GPIB commands.

- Commands to use when in the AQ6319 mode (invalid in the AQ6317-compatible mode)

Control command:

:SYSTem:COMMunicate:CFORmat<wsp><mode>

<mode> : AQ6317 | AQ6319 | 0 | 1

AQ6317 | 0 Switches to the AQ6317-compatible mode

AQ6319 | 1 Switches to the AQ6319 mode

Query command:

:SYSTem:COMMunicate:CFORmat?

A return value: Same as the above

- Commands to use when in the AQ6317-compatible mode (a command error occurs if used in the AQ6319 mode.)

Control command:

CFORM\* (\*: 0 = AQ6317-compatible mode, 1 = AQ6319 mode)

Query command:

CFORM?

A return value: Same as the above

### 2.7.4 AQ6317 Status Byte

Status Bytes of AQ6317-compatible Mode, operates like Status Byte in AQ6317.

(☛ Table:2-13 Contents of Status Byte)

Refer to the Manual of AQ6317-Series for the details of GP-IB.

Table:2-13 Contents of Status Byte

| Bit   | Function and setting condition                                                                                                                                 | Clear timing                                                                                                                                                                                                         |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bit 7 | 0                                                                                                                                                              |                                                                                                                                                                                                                      |
| Bit 6 | Send an SRQ signal.                                                                                                                                            | <ul style="list-style-type: none"> <li>• Upon execution of serial pulling</li> <li>• Upon receipt of DCL or SDC</li> </ul>                                                                                           |
| Bit 5 | When receiving data exceeding the receive buffer capacity of 512 bytes "1" is set.                                                                             | <ul style="list-style-type: none"> <li>• Upon execution of serial pulling</li> <li>• Upon receipt of DCL or SDC</li> <li>• At a start of measurement</li> </ul>                                                      |
| Bit 4 | 0                                                                                                                                                              |                                                                                                                                                                                                                      |
| Bit 3 | When a command data error occurs, "1" is set.                                                                                                                  | <ul style="list-style-type: none"> <li>• Upon execution of serial pulling</li> <li>• Upon receipt of DCL or SDC</li> <li>• At a start of measurement</li> </ul>                                                      |
| Bit 2 | When a warning error(including program execute error)occurs,"1" is set. For the contents of the warning its number can be output by the talker command "WARN?" | <ul style="list-style-type: none"> <li>• When the warning error display disappears</li> <li>• Upon execution of serial pulling</li> <li>• Upon receipt of DCL or SDC</li> <li>• At a start of measurement</li> </ul> |
| Bit 1 | When the execution of the FD copy program terminates or when a setup operation for optical output is completed, "1" is set.                                    | <ul style="list-style-type: none"> <li>• Upon execution of serial pulling</li> <li>• Upon receipt of DCL or SDC</li> <li>• At a start of measurement</li> </ul>                                                      |
| Bit 0 | When a sweep is completed, "1" is set.                                                                                                                         | <ul style="list-style-type: none"> <li>• Upon execution of serial pulling</li> <li>• Upon receipt of DCL or SDC</li> <li>• At a start of measurement</li> </ul>                                                      |

### 2.7.5 List of the AQ6317-compatible Commands

By setting this unit to "AQ6317-compatible mode," remote control is possible using the GPIB commands provided in the AQ6317 optical spectrum analyzer. However, there are some commands unavailable in this unit or some commands whose parameter ranges are different. For compatibility with the AQ6319, see the following table, AQ6317-compatible Commands.

Table 2-14: AQ6317-compatible Commands

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                                      | Remarks                 |
|-------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------|
| 3D                            | ×                                | -                                                                                                                   |                         |
| 3DRCL                         | ×                                | -                                                                                                                   |                         |
| A+BCL                         | ○                                | :CALCulate:MATH:TRC<wsp>A+B(LIN)                                                                                    |                         |
| A=B                           | ○                                | :TRACe:COPIY<wsp>TRB,TRA                                                                                            |                         |
| A=C                           | ○                                | :TRACe:COPIY<wsp>TRC,TRA                                                                                            |                         |
| A·BC                          | ○                                | :CALCulate:MATH:TRC<wsp>A·B(LOG)                                                                                    |                         |
| A·BCL                         | ○                                | :CALCulate:MATH:TRC<wsp>A·B(LIN)                                                                                    |                         |
| ACTV*                         | ○                                | :TRACe:ACTive<wsp><trace name><br><trace name>= TRA   TRB   TRC                                                     |                         |
| ANA?                          | ○                                | :CALCulate:DATA?                                                                                                    | Different talker format |
| ANGL***                       | ×                                | -                                                                                                                   |                         |
| AREA*                         | ×                                | -                                                                                                                   |                         |
| ARES?                         | ×                                | -                                                                                                                   |                         |
| ARESDSP*                      | ×                                | -                                                                                                                   |                         |
| ATANA*                        | ○                                | :CALCulate[:IMMediate]:AUTO<wsp>OFF ON 0 1                                                                          |                         |
| ATCTR*                        | ○                                | :CALCulate:MARKer:MAXimum:SCENter:AUTO<wsp><br>> OFF ON 0 1                                                         |                         |
| ATOFS*                        | ○                                | :CALibration:ZERO[:AUTO]<wsp>OFF ON 0 1                                                                             |                         |
| ATREF*                        | ○                                | :CALCulate:MARKer:MAXimum:SRLevel:AUTO                                                                              |                         |
| ATSCL*                        | ○                                | :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:AUTO<wsp><br>> OFF ON 0 1                                                        |                         |
| ATSR*                         | ○                                | :CALCulate:MARKer:AUTO<wsp>OFF ON 0 1                                                                               |                         |
| AUTO                          | ○                                | :INITiate:SMODE<wsp>AUTO 3;INITiate                                                                                 |                         |
| AVG****                       | △                                | :SENSe:AVERage:COUNT<wsp><integer>                                                                                  | Parameter range         |
| B=A                           | ○                                | :TRACe:COPIY<wsp>TRA,TRB                                                                                            |                         |
| B=C                           | ○                                | :TRACe:COPIY<wsp>TRC,TRB                                                                                            |                         |
| B·AC                          | ○                                | :CALCulate:MATH:TRC<wsp>B·A(LOG)                                                                                    |                         |
| B·ACL                         | ○                                | :CALCulate:MATH:TRC<wsp>B·A(LIN)                                                                                    |                         |
| BASL***.*                     | ○                                | :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing<wsp>LINear 1;<br>:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:BLEVel<wsp><NRF>[MW] |                         |
| BD*                           | ○                                | -                                                                                                                   |                         |



| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                        | Remarks                 |
|-------------------------------|----------------------------------|---------------------------------------------------------------------------------------|-------------------------|
| BLKA                          | ○                                | :TRACe:STATe:TRA<wsp>OFF 0                                                            |                         |
| BLKB                          | ○                                | :TRACe:STATe:TRB<wsp>OFF 0                                                            |                         |
| BLKC                          | ○                                | :TRACe:STATe:TRC<wsp>OFF 0                                                            |                         |
| BTSR                          | ○                                | :CALCulate:MARKer:MINimum                                                             |                         |
| BZCLK*                        | ○                                | :SYSTem:BUZZer:CLICk<wsp>OFF ON 0 1                                                   |                         |
| BZWRN*                        | ○                                | :SYSTem:BUZZer:WARNing<wsp>OFF ON 0 1                                                 |                         |
| C=A                           | ○                                | :TRACe:COPIY<wsp>TRA,TRC                                                              |                         |
| C=B                           | ○                                | :TRACe:COPIY<wsp>TRC,TRB                                                              |                         |
| CLMES                         | ×                                | -                                                                                     |                         |
| CLR                           | ○                                | :TRACe:DELeTe<wsp>TRA;<br>:TRACe:DELeTe<wsp>TRB;<br>:TRACe:DELeTe<wsp>TRC             |                         |
| CNDDT*                        | ○                                | :MMEMory:STORe:DATA:ITEM<wsp>CONDition,OFF ON 0 1                                     |                         |
| COPY*                         | ○                                | :HCOPY[:IMMediate]                                                                    |                         |
| CRS*                          | ○                                | -                                                                                     |                         |
| CTR=M                         | ○                                | :CALCulate:MARKer:SCENter                                                             |                         |
| CTR=P                         | ○                                | :CALCulate:MARKer:MAXimum:SCENter                                                     |                         |
| CTRF***,***                   | △                                | :SENSe:WAVelength:CENTer<wsp><NRf>[HZ]                                                | Parameter range         |
| CTRWL****,**                  | △                                | :SENSe:WAVelength:CENTer<wsp><NRf>[M]                                                 | Parameter range         |
| CVFTC**                       | ×                                | -                                                                                     | Shifted to trace G      |
| CVPKC**                       | ×                                | -                                                                                     | Shifted to trace G      |
| CWPLS?                        | ×                                | -                                                                                     |                         |
| D&TDT*                        | ○                                | :MMEMory:STORe:DATA:ITEM<wsp>DATE,OFF ON 0 1                                          |                         |
| DATE?                         | ○                                | :SYSTem:DATE?                                                                         | Different talker format |
| DATE YR.MO.DY<br>TIME HH:MM   | ○                                | :SYSTem:DATE<wsp><year>,<month>,<day>;<br>:SYSTem:TIME<wsp><hour>,<minutes>,<seconds> |                         |
| DEFCL*                        | △                                | :DISPlay:COLor<wsp><mode><br><mode>= 0:B&W, 1 to 5:mode 1 to 5                        | Display color           |
| DEL'@@@@.***'                 | ○                                | :MMEMory:DELeTe<wsp><"file name">,FLOPPy                                              |                         |
| DFBAN                         | ○                                | :CALCulate:CATegory<wsp>DFBLd 4                                                       |                         |
| DFBLD○;□;△;****               | ×                                | -                                                                                     |                         |
| DIR?                          | ×                                | -                                                                                     |                         |
| DISP?                         | ○                                | -                                                                                     |                         |
| DSPA                          | ○                                | :TRACe:STATe:TRA<wsp>ON 1                                                             |                         |
| DSPB                          | ○                                | :TRACe:STATe:TRB<wsp>ON 1                                                             |                         |
| DSPA?                         | ○                                | :TRACe:STATe:TRA?                                                                     |                         |
| DSPB?                         | ○                                | :TRACe:STATe:TRB?                                                                     |                         |
| DSPC                          | ○                                | :TRACe:STATe:TRC<wsp>ON 1                                                             |                         |
| DSPC?                         | ○                                | :TRACe:STATe:TRC?                                                                     |                         |
| DTAD*                         | ○                                | :MMEMory:STORe:DATA:MODE<wsp>ADD OVER 0 1                                             |                         |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command       | Remarks         |
|-------------------------------|----------------------------------|----------------------------------------------------------------------|-----------------|
| DTARA*                        | ○                                | :MMEMory:STORe:DATA:ITEM<wsp>DATA,OFF ON 0 1                         |                 |
| DUTCH***;####.##              | ×                                | -                                                                    |                 |
| DUTCHF***;###.###             | ×                                | -                                                                    |                 |
| DUTLEV**.*                    | ×                                | -                                                                    |                 |
| DUTSNR**.*                    | ×                                | -                                                                    |                 |
| EDFCVF*                       | ×                                | -                                                                    |                 |
| EDFTH**.*                     | ×                                | -                                                                    |                 |
| EDNF                          | ×                                | -                                                                    |                 |
| ENVK**.*                      | ○                                | :CALCulate:PARAmeter[:CATegory]:SWENvelope:K<wsp><NRf>               |                 |
| ENV1**.*                      | ○                                | :CALCulate:PARAmeter[:CATegory]:SWENvelope:TH1<wsp><NRf>[DB]         |                 |
| ENV2**.*                      | ○                                | :CALCulate:PARAmeter[:CATegory]:SWENvelope:TH2<wsp><NRf>[DB]         |                 |
| EXEC**                        | ○                                | :PROGram:EXECute<wsp><integer>                                       |                 |
| EXTRG                         | ○                                | :TRIGger[:SEQuence]:STATe<wsp>OFF ON 0 1                             |                 |
| FIG*                          | ○                                | :UNIT:POWer:DIGit<wsp>1 2 3                                          |                 |
| FILBTM○;□;△;***               | ○                                | :CALCulate:PARAmeter[:CATegory]:FILBtm<wsp><item>,<paramater>,<data> |                 |
| FILBTMAN                      | ○                                | :CALCulate:CATegory<wsp>FILBtm   14                                  |                 |
| FILPK○;□;△;***                | ○                                | :CALCulate:PARAmeter[:CATegory]:FILPk<wsp><item>,<paramater>,<data>  |                 |
| FILPKAN                       | ○                                | :CALCulate:CATegory<wsp>FILPk   13                                   |                 |
| FIXA                          | ○                                | :TRACe:ATTRibute:TRA<wsp>FIX   1                                     |                 |
| FIXB                          | ○                                | :TRACe:ATTRibute:TRB<wsp>FIX   1                                     |                 |
| FIXC                          | ○                                | :TRACe:ATTRibute:TRC<wsp>FIX   1                                     |                 |
| FMKR***,****                  | △                                | :CALCulate:MARKer:X<wsp>0,<NRf>[HZ]                                  | Parameter range |
| FPAN                          | ○                                | :CALCulate:CATegory<wsp>FPLD   5                                     |                 |
| FPLD;○;□;△;****               | ○                                | :CALCulate:PARAmeter[:CATegory]:FPLD<wsp><item>,<paramenter>,<data>  |                 |
| GP2ADR**                      | ○                                | :SYSTem:COMMunication:GPIB2:ADDReSS<wsp><integer>                    |                 |
| GRCOL*                        | ○                                | -                                                                    |                 |
| GRFMT*                        | ○                                | -                                                                    |                 |
| HD*                           | ○                                | -                                                                    |                 |
| HELP*                         | ×                                | -                                                                    |                 |
| *IDN?                         | ○                                | *IDN?                                                                |                 |
| INIT                          | ○                                | :SYSTem:PRESet                                                       |                 |
| KABC                          | ○                                | :CALCulate:MATH:TRC<wsp>1-K(A/B)                                     |                 |
| KABCK*****.****               | ○                                | :CALCulate:MATH:TRC:K<wsp><NRf>                                      |                 |
| KBAC                          | ○                                | :CALCulate:MATH:TRC<wsp>1-K(B/A)                                     |                 |
| KYDNE                         | ×                                | -                                                                    |                 |
| L1FMK***,****                 | △                                | :CALCulate:LMarker:X<wsp>1,<NRf>[HZ]                                 | Parameter range |

| AQ6317 Series Control Command                                                                                                                                                                                                                   | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                                                                                                        | Remarks                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| L1MK****.*                                                                                                                                                                                                                                      | △                                | :CALCulate:LMARker:X<wsp>1,<NRf>[M]                                                                                                                                                   | Parameter range                           |
| L1MK?                                                                                                                                                                                                                                           | △                                | :CALCulate:LMARker:X?<wsp>1                                                                                                                                                           | Parameter range                           |
| L2FMK****.****                                                                                                                                                                                                                                  | △                                | :CALCulate:LMARker:X<wsp>2,<NRf>[HZ]                                                                                                                                                  | Parameter range                           |
| L2MK****.*                                                                                                                                                                                                                                      | △                                | :CALCulate:LMARker:X<wsp>2,<NRf>[M]                                                                                                                                                   | Parameter range                           |
| L2MK?                                                                                                                                                                                                                                           | △                                | :CALCulate:LMARker:X?<wsp>2                                                                                                                                                           | Parameter range                           |
| L3DB****.**                                                                                                                                                                                                                                     | △                                | :CALCulate:LMARker:Y<wsp>3,<NRf>[DB]                                                                                                                                                  | Parameter range                           |
| L3DBM****.**                                                                                                                                                                                                                                    | △                                | :CALCulate:LMARker:Y<wsp>3,<NRf>[DBM]                                                                                                                                                 | Parameter range                           |
| L3LN*.***E±**                                                                                                                                                                                                                                   | △                                | :CALCulate:LMARker:Y<wsp>3,<NRf>                                                                                                                                                      | Parameter range                           |
| L3MK?                                                                                                                                                                                                                                           | △                                | :CALCulate:LMARker:Y?<wsp>3                                                                                                                                                           | Parameter range                           |
| L4DB****.**                                                                                                                                                                                                                                     | △                                | :CALCulate:LMARker:Y<wsp>4,<NRf>[DB]                                                                                                                                                  | Parameter range                           |
| L4DBM****.**                                                                                                                                                                                                                                    | △                                | :CALCulate:LMARker:Y<wsp>4,<NRf>[DBM]                                                                                                                                                 | Parameter range                           |
| L4LN*.***E±**                                                                                                                                                                                                                                   | △                                | :CALCulate:LMARker:Y<wsp>4,<NRf>                                                                                                                                                      | Parameter range                           |
| L4MK?                                                                                                                                                                                                                                           | △                                | :CALCulate:LMARker:Y?<wsp>4                                                                                                                                                           | Parameter range                           |
| LBL '*****'                                                                                                                                                                                                                                     | △                                | :DISPlay[:WINDow]:TEXT:DATA<wsp><string>                                                                                                                                              | Number of characters                      |
| LBLCL                                                                                                                                                                                                                                           | ○                                | :DISPlay[:WINDow]:TEXT:CLEAr                                                                                                                                                          |                                           |
| LBLDT*                                                                                                                                                                                                                                          | ○                                | :MMEMory:STORe:DATA:ITEM<wsp>LABel,OFF   ON   0   1                                                                                                                                   |                                           |
| LCALT****.#.###                                                                                                                                                                                                                                 | △                                | :CALibration:POWer:OFFSet:TABLE<wsp><integer> <NRf> [DB]                                                                                                                              | Parameter range                           |
| LDATA<br>R****-R****<br>LTATB<br>R****-R****<br>LDATC<br>R****-R****<br>LMEM\$\$<br>R****-R****<br>WDATA<br>R****-R****<br>WDATB<br>R****-R****<br>WDATC<br>R****-R****<br>WMEM\$\$<br>R****-R****<br>DTNUM A<br>DTNUM B<br>DTNUM C<br>DTNUM ** | ○                                | :TRACe[:DATA]:X?<wsp><trace name> [,<start point>,<stop point>]<br><br>:TRACe[:DATA]:Y?<wsp><trace name> [,<start point>,<stop point>]<br><br>:TRACe[:DATA]:SNUMber?<wsp><trace name> | (LMEM\$\$ and WMEM\$\$ are not supported) |
| LDTDIG*                                                                                                                                                                                                                                         | ○                                | UNIT:POWer:DIGit<wsp>1   2   3                                                                                                                                                        |                                           |
| LED○;□;△;****                                                                                                                                                                                                                                   | ○                                | :CALCulate:PARAmeter[:CATegory]:LED<wsp><item>,<paramater>,<data>                                                                                                                     |                                           |
| LEDAN                                                                                                                                                                                                                                           | ○                                | :CALCulate:CATegory<wsp>LED   6                                                                                                                                                       |                                           |
| LHLD*                                                                                                                                                                                                                                           | ○                                | :DISPlay[:WINDow]:SPLit<wsp>ON   1;<br>:DISPlay[:WINDow]:SPLit:HOLD:LOWer<wsp>OFF   ON   0   1                                                                                        |                                           |
| LMKCL                                                                                                                                                                                                                                           | ○                                | :CALCulate:LMARker:AOFF                                                                                                                                                               |                                           |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                                                     | Remarks |
|-------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------|
| LNGT**.*                      | ○                                | :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:LENGth<wsp><NRf>[KM]                                                                            |         |
| LOFSKM***.*                   | ○                                | :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:OLEVel<wsp><NRf>[DB/KM]                                                                         |         |
| LOFST***.*                    | ○                                | :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:OLEVel<wsp><NRf>[DB]                                                                            |         |
| LOGLMT***                     | ×                                | -                                                                                                                                  |         |
| LPF                           | ×                                | -                                                                                                                                  |         |
| LSCL**.*                      | ○                                | :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing<wsp>LOGarighmic   0;<br>:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:PDIVision<wsp><integer>[DIV] |         |
| LSUNT*                        | ○                                | :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:UNIT<wsp>DBM   DBM/NM                                                                           |         |
| LTABS                         | ×                                | -                                                                                                                                  |         |
| LTALM?                        | ×                                | -                                                                                                                                  |         |
| LTALMDT?                      | ×                                | -                                                                                                                                  |         |
| LTATSCL*                      | ×                                | -                                                                                                                                  |         |
| LTATSET                       | ×                                | -                                                                                                                                  |         |
| LTCH***                       | ×                                | -                                                                                                                                  |         |
| LTCHCUR***                    | ×                                | -                                                                                                                                  |         |
| LTINTVL****.*                 | ×                                | -                                                                                                                                  |         |
| LTL                           | ×                                | -                                                                                                                                  |         |
| LTLHI***.**                   | ×                                | -                                                                                                                                  |         |
| LTLLOW***.**                  | ×                                | -                                                                                                                                  |         |
| LTLVLCTR***.**                | ×                                | -                                                                                                                                  |         |
| LTLVLSCL**.*                  | ×                                | -                                                                                                                                  |         |
| LTREFINI                      | ×                                | -                                                                                                                                  |         |
| LTREFSET                      | ×                                | -                                                                                                                                  |         |
| LTREL                         | ×                                | -                                                                                                                                  |         |
| LTSNR                         | ×                                | -                                                                                                                                  |         |
| LTSNRCTR***.**                | ×                                | -                                                                                                                                  |         |
| LTSNRLIM***.**                | ×                                | -                                                                                                                                  |         |
| LTSNRSCL**.*                  | ×                                | -                                                                                                                                  |         |
| LTSWP                         | ×                                | -                                                                                                                                  |         |
| LTIME****                     | ×                                | -                                                                                                                                  |         |
| LTMCUR****                    | ×                                | -                                                                                                                                  |         |
| LTWL                          | ×                                | -                                                                                                                                  |         |
| LTWLCTR****.**                | ×                                | -                                                                                                                                  |         |
| LTWLLIM***.**                 | ×                                | -                                                                                                                                  |         |
| LTWLSPN****.*                 | ×                                | -                                                                                                                                  |         |
| LVSFT***.**                   | ○                                | :SENSe:CORRection:LEVel:SHIFt<wsp><NRf>[DB]                                                                                        |         |
| MAXA                          | ○                                | :TRACe:ATTRibute:TRA<wsp>MAX   2                                                                                                   |         |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command     | Remarks            |
|-------------------------------|----------------------------------|--------------------------------------------------------------------|--------------------|
| MCLR***                       | △                                | :CALCulate:MARKer[:STATe]<wsp><marker>,OFF 0                       | Parameter range    |
| MEM*                          | ×                                | -                                                                  |                    |
| MESWL*                        | ○                                | :SENSe:CORRection:RVELOCITY:MEDium<wsp>AIR VACuum 0 1              |                    |
| MIMSK**.*                     | ×                                | -                                                                  |                    |
| MINB                          | ○                                | :TRACe:ATTRibute:TRB<wsp>MIN 3                                     |                    |
| MKCL                          | ○                                | :CALCulate:MARKer:AOff                                             |                    |
| MKR***                        | △                                | :CALCulate:MARKer[:STATe]<wsp><marker>, ON 1                       | Parameter range    |
| MKR?                          | ○                                | :CALCulate:MARKer:X?<wsp>0                                         |                    |
| MKR?****                      | ×                                | :CALCulate:MARKer:X?<wsp><marker>                                  |                    |
| MKR1                          | ○                                | :CALCulate:MARKer[:STATe]<wsp>1, ON 1                              |                    |
| MKR1?                         | ○                                | :CALCulate:MARKer:X?<wsp>1                                         |                    |
| MKR2                          | ○                                | :CALCulate:MARKer[:STATe]<wsp>2, ON 1                              |                    |
| MKR2?                         | ○                                | :CALCulate:MARKer:X?<wsp>2                                         |                    |
| MKROS*                        | ○                                | :CALCulate:MARKer:FUNCTion:FORMat<wsp>OFFSet SPACing 0 1           |                    |
| MKRPRt                        | ○                                | :HCOPY[:IMMediate]:FUNCTion:MARKer:LIST                            |                    |
| MKRUP*                        | ○                                | :CALCulate:MARKer:FUNCTion:UPDate<wsp>OFF ON 0 1                   |                    |
| MKUNT*                        | ○                                | :CALCulate:MARKer:UNIT<wsp>WAVElength FREQuency 0 1                |                    |
| MLTMKR*                       | ×                                | -                                                                  |                    |
| MODFT*                        | ○                                | :CALCulate:PARAmeter[:CATegory]:SWTHresh:MFI T<wsp>OFF ON 0 1      |                    |
| MODIF**.*                     | ○                                | :CALCulate:PARAmeter:COMMon:MDIFF<wsp><NRf>[DB]                    |                    |
| MSKL*                         | ○                                | :DISPlay[:WINDow]:TRACe:YNMAStYPE<wsp>VERTical HORIZontal 0 1      |                    |
| NCHMOD*                       | ○                                | :CALCulate:PARAmeter[:CATegory]:NOTCh:TYPE<wsp>PEAK BOTTom 0 1     |                    |
| NCHTH**.*                     | ○                                | :CALCulate:PARAmeter[:CATegory]:NOTCh:TH<wsp><NRf>[DB]             |                    |
| NMSK****                      | △                                | :DISPlay[:WINDow]:TRACe:YNMASt<wsp><NRf>[DB]                       | Parameter range    |
| NORMC                         | ×                                | -                                                                  | Shifted to trace G |
| NORMD                         | ○                                | :DISPlay[:WINDow]:SPLit<wsp>OFF 0                                  |                    |
| NSR                           | ○                                | :CALCulate:MARKer:MAXimum:NEXT or :CALCulate:MARKer:MINimum:NEXT   |                    |
| NSRL                          | ○                                | :CALCulate:MARKer:MAXimum:LEFT or :CALCulate:MARKer:MINimum:LEFT   |                    |
| NSRR                          | ○                                | :CALCulate:MARKer:MAXimum:RIGHT or :CALCulate:MARKer:MINimum:RIGHT |                    |
| OFIN**.*                      | ×                                | -                                                                  |                    |
| OFOU**.*                      | ×                                | -                                                                  |                    |
| OPALIGN                       | ○                                | :CALibration:ALIGN[:IMMediate]                                     |                    |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                   | Remarks                     |
|-------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------|
| PKHLD****                     | ×                                | -                                                                                                |                             |
| PKSR                          | ○                                | :CALCulate:MARKer:MAXimum                                                                        |                             |
| PKSR?                         | ×                                | -                                                                                                |                             |
| PLMES                         | ×                                | -                                                                                                |                             |
| PLMOD?                        | ×                                | -                                                                                                |                             |
| PLMSK**. **                   | ×                                | -                                                                                                |                             |
| PMD                           | ○                                | :CALCulate:CATegory<wsp>PMD   9                                                                  |                             |
| PMDTH**. **                   | ○                                | :CALCulate:PARAmeter[:CATegory]:PMD:TH<wsp><NRf>[DB]                                             |                             |
| PMRPT                         | ×                                | -                                                                                                |                             |
| PMRST                         | ×                                | -                                                                                                |                             |
| PMSGL                         | ×                                | -                                                                                                |                             |
| PMSTP                         | ×                                | -                                                                                                |                             |
| PMST?                         | ×                                | -                                                                                                |                             |
| PMUNT*                        | ×                                | -                                                                                                |                             |
| POFS**. **                    | ○                                | :CALCulate:PARAmeter[:CATegory]:POWER:OFFSet<wsp><NRf>[DB]                                       |                             |
| PRDEL**                       | ○                                | -                                                                                                |                             |
| PREXT                         | ○                                | -                                                                                                |                             |
| PRFED**                       | △                                | :HCOPY[:IMMediate]:FEED                                                                          | Amount of feed              |
| PRMK**. **                    | ○                                | :CALCulate:PARAmeter[:CATegory]:SWPKrms:K<wsp><NRf>                                              |                             |
| PRMTH**. *                    | ○                                | :CALCulate:PARAmeter[:CATegory]:SWPKrms:TH<wsp><NRf>[DB]                                         |                             |
| PWR                           | ○                                | :CALCulate:CATegory<wsp>POWER   8                                                                |                             |
| RAVA***                       | ○                                | :TRACe:ATTRibute:RAVG[:TRA]<wsp><integer>                                                        |                             |
| RAVB***                       | ○                                | :TRACe:ATTRibute:RAVG:TRB<wsp><integer>                                                          |                             |
| RCLA**                        | △                                | :MEMory:LOAD<wsp><integer>,TRA                                                                   | Parameter range             |
| RCLB**                        | △                                | :MEMory:LOAD<wsp><integer>,TRB                                                                   | Parameter range             |
| RCLC**                        | △                                | :MEMory:LOAD<wsp><integer>,TRC                                                                   | Parameter range             |
| RD* '@@@@'                    | ○                                | :MMEMory:LOAD:TRACe<wsp><trace name>,<"file name">,FLOppy<br><trace name>= TRA   TRB   TRC       |                             |
| RD3D* '@@@@'                  | ×                                | -                                                                                                |                             |
| RDDT '@@@@'                   | ○                                | :MMEMory:LOAD:DATA<wsp><"file name">,FLOppy                                                      |                             |
| RDLT '@@@@'                   | ×                                | -                                                                                                |                             |
| RDMEM** '@@@@'                | ○                                | :MMEMory:LOAD:MEMory<wsp><integer>,<"file name">,FLOppy                                          | Increase in memory numbers  |
| RDPRG** '@@@@'                | ○                                | :MMEMory:LOAD:PROGram<wsp><program number>,<"file name">,FLOppy                                  | Increase in program numbers |
| RDSET '@@@@'                  | ○                                | :MMEMory:LOAD:SETTing<wsp><"file name">,FLOppy                                                   |                             |
| RDTMP '@@@@'                  | ○                                | :MMEMory:LOAD:TEMPlate<wsp><template>,<"file name">,FLOppy<br><template>= UPPER   LOWER   TARGet | Line type can be selected   |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                                                      | Remarks                 |
|-------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| REF=M                         | ○                                | :CALCulate:MARKer:SRLevel                                                                                                           |                         |
| REF=P                         | ○                                | :CALCulate:MARKer:MAXimum:SRLevel                                                                                                   |                         |
| REFL***.*                     | △                                | :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing<wsp><br>LOGarithmic   0;<br>:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RLEVel<wsp><br><NRf>[DBM] | Parameter range         |
| REFLM*.*                      | ○                                | :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing<wsp><br>LINear   1;<br>:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RLEVel<wsp><br><NRf>[MW]       |                         |
| REFLN*.*                      | ○                                | :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing<wsp><br>LINear   1;<br>:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RLEVel<wsp><br><NRf>[NW]       |                         |
| REFLP*.*                      | ×                                | -                                                                                                                                   |                         |
| REFLU*.*                      | ○                                | :DISPlay[:WINDow]:TRACe:Y1[:SCALe]:SPACing<wsp><br>LINear   1;<br>:DISPlay[:WINDow]:TRACe:Y1[:SCALe]:RLEVel<wsp><br><NRf>[UW]       |                         |
| REFL?                         | △                                | :DISPlay[:WINDow]:Y1[:SCALe]:RLEVel?                                                                                                | Parameter range         |
| REL*                          | ×                                | -                                                                                                                                   |                         |
| RESCOR*                       | ×                                | -                                                                                                                                   |                         |
| RESLN*.*                      | △                                | :SENSe:BANDwidth   :BWIDTH[:RESolution]<wsp><br><NRf>[M]                                                                            | Parameter range         |
| RESLNF***                     | ×                                | -                                                                                                                                   |                         |
| RMSK*.*                       | ○                                | :CALCulate:PARAmeter[:CATegory]:RMS:K<wsp><br><NRf>                                                                                 |                         |
| RMSTH*.*                      | ○                                | :CALCulate:PARAmeter[:CATegory]:RMS:TH<wsp><br><NRf>[DB]                                                                            |                         |
| RPT                           | ○                                | :INITiate:SMODE<wsp>REPeat   2;INITiate                                                                                             |                         |
| *RST                          | △                                | *RST                                                                                                                                | Different action        |
| SAVEA**                       | △                                | :MEMory:STORe<wsp><integer>,TRA                                                                                                     | Parameter range         |
| SAVEB**                       | △                                | :MEMory:STORe<wsp><integer>,TRB                                                                                                     | Parameter range         |
| SAVEC**                       | △                                | :MEMory:STORe<wsp><integer>,TRC                                                                                                     | Parameter range         |
| SENS?                         | ○                                | :SENSe:SENSe?                                                                                                                       | Different talker format |
| SD*                           | ○                                | -                                                                                                                                   |                         |
| SEGP****                      | △                                | :SENSe:SWEep:SEGMENT:POINTs<wsp><integer>                                                                                           | Parameter range         |
| SGL                           | ○                                | :INITiate:SMODE<wsp>SINGle   1;INITiate                                                                                             |                         |
| SHI1                          | △                                | :SENSe:SENSe<wsp>HIGH1   3;<br>:SENSe:CHOPer<wsp>OFF   0                                                                            | Chopper not used        |
| SHI2                          | △                                | :SENSe:SENSe<wsp>HIGH2   4;<br>:SENSe:CHOPer<wsp>OFF   0                                                                            | Chopper not used        |
| SHI3                          | △                                | :SENSe:SENSe<wsp>HIGH3   5;<br>:SENSe:CHOPer<wsp>OFF   0                                                                            | Chopper not used        |
| SKM*.*                        | ○                                | DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT<wsp><br>DB/KM   2                                                                            |                         |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                                         | Remarks         |
|-------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------|
|                               |                                  | DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<wsp><br>><br><NRf>[DB/KM]                                                  |                 |
| SLIN*.***                     | ○                                | DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT<wsp><br>LINear   1<br>DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<br><wsp><NRf> |                 |
| SLOG**.*                      | ○                                | DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT<wsp><br>DB   0<br>DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<br><wsp><NRf>[DB] |                 |
| SMEAS                         | ○                                | :INITiate:SMODE<wsp>SEGment   4                                                                                        |                 |
| SMID                          | ○                                | :SENSe:SENSe<wsp>MID   2                                                                                               |                 |
| SMIN***.*                     | ○                                | :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:SMINimum<br><wsp><NRf>                                                              |                 |
| SMINP***.*                    | ○                                | :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:SMINimum<br><wsp><NRf>[%]                                                           |                 |
| SMPL****                      | △                                | :SENSe:SWEep:POINts<wsp><integer>                                                                                      | Parameter range |
| SMSR*                         | ○                                | :CALCulate:PARAmeter[:CATegory]:SMSR:MODE<wsp><br>SMSR1   SMSR2                                                        |                 |
| SNAT                          | ○                                | :SENSe:SENSe<wsp>NAUT   1                                                                                              |                 |
| SNHD                          | ○                                | :SENSe:SENSe<wsp>NHLD   0                                                                                              |                 |
| SP=LM                         | ○                                | :CALCulate:LMARker:SSPan                                                                                               |                 |
| SPAN****.*                    | △                                | :SENSe:WAVelength:SPAN<wsp><NRf>[M]                                                                                    | Parameter range |
| SPANF***.***                  | ○                                | -                                                                                                                      |                 |
| SPLIT                         | ○                                | :DISPlay[:WINDow]:SPLit<wsp>ON   1                                                                                     |                 |
| SPN=W                         | ○                                | -                                                                                                                      |                 |
| SPS***.*                      | ○                                | DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT<wsp><br>%   3<br>DISPlay[:WINDow]:TRACe:Y2[:SCALe]:PDIVision<br><wsp><NRf>[%]   |                 |
| SRLMK*                        | ○                                | :CALCulate:LMARker:SRANge<wsp>OFF   ON   0   1                                                                         |                 |
| SRMSK***                      | ○                                | -                                                                                                                      |                 |
| SRQ*                          | ○                                | *SRE<wsp><integer>                                                                                                     |                 |
| SSE*                          | ×                                | -                                                                                                                      |                 |
| SSMSK**. **                   | ○                                | :CALCulate:PARAmeter[:CATegory]:SMSR:MASK<wsp><br><NRf>[M]                                                             |                 |
| SSUNT?                        | ○                                | :DISPlay[:WINDow]:TRACe:Y2[:SCALe]:UNIT?                                                                               |                 |
| STAF***.***                   | △                                | :SENSe:WAVelength:STARt<wsp><NRf>[HZ]                                                                                  | Parameter range |
| STAWL****. **                 | △                                | :SENSe:WAVelength:STARt<wsp><NRf>[M]                                                                                   | Parameter range |
| STP                           | ○                                | :ABORt                                                                                                                 |                 |
| STPF***.***                   | △                                | :SENSe:WAVelength:STOP<wsp><NRf>[HZ]                                                                                   | Parameter range |
| STPWL****. **                 | △                                | :SENSe:WAVelength:STOP<wsp><NRf>[M]                                                                                    | Parameter range |
| SW*                           | ○                                | :CALCulate:CATegory<wsp>SWThresh   0                                                                                   |                 |
| SWDSP*                        | ×                                | -                                                                                                                      |                 |
| SWENV**. **                   | ○                                | :CALCulate:PARAmeter[:CATegory]:SWENvelope:TH1<br><wsp><NRf>[DB]                                                       |                 |
| SWEEP?                        | ○                                | -                                                                                                                      |                 |



| AQ6317 Series Control Command  | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                           | Remarks                     |
|--------------------------------|----------------------------------|------------------------------------------------------------------------------------------|-----------------------------|
| SWPI*****                      | ○                                | :SENSe:SWEep:TIME:INterval<wsp><integer>[SE C]                                           |                             |
| SWPM*                          | ○                                | :SENSe:WAVelength:SRANge<wsp>OFF ON 0 1                                                  |                             |
| SWPRM**.**                     | ○                                | :CALCulate:PARAmeter[:CATegory]:SWPKrms:TH<wsp><NRf>[DB]                                 |                             |
| SWRMS**.**                     | ○                                | :CALCulate:PARAmeter[:CATegory]:RMS:TH<wsp><NRf>[DB]                                     |                             |
| SWTHR**.**                     | ○                                | :CALCulate:PARAmeter[:CATegory]:SWThresh:TH<wsp><NRf>[DB]                                |                             |
| THRK**.**                      | ○                                | :CALCulate:PARAmeter[:CATegory]:SWThresh:K<wsp><NRf>                                     |                             |
| THRTH**.**                     | ○                                | :CALCulate:PARAmeter[:CATegory]:SWThresh:TH<wsp><NRf>[DB]                                |                             |
| TIME?                          | ○                                | -                                                                                        |                             |
| TLDAT*****;**<br>***.**;***.** | ×                                | :TRACe:TEMPlate:DATA<wsp><template>,<wavelength>[M],<level>[DB]                          |                             |
| TLDATCLR                       | ○                                | :TRACe:TEMPlate:DATA:ADElete<wsp><template><template>= UPPER   LOWEr   TARGet            | Line type can be specified. |
| TLDISP*                        | ○                                | :TRACe:TEMPlate:DISPlay                                                                  |                             |
| TLEXTRA*                       | ×                                | :TRACe:TEMPlate:EDIT:ETYPe                                                               |                             |
| TLGONO*                        | ○                                | :TRACe:TEMPlate:GONOG                                                                    |                             |
| TLLVSFT***.**                  | ○                                | :TRACe:TEMPlate:WAVelength:SHIFt                                                         |                             |
| TLRESLT?                       | ○                                | :TRACe:TEMPlate:RESult?                                                                  |                             |
| TLSADR**                       | ○                                | :SYSTem:COMMunicate:GPIB2:TLS:ADDReSS<wsp><integer>                                      |                             |
| TLSSYNc *                      | ○                                | :SENSe:SWEep:TLSSync<wap> OFF ON 0 1                                                     |                             |
| TLTYPE*                        | ○                                | :TRACe:TEMPlate:TTPe                                                                     |                             |
| TLWLSFT***.**                  | ○                                | :TRACe:TEMPlate:WAVelength:SHIFt                                                         |                             |
| TRA?                           | ○                                | :TRACe:ATTRibute:TRA?                                                                    | Different talker format     |
| TRB?                           | ○                                | :TRACe:ATTRibute:TRB?                                                                    | Different talker format     |
| TRC?                           | ○                                | :TRACe:ATTRibute:TRB?                                                                    | Different talker format     |
| TRFMT*                         | ○                                | -                                                                                        |                             |
| UCWRN*                         | ○                                | :SYSTem:DISPlay:UNCal<wsp>OFF ON 0 1                                                     |                             |
| UHLD*                          | ○                                | :DISPlay[:WINDow]:SPLit<wsp>ON 1;<br>:DISPlay[:WINDow]:SPLit:HOLD:UPPER<wsp>OFF ON 0 1   |                             |
| ULTRA*                         | ○                                | :DISPlay[:WINDow]:SPLit<wsp>ON 1;<br>:DISPlay[:WINDow]:SPLit:POSition<wsp>TRA,UP LOW 0 1 |                             |
| ULTRB*                         | ○                                | :DISPlay[:WINDow]:SPLit<wsp>ON 1;<br>:DISPlay[:WINDow]:SPLit:POSition<wsp>TRB,UP LOW 0 1 |                             |
| ULTRC*                         | ○                                | :DISPlay[:WINDow]:SPLit<wsp>ON 1;<br>:DISPlay[:WINDow]:SPLit:POSition<wsp>TRC,UP LOW 0 1 |                             |
| WARN?                          | △                                | :SYSTem:ERRor[:NEXT]?                                                                    |                             |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                                                                                                                                                                       | Remarks         |
|-------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| WCAL****.***                  | △                                | :CALibration:WAVelength:EXTernal:SOURce<wsp>LASer   0;<br>:CALibration:WAVelength:EXTernal:WAVelength<wsp><NRf>[M]                                                                                                                                   | Parameter range |
| WCALG****.***                 | △                                | :CALibration:WAVelength:EXTernal:SOURce<wsp>GASCell   1;<br>:CALibration:WAVelength:EXTernal:WAVelength<wsp><NRf>[M]                                                                                                                                 | Parameter range |
| WCALS                         | ○                                | :CALibration:WAVelength:INTernal[:IMMediate]                                                                                                                                                                                                         |                 |
| WCALT****;#.###               | △                                | :CALibration:WAVelength:OFFSet:TABLE<wsp><integer>,<NRf> [DB]                                                                                                                                                                                        | Parameter range |
| WDMAN                         | ○                                | :CALCulate:CATegory<wsp>WDM   10                                                                                                                                                                                                                     |                 |
| WDMCHAUT*                     | ×                                | -                                                                                                                                                                                                                                                    |                 |
| WDMCHSW***;#                  | ×                                | -                                                                                                                                                                                                                                                    |                 |
| WDMDIFF**.**                  | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:MDIFF<wsp><NRf>[DB]                                                                                                                                                                                              |                 |
| WMDISP*                       | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:DTYPE<wsp><display type><br><display type>=ABSolute   0 , RELatibe   1 ,<br>MDRift   2 , GDRift   3                                                                                                              |                 |
| WMDSPMSK***                   | △                                | :CALCulate:PARAmeter[:CATegory]:WDM:DMASK<wsp><NRf>[DB]                                                                                                                                                                                              | Parameter range |
| WMDUAL*                       | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:DUAL<wsp>OFF   ON   0   1                                                                                                                                                                                        |                 |
| WDMMAX***                     | ×                                | -                                                                                                                                                                                                                                                    |                 |
| WDMMR                         | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:MMReset                                                                                                                                                                                                          |                 |
| WDMNOI*                       | ○                                | [NOISE_ALGO が<br>AutoCenter]=:CALCulate:PARAmeter[:CATegory]<br>:WDM::NALGo<wsp>ACENter   2<br>[NOISE_ALGOAutoFix]<br>=:CALCulate:PARAmeter[:CATegory]<br>:WDM::NALGo<wsp>MFIx   1;<br>:CALCulate:PARAmeter[:CATegory]<br>:WDM:FALGo<wsp>LINear   0; |                 |
| WDMNOIBW****                  | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:NBW<wsp><NRf>[M   HZ]                                                                                                                                                                                            |                 |
| WDMNOIP**.**                  | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:FALGo<wsp>LINear   0;<br>:CALCulate:PARAmeter[:CATegory]:WDM:NBW<wsp><NRf>[M]                                                                                                                                    |                 |
| WDMOS*                        | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:RELation<wsp>OFFSet   SPACing   0   1                                                                                                                                                                            |                 |
| WDMREF*                       | ×                                | -                                                                                                                                                                                                                                                    |                 |
| WDMREFDAT*                    | ×                                | -                                                                                                                                                                                                                                                    |                 |
| WDMRH                         | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:RCH<wsp>>0                                                                                                                                                                                                       |                 |
| WDMRN***                      | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:RCH<wsp>>                                                                                                                                                                                                        |                 |

| AQ6317 Series Control Command | Action in AQ6317-compatible mode | AQ6319 Control Command Corresponding to AQ6317 Control Command                                                                    | Remarks                     |
|-------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
|                               |                                  | <integer>                                                                                                                         |                             |
| WDMSLOPE*                     | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:OSLope<wsp><br>p><br>OFF   ON   0   1                                                         |                             |
| WDMTCOPY                      | ○                                | :HCOPY[:IMMediate]:FUNCTion:CALCulate:LIST                                                                                        |                             |
| WDMTH**.*                     | ○                                | :CALCulate:PARAmeter[:CATegory]:WDM:TH<wsp><br><NRf>[DB]                                                                          |                             |
| WDMUNT*                       | ○                                | :CALCulate:MARKer:UNIT<wsp>WAVelength  <br>FREQuency   0   1                                                                      |                             |
| WLSFT**.*                     | ○                                | :SENSe:CORRection:WAVelength:SHIFt<wsp><NRf><br>><br>[M]                                                                          |                             |
| WMKR****.*                    | △                                | :CALCulate:MARKer:X<wsp>0,<NRf>[M]                                                                                                | Parameter range             |
| WNFAN                         | ○                                | :CALCulate:CATegory<wsp>NF   11                                                                                                   |                             |
| WNFCVF*                       | ×                                | :CALCulate:PARAmeter[:CATegory]:NF:FALGo<wsp><br>><algorhythm><br><algorhythm>=AFIX   0 , MFIx   1 , ACENter   2 ,<br>MCENter   3 |                             |
| WNFFA**.*                     | ○                                | :CALCulate:PARAmeter[:CATegory]:NF:FARea<wsp><br>><br><NRf>[M   HZ]                                                               |                             |
| WNFNP**.*                     | ○                                | :CALCulate:PARAmeter[:CATegory]:NF:MARea<wsp><br>><br><NRf>[M   HZ]                                                               |                             |
| WNFOFI***.**                  | ○                                | :CALCulate:PARAmeter[:CATegory]:NF:IOFFset<wsp><br>><br><NRf>[DB]                                                                 |                             |
| WNFOFO***.**                  | ○                                | :CALCulate:PARAmeter[:CATegory]:NF:OOFFset<wsp><br>><NRf>[DB]                                                                     |                             |
| WNFSSE*                       | ×                                | -                                                                                                                                 |                             |
| WR* '@@@@'                    | △                                | :MMEMory:STORe:TRACe<wsp><trace name>,BIN   CSV,<br><"file name">,FLOPpy<br><trace name>= TRA   TRB   TRC                         | "BIN   CSV" can be selected |
| WR3D* '@@@@'                  | ×                                | -                                                                                                                                 |                             |
| WRDT '@@@@'                   | ○                                | :MMEMory:STORe:DATA<wsp><"file name">,FLOPpy                                                                                      |                             |
| WRGR '@@@@'                   | △                                | :MMEMory:STORe:GRAPhics<wsp>B&W  <br>COLor,BMP   TIFF,<"file name">,FLOPpy                                                        | File type can be selected.  |
| WRMEM** '@@@@'                | ○                                | :MMEMory:STORe:MEMory<wsp><integer>,BIN  <br>CSV,<"file name">,FLOPpy                                                             | File type can be selected.  |
| WRPRG** '@@@@'                | ○                                | :MMEMory:STORe:PROGram<wsp><integer>,<br><"file name">,FLOPpy                                                                     |                             |
| WRSET '@@@@'                  | ○                                | :MMEMory:STORe:SETTing<wsp><br><"file name">,FLOPpy                                                                               |                             |
| WRTA                          | ○                                | :TRACe:ATTRibute:TRA<wsp>WRITe   0                                                                                                |                             |
| WRTB                          | ○                                | :TRACe:ATTRibute:TRB<wsp>WRITe   0                                                                                                |                             |
| WRTC                          | ○                                | :TRACe:ATTRibute[:TRC]<wsp>WRITe   0                                                                                              |                             |
| WRTLt '@@@@'                  | ×                                | -                                                                                                                                 |                             |
| XUNT*                         | ○                                | :UNIT:X<wsp>WAVelength   FREQuency   0   1                                                                                        |                             |
| ZSCL**                        | ×                                | -                                                                                                                                 |                             |
| ZSWPT**                       | △                                | :SENSe:SWEEp:TIME:0NM<wsp><integer>[SEC]                                                                                          | Parameter range             |

## Legend of the Symbols Used:

○: Compatible, △: Partly compatible, ×: Not compatible

**2.7.6 HIGH1,HIGH2,HIGH3 of measurement sensitivity**

When measurement sensitivity of the unit is set the HIGH1 or HIGH2 or HIGH3 , unless set the <CHOP> or <SWITCH>key in the <CHOP MODE>key ,the chopper cannot operate.

However , optical spectrum analyzer of AQ6317 series is designed to always be operated the chopper which removed monochromator stray light.

Even if the unit is set the AQ6317-compatible mode , it has a remote command for chopper setting that need to be changed.

○Commands to use when in the AQ6317-compatible mode(a command error occurs if used in the AQ6319 mode.)

Control command:

CHOP\* (\*: 0 =Chopper OFF , 1 = CHOP mode , 2 = SWITCH mode )

Query command

CHOP?

A return value: Same as the above

**NOTE**

- If you are changed measurement sensitivity by the SHI1,2,3 commands after chopper mode's state is CHOP or SWITCH, chopper is OFF. Please send the CHOP command after changed measurement sensitivity.  
(Ex) When you want to change HIGH1/CHOP.  
"SHI1,CHOP1"

## 2.8 GP-IB2 Port

### 2.8.1 GP-IB2 Port's Interface Functions

Table 2-14 shows the GP-IB2 port's interface functions.

Table 2-15: GP-IB2 Port's Interface Functions

| Code | Interface Function                     |
|------|----------------------------------------|
| SH1  | All capabilities of send function      |
| AH1  | All capabilities of receive function   |
| T4   | Basic talker function                  |
| L2   | Basic listener function                |
| SR0  | Service request function not provided  |
| RL0  | Local lockout function not provided    |
| PP0  | Parallel polling function not provided |
| DC0  | Device clear function not provided     |
| DT0  | Device trigger function not provided   |
| C1   | System controller IFC transmission     |
| C2   | Controller in charge                   |
| C3   | REN transmission                       |
| C28  | Interface message transmission         |
| E1   | Open collector driver                  |

### 2.8.2 Connecting to a GPIB System

Connect a Tunable Laser Source and/or a GPIB device controlled using the program function to this GP-IB2 port.

Turn OFF all the power switches of the AQ6319 and the device to be connected to it. Using the GPIB cable, connect the GP-IB2 connector on the rear of this unit to the device. Make sure to tighten the connector-fixing screws firmly.

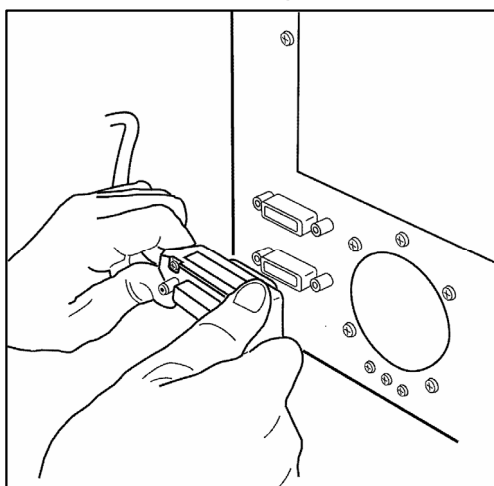


Figure 2-10: Connecting the GPIB Connector



### Caution

- Connecting/disconnecting the GPIB cable with the power switch turned ON may cause the AQ6319 to malfunction or fail.

### 2.8.3 Setting the GP-IB2 Port Address

The address of the GP-IB2 port can be selected in the range of 0 to 30, using the <GP-IB2 PORT ADDRESS> key. The address of the device to be controlled by this unit can also be selected from the range of 0 to 30.

### 2.8.4 Setting the System Controller

To have this unit act as a controller using the GP-IB2 port, press the [SYSTEM] switch and then the <GP-IB SETTING> key to set <SYSTEM CONTROLLER OFF/ON> to ON.

If <SYSTEM CONTROLLER OFF/ON> is OFF, this unit cannot remote control the external device.

(☛ Section 6.16, SYSTEM in the Instruction Manual)

### 2.8.5 Control of the External Device

Using the program functions, the AQ6319 can remote control the external device connected to the GP-IB2 port. For more information on external device control, see Chapter 1, Program Functions.



#### Caution

- A controller such as a PC that is connected to the GP-IB2 port cannot remotely control the AQ6319.
- Even if an external device to be controlled by the AQ6319 using program functions or a wavelength tunable light source is connected to the GP-IB1 port, it cannot remote control the AQ6319.
- The two GP-IB ports are independent of each other. Thus, a controller connected to the GP-IB1 port cannot directly send a message to an external device connected to the GP-IB2 port.
- Also, with a controller connected to the GP-IB1 port, connecting the GP-IB1 port and the GP-IB2 port results in improper operations.
- Do not connect a cable between these ports, or press [SYSTEM] and then <GP-IB SETTING> to set <SYSTEM CONTROLLER> to OFF. The default is ON.  
(☛ Section 6.16, SYSTEM in the Instruction Manual)

## 2.9 Remote Control via the RS-232C Port

The AQ6319 has one RS-232C port, allowing remote control using an external PC.

### 2.9.1 Connection to the RS-232C System

Turn OFF the power of this unit.

Connect a dedicated cable to the RS-232C interface at the rear of this unit. Be sure to securely tighten the connector-fixing screws.

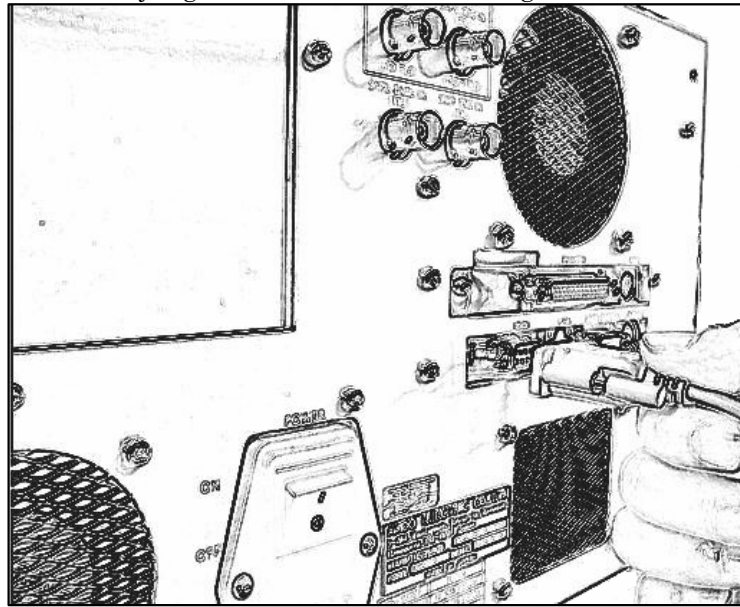


Fig. 2-11 Connecting the RS-232C Connector



### Caution

- Attaching or detaching the RS-232C cable with the power of this unit turned ON may cause this unit to malfunction or to be damaged.

### 2.9.2 Setting up the Interface

The parameters of the RS-232C port can be set by pressing the [SYSTEM] switch and then the <RS-232C SETTING> key.

The settable parameters are as follows:

- Baud rate setting (BPS)  
115200, 57600, 38400, 19200, 9600 (default), 4800, 2400, 1200
- Flow control  
Xon/Xoff  
Hardware  
NONE (default)
- Parity bit  
NONE (default)  
ODD  
EVEN

Of the RS-232C port's setting parameters, the user cannot modify the following parameters:

- Data bit length  
Fixed to 8 bits
- Stop bit length  
Fixed to 1 bit

### 2.9.3 Remote Control Using the Remote Commands

The AQ6319 can be remote controlled using the RS-232C port.

In this case, use a cross cable to connect this unit's RS-232C port and an external PC.

For remote commands, use the same commands as those for control via the GPIB interface.

#### (1) Switching among the GPIB and RS-232C and ETHERNET Interfaces

Select either GPIB or RS-232C or ETHERNET as an interface to use for remote control.

To set the interface, press the [SYSTEM] switch and then the <REMOTE INTERFACE> key.

#### (2) About Remote Commands

As with GPIB-based remote control, you can select the command format from the AQ6319 mode or from the AQ6317-compatible mode.

To select the command format, press the [SYSTEM] switch, the <RS-232C SETTING> key, and then the <COMMAND FORMAT> key.

#### (3) Interrupt by SRQ

An SRQ-caused interrupt does not occur during RS-232C-based remote control.

#### (4) Status Registers

The status registers operate in the same manner as in remote control via the GPIB interface.

Using the "<STB?>" command or "<SPOOL?>" command dedicated allows you to read the status registers, as in the case with serial polling via the GPIB interface.

#### (5) Delimiter

A delimiter for RS-232C-based remote control is fixed to CR + LF.

#### (6) Transmission of Talker Data

When this unit is received talker data from external PC, sending the data to external PC's buffer. Receiving the external PC's buffer data and the query data is stored.



## 2.10 Remote Control via the LAN port

The AQ6319 has one LAN port, allowing remote control using an external PC.

### 2.10.1 Connection to the LAN system

Turn OFF the power of this unit.

Connect a dedicated cable to the LAN interface at the rear of this unit.

Mate the terminals of the cable and the unit with each other, as shown in Fig 2-14, and ensure that the cable is securely connected.

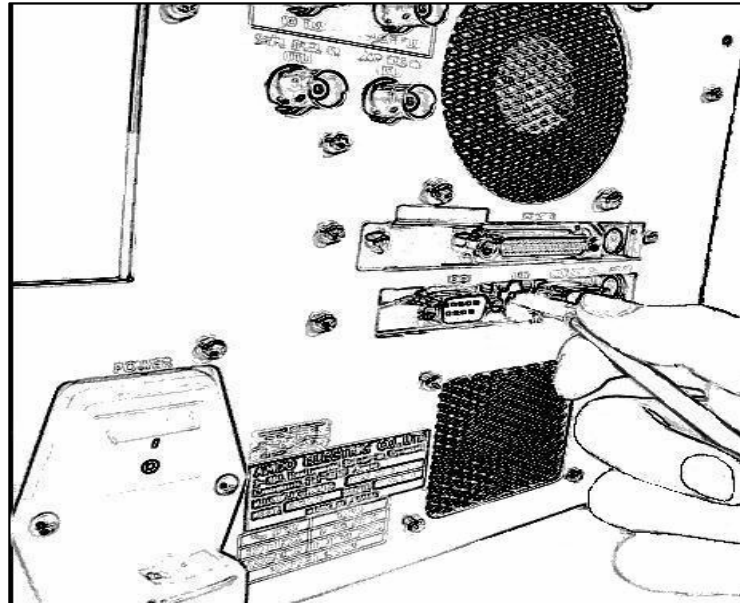


Fig 2-14 Connecting the LAN connector

### 2.10.2 Setting up the Interface

The parameter of the LAN port can be set by pressing the [SYSTEM] switch and then the <NETWORK SETTING> key.

The settable parameters are as follows:

- COMPUTER NAME

The factory default setting is as follows.

"AQ6319@@@@@@@@@" (where "@@@@@@@@@" is the serial number)

(The machine number is an 9-digit alphanumeric number on the back of the unit.)

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•TCP/IP SETTING

It is necessary to set up the IP address of the unit correctly so that the remote control with [LAN] port.

If a DHCP server is provided on the network to which this unit is connected, the IP address given to the unit is automatically set. Thus, set the item IP ADDRESS SETTING in the [SYSTEM] →<NETWORK SETTING>→<TCP/IP SETTING> to "☒AUTO (DHCP)".

Please ask the network administrator of your company about the details of a network.

•REMOTE PORT NO.

Sets the port number for remote control. (Default: 10001)

### 2.10.3 Remote Control Using the Remote Commands

The AQ6319 can be remote controlled using the LAN port.

In this case, use a LAN cable to connect this unit's LAN port and an external PC.

For remote commands, use the same commands as those for control via the GPIB interface.

(1) Switching among the GPIB and RS-232C and ETHERNET Interfaces

Select either GPIB or RS-232C or ETHERNET as an interface to use for remote control.

To set the interface, press the [SYSTEM] switch and then the <REMOTE INTERFACE> key.

When you set the "ETHERNET", connection state of LAN remote is reset.

Otherwise, it is kept connection state unless controller has performed the disconnect transaction.

(2) About Remote Commands

As with GPIB-based remote control, you can select the command format from the AQ6319 mode or from the AQ6317-compatible mode.

To select the command format, press the [SYSTEM] switch, the <GP-IB Setting> key, and then the <COMMAND FORMAT> key.

(3) Interrupt by SRQ

An SRQ-caused interrupt does not occur during LAN-based remote control.

(4) Status Registers

The status registers operate in the same manner as in remote control via the GPIB interface.

Using the "SPOOL?" command dedicated for remote control using the LAN port allows you to read the status registers, as in the case with serial polling via the GPIB interface.

(5) Delimiter

A delimiter for LAN-based remote control is fixed to CR + LF.

(6) Transmission of Talker Data

When this unit is recieved talker data form external PC, sending the data to external PC's buffer. Recieving the external PC's buffer data and the query data is stored.

(7) Connecting to a LAN system

This unit can be connected only one contoroller (external PC or the like)既に  
If this unit occurs to the connection request from other controller during  
connection state, new connection is not performed and it is kept on doing  
the connection state currently.

## 2.11 Programming Precautions

- (1) A program code corresponding to a function not available in manual operations will be ignored. In this case, a warning message appears on the screen.
- (2) If a program code that performs a sweep, printing, or plotting function and other program codes are sent in succession, the successive program codes are immediately executed without waiting for the sweep, printing, or plotting function to end. Be aware that for printing or plotting, particularly, most of the commands become invalid. To determine the end of a sweep, printing, or plotting function, use the service request function.
- (3) Sending an invalid talker command to the AQ6319 and then receiving data with this unit specified as a talker causes the GPIB bus to freeze because this unit has no data to send. In this case, a GPIB timeout occurs.

## 2.12 Example of a Program

### 2.12.1 Remote Control Sample Program using GP-IB Interface

The following shows an example of remote controlling the AQ6319 using the GPIB port. This program example uses Visual Basic 6.0 (called VB here) as the programming language.

Further, a GPIB board of National Instruments (called NI here) is used as a GPIB controller and the NI-supplied driver is used as a library.

[VB Sample]

The program sets the measurement conditions (center wavelength, span, sensitivity, and the sampling number) and then performs a sweep.

After completing this sweep, the program executes THRESH-based spectrum width analysis and then outputs the results to the screen.

```

Dim osa As Integer '
Dim strData As String * 1024 '
Dim dblMeanWl As Double '
Dim dblSpecWd As Double '

Call ibclr(osa) ' Device clear
Call ibwrt(osa, ":sens:wav:cent 1550nm") ' Set center measurement wavelength
Call ibwrt(osa, ":sens:wav:span 10nm") ' Set measurement span
Call ibwrt(osa, ":sens:sense mid") ' Set measuring sensitivity: MID
Call ibwrt(osa, ":sens:sweep:points:auto on") ' SMPL:AUTO

Call ibwrt(osa, ":init:smode 1") ' Sweep mode:SINGLE
Call ibwrt(osa, ":init: *opc?") ' Make a sweep
Call ibrd(osa, strData) ' Wait for *OPC? Data (wait for a sweep to end)

Call ibwrt(osa, ":calc:category swth") ' THRESH analysis mode
Call ibwrt(osa, ":calc") ' Perform analysis
Call ibwrt(osa, ":calc:data?") ' Request analytical result output
Call ibrd(osa, strData) ' Load analytical results

dblMeanWl = val(Left(strData, 16)) ' Capture center wavelength
dblSpecWd = val(Mid(strData, 18, 16)) ' Capture spectrum width

MsgBox ("MEAN WL: " & dblMeanWl * 1000000000# & "nm" & vbCrLf & _
 "SPEC WD: " & dblSpecWd * 1000000000# & "nm") ' Output the result to the screen

ibloc (osa) ' GOTO LOCAL

```

### 2.12.2 Remote Control Sample Program using LAN Interface

The following shows an example of remote controlling the AQ6319 using the LAN port. [VB Sample]

The program sets the measurement conditions (center wavelength, span, sensitivity, and the sampling number) and then performs a sweep.

After completing this sweep, the program executes THRESH-based spectrum width analysis and then outputs the results to the screen

(The program performs same operation as 2.12.1 Remote Control Sample Program Using GP-IB Interface)

```

Dim intData As Integer
Dim dblMeanWL As Double
Dim dblSpecWd As Double
Dim strData As String

' === Connect ===
With Winsock1
 .RemoteHost = "AQ6319-12345678" ' Set the unit's computer name or IP address
 .RemotePort = 10001 ' Set port number using remote control
End With

Winsock1.Connect ' Connect

' === Wait to connect complete ===
While (Winsock1.State <> sckConnected) ' Watch for connect complete
 DoEvents
Wend

' === Set the measurement parameter ===
SendLan ":sens:wav:cent 1550nm" ' Set measurement center wavelength
SendLan ":sens:wav:span 10nm" ' Set measurement span
SendLan ":sens:sens mid" ' Set measuring sensitivity: MID
SendLan ":sens:sweep:points:auto on" ' Sampling Point = AUTO

' === Sweep execute ===
SendLan ":init:smode 1" ' Sweep mode : Single
SendLan ":init" ' Make a sweeo

' === Wait for *OPC? data (wait for a sweep to end) ===
Do ' Wait for *OPC? Data (wait for a sweep to end)
 SendLan "*OPC?"
 RecieveLan strData
 intData = Val(strData)
 Loop While ((intData And 1) <> 1)

' === Analysis ===
SendLan ":calc:category swth" ' THRESH analysis mode
SendLan ":calc" ' Peform analysis
SendLan ":calc:data?" ' Request analytical result output
RecieveLan strData ' Load analytical results

' === Capture analytical results ===
dblMeanWL = Val(Left(strData, 16)) ' Capture center wavelength
dblSpecWd = Val(Mid(strData, 18, 16)) ' Capture spectrum width

' === Output the result to the screen ===
MsgBox ("MEAN WL: " & dblMeanWL * 10000000000# & " nm" & vbCrLf &
"SPEC WD: " & dblSpecWd * 10000000000# & " nm") ' Output the result to the screen

' === Disconnect ===
Winsock1.Close ' Disconnect transaction

'Wait to disconnect complete
While (Winsock1.State <> sckClosed)

```

```

 DoEvents
 Wend

End

'=====
' Sub routine
' Send Remote Command
'=====
Sub SendLan(strData As String)

 Winsock1.SendData strData & vbCrLf ' Send the remote command after add delimiter(CR+LF)
 DoEvents

End Sub

'=====
' Sub routine
' Recieve query data
'=====
Sub RecieveLan(strData As String)
Dim strData2 As String

 strData = ""

 Do ' Watch for delimiter and caputer the query data
 Winsock1.GetData strData2, vbString
 strData = strData + strData2
 DoEvents

 Loop While (Right(strData, 1) <> vbCrLf)

End Sub

```

